

Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

Bastien Padeloup

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Ph.D. director: Dominique Pastor

Ph.D. advisor: Vincent Gripon

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UNIVERSITE

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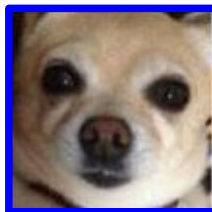
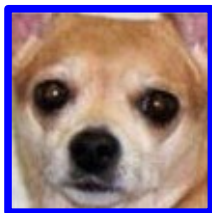
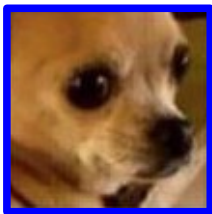
Introduction — The problem of signals classification



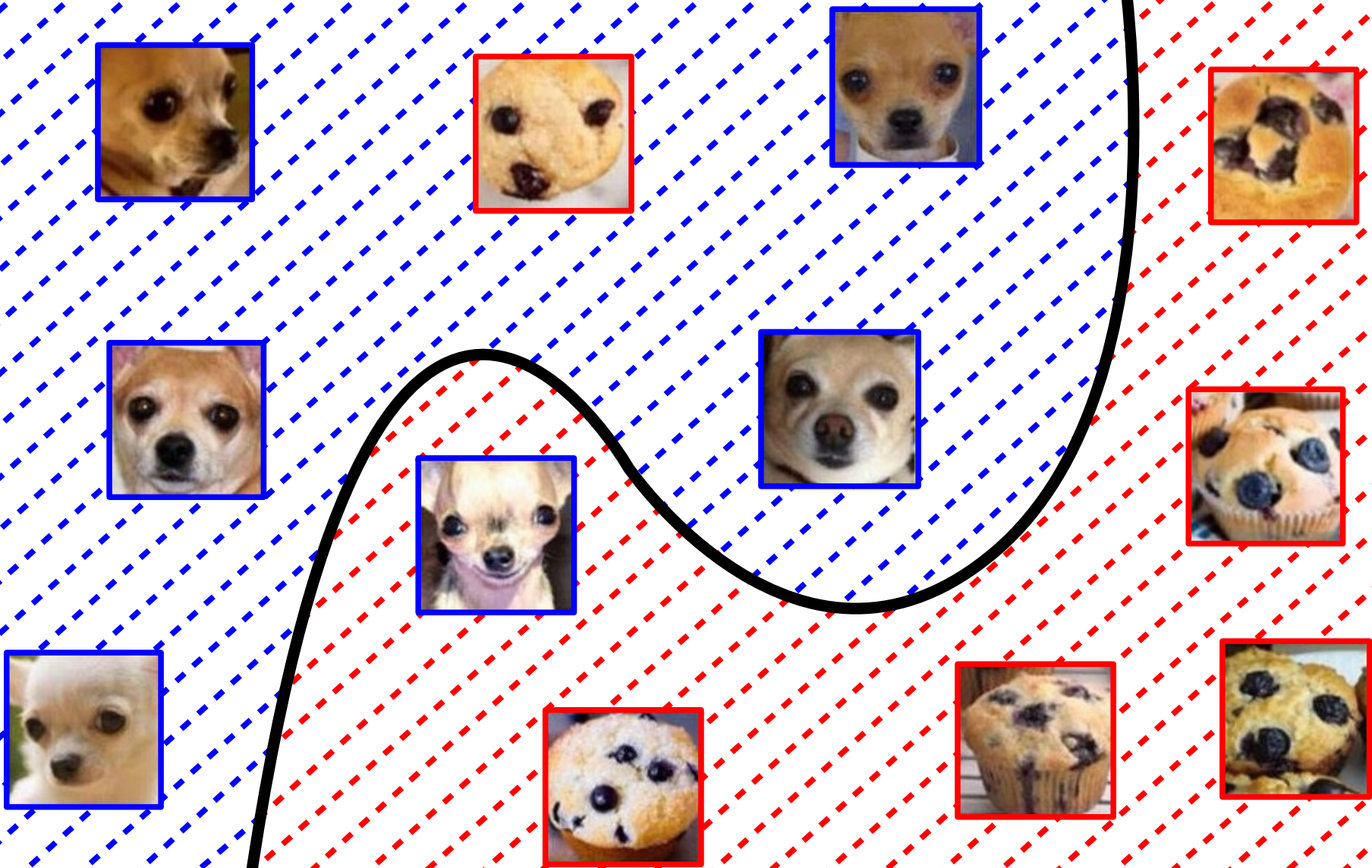
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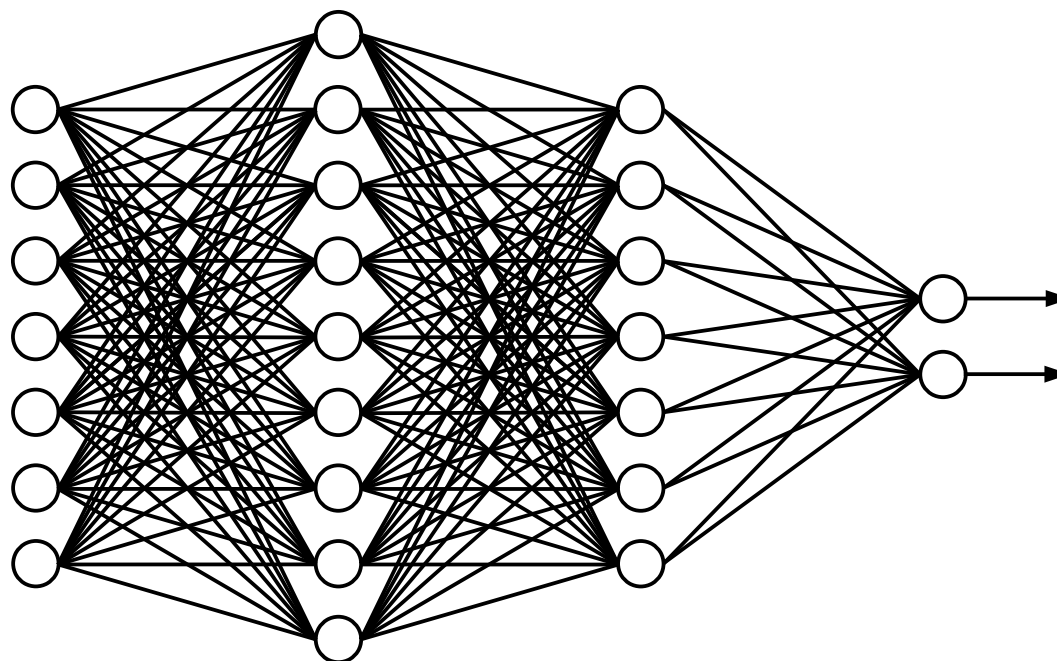
Introduction — Supervised classification (*eg.*: SVM)



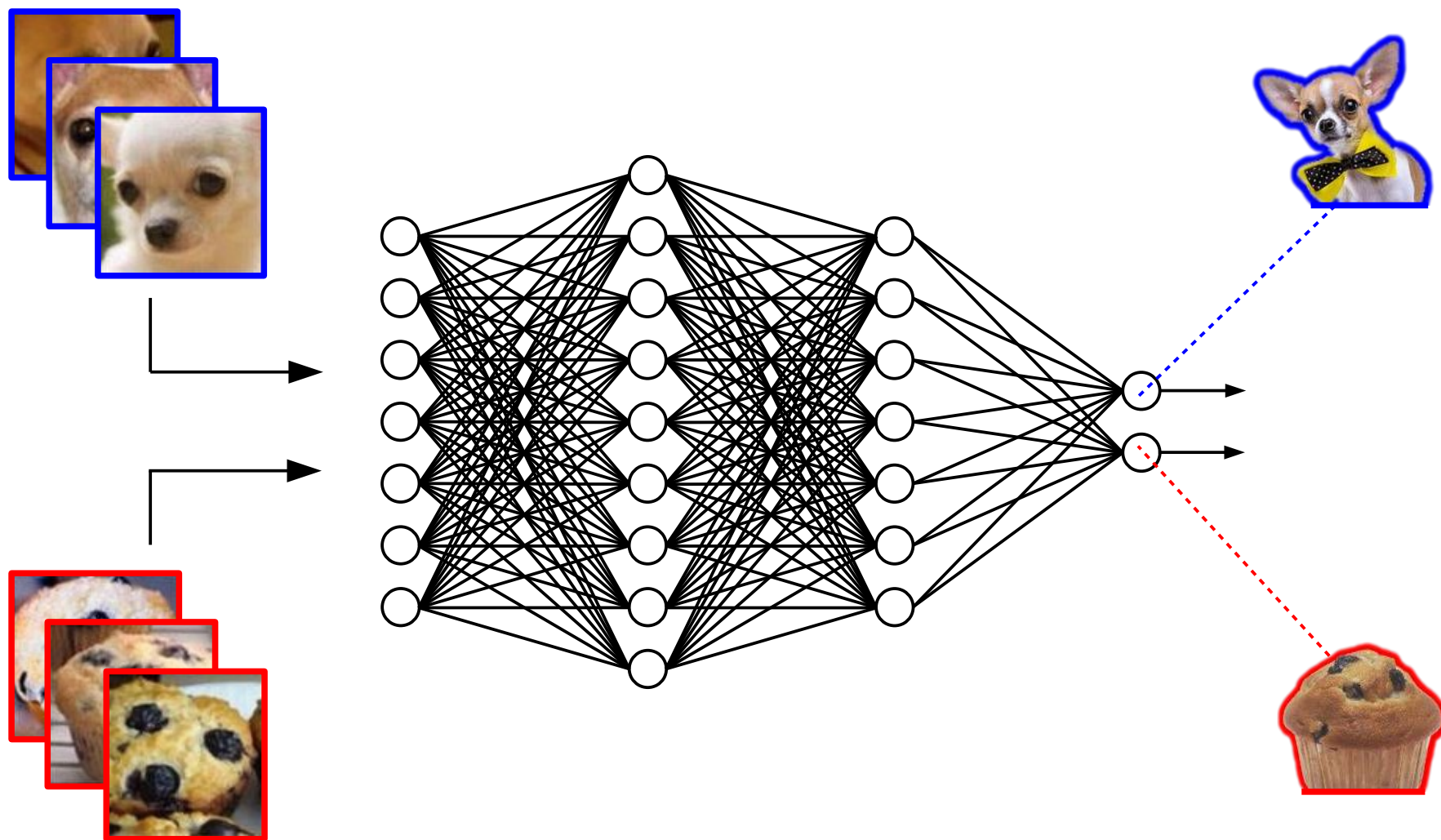
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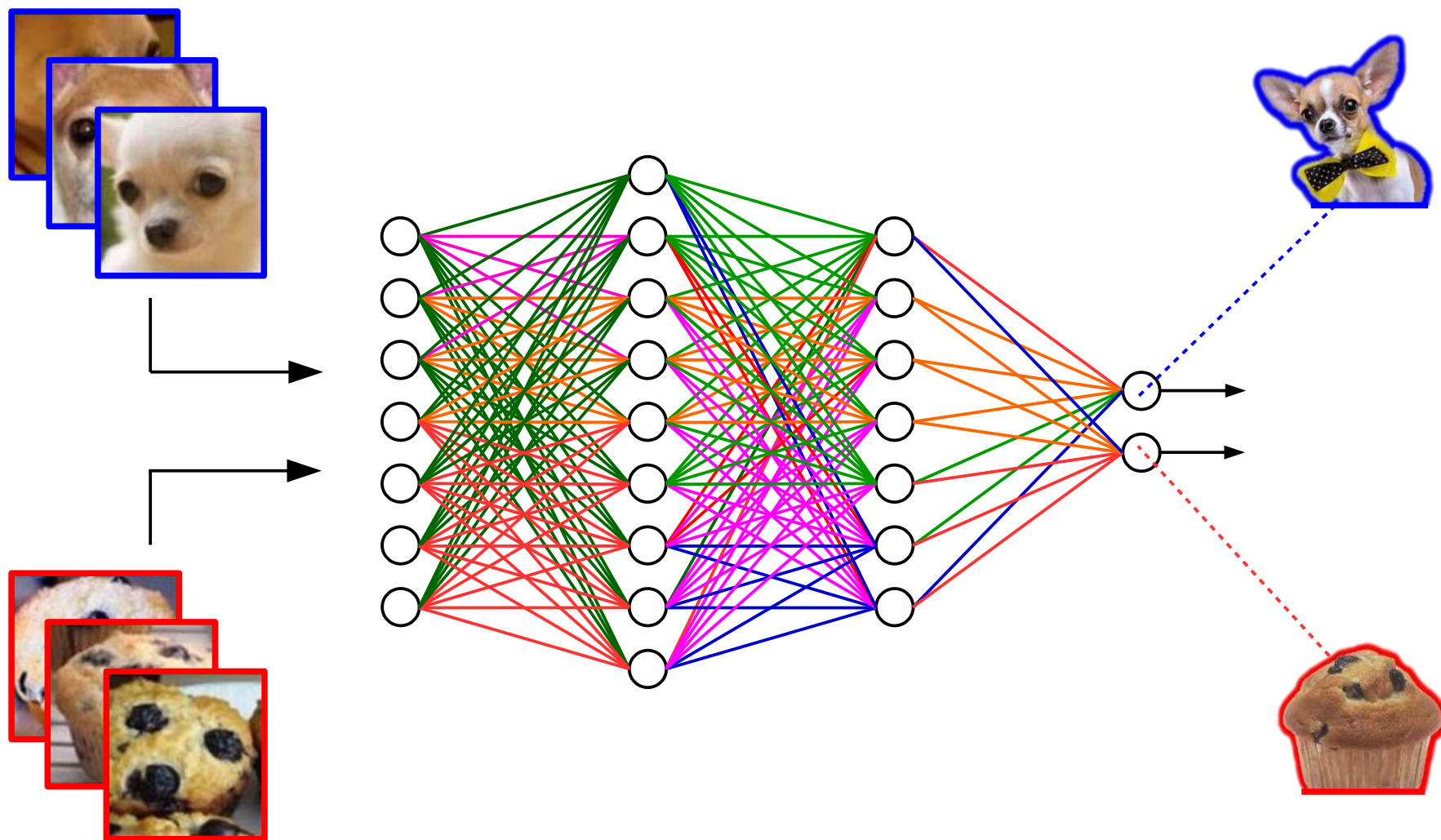
Introduction — Supervised classification (*eg.*: NN)



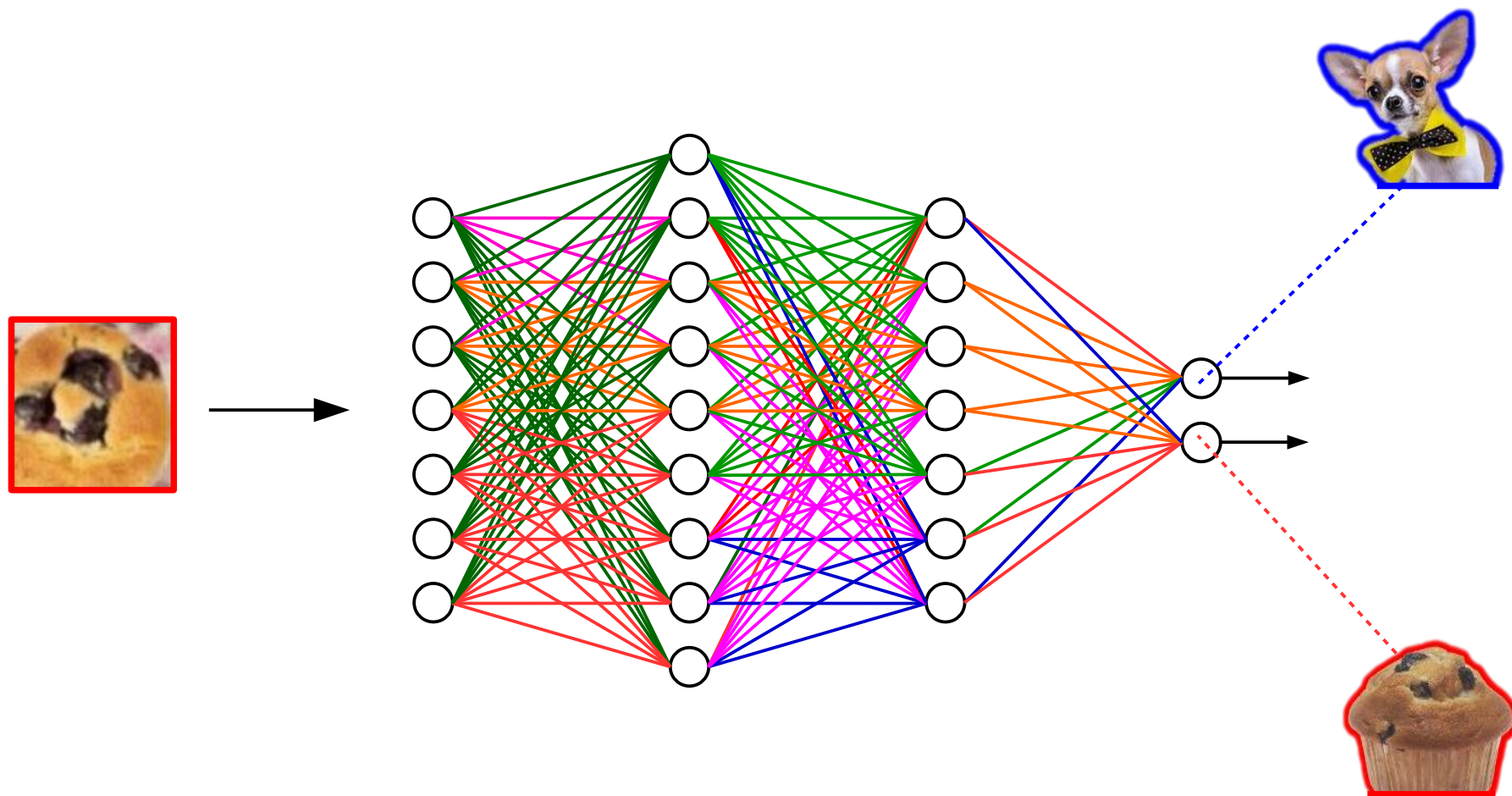
Introduction — Supervised classification (eg.: NN)



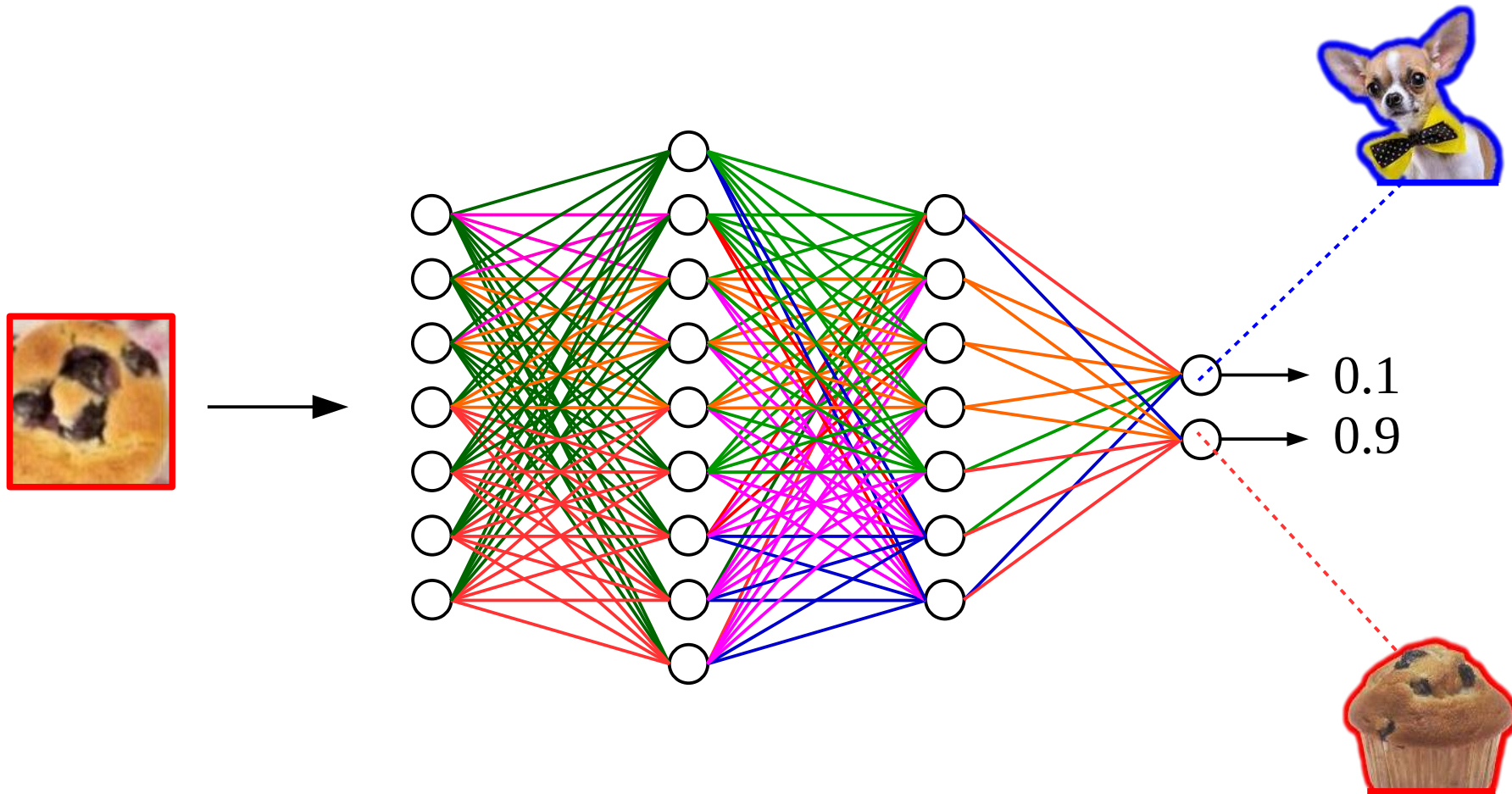
Introduction — Supervised classification (eg.: NN)



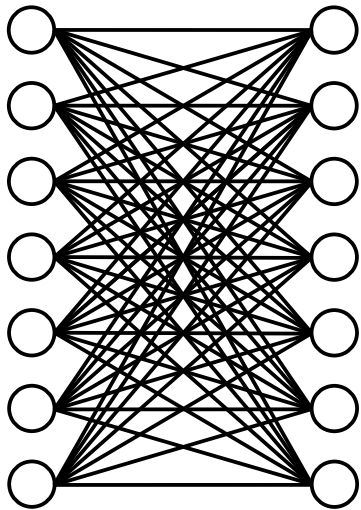
Introduction — Supervised classification (eg.: NN)



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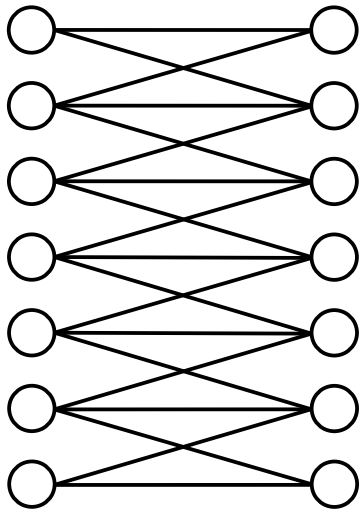


Introduction — From dense to convolutional layers



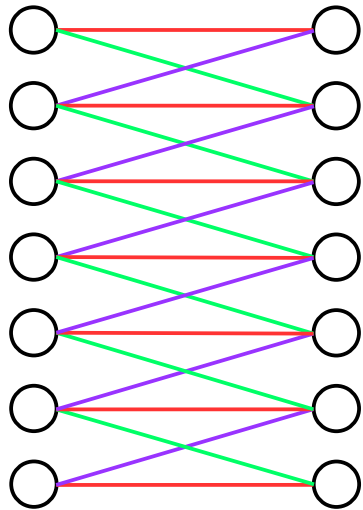
$$\begin{bmatrix} w_{11} & w_{12} & w_{13} & w_{14} & w_{15} & w_{16} & w_{17} \\ w_{21} & w_{22} & w_{23} & w_{24} & w_{25} & w_{26} & w_{27} \\ w_{31} & w_{32} & w_{33} & w_{34} & w_{35} & w_{36} & w_{37} \\ w_{41} & w_{42} & w_{43} & w_{44} & w_{45} & w_{46} & w_{47} \\ w_{51} & w_{52} & w_{53} & w_{54} & w_{55} & w_{56} & w_{57} \\ w_{61} & w_{62} & w_{63} & w_{64} & w_{65} & w_{66} & w_{67} \\ w_{71} & w_{72} & w_{73} & w_{74} & w_{75} & w_{76} & w_{77} \end{bmatrix}$$

Introduction — From dense to convolutional layers



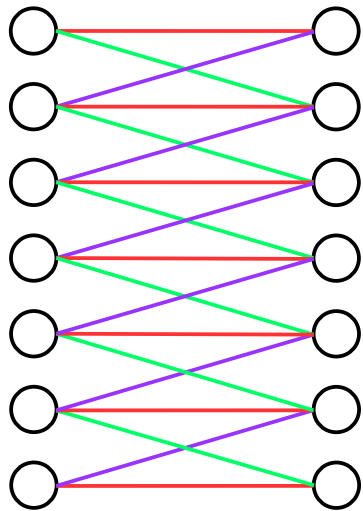
$$\begin{bmatrix} w_{11} & w_{12} & 0 & 0 & 0 & 0 & 0 \\ w_{21} & w_{22} & w_{23} & 0 & 0 & 0 & 0 \\ 0 & w_{32} & w_{33} & w_{34} & 0 & 0 & 0 \\ 0 & 0 & w_{43} & w_{44} & w_{45} & 0 & 0 \\ 0 & 0 & 0 & w_{54} & w_{55} & w_{56} & 0 \\ 0 & 0 & 0 & 0 & w_{65} & w_{66} & w_{67} \\ 0 & 0 & 0 & 0 & 0 & w_{76} & w_{77} \end{bmatrix}$$

Introduction — From dense to convolutional layers



$$\begin{bmatrix} w_{11} & w_{12} & 0 & 0 & 0 & 0 & 0 \\ w_{21} & w_{22} & w_{23} & 0 & 0 & 0 & 0 \\ 0 & w_{32} & w_{33} & w_{34} & 0 & 0 & 0 \\ 0 & 0 & w_{43} & w_{44} & w_{45} & 0 & 0 \\ 0 & 0 & 0 & w_{54} & w_{55} & w_{56} & 0 \\ 0 & 0 & 0 & 0 & w_{65} & w_{66} & w_{67} \\ 0 & 0 & 0 & 0 & 0 & w_{76} & w_{77} \end{bmatrix}$$

Introduction — From dense to convolutional layers



w_{11}	w_{12}	0	0	0	0	0
w_{21}	w_{22}	w_{23}	0	0	0	0
0	w_{32}	w_{33}	w_{34}	0	0	0
0	0	w_{43}	w_{44}	w_{45}	0	0
0	0	0	w_{54}	w_{55}	w_{56}	0
0	0	0	0	w_{65}	w_{66}	w_{67}
0	0	0	0	0	w_{76}	w_{77}

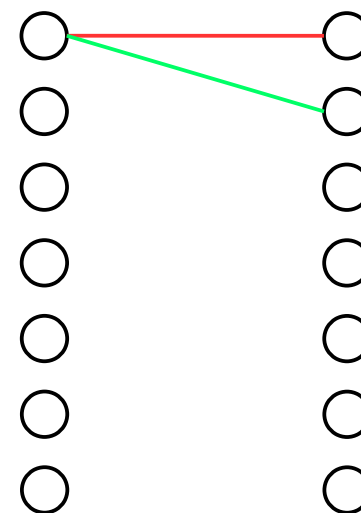
Locality: *objects* lie on adjacent vertices

Weight-sharing: *objects* are similar wherever they are

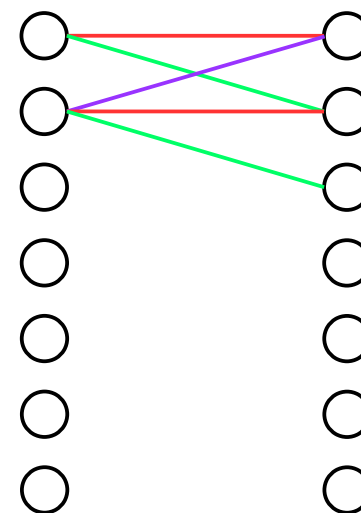
Introduction — The idea behind convolutional layers



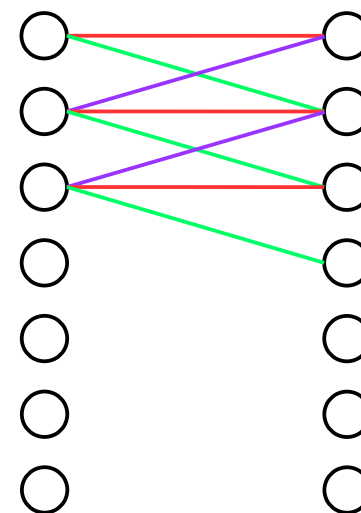
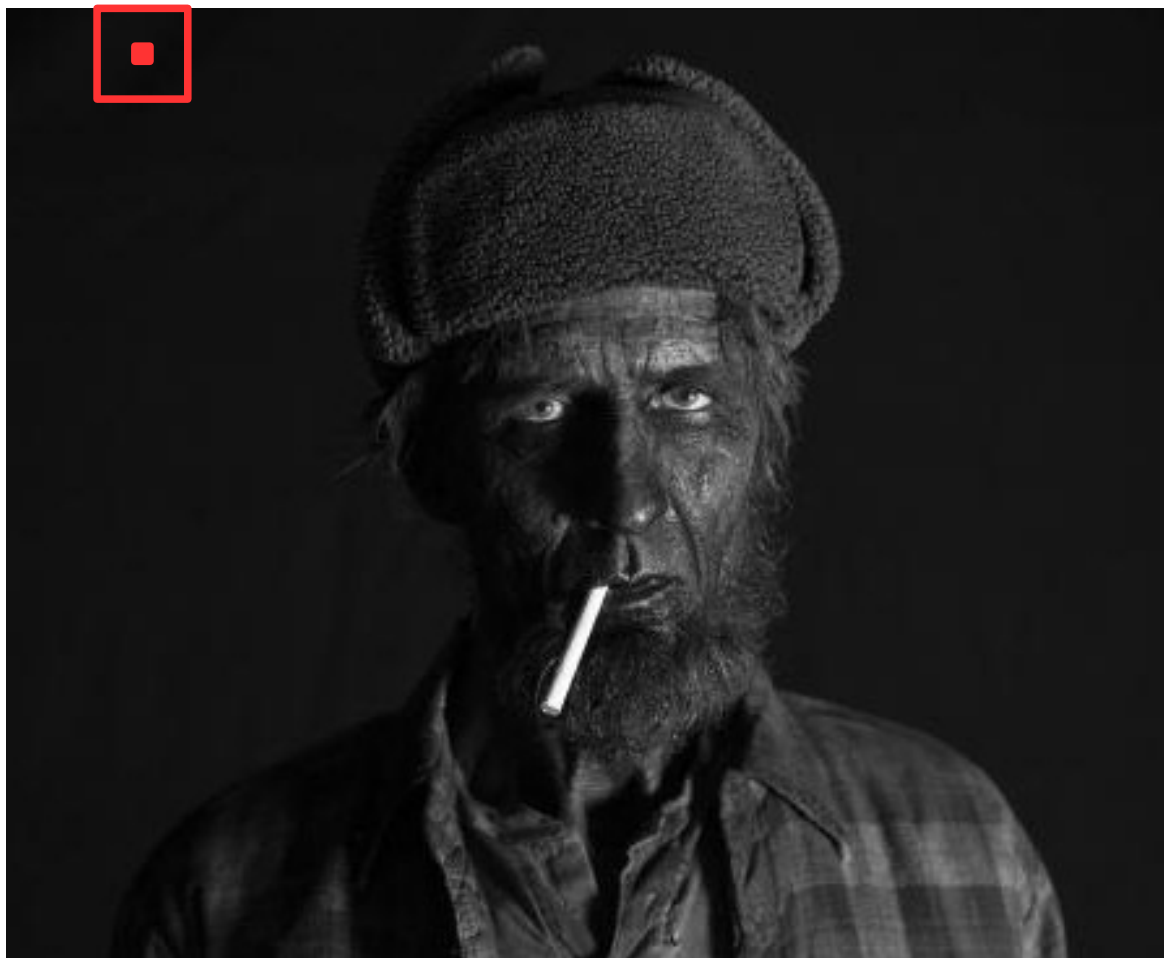
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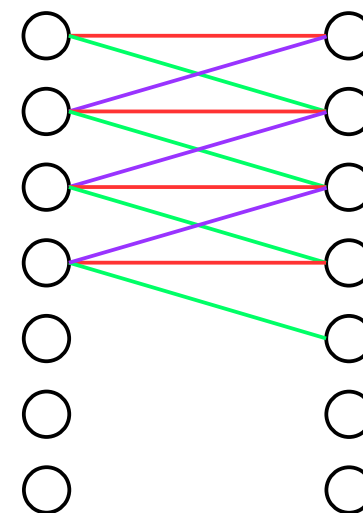
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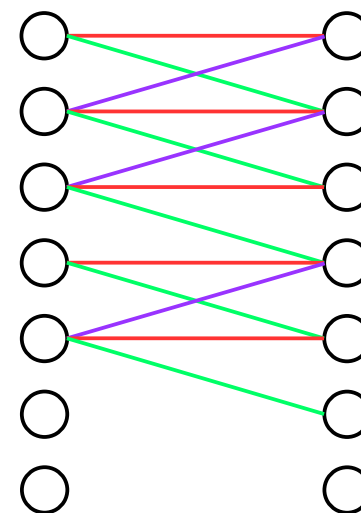
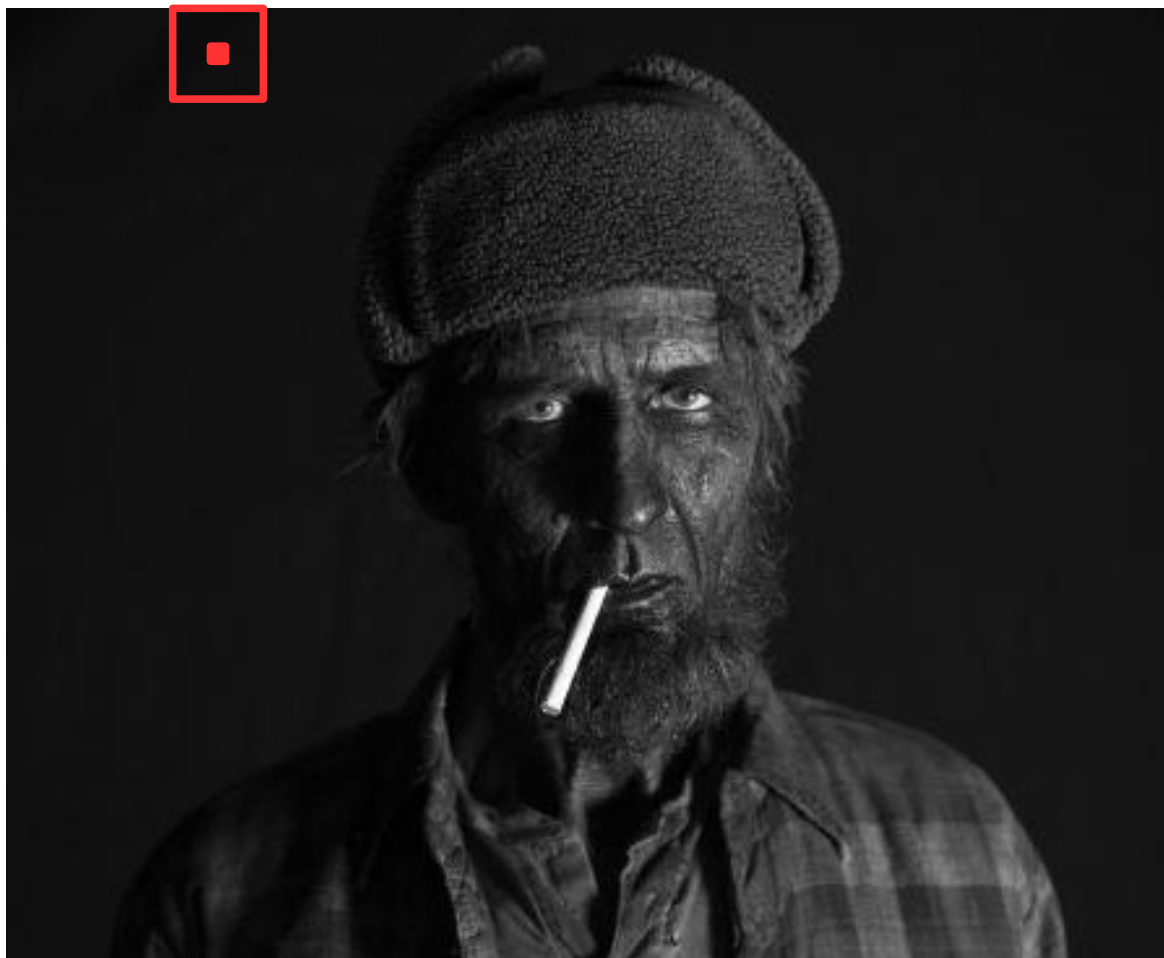
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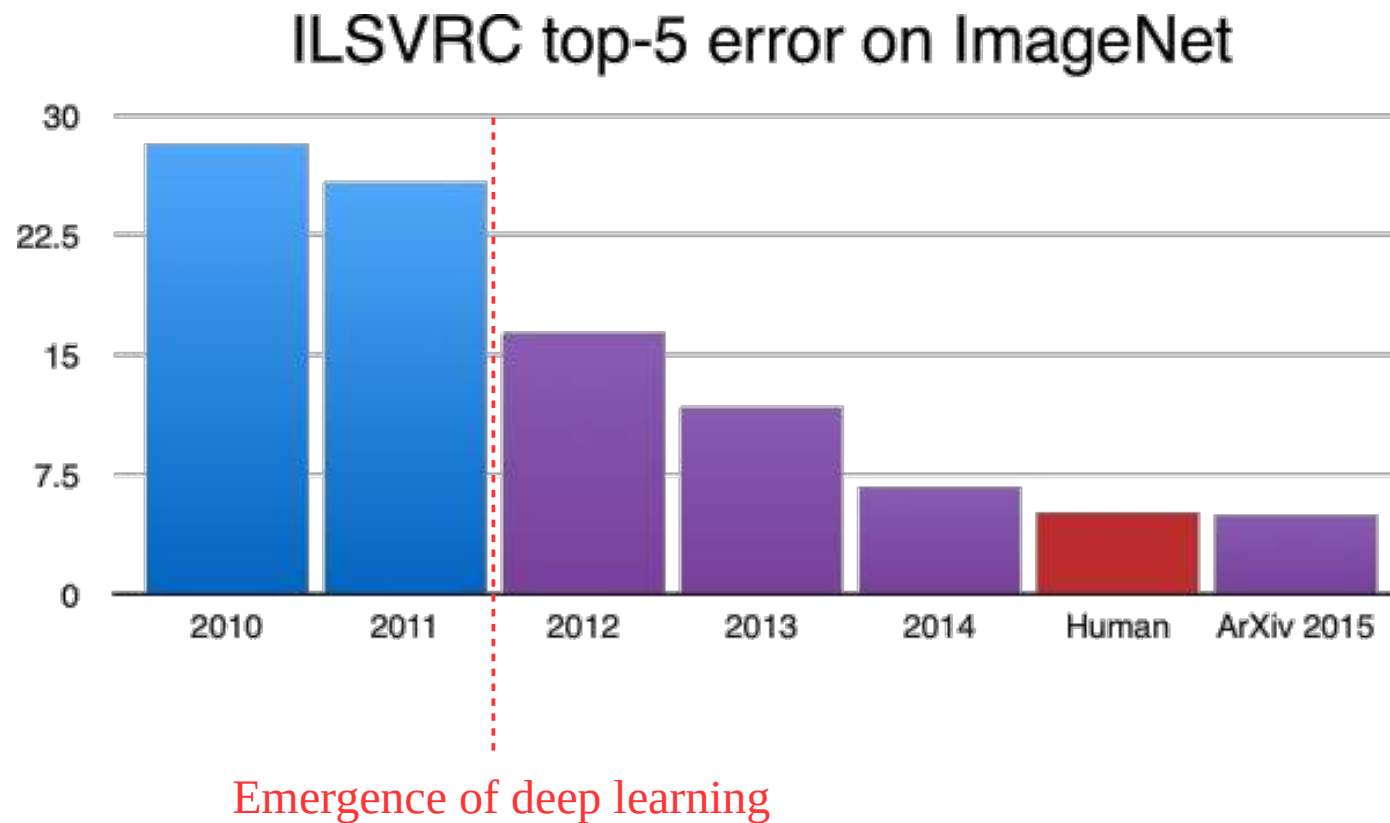
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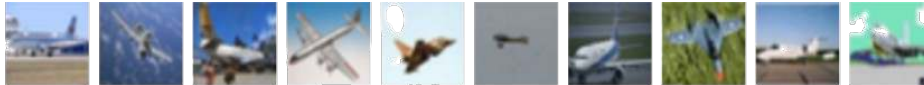
Introduction — Outstanding performance of deep neural networks



<https://devblogs.nvidia.com/parallelforall/mocha-jl-deep-learning-julia/>

Introduction — Outstanding performance of CNNs

Airplane



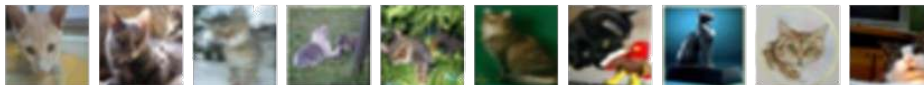
Automobile



Bird



Cat



Deer



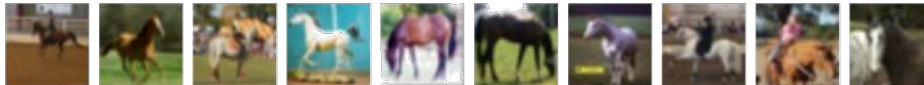
Dog



Frog



Horse



Ship



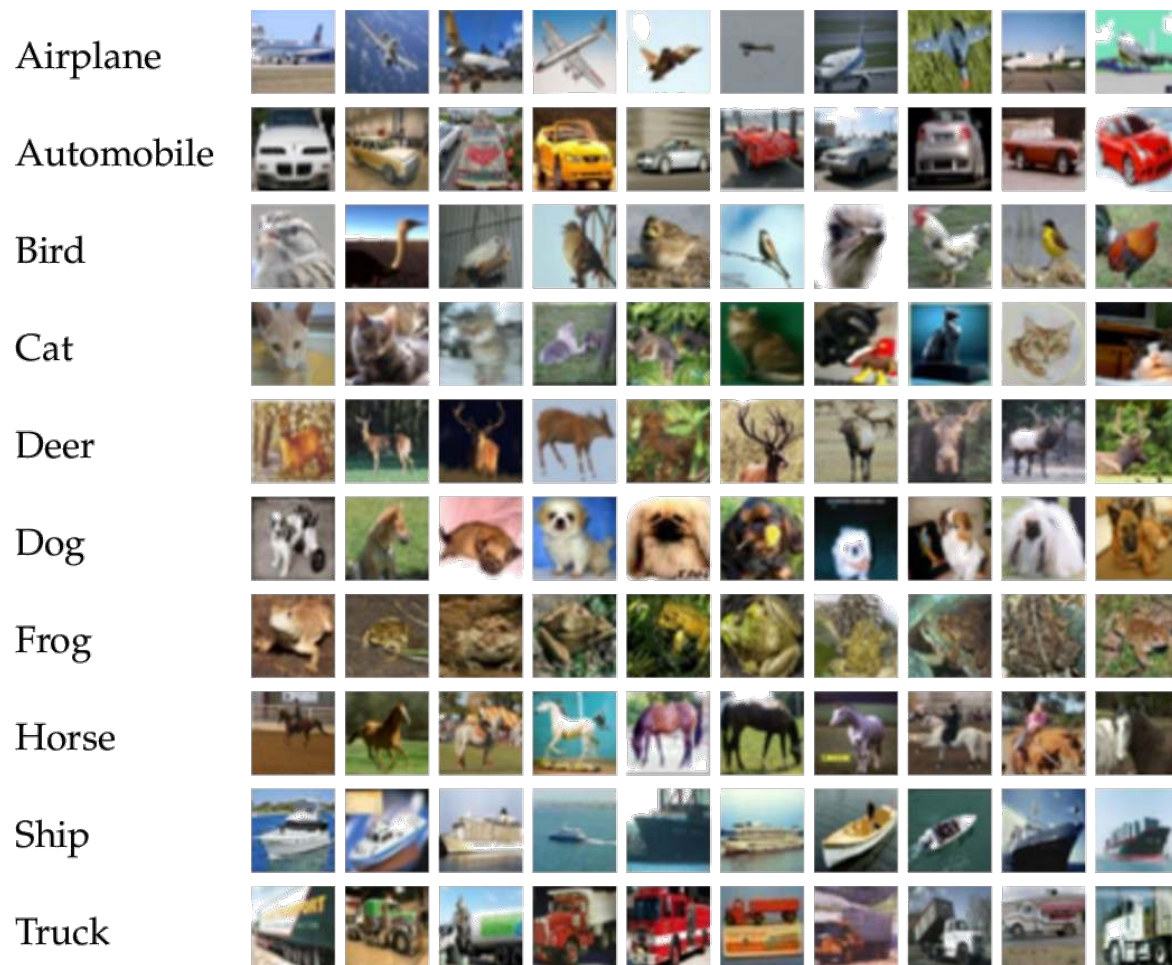
Truck



Dense
72.7 %

Cheng *et al.* — Deep Haar scattering networks (2016)

Introduction — Outstanding performance of CNNs

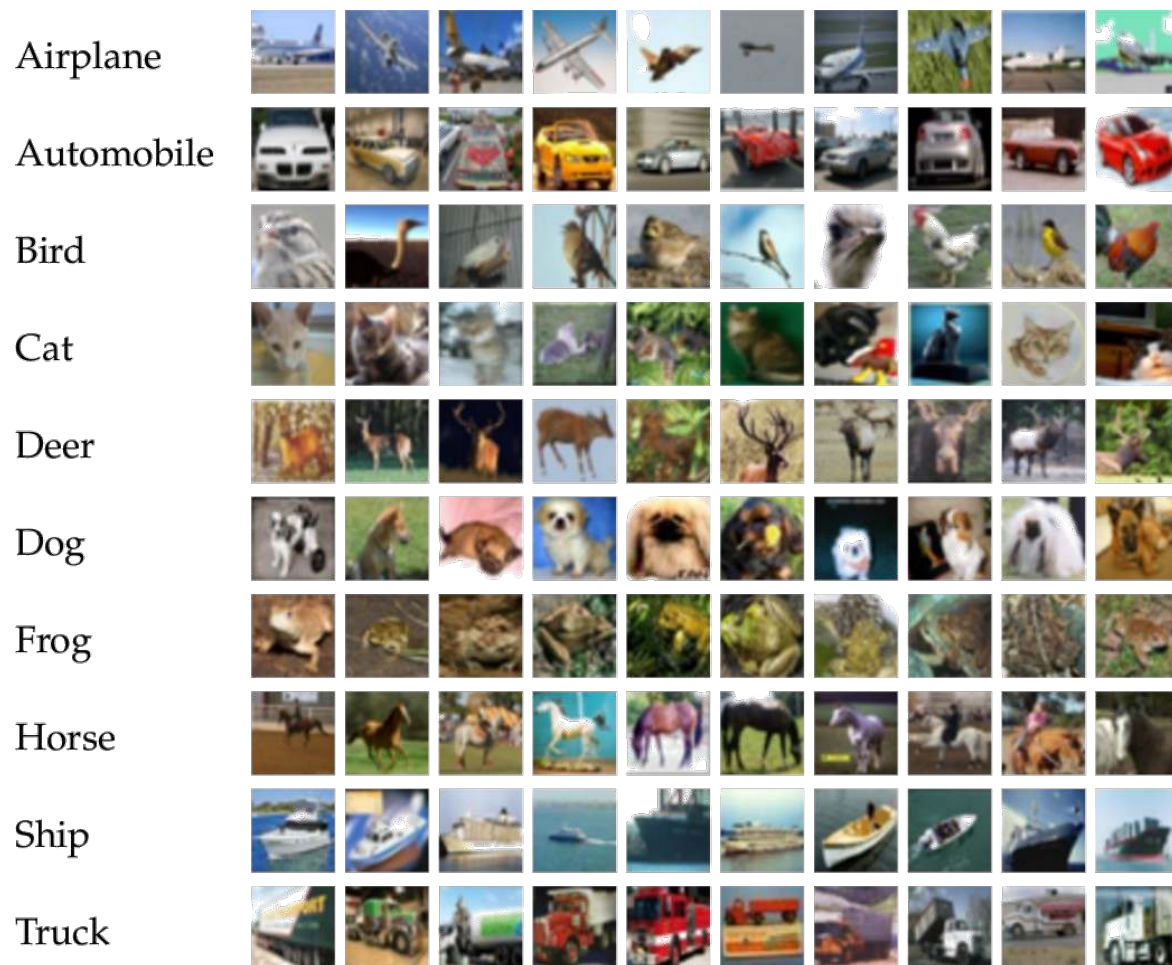


Dense
72.7 %

Convolutional
94.81 %

Huang *et al.* — Densely connected convolutional networks (2016)

Introduction — Outstanding performance of CNNs



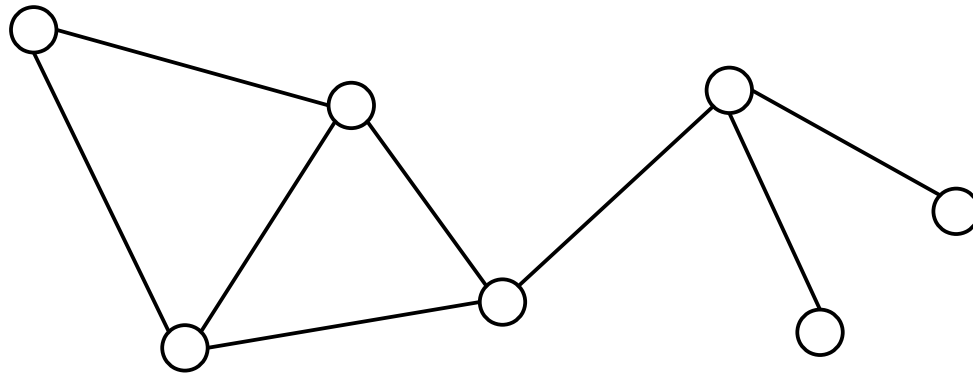
Dense
72.7 %

Convolutional
94.81 %

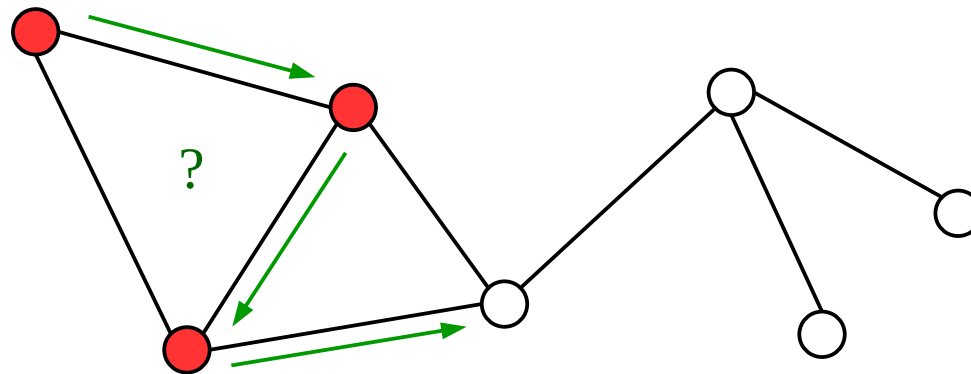
Data augmentation
97.28 %

Dutt *et al.* — Coupled ensembles of neural networks (2017)

Introduction — What about irregular data?



Introduction — What about irregular data?



Outline

Existing methods & required notions

The method we propose

- Global idea of the method —
- Graph inference from data —
- Identification of translations —

Results

Outline

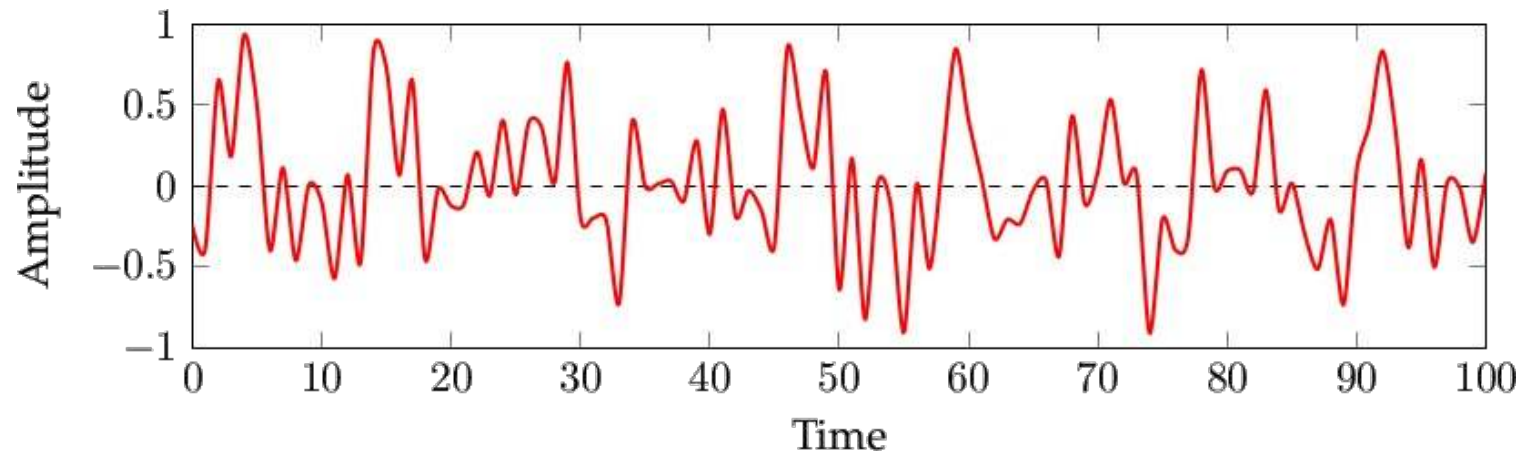
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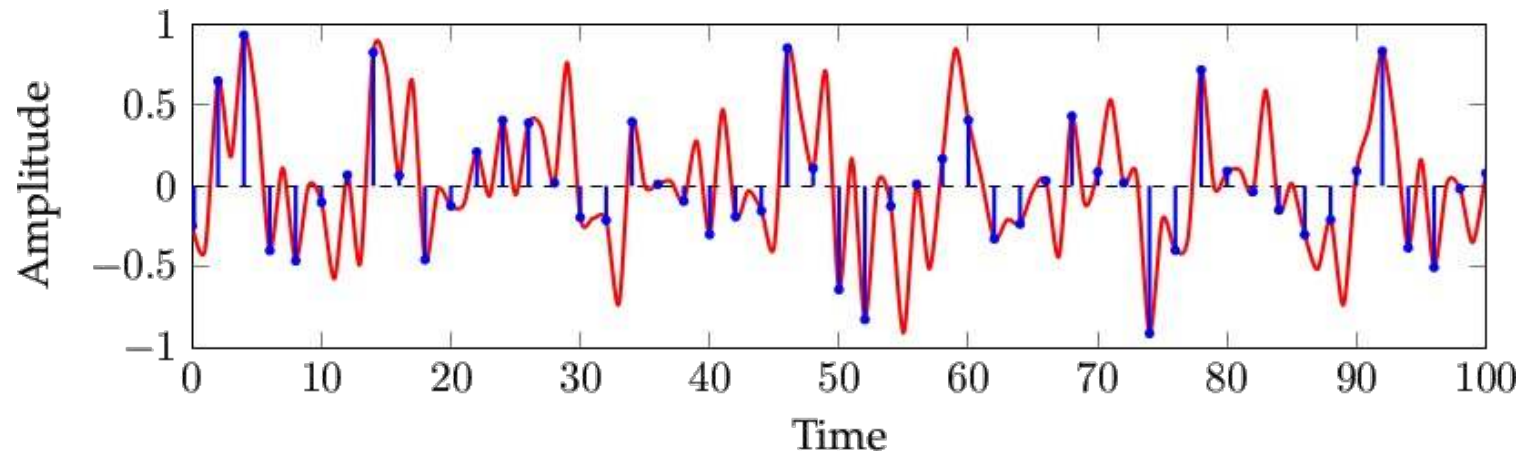
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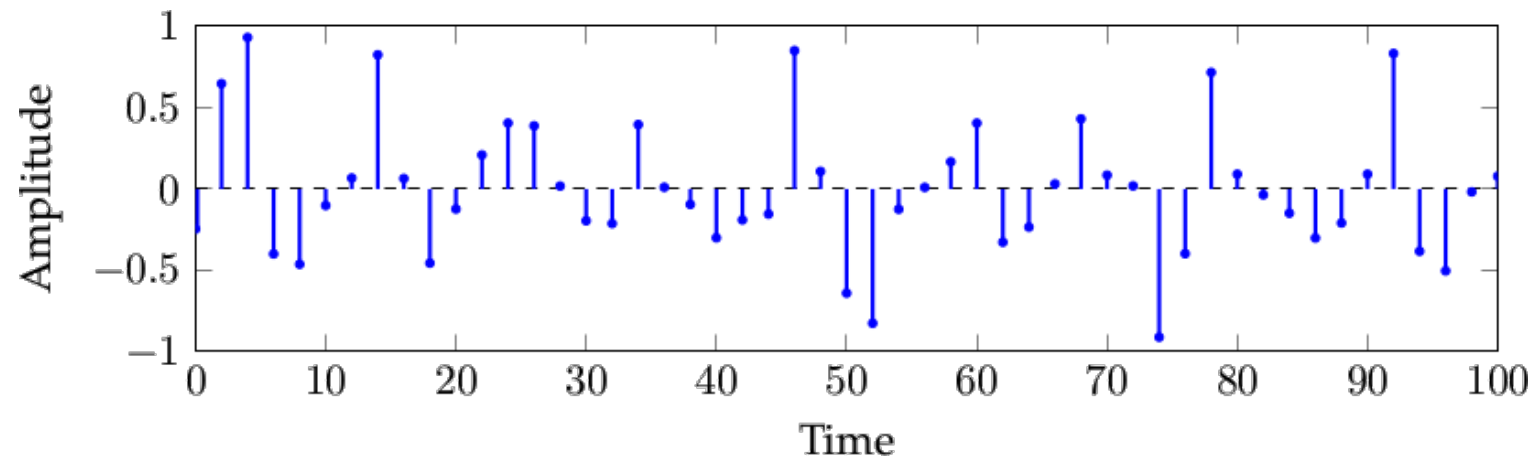
Methods & notions — The origins of GSP



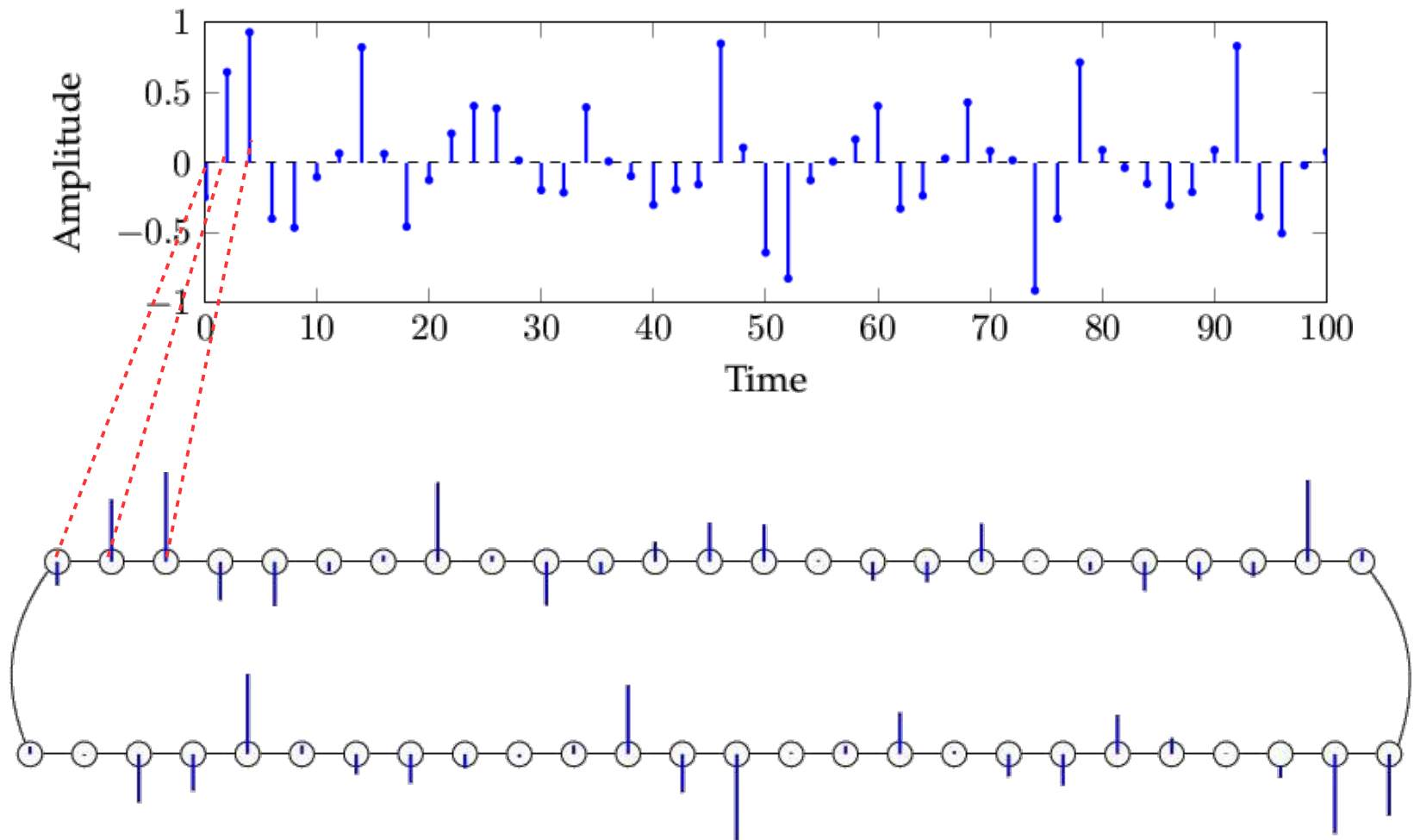
Methods & notions — The origins of GSP



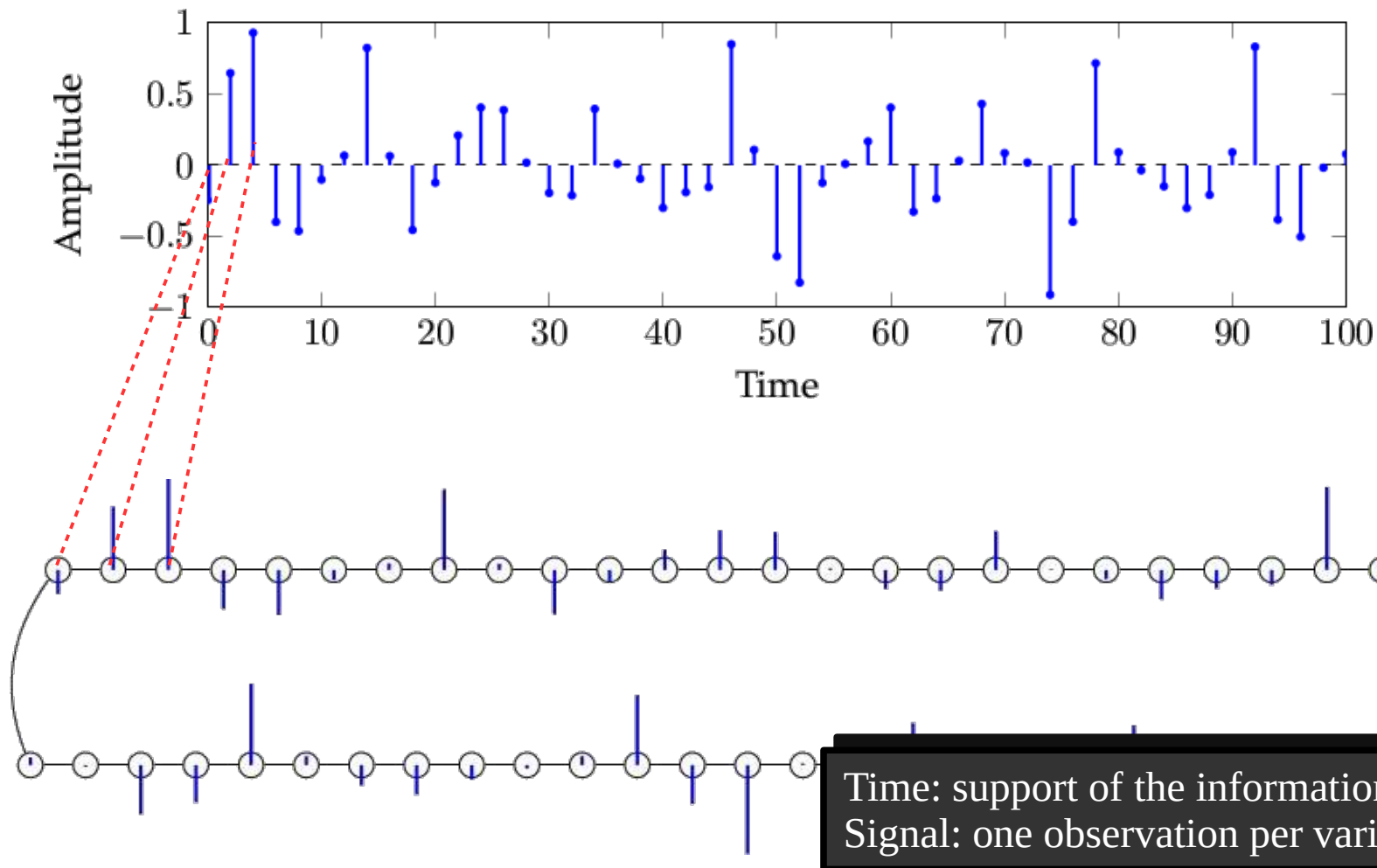
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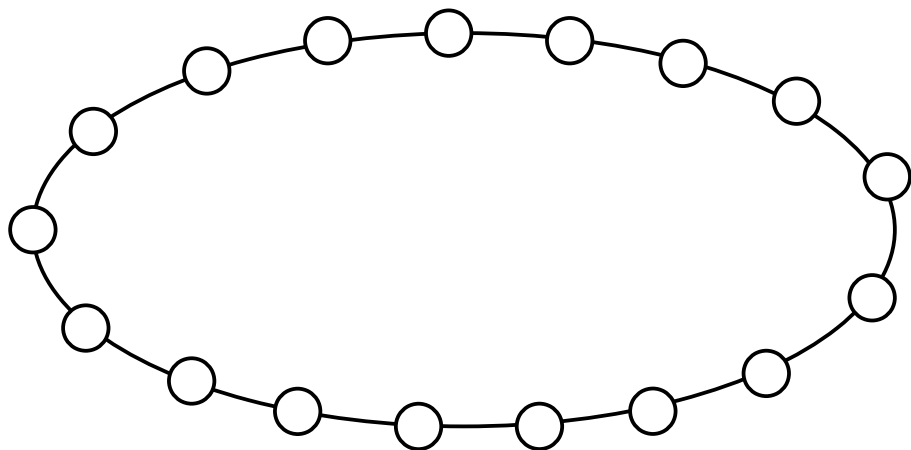
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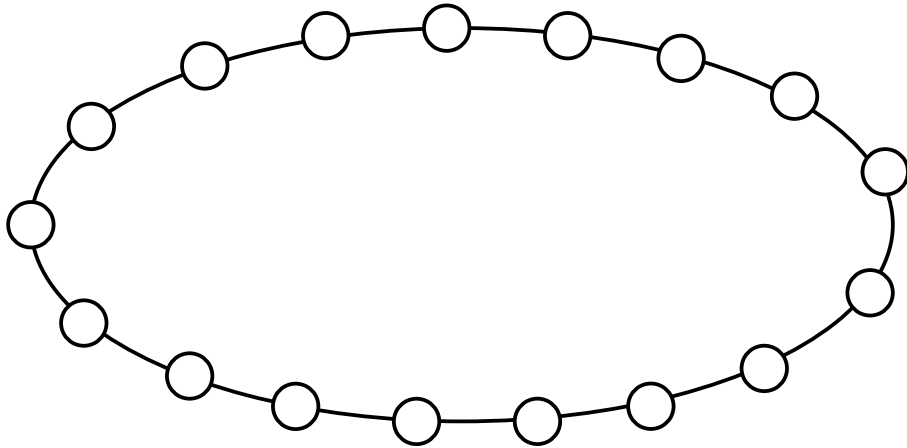
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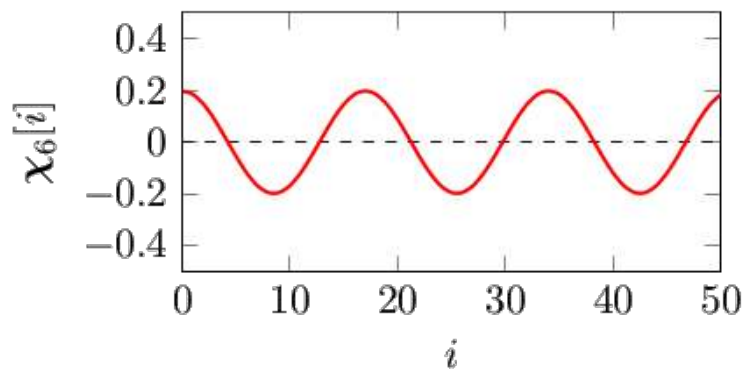
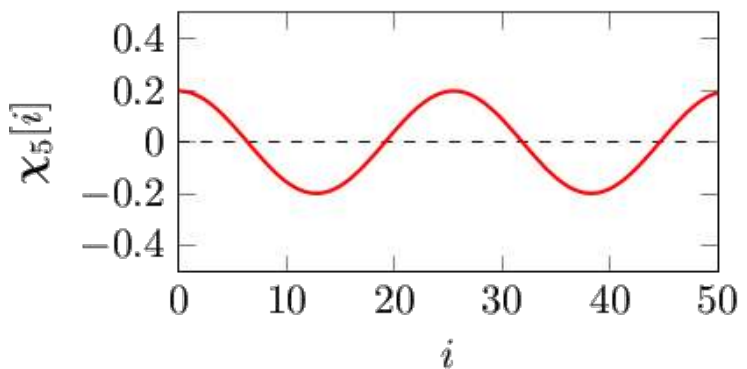
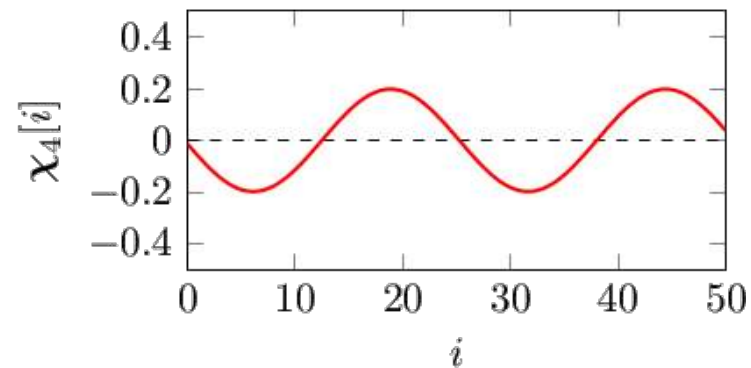
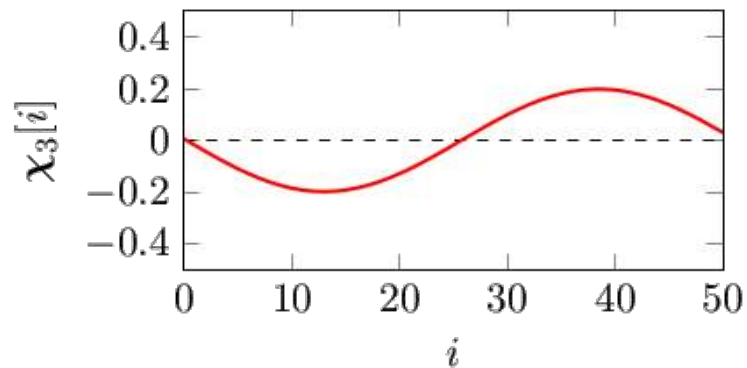
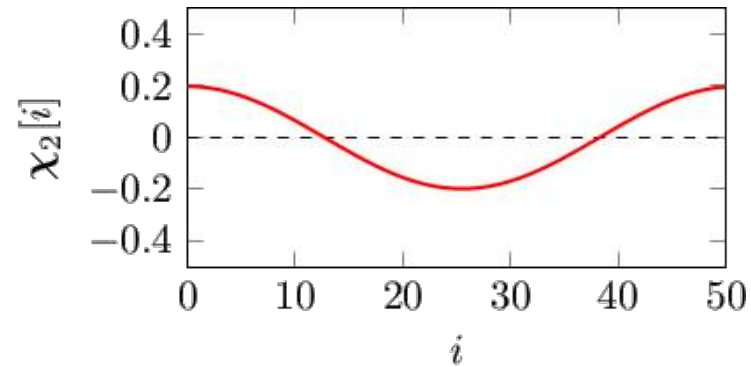
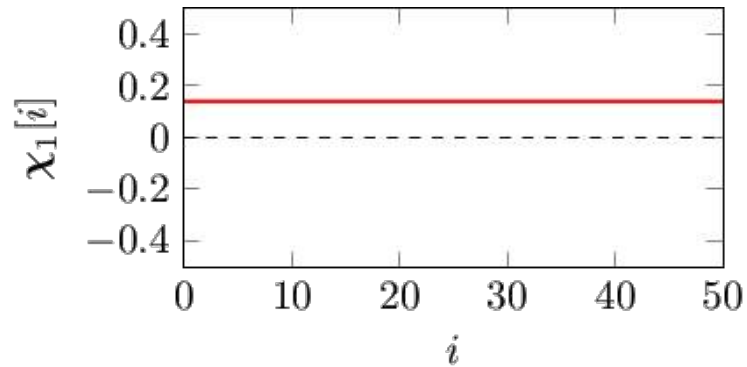


Methods & notions — The origins of GSP

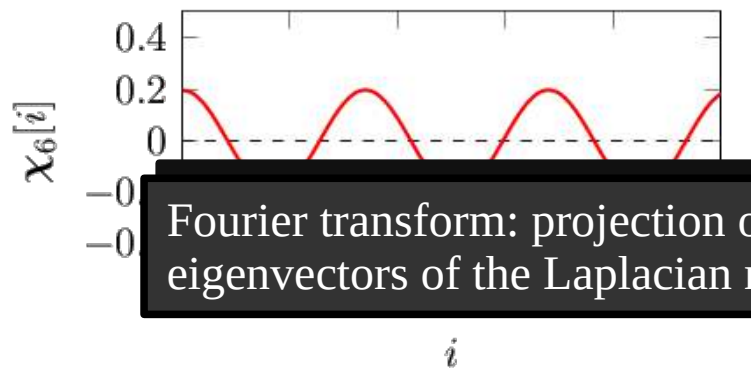
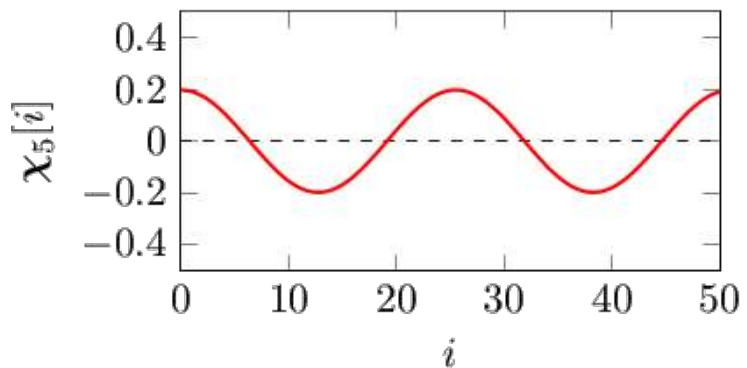
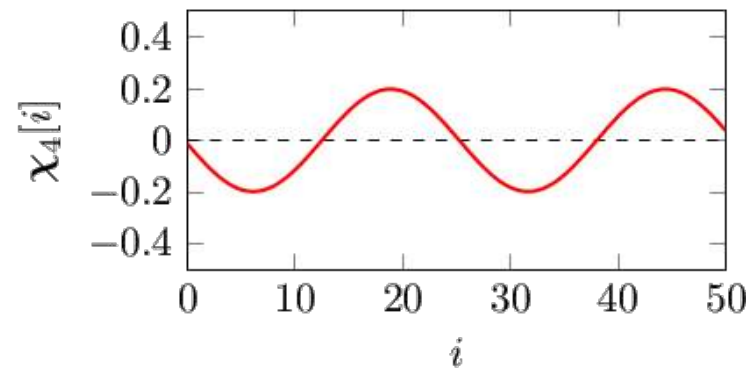
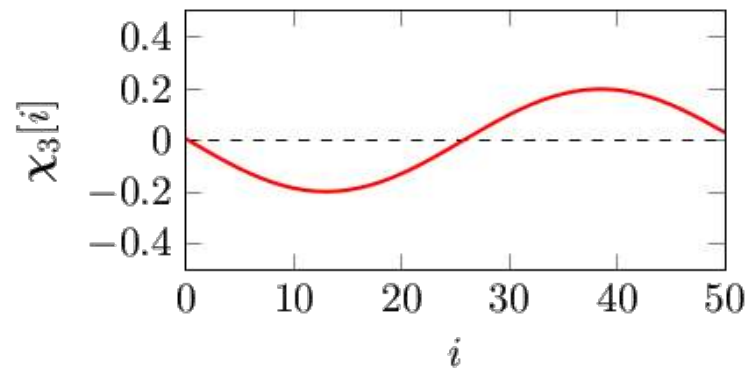
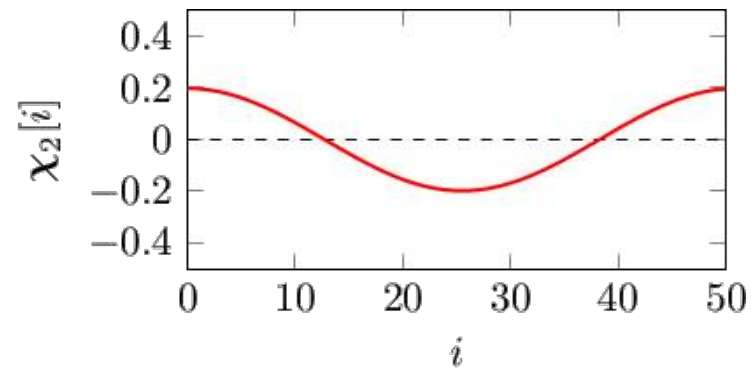
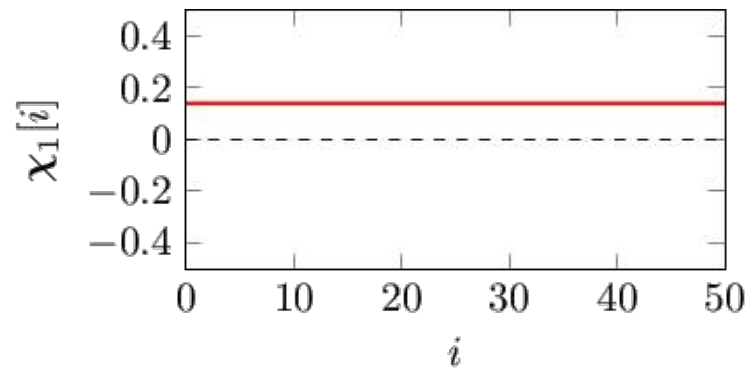


$$\begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 2 & 2 \end{bmatrix}$$

Methods & notions — The origins of GSP



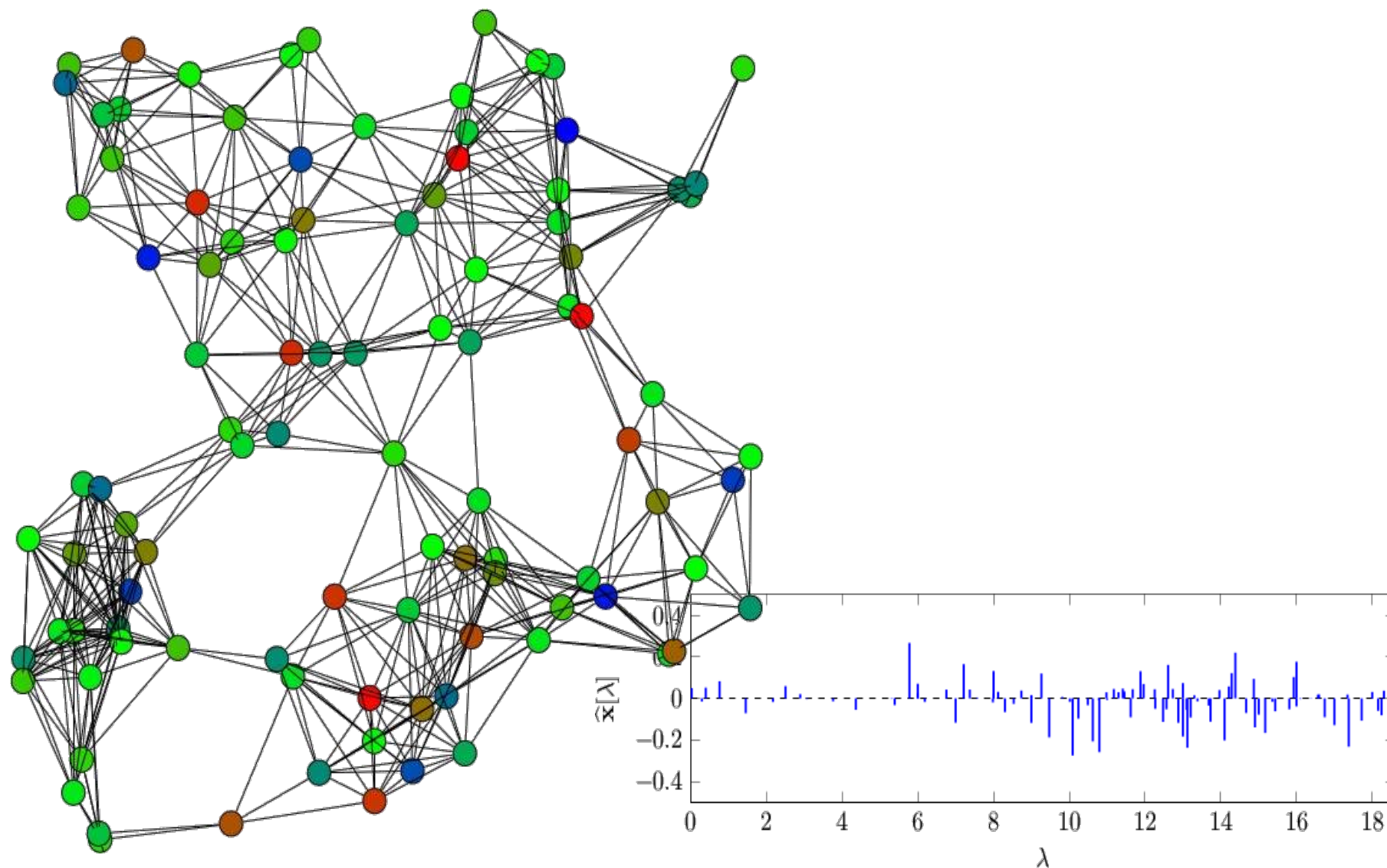
Methods & notions — The origins of GSP



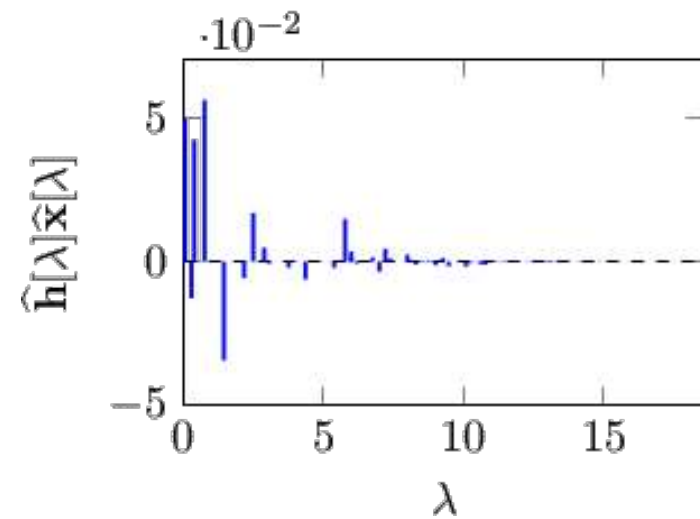
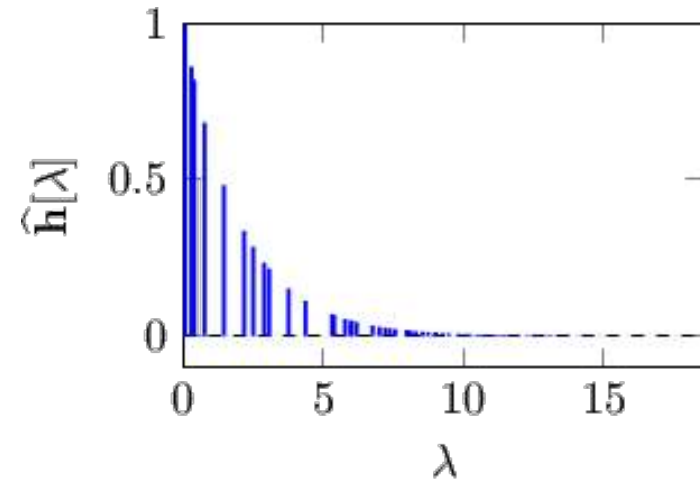
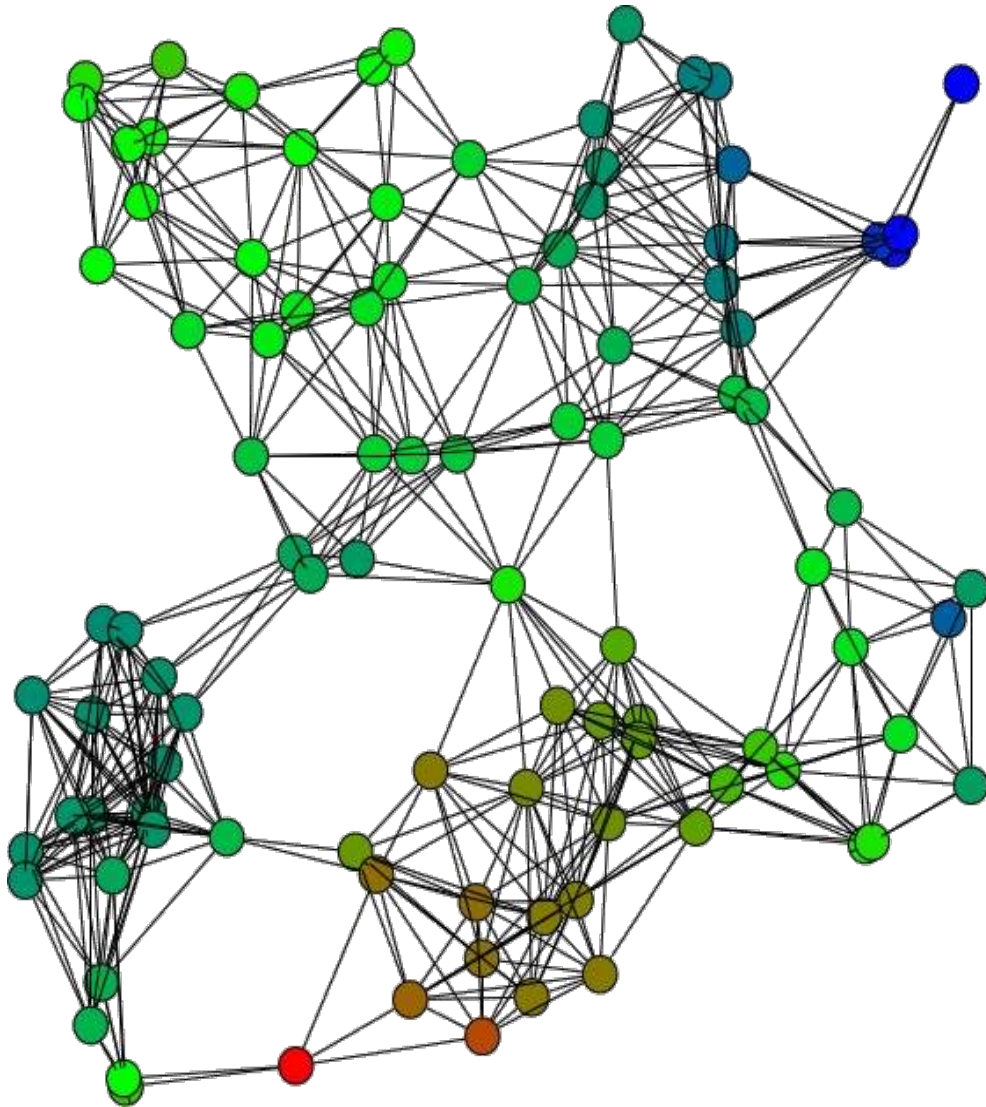
Fourier transform: projection on eigenvectors of the Laplacian matrix

!

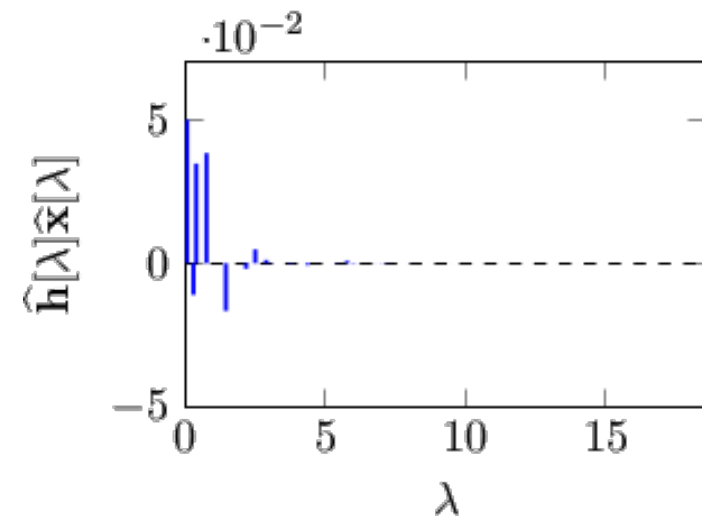
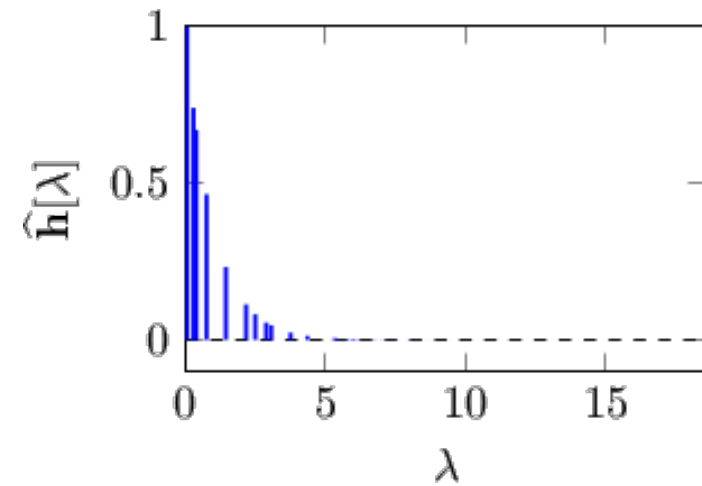
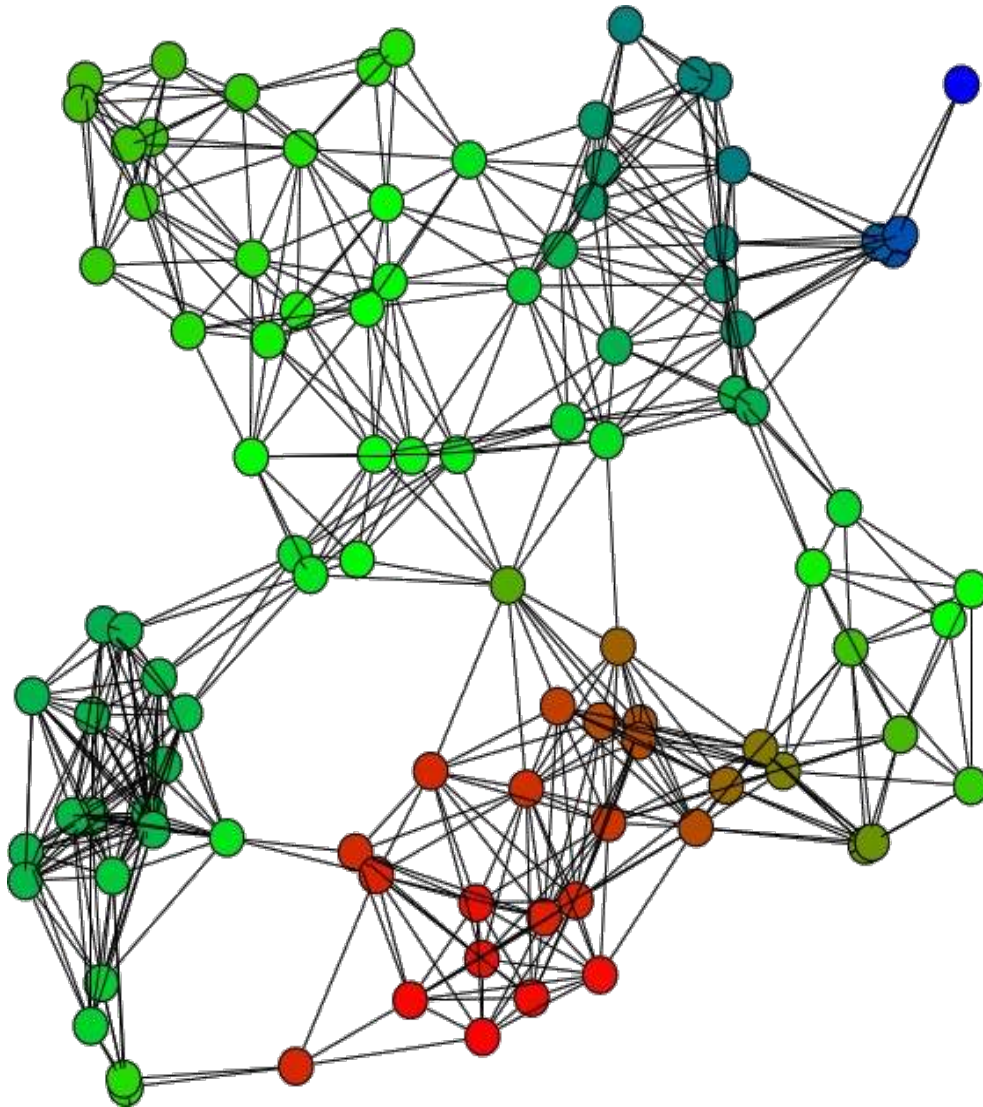
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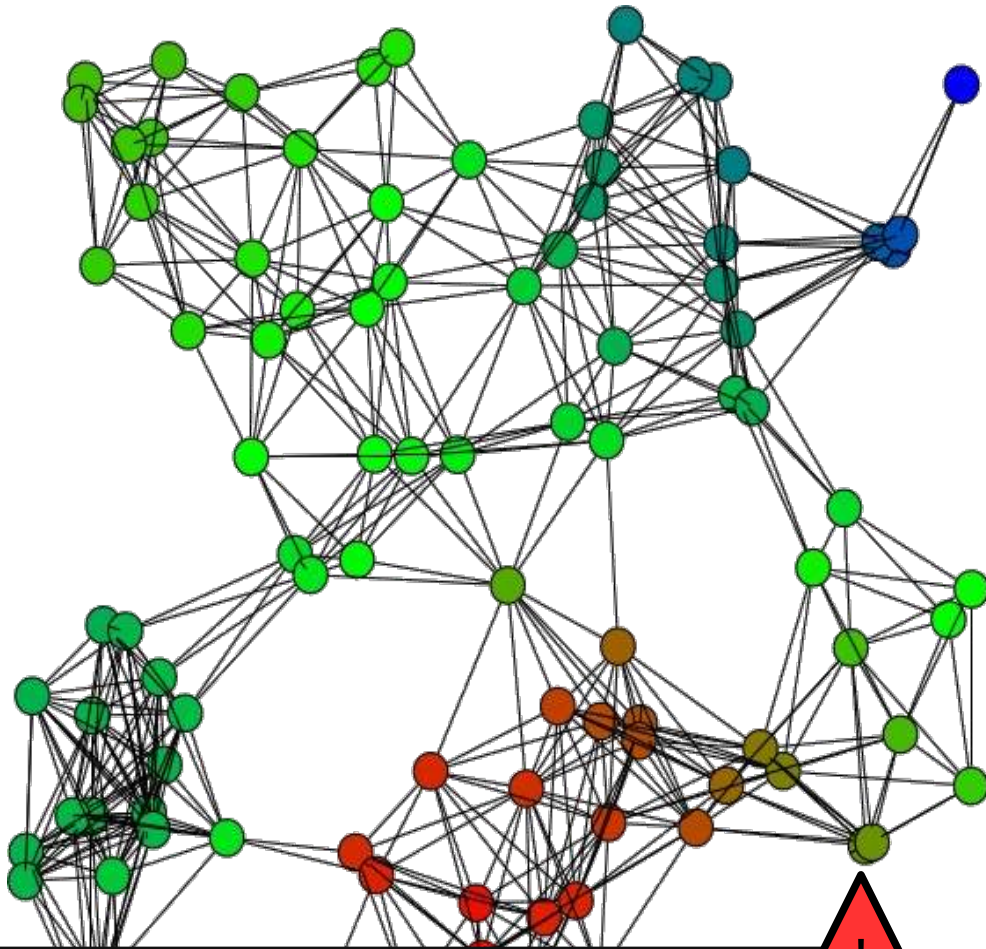
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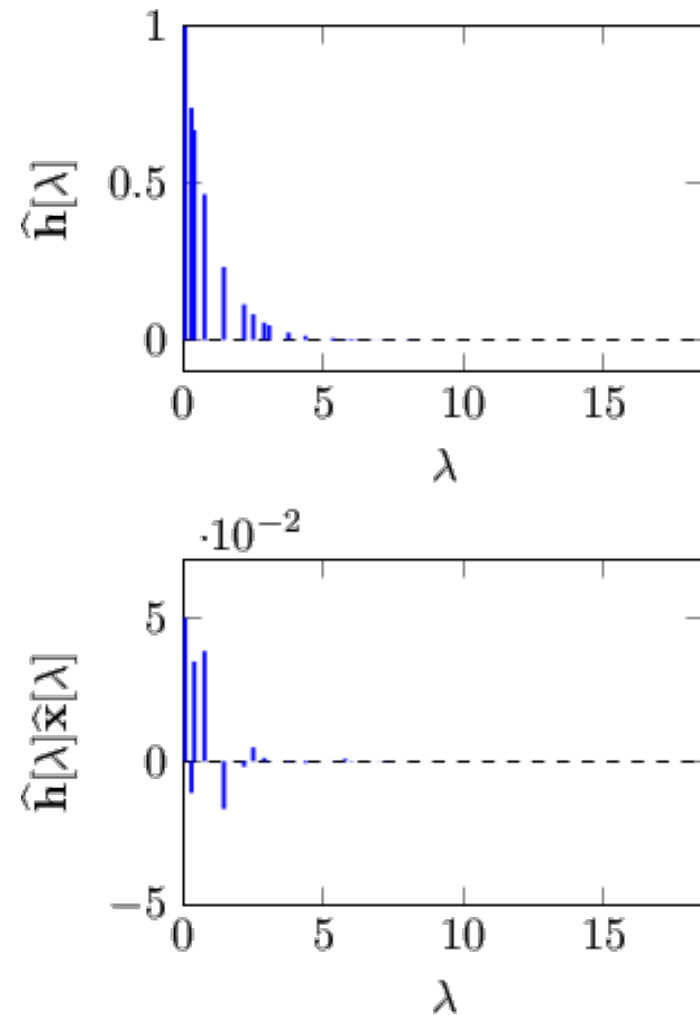
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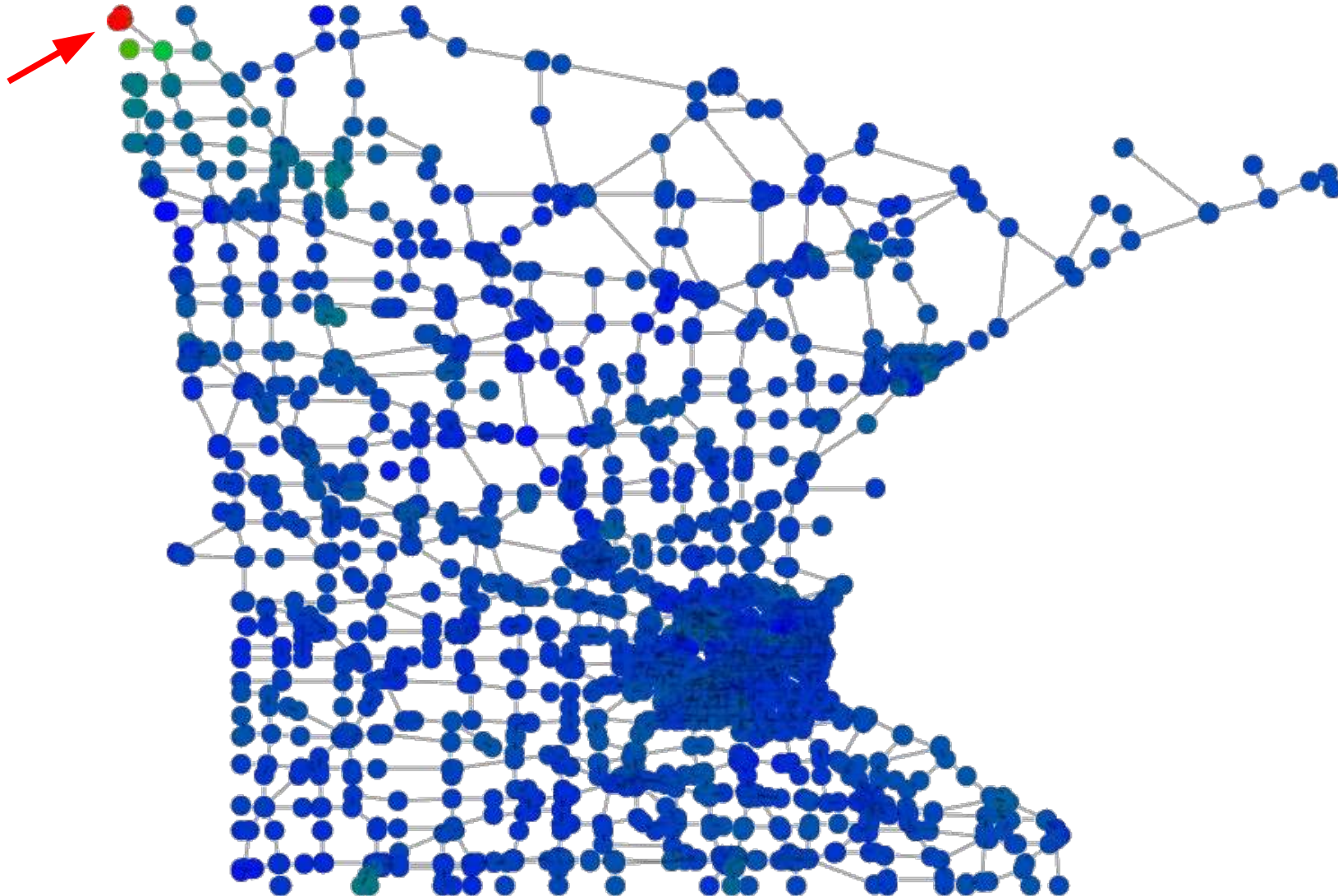
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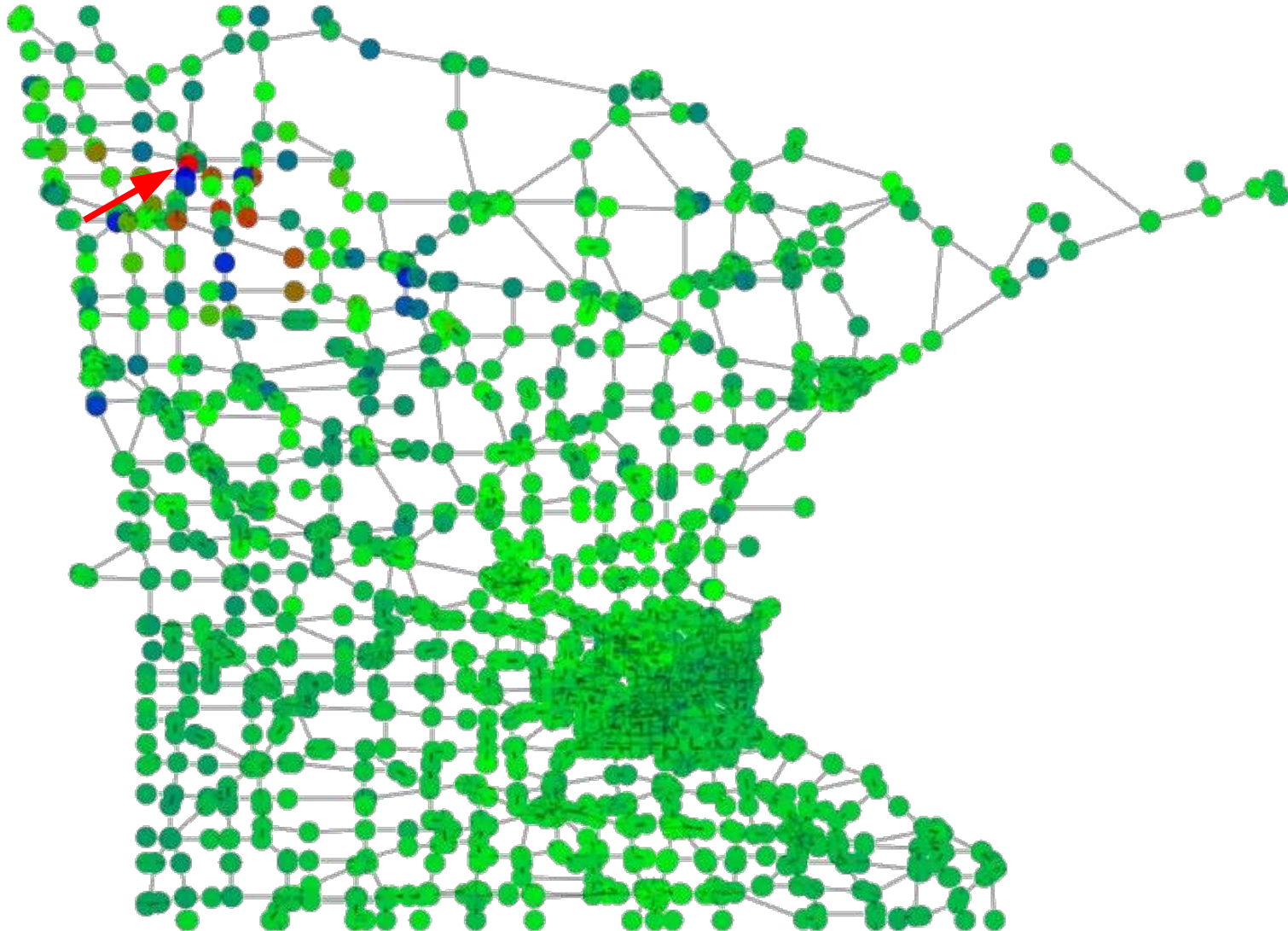
Eigenvalues are analogous to frequencies
This generalizes to all graphs



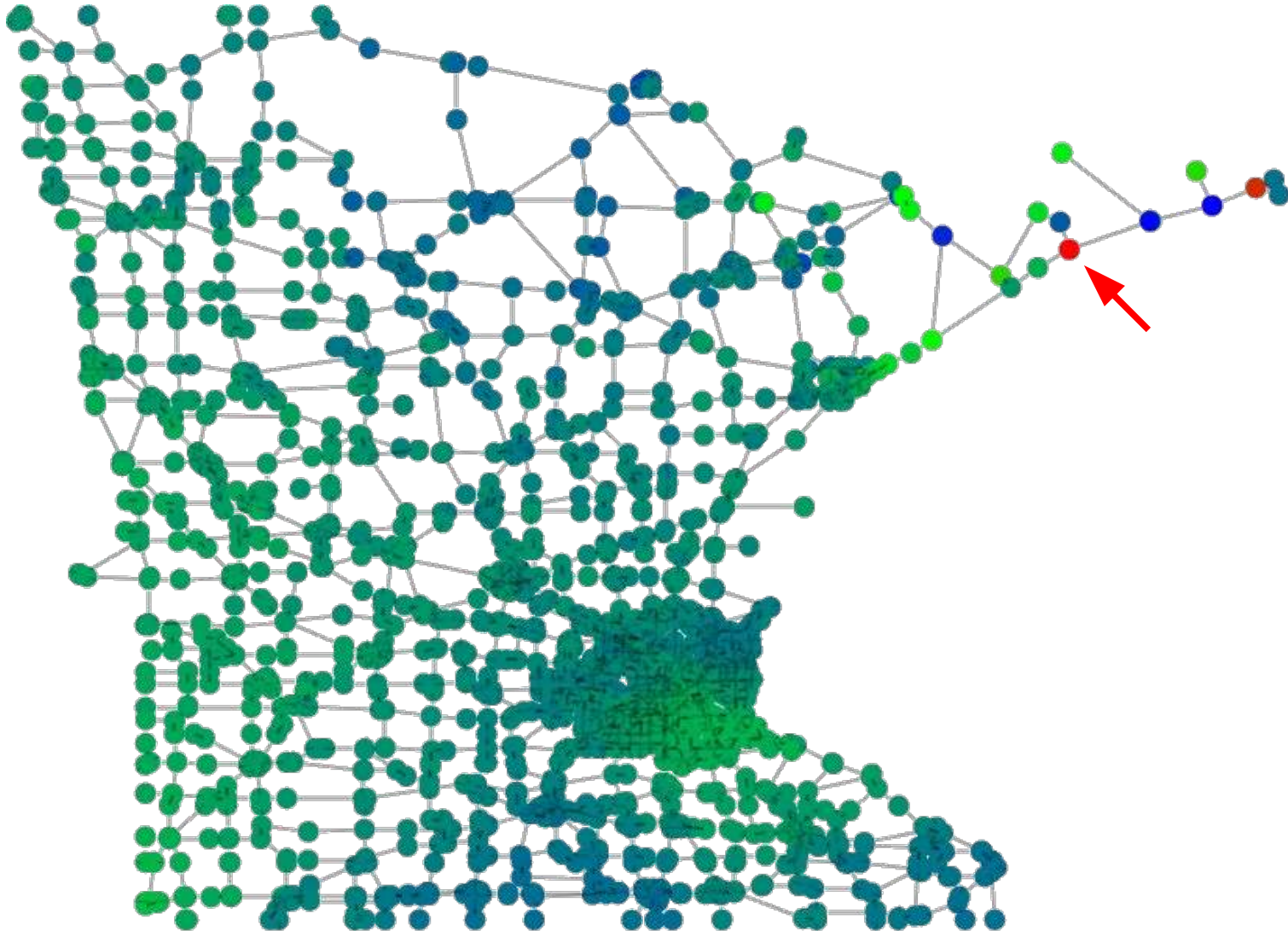
Methods & notions — Translation in the GSP framework



Methods & notions — Translation in the GSP framework



Methods & notions — Translation in the GSP framework



Methods & notions — Two families of methods

λ

Bruna *et al.* – Spectral networks and locally connected networks on graphs (2013)

Henaff *et al.* – Deep convolutional networks on graph-structured data (2015)

Defferrard *et al.* – Convolutional neural networks on graphs with fast localized spectral filtering (2016)

Atwood *et al.* – Search-convolutional neural networks (2015)

Puy *et al.* – Unifying local and non-local signal processing with graph CNNs (2017)

Hechtlinger *et al.* – A Generalization of convolutional neural networks to graph-structured data (2017)

Vialatte *et al.* – Learning local receptive fields and their weight sharing scheme on graphs (2017)

$\mathcal{G}$

Outline

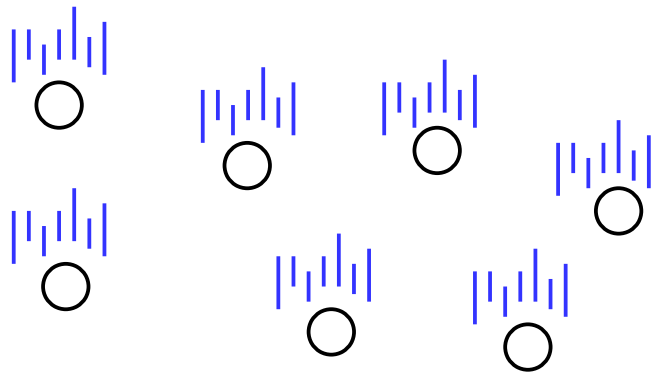
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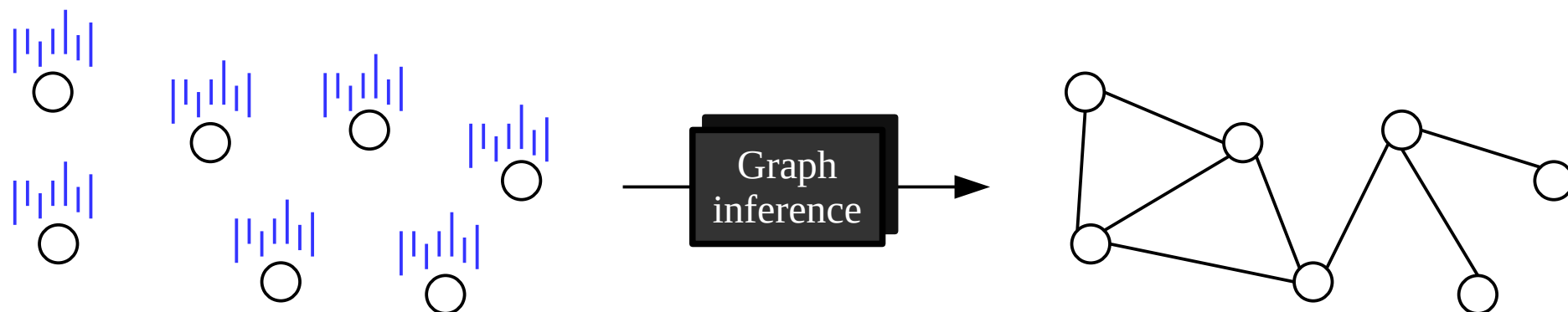
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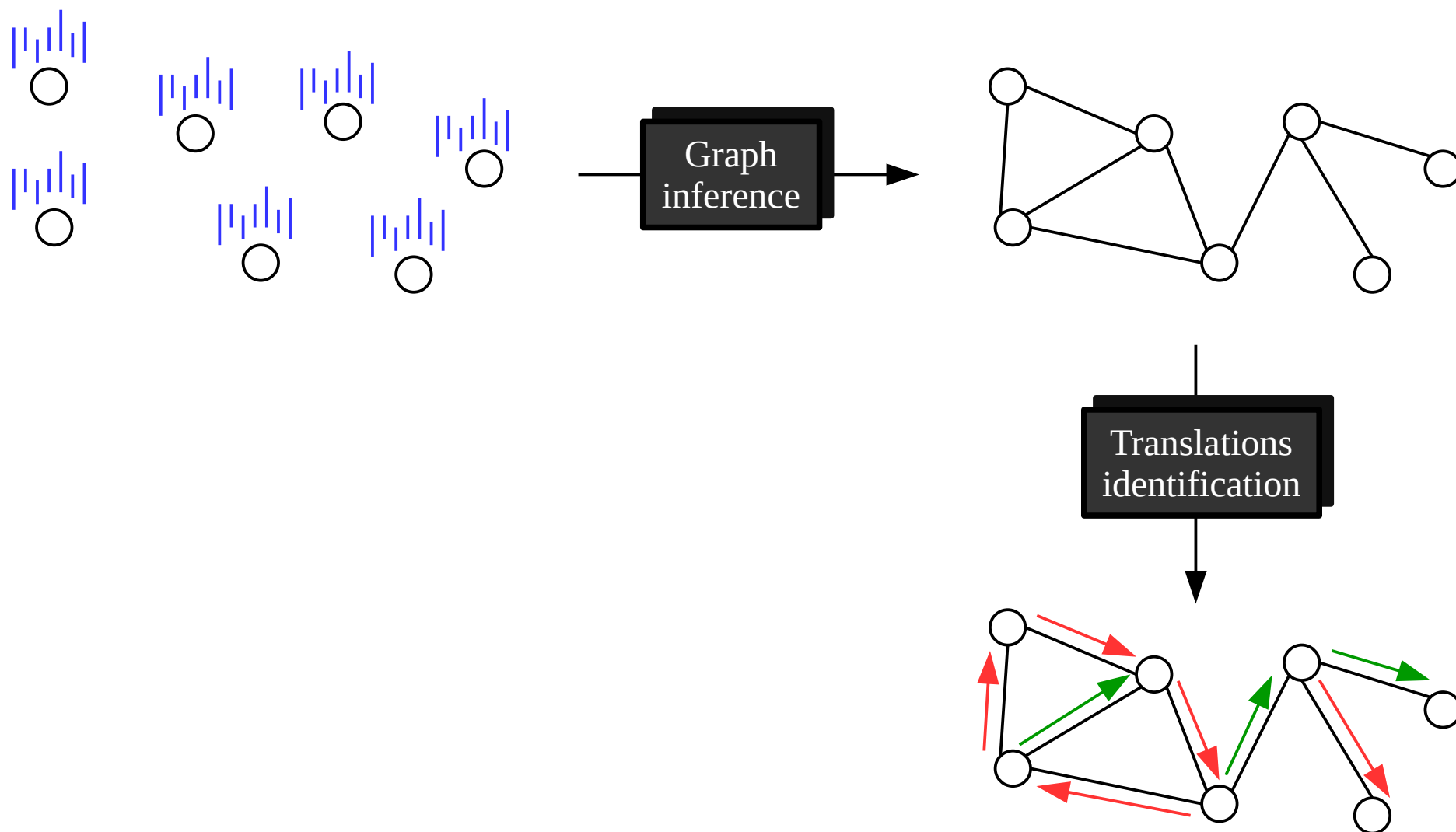
Our method — Global idea — From signals to CNNs



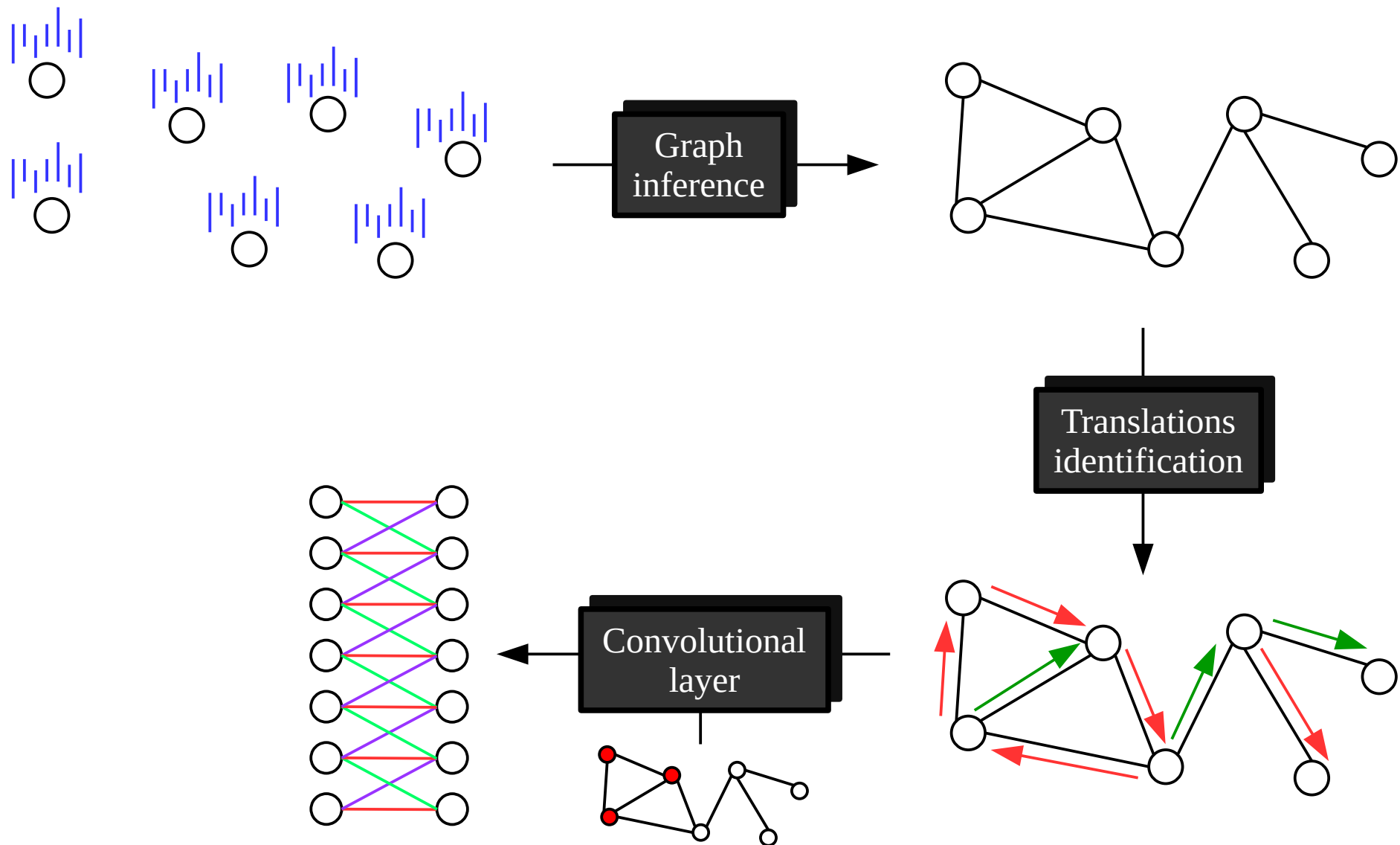
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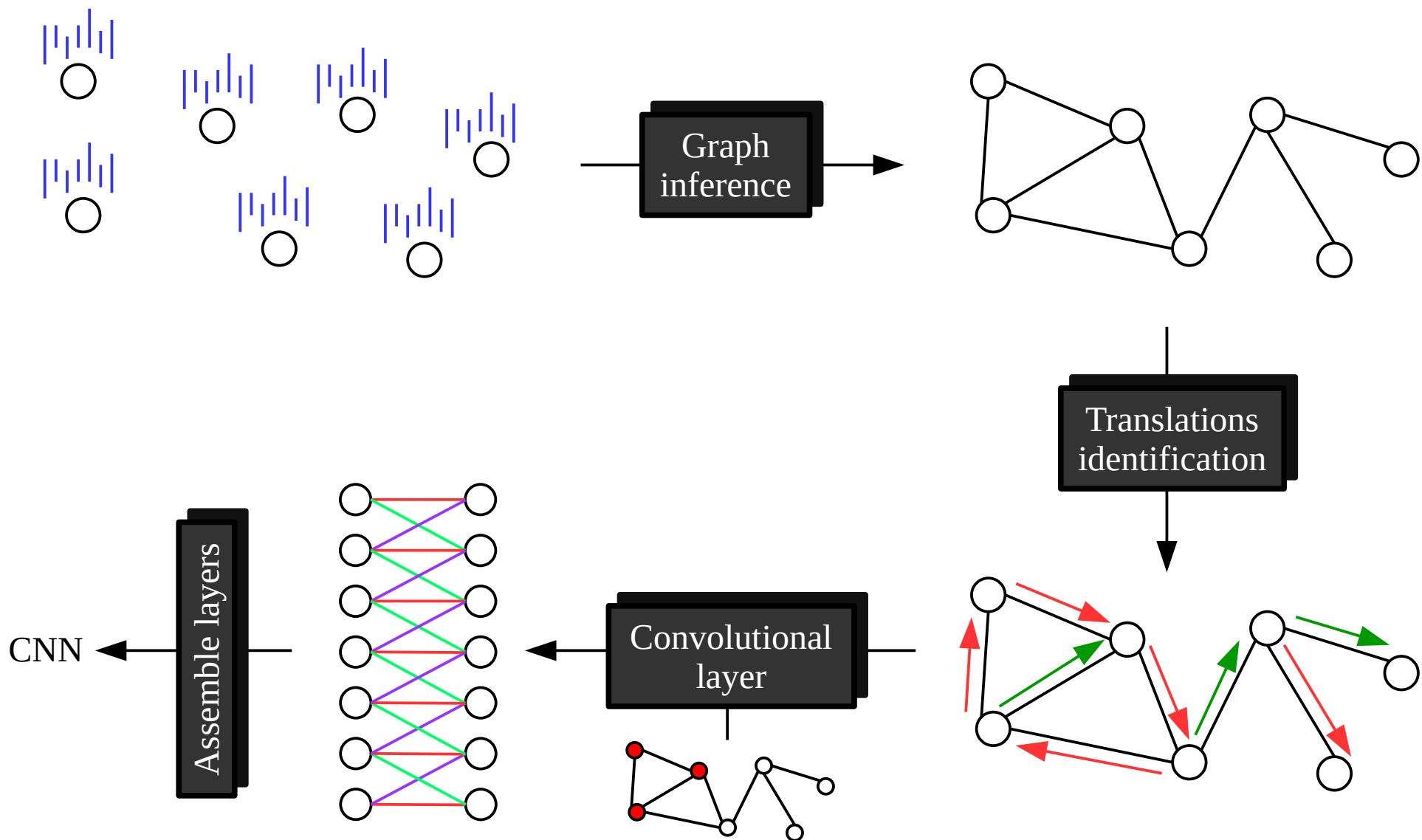
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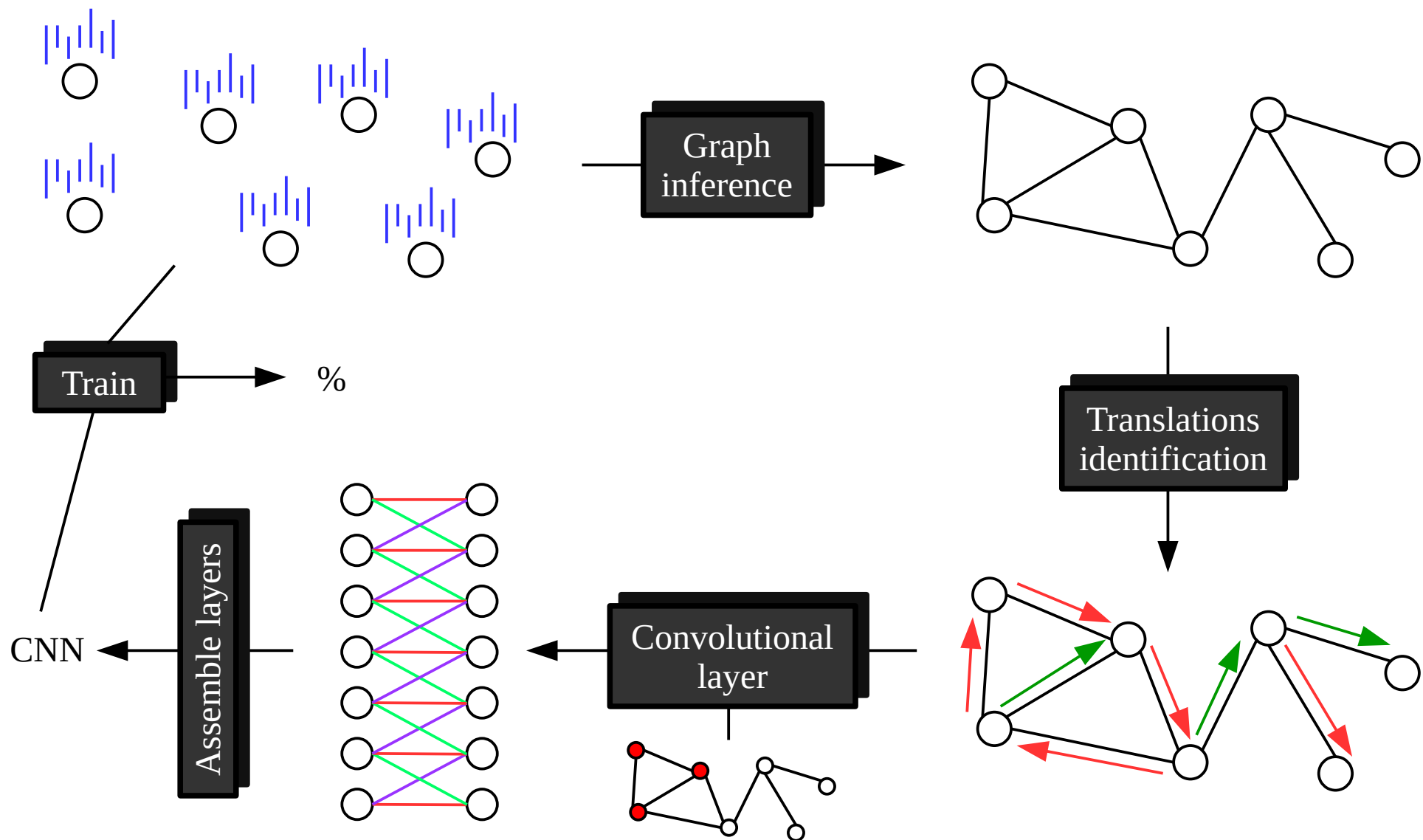
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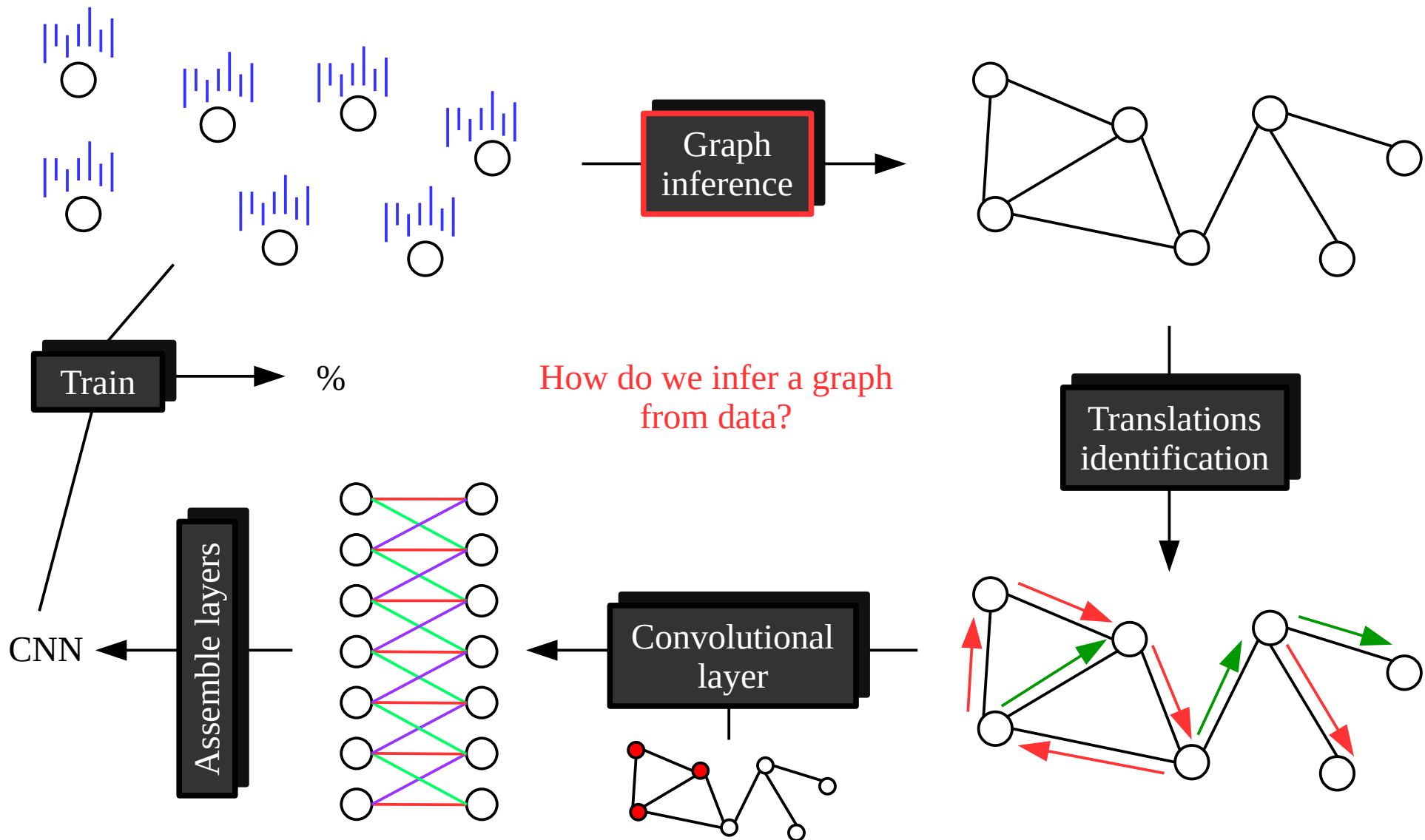
Our method — Global idea — From signals to CNNs



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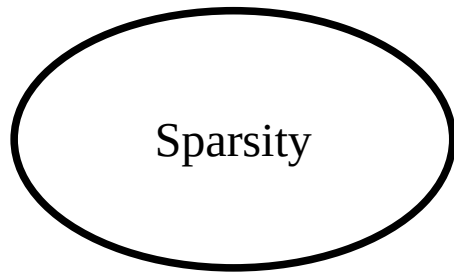
Our method — Graph inference



Our method — Graph inference — An ill-posed problem

$$2 \frac{N(N-1)}{2}$$

Our method — Graph inference — An ill-posed problem



$$2 \frac{N(N-1)}{2}$$

Friedman *et al.* — Sparse inverse covariance estimation with the graphical lasso (2008)

Pavez *et al.* — Generalized Laplacian precision matrix estimation for graph signal processing (2006)

Our method — Graph inference — An ill-posed problem

Smoothness

Sparsity

$$2^{\frac{N(N-1)}{2}}$$

Lake *et al.* — Discovering structure by Learning sparse graphs (2010)

Kalofolias — How to learn a graph from smooth signals (2016)

Our method — Graph inference — An ill-posed problem

Smoothness

Stationarity /
Diffusion

Sparsity

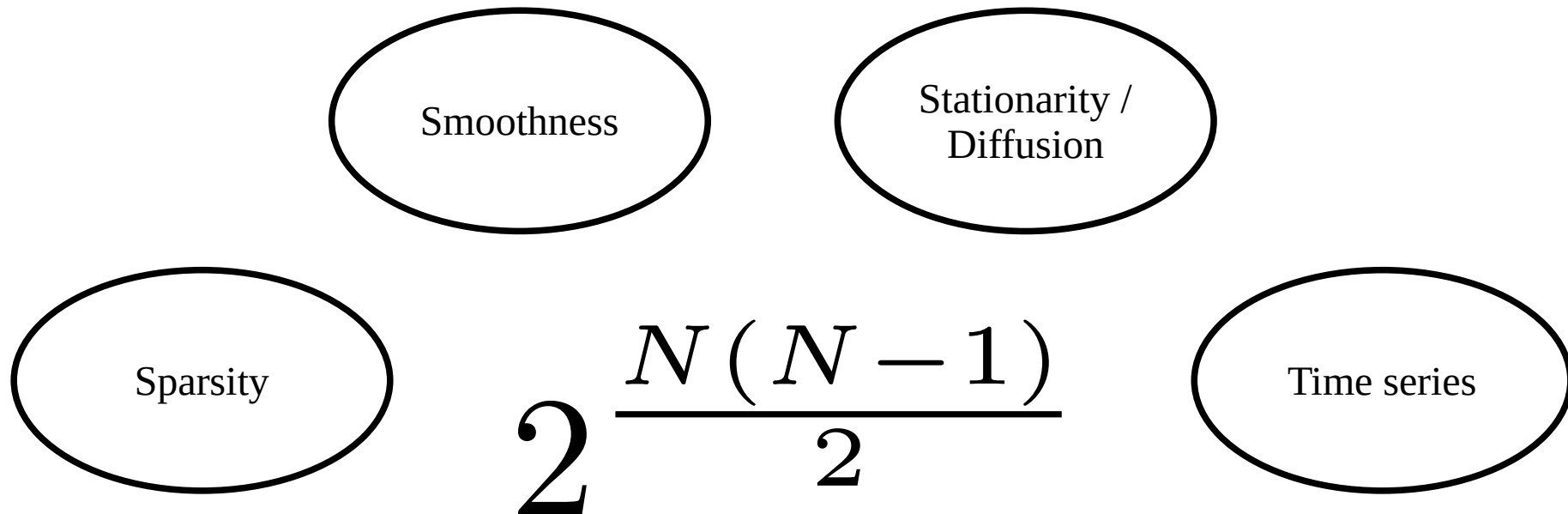
$$2^{\frac{N(N-1)}{2}}$$

Pasdeloup *et al.* — Graph reconstruction from the observation of diffused signals (2015)

Pasdeloup *et al.* — Characterization and inference of graph diffusion processes from observations of stationary signals (2017)

Segarra *et al.* — Network topology identification from spectral templates (2016)

Our method — Graph inference — An ill-posed problem



Mei *et al.* — Signal processing on graphs:
Estimating the structure of a graph (2015)

Shen *et al.* — Topology inference of directed
graphs using nonlinear structural vector
autoregressive models (2017)

Our method — Graph inference — An ill-posed problem

Smoothness

Stationarity /
Diffusion

Sparsity

$$2^{\frac{N(N-1)}{2}}$$

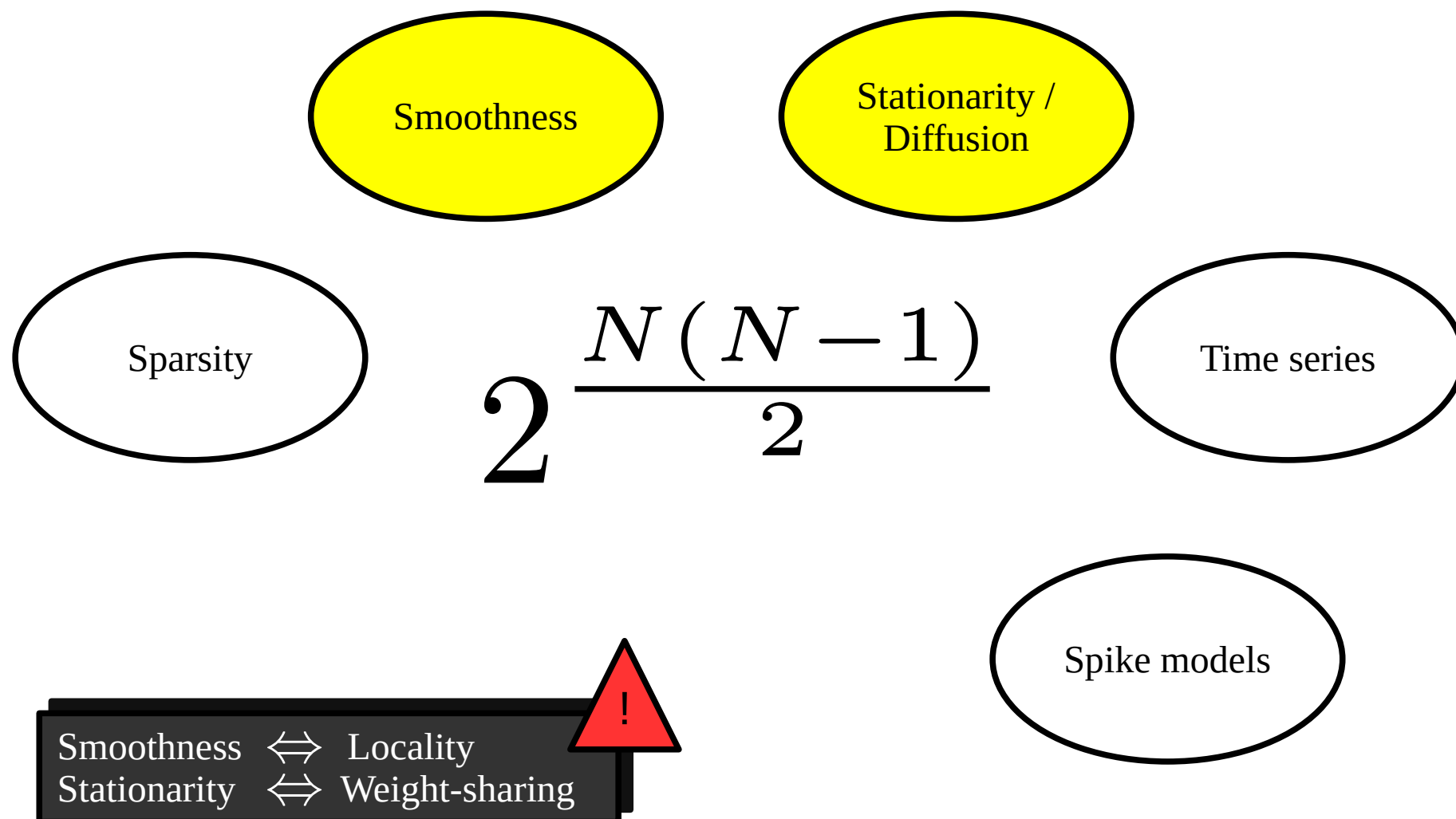
Time series

Spike models

Thanou *et al.* — Learning heat diffusion graphs (2016)

Petric Maretic *et al.* — Graph learning under sparsity priors (2017)

Our method — Graph inference — An ill-posed problem

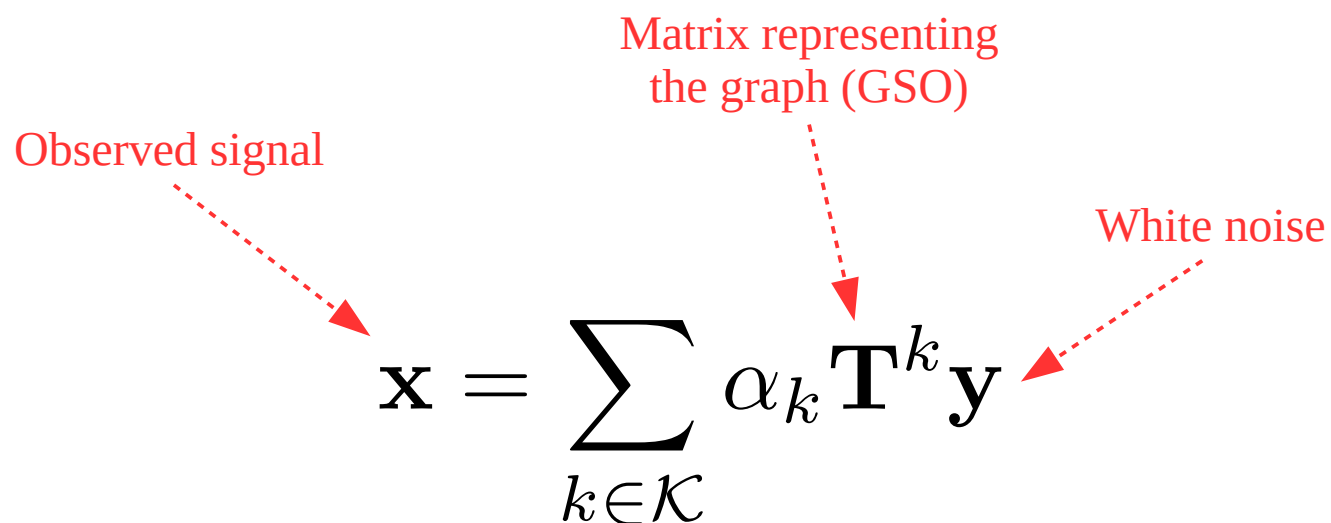


Our method — Graph inference — A model of stationary signals

Observed signal

Matrix representing the graph (GSO)

White noise

$$\mathbf{x} = \sum_{k \in \mathcal{K}} \alpha_k \mathbf{T}^k \mathbf{y}$$


Our method — Graph inference — A model of stationary signals

Observed signal

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$$\mathbf{x} = \sum_{k \in \mathcal{K}} \alpha_k \mathbf{T}^k \mathbf{y}$$

$$\begin{aligned} \Sigma &= \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \\ &= \mathbb{E}\left[\sum_{k \in \mathcal{K}} \beta_k \mathbf{T}^k \mathbf{y}\mathbf{y}^\top\right] \\ &\propto \sum_{k \in \mathcal{K}} \beta_k \mathbf{T}^k \end{aligned}$$

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Same eigenvectors for Σ and $\mathbf{T}$
Hard to obtain in practice



Our method — Graph inference — The set of admissible matrices

$$\mathbf{T} = \mathcal{X} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathcal{X}^\top$$

Our method — Graph inference — The set of admissible matrices

$$\mathbf{T} = \mathcal{X} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathcal{X}^\top$$

$$\forall i, j : \mathbf{T}[i, j] = \alpha \lambda_1 + \beta \lambda_2 + \dots + \gamma \lambda_N$$

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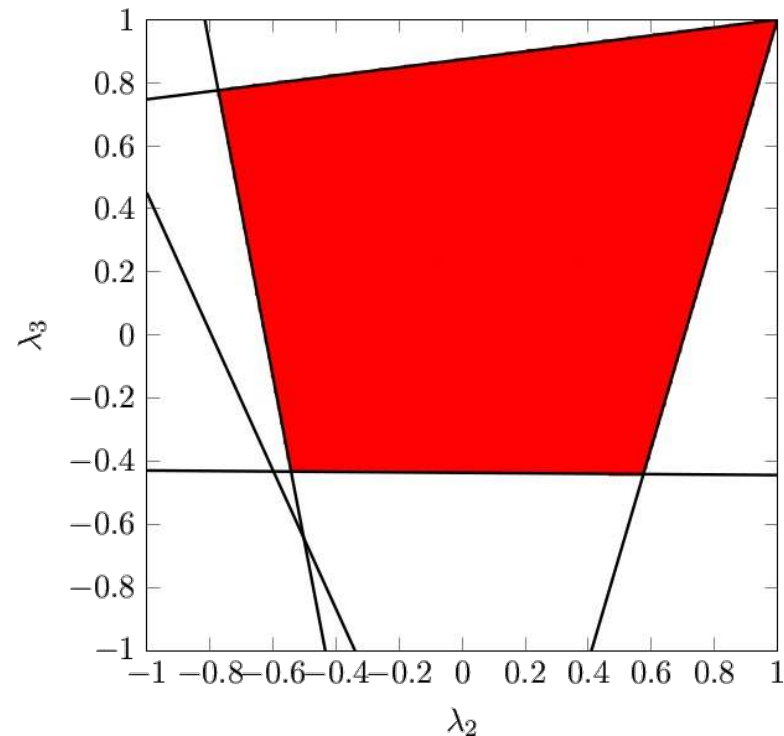
$$\forall i, j : \mathbf{T}[i, j] \geq 0$$

Our method — Graph inference — The set of admissible matrices

$$\mathbf{T} = \mathcal{X} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathcal{X}^\top$$

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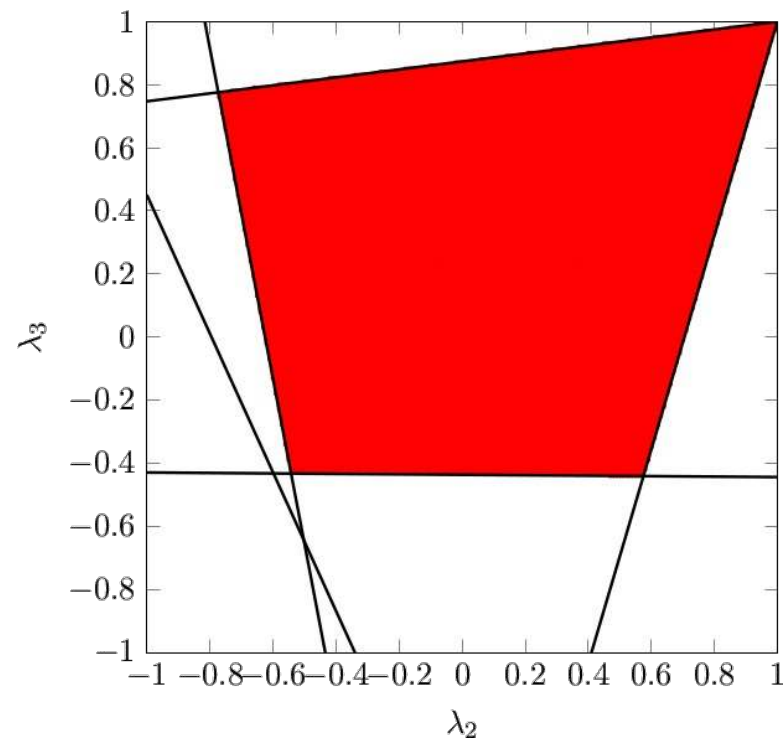
Our method — Graph inference — The set of admissible matrices

$$\mathbf{T} = \mathcal{X} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathcal{X}^\top$$

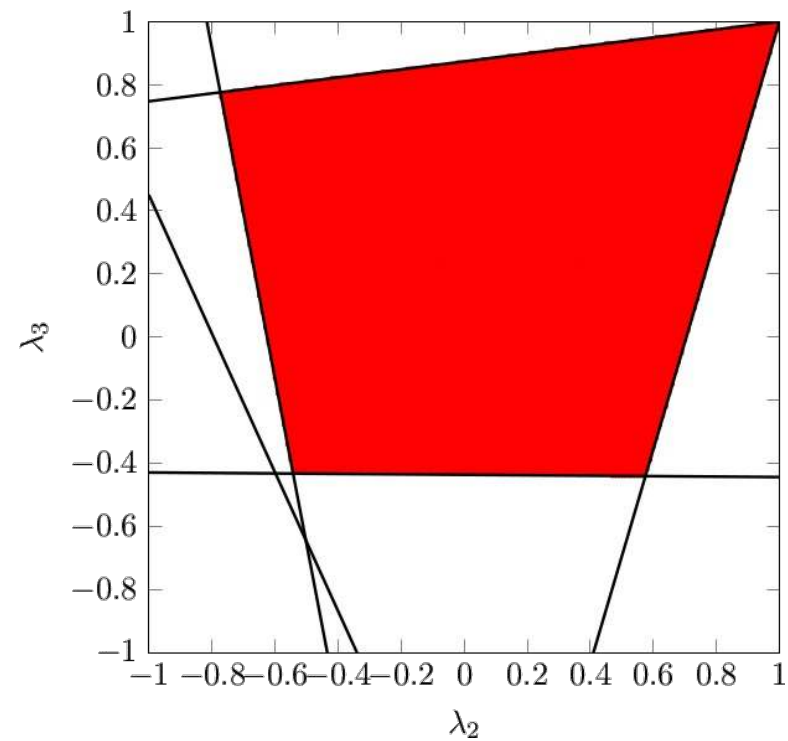
$$\forall i, j : \mathbf{T}[i, j] = \alpha\lambda_1 + \beta\lambda_2 + \dots + \gamma\lambda_N$$

$$\forall i, j : \mathbf{T}[i, j] \geq 0$$

Admissible matrices lie in a convex set
Need to select a vector of eigenvalues

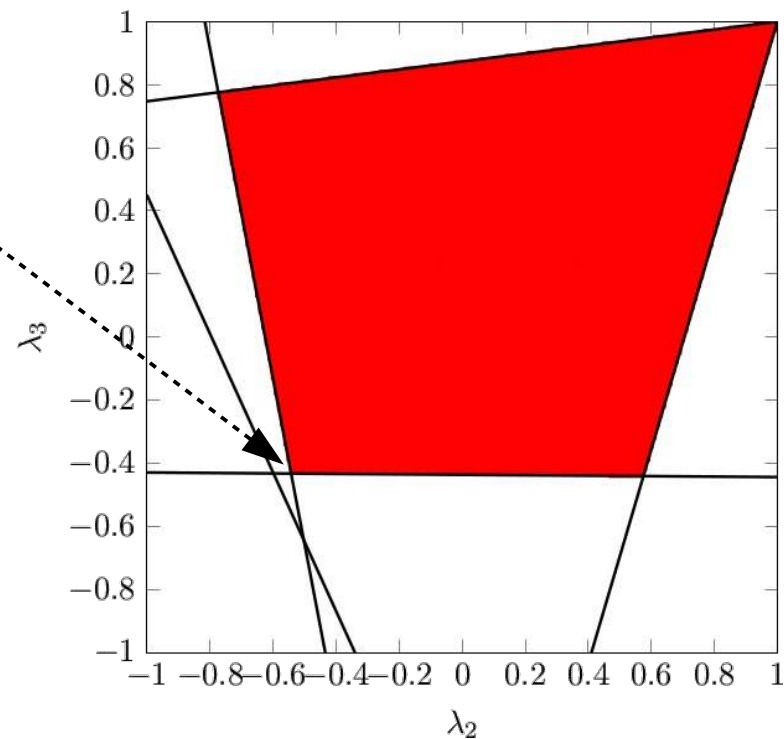


Our method — Graph inference — Eigenvalues selection



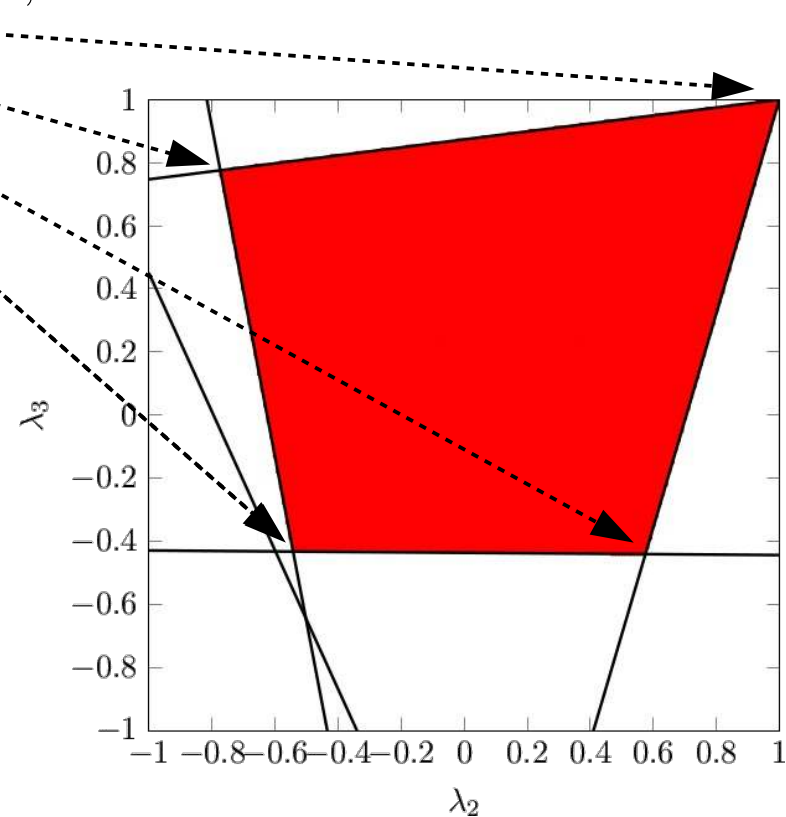
Our method — Graph inference — Eigenvalues selection

$$\arg \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i$$



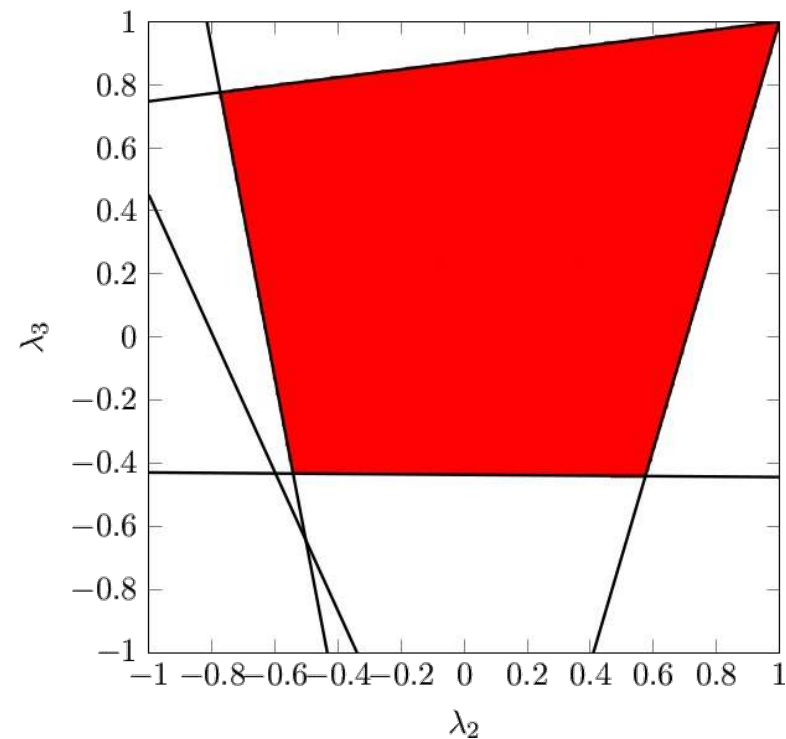
Our method — Graph inference — Eigenvalues selection

$$\arg \min_{\lambda_1, \dots, \lambda_N} \left\| \mathcal{X} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathcal{X}^\top \right\|_{1,1}$$



Our method — Graph inference — Eigenvalues selection

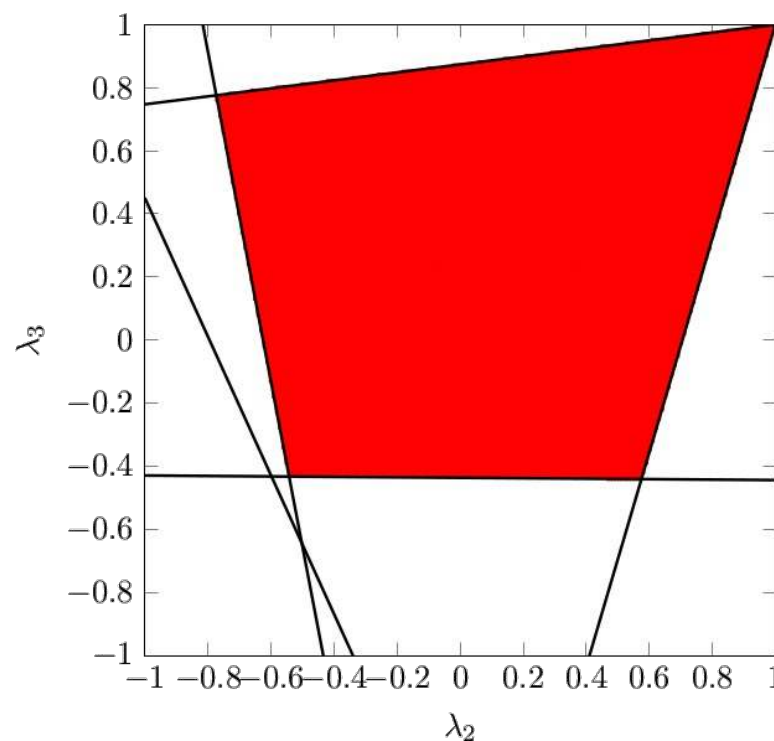
$$\mathbf{T} = \mathcal{X}_{\mathbf{T}} \begin{bmatrix} \lambda_{\mathbf{T}1} & 0 & 0 \\ 0 & \lambda_{\mathbf{T}2} & 0 \\ 0 & 0 & \lambda_{\mathbf{T}3} \end{bmatrix} \mathcal{X}_{\mathbf{T}}^{\top}$$



Our method — Graph inference — Eigenvalues selection

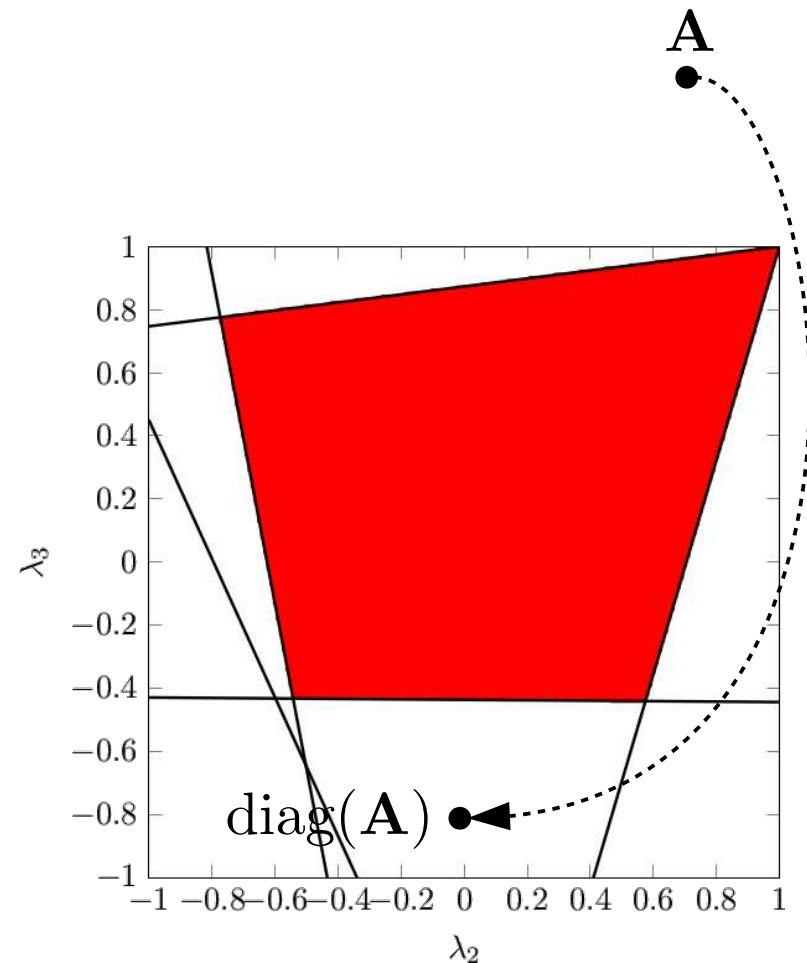
$$\begin{aligned}\mathbf{T} &= \mathcal{X}_{\mathbf{T}} \begin{bmatrix} \lambda_{\mathbf{T}1} & 0 & 0 \\ 0 & \lambda_{\mathbf{T}2} & 0 \\ 0 & 0 & \lambda_{\mathbf{T}3} \end{bmatrix} \mathcal{X}_{\mathbf{T}}^{\top} \\ &= \mathcal{X} \mathbf{A} \mathcal{X}^{\top}\end{aligned}$$

$\mathbf{A}$
•



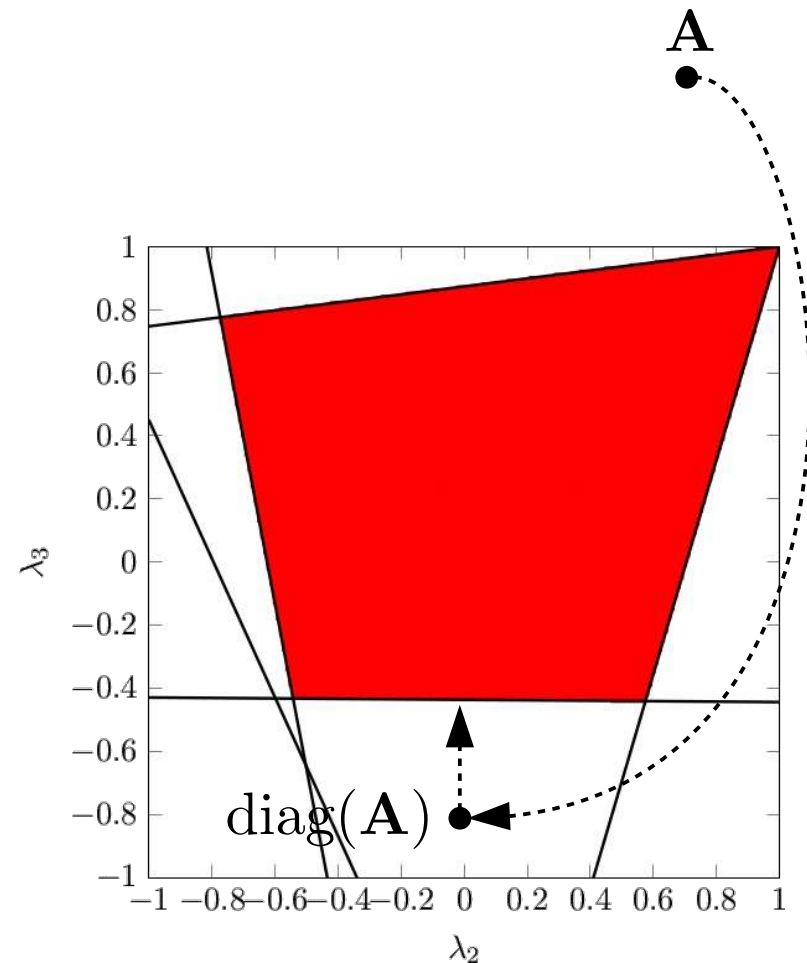
Our method — Graph inference — Eigenvalues selection

$$\begin{aligned}\mathbf{T} &= \mathcal{X}_{\mathbf{T}} \begin{bmatrix} \lambda_{\mathbf{T}1} & 0 & 0 \\ 0 & \lambda_{\mathbf{T}2} & 0 \\ 0 & 0 & \lambda_{\mathbf{T}3} \end{bmatrix} \mathcal{X}_{\mathbf{T}}^{\top} \\ &= \mathcal{X} \mathbf{A} \mathcal{X}^{\top} \\ &\approx \mathcal{X} \text{diag}(\mathbf{A}) \mathcal{X}^{\top}\end{aligned}$$



Our method — Graph inference — Eigenvalues selection

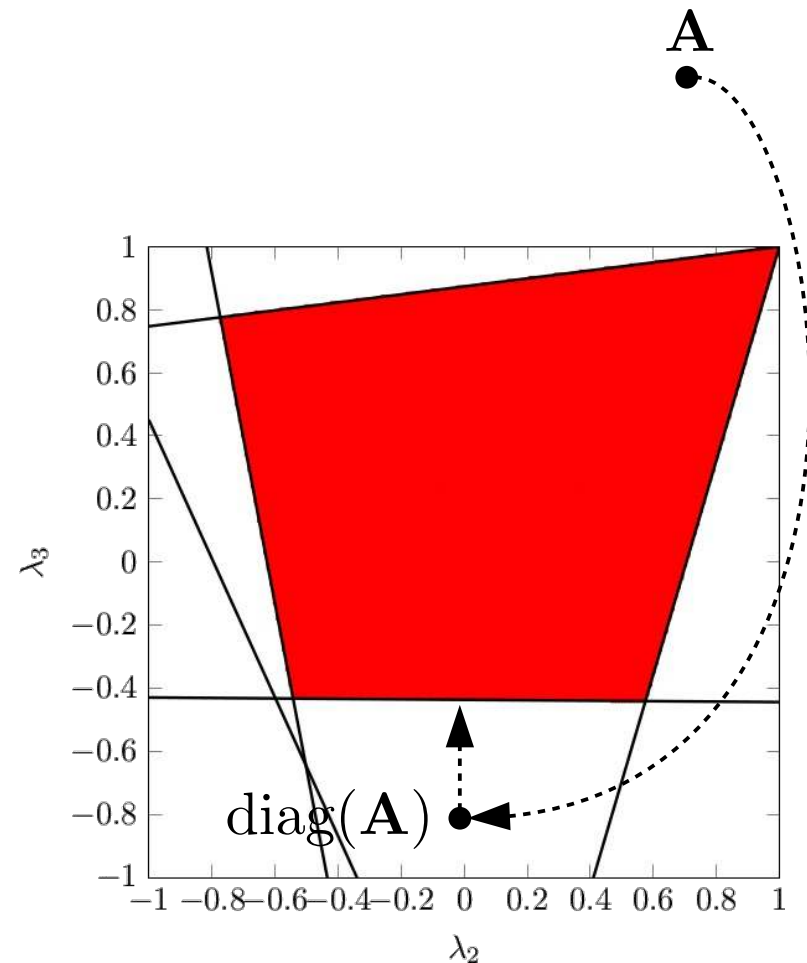
$$\begin{aligned}\mathbf{T} &= \mathcal{X}_{\mathbf{T}} \begin{bmatrix} \lambda_{\mathbf{T}1} & 0 & 0 \\ 0 & \lambda_{\mathbf{T}2} & 0 \\ 0 & 0 & \lambda_{\mathbf{T}3} \end{bmatrix} \mathcal{X}_{\mathbf{T}}^{\top} \\ &= \mathcal{X} \mathbf{A} \mathcal{X}^{\top} \\ &\approx \mathcal{X} \text{diag}(\mathbf{A}) \mathcal{X}^{\top} \\ &\approx \mathcal{X} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathcal{X}^{\top}\end{aligned}$$



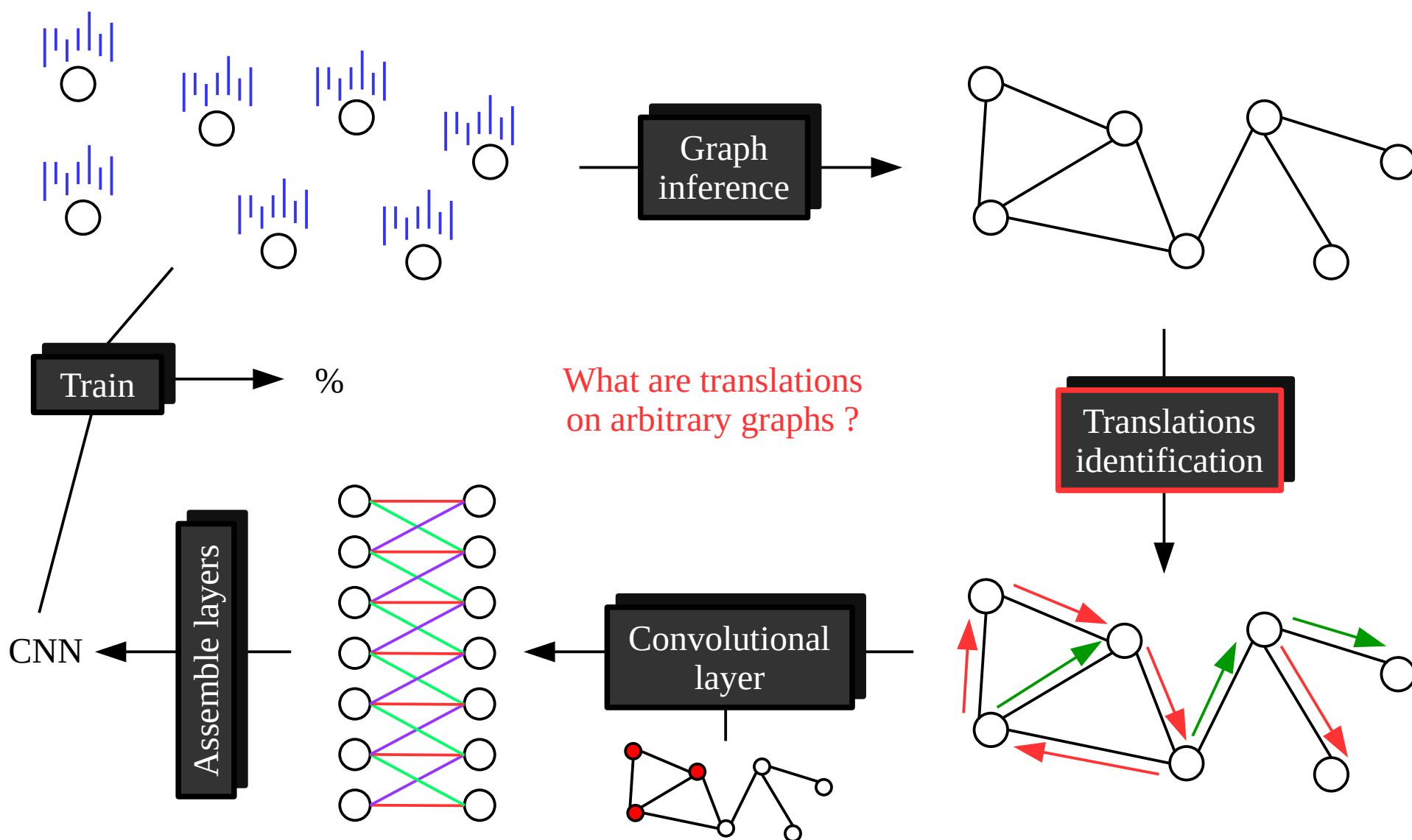
Our method — Graph inference — Eigenvalues selection

$$\begin{aligned}\mathbf{T} &= \mathcal{X}_{\mathbf{T}} \begin{bmatrix} \lambda_{\mathbf{T}1} & 0 & 0 \\ 0 & \lambda_{\mathbf{T}2} & 0 \\ 0 & 0 & \lambda_{\mathbf{T}3} \end{bmatrix} \mathcal{X}_{\mathbf{T}}^{\top} \\ &= \mathcal{X} \mathbf{A} \mathcal{X}^{\top} \\ &\approx \mathcal{X} \text{diag}(\mathbf{A}) \mathcal{X}^{\top} \\ &\approx \mathcal{X} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \mathcal{X}^{\top}\end{aligned}$$

Countless other strategies
Depends on the desired application



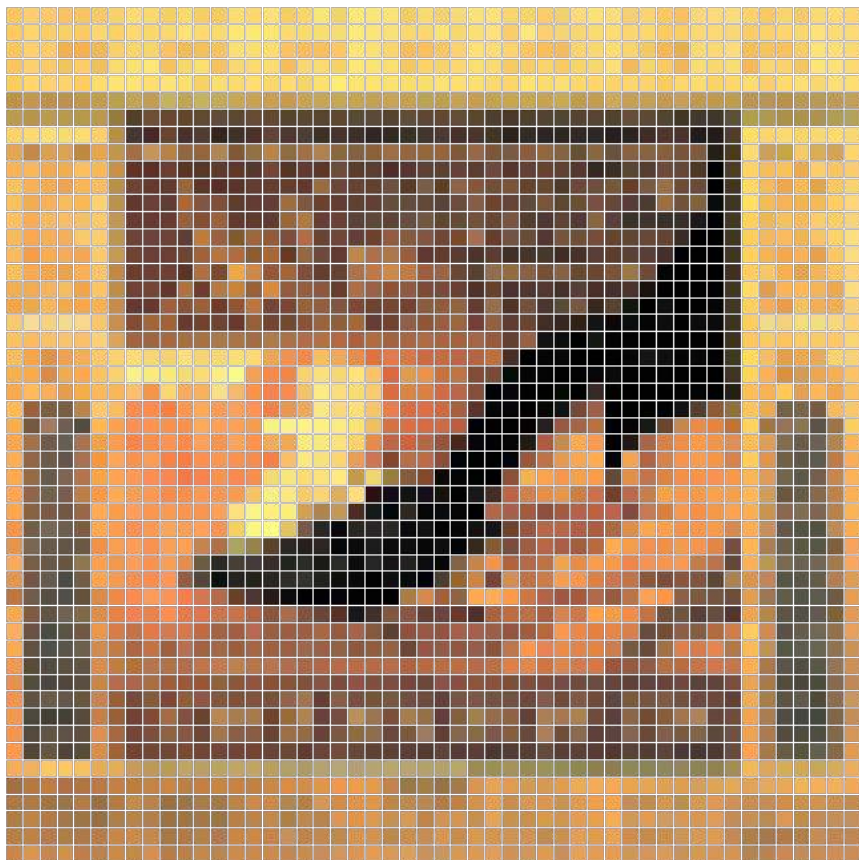
Our method — Translations identification



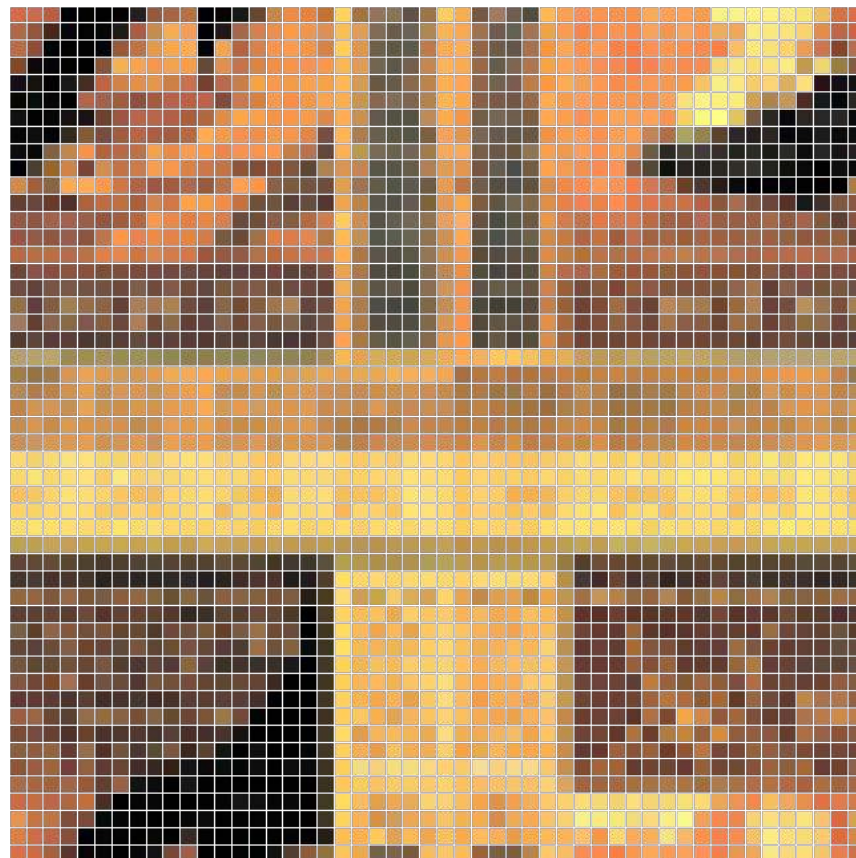
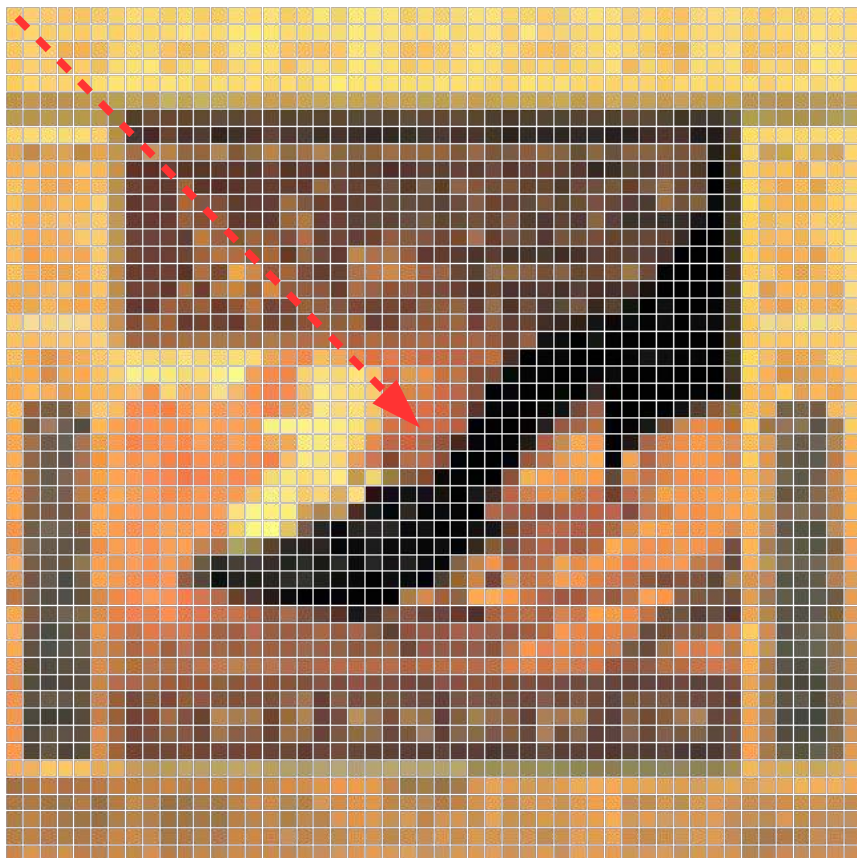
Our method — Translations identification — The case of images



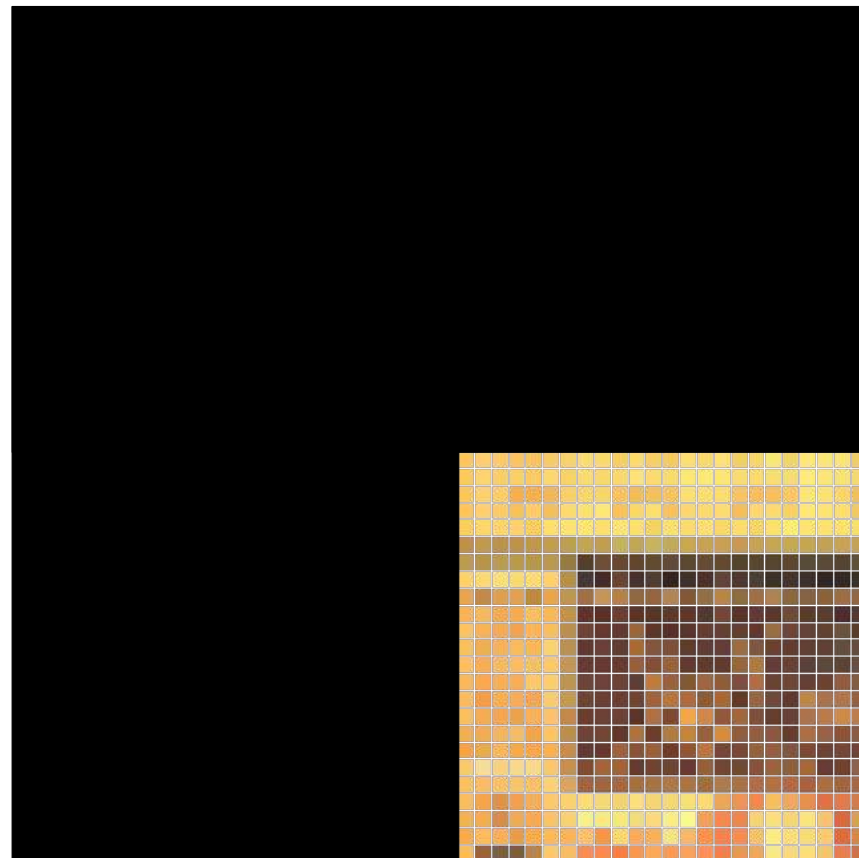
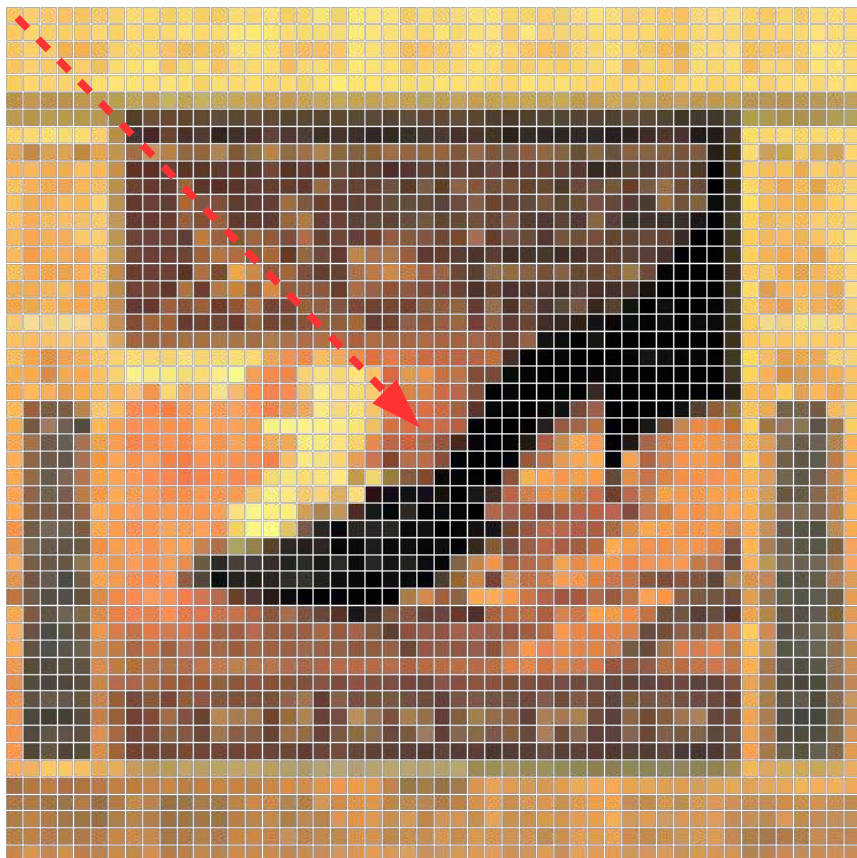
Our method — Translations identification — The case of images



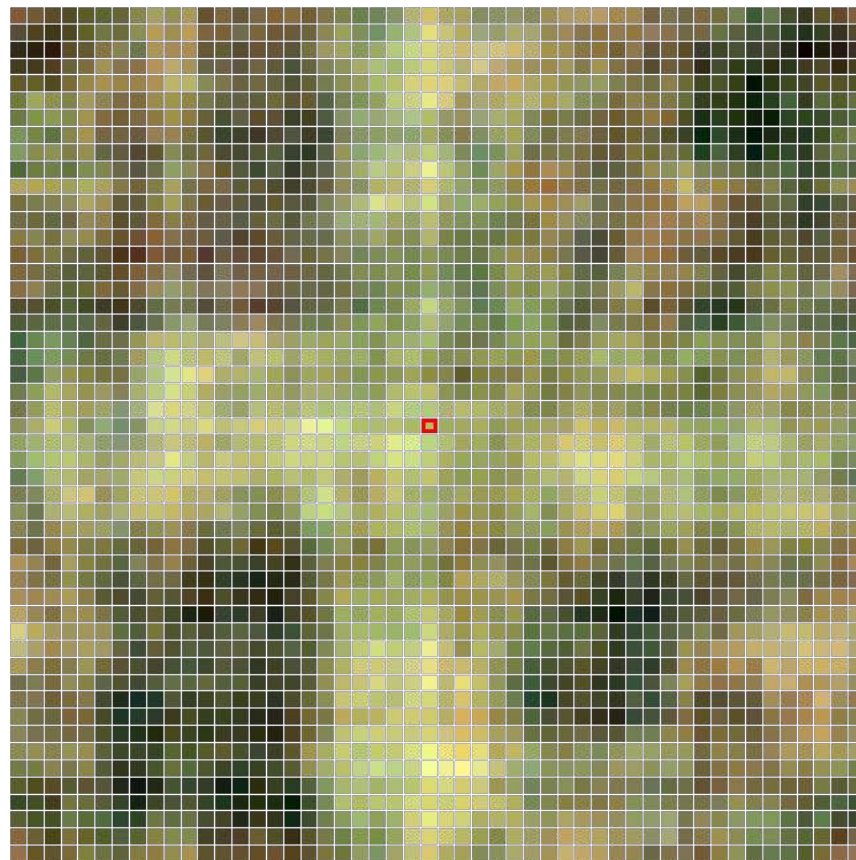
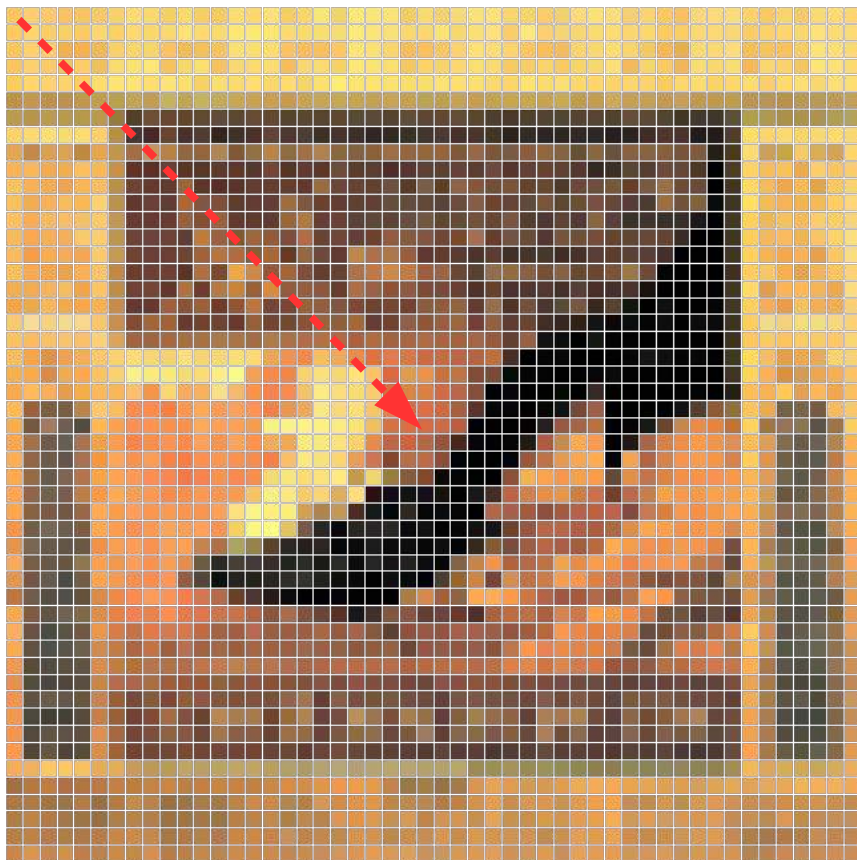
Our method — Translations identification — The case of images



Our method — Translations identification — The case of images



Our method — Translations identification — The case of images



Our method — Translations identification — Three properties

Injectivity

Translation operator

« Black hole »

$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

Our method — Translations identification — Three properties

Injectivity

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« Black hole »

$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

Restriction to the edges (EC)

$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

Our method — Translations identification — Three properties

Injectivity

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Restriction to the edges (EC)

$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

Preservation of the neighborhood (SNP)

$$\begin{aligned} \forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \\ \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp) \end{aligned}$$

Our method — Translations identification — Three properties

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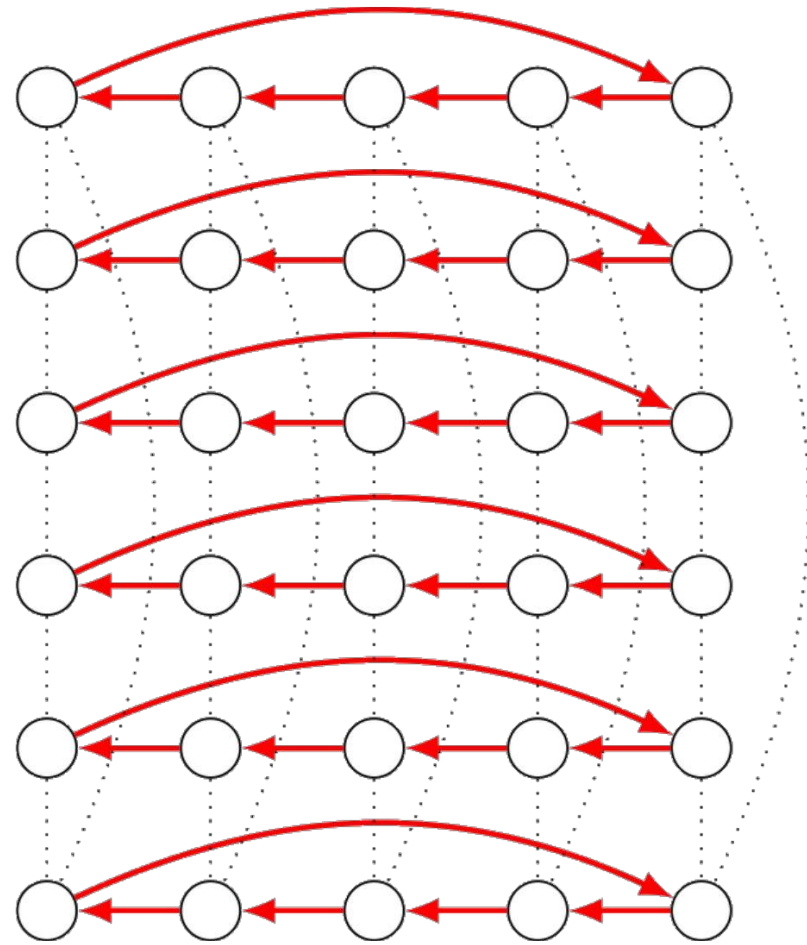
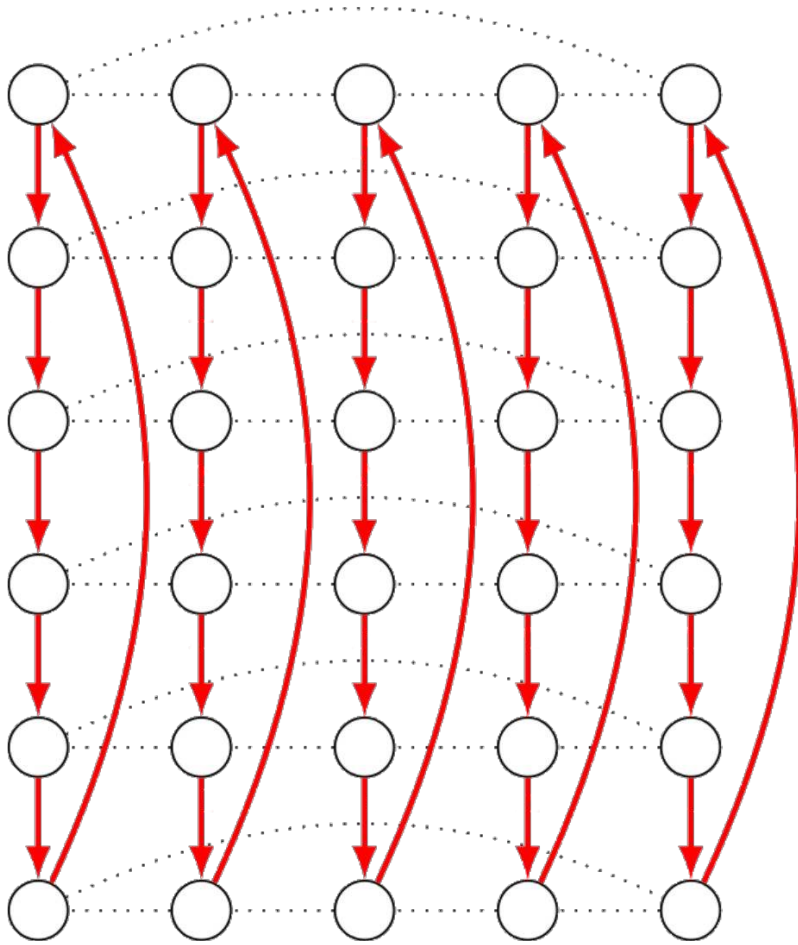
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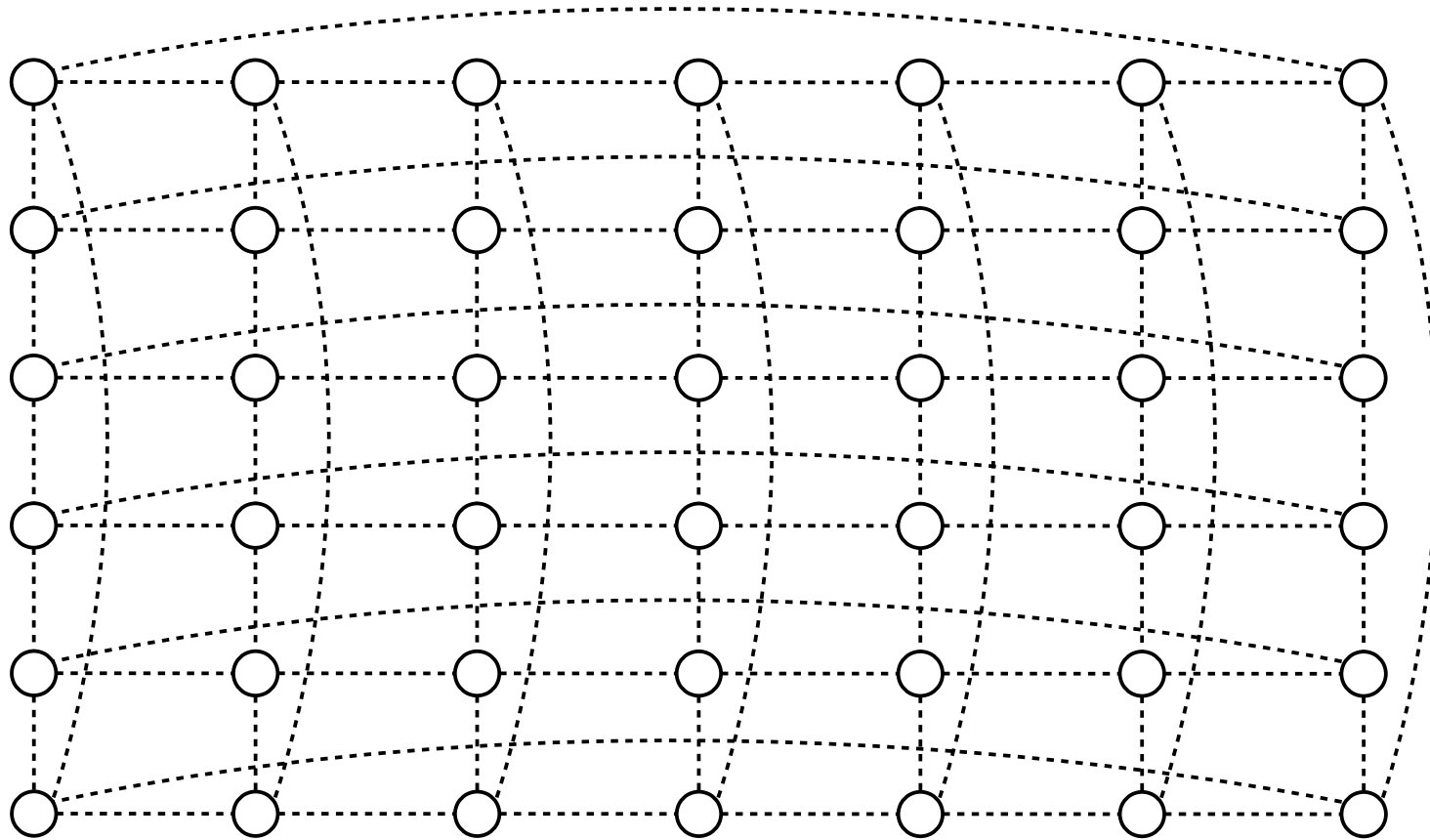
$$\forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp)$$

Preserves locality/structure of a signal
Perfect for convolutional kernels

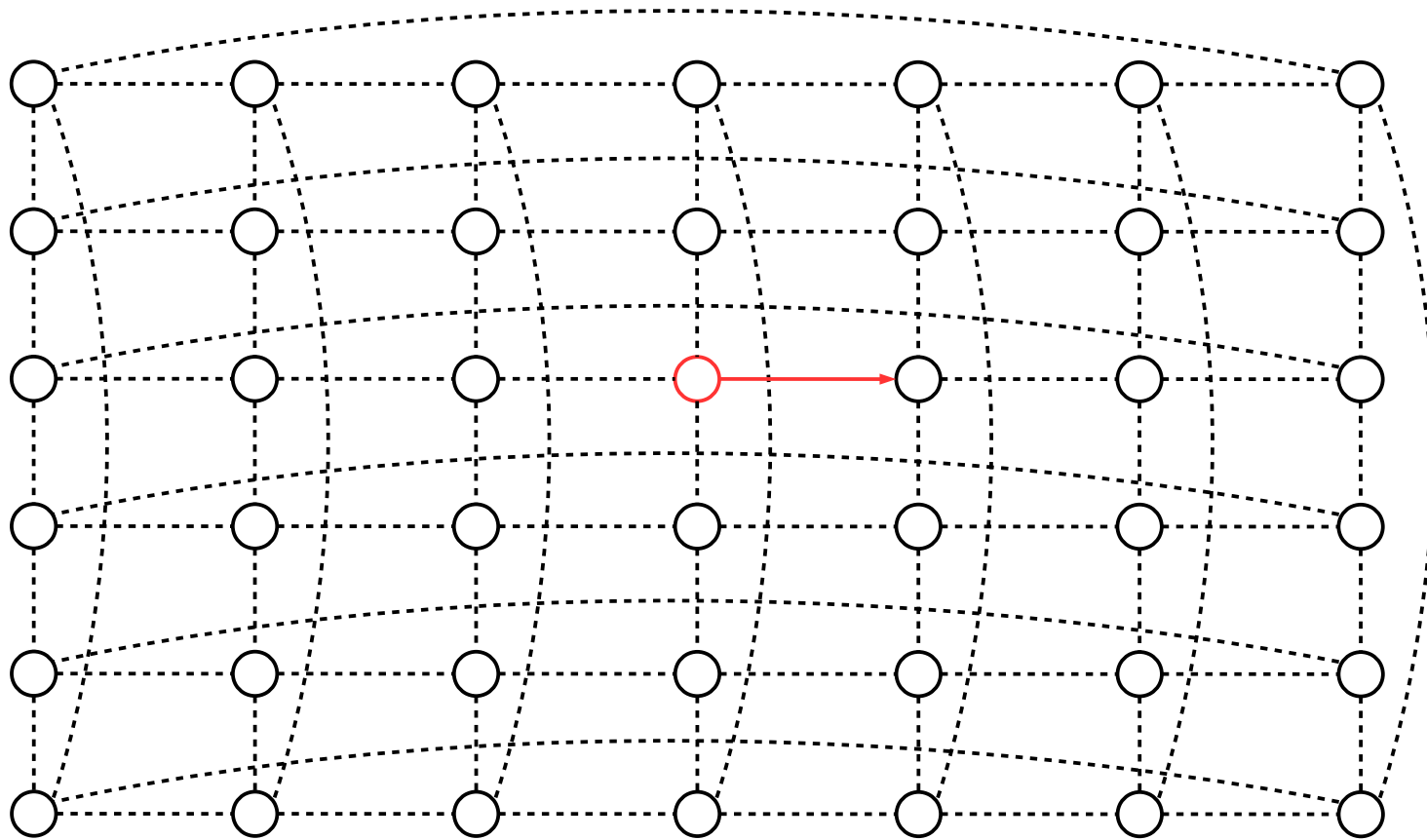
Our method — Translations identification — On the torus graph



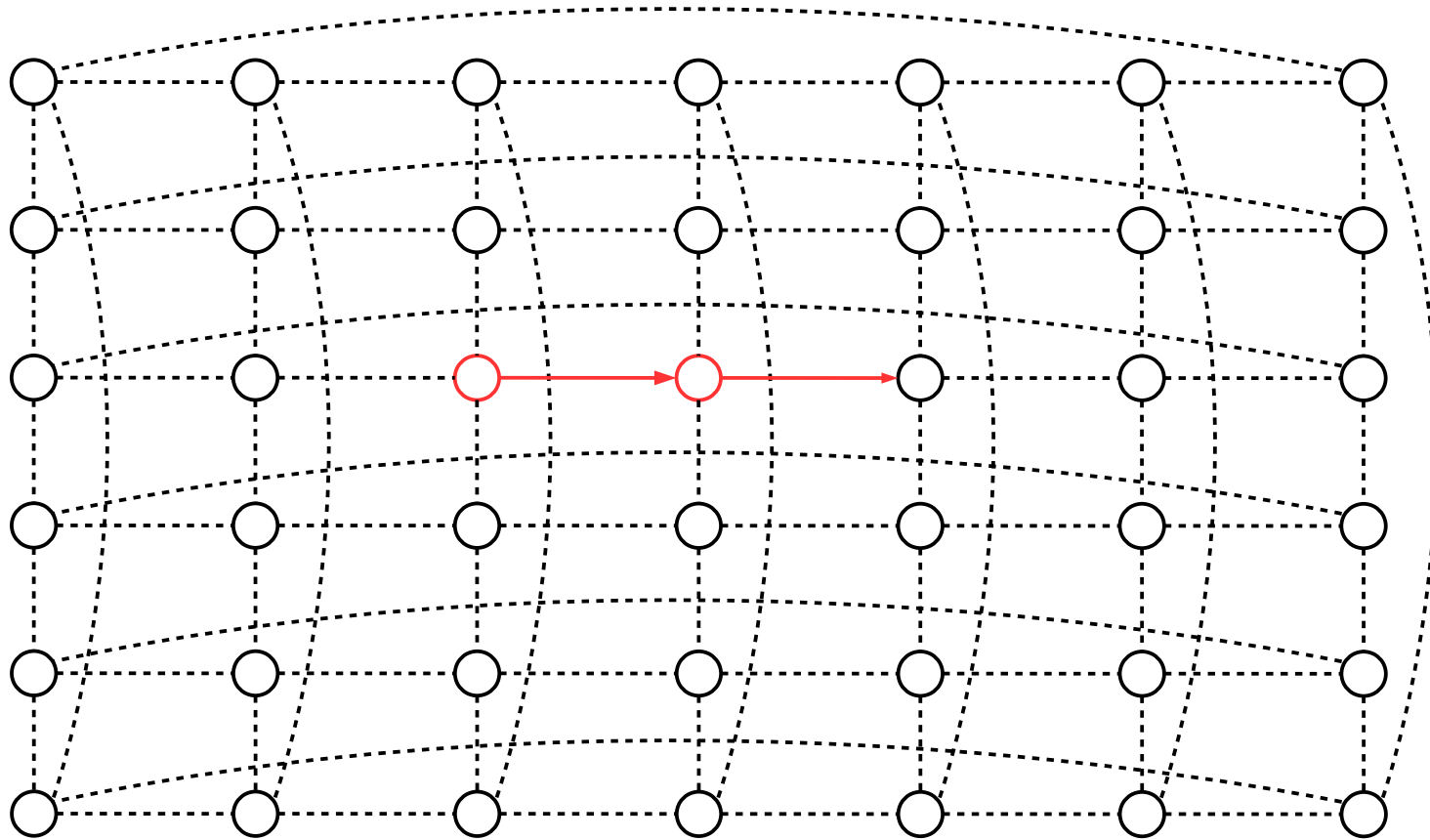
Our method — Translations identification — Sketch of the proof



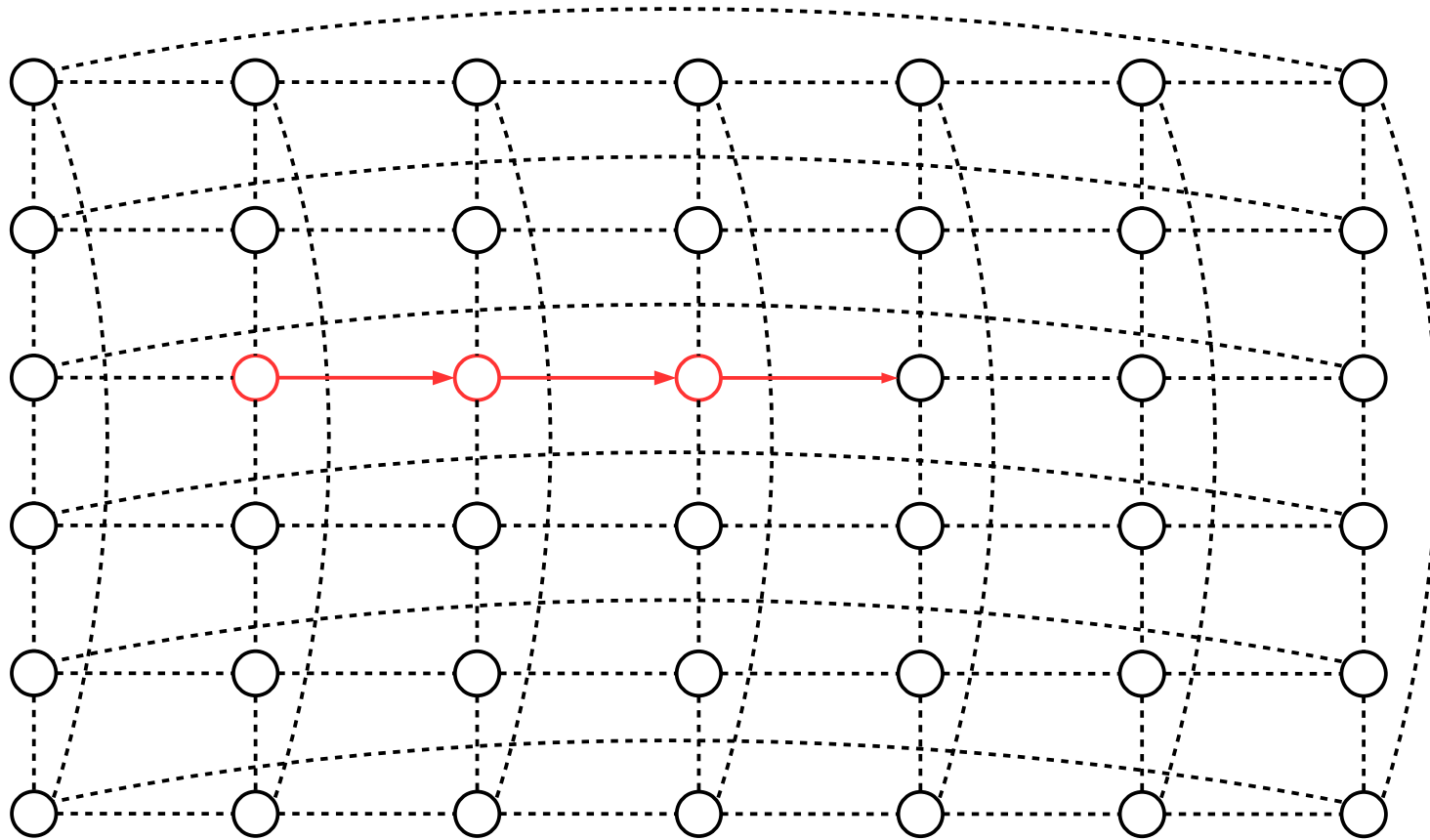
Our method — Translations identification — Sketch of the proof



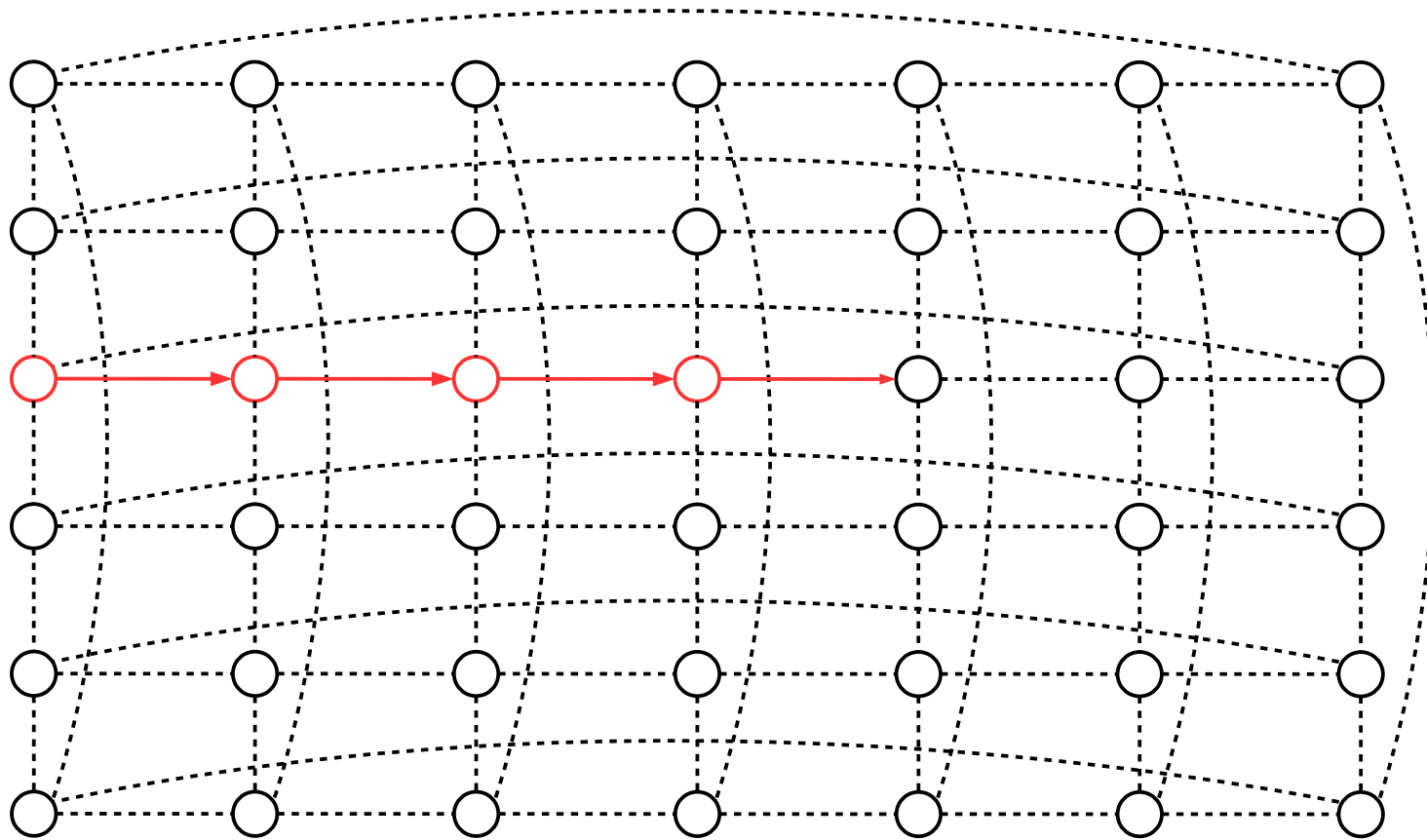
Our method — Translations identification — Sketch of the proof



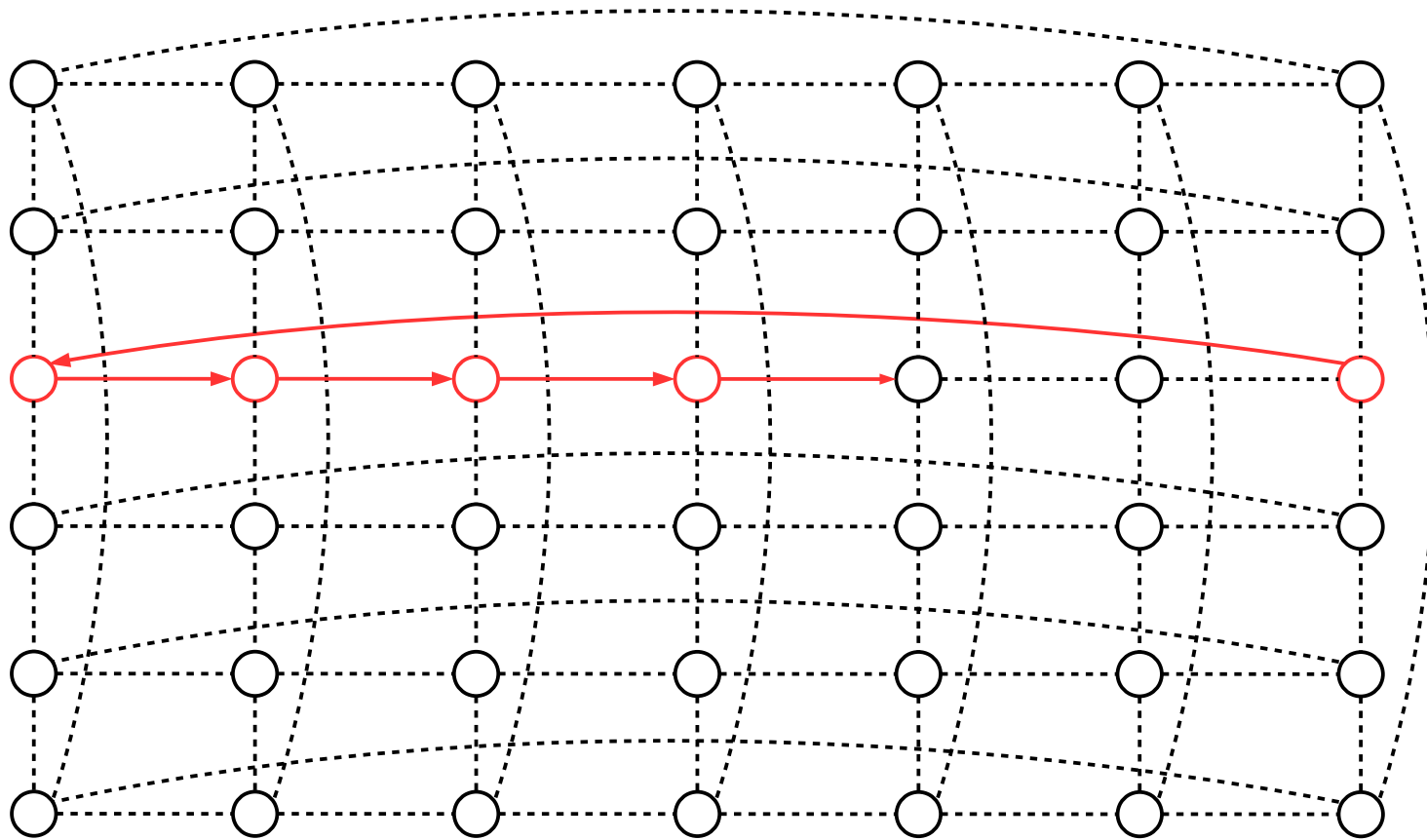
Our method — Translations identification — Sketch of the proof



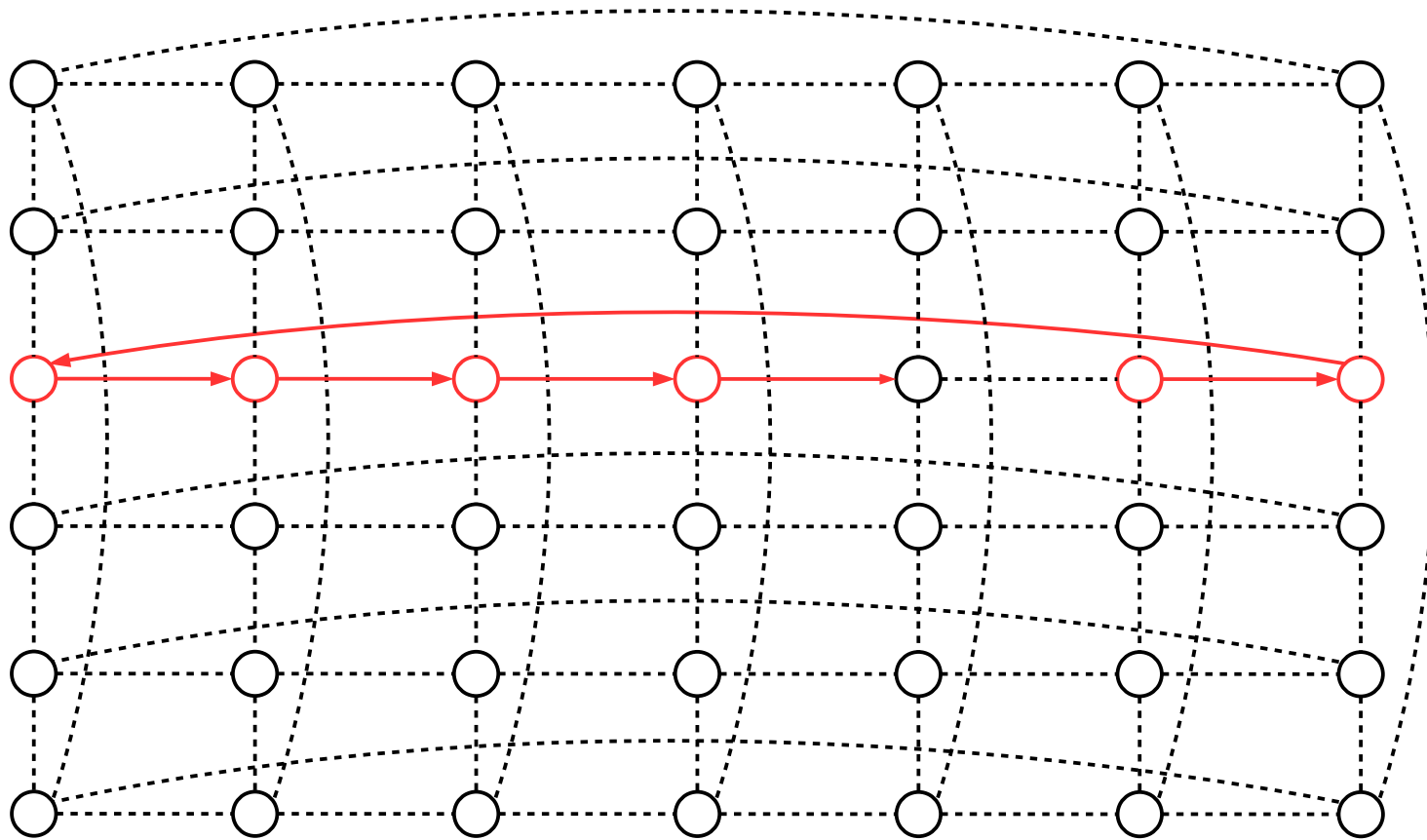
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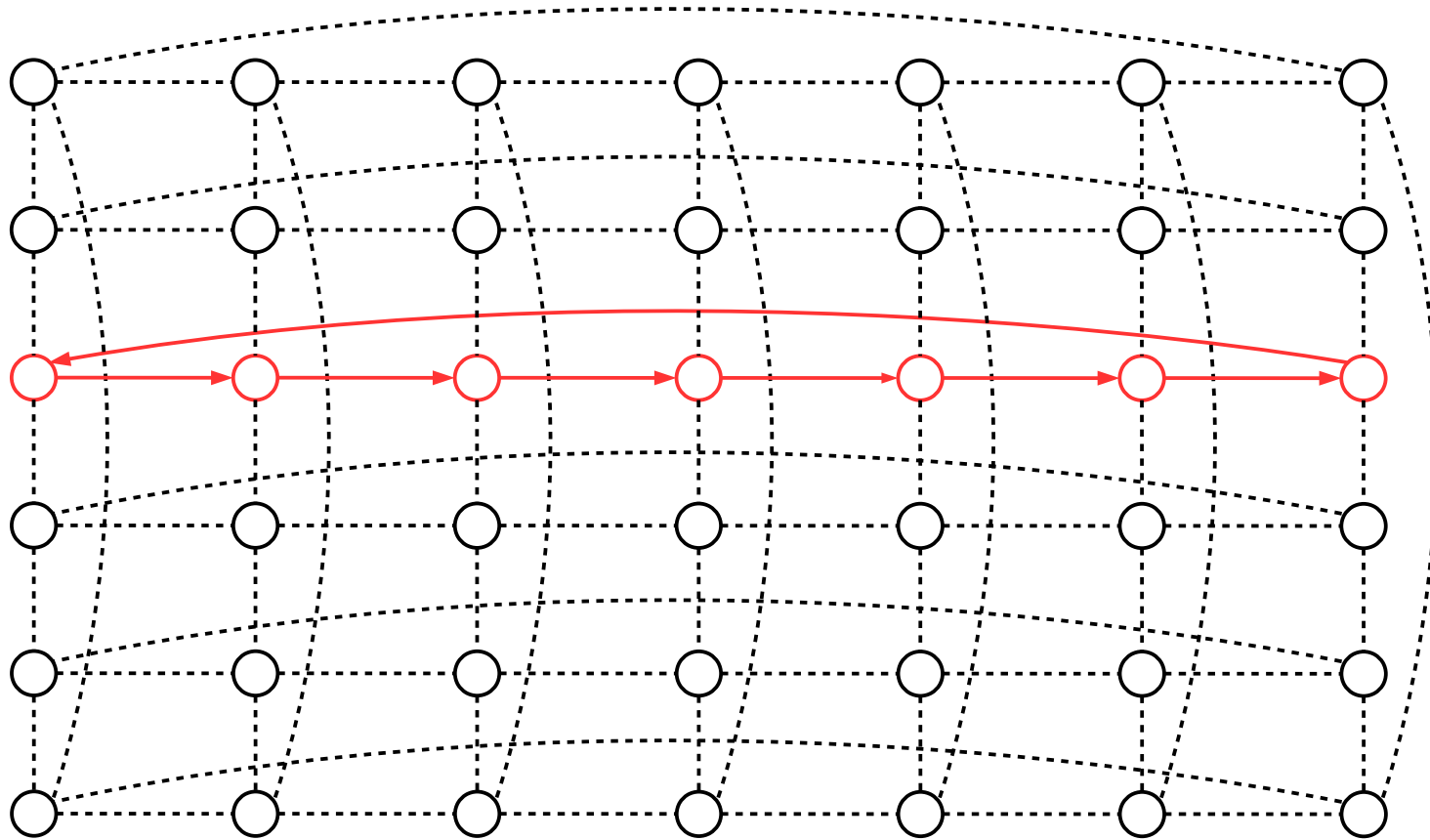
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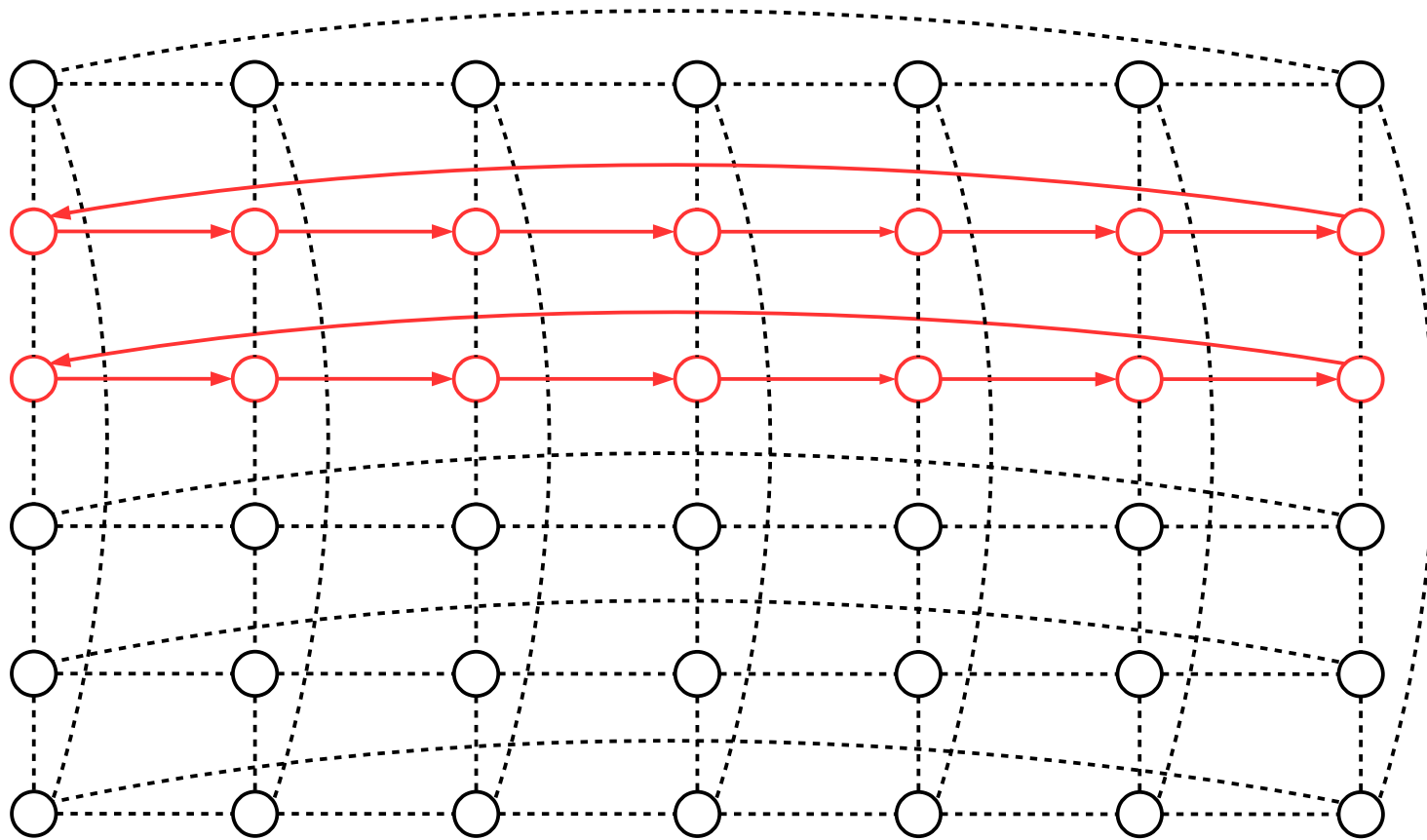
Our method — Translations identification — Sketch of the proof



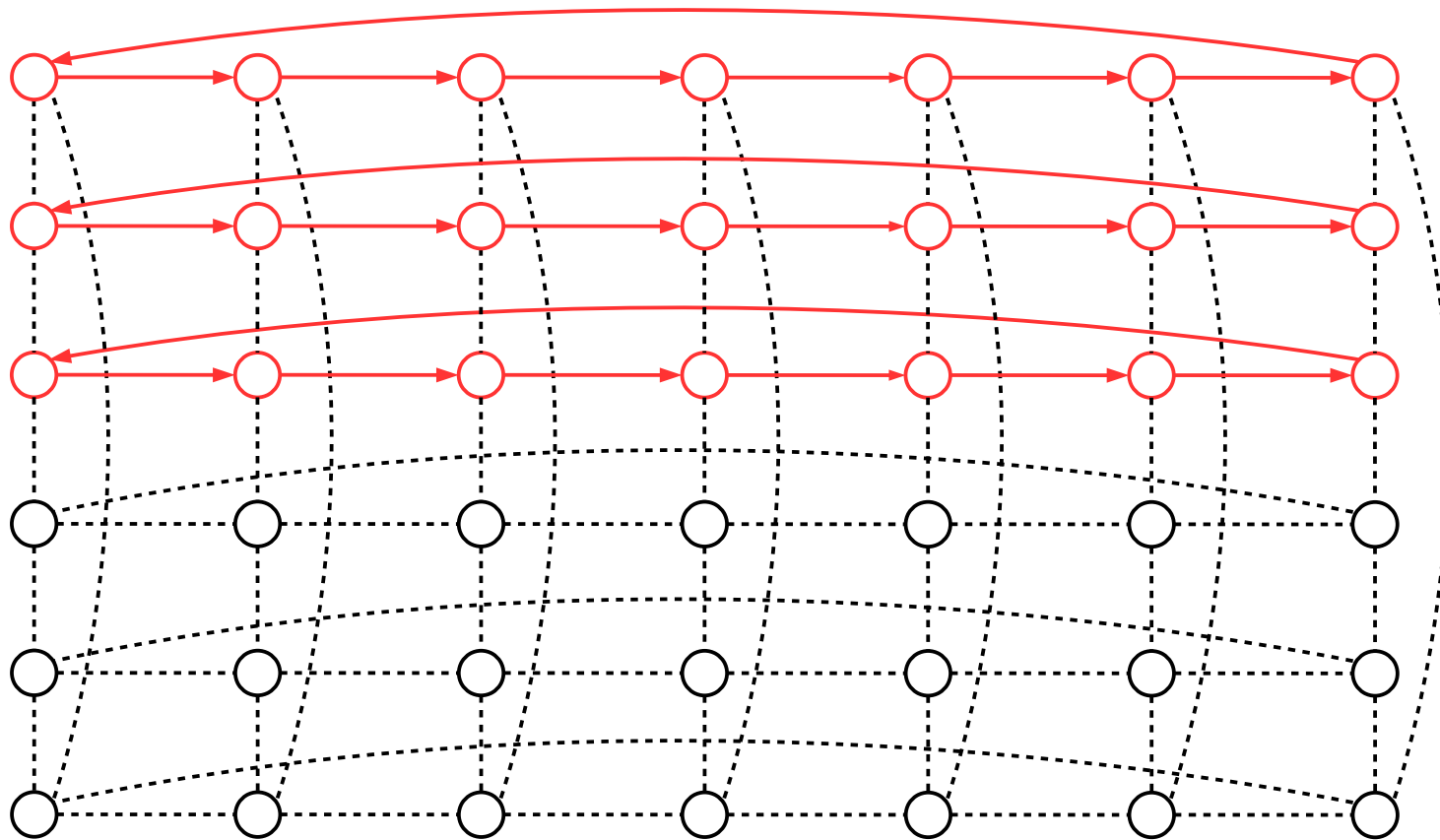
Our method — Translations identification — Sketch of the proof



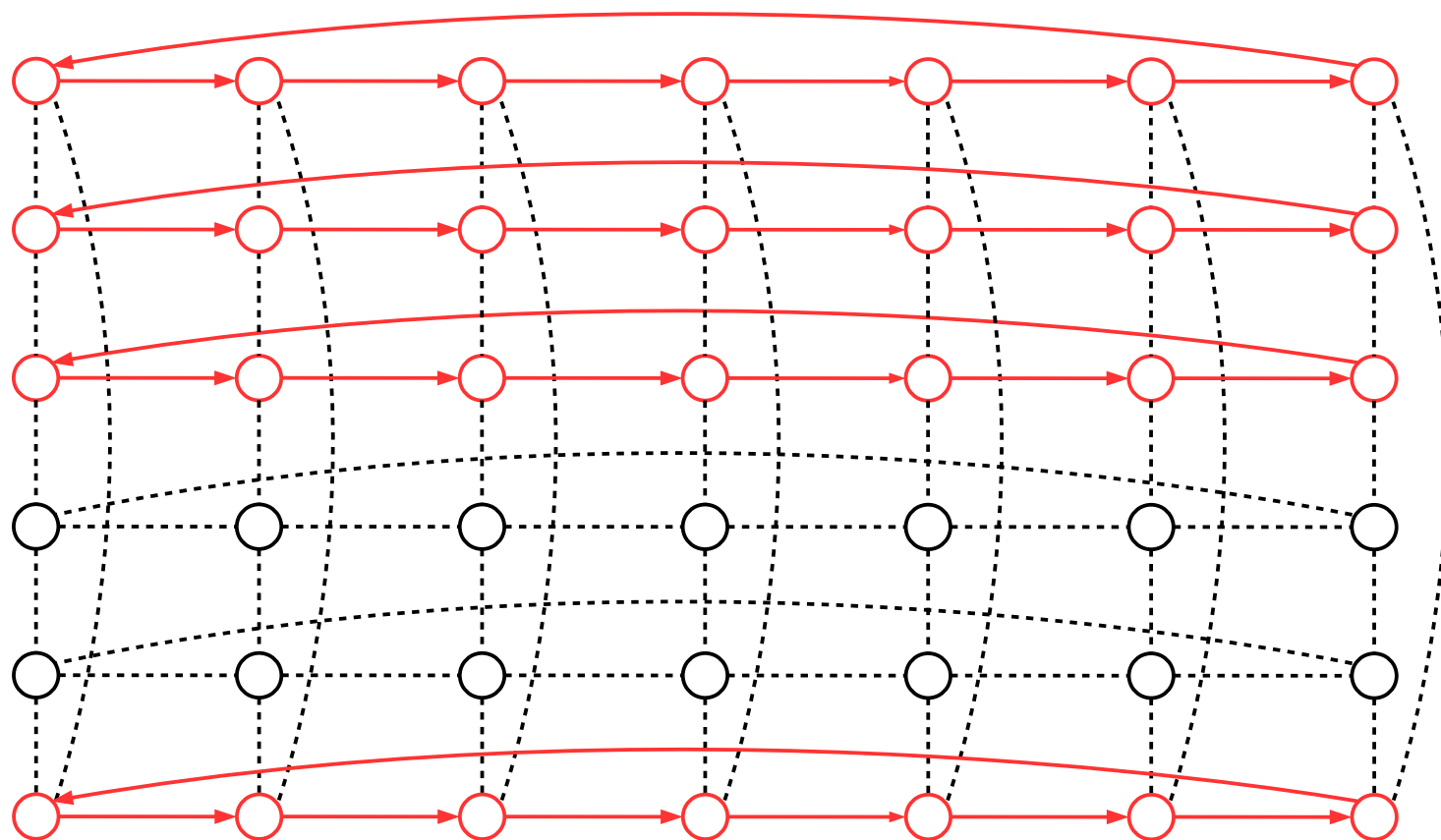
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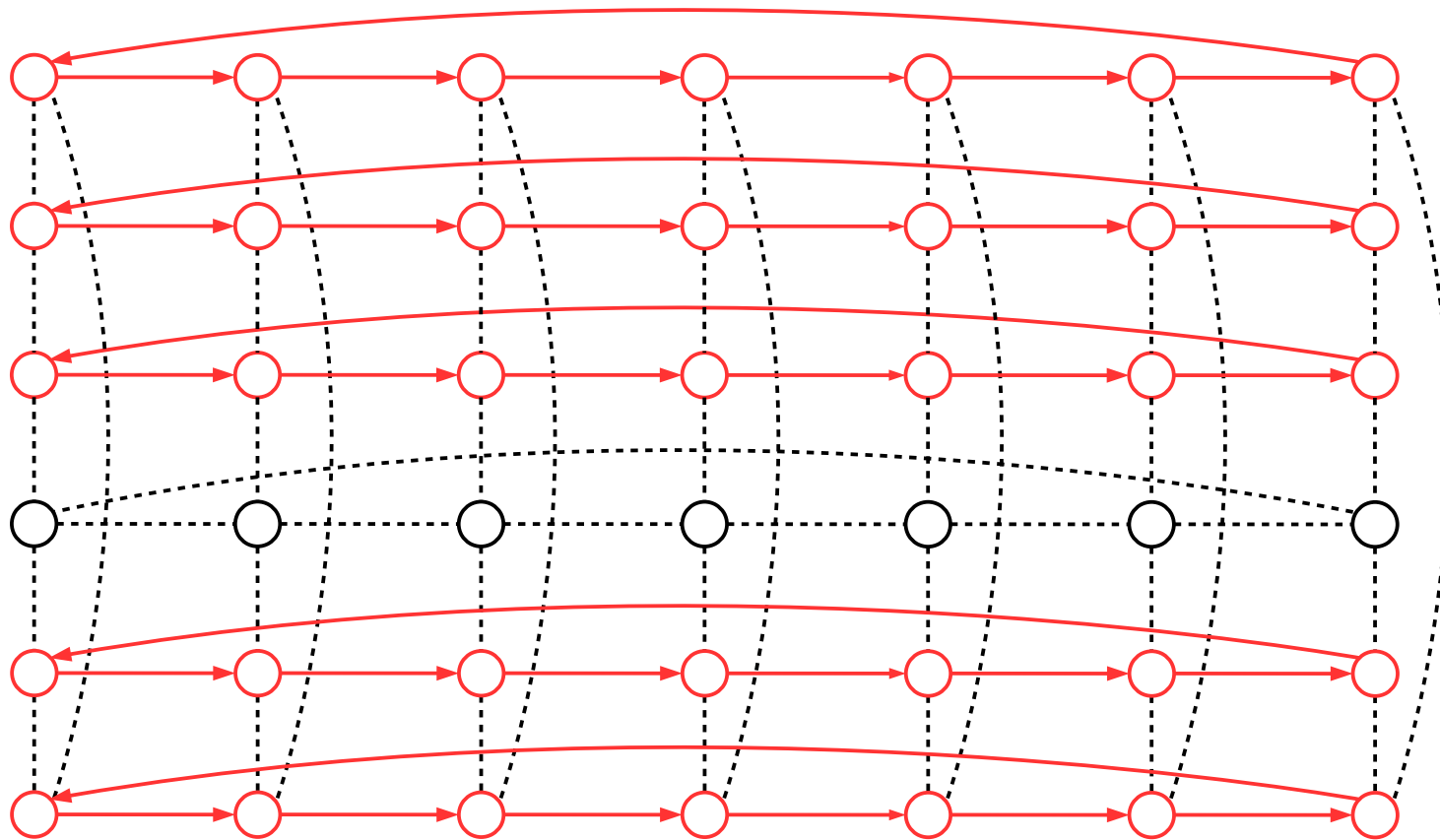
Our method — Translations identification — Sketch of the proof



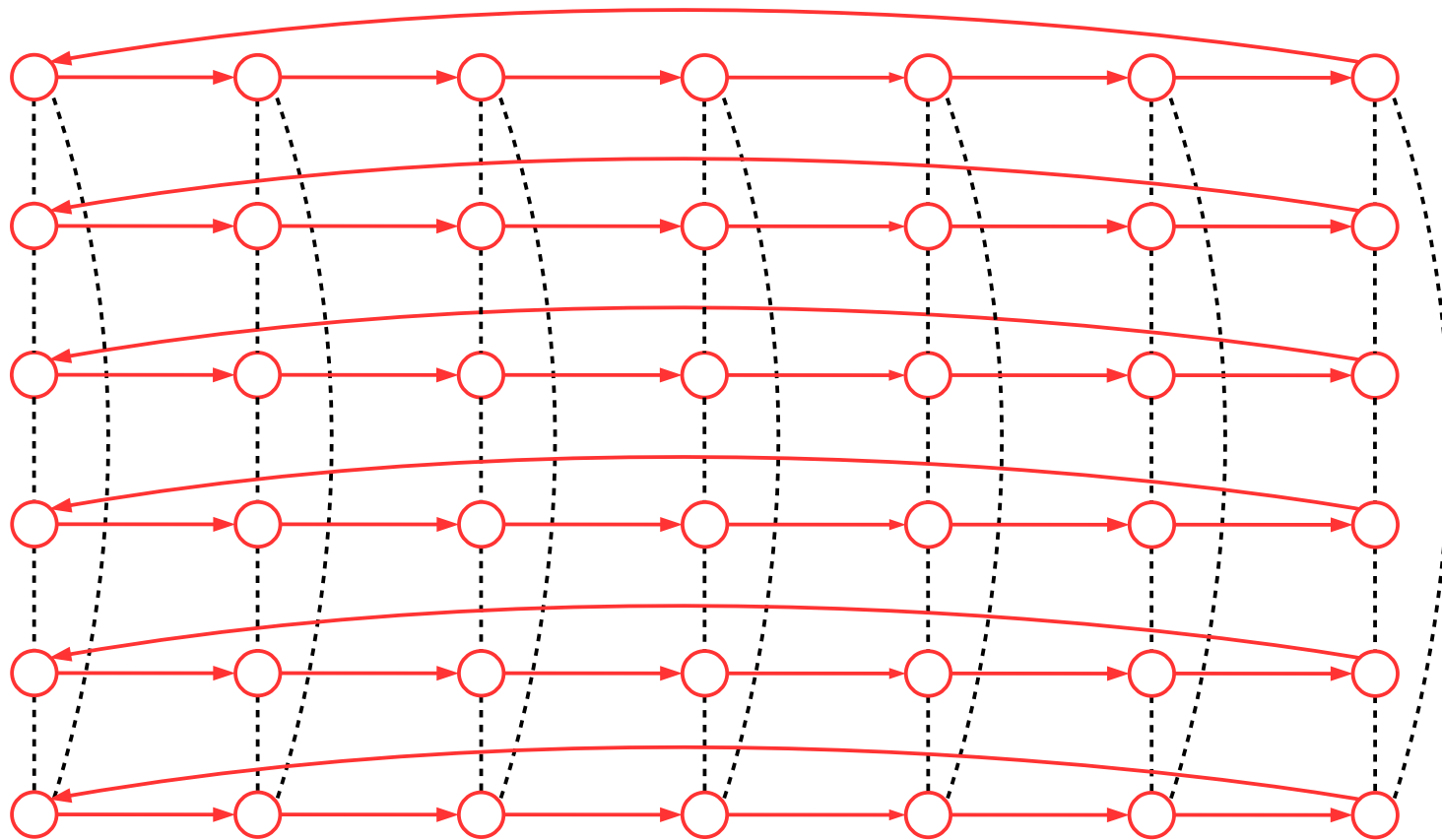
Our method — Translations identification — Sketch of the proof



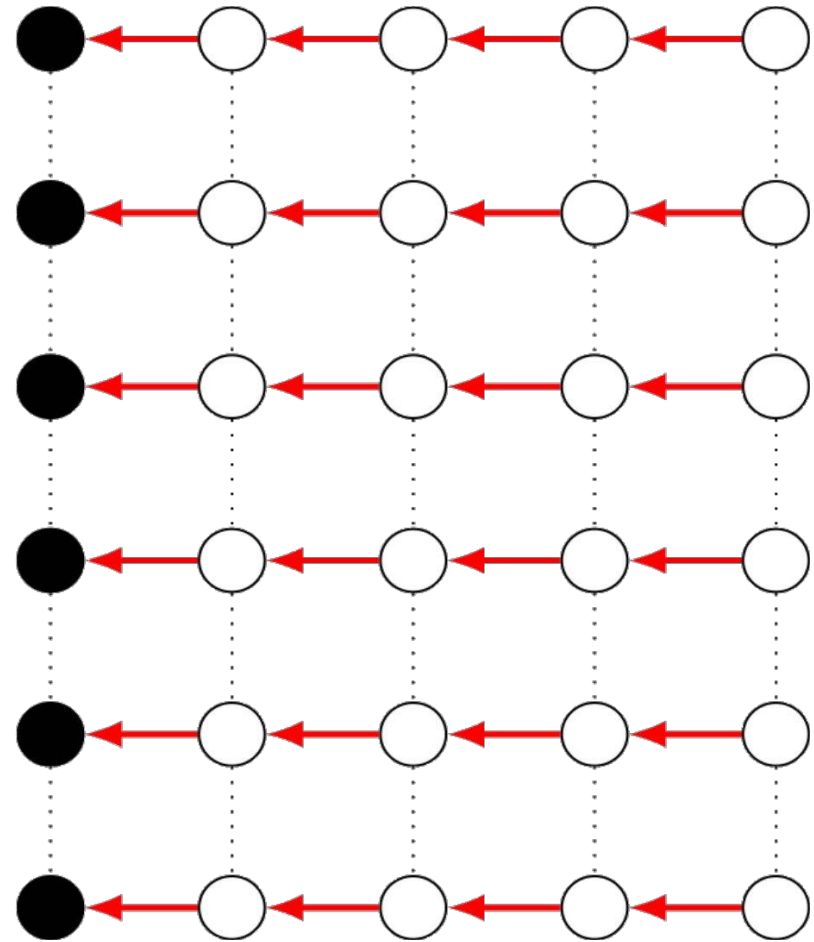
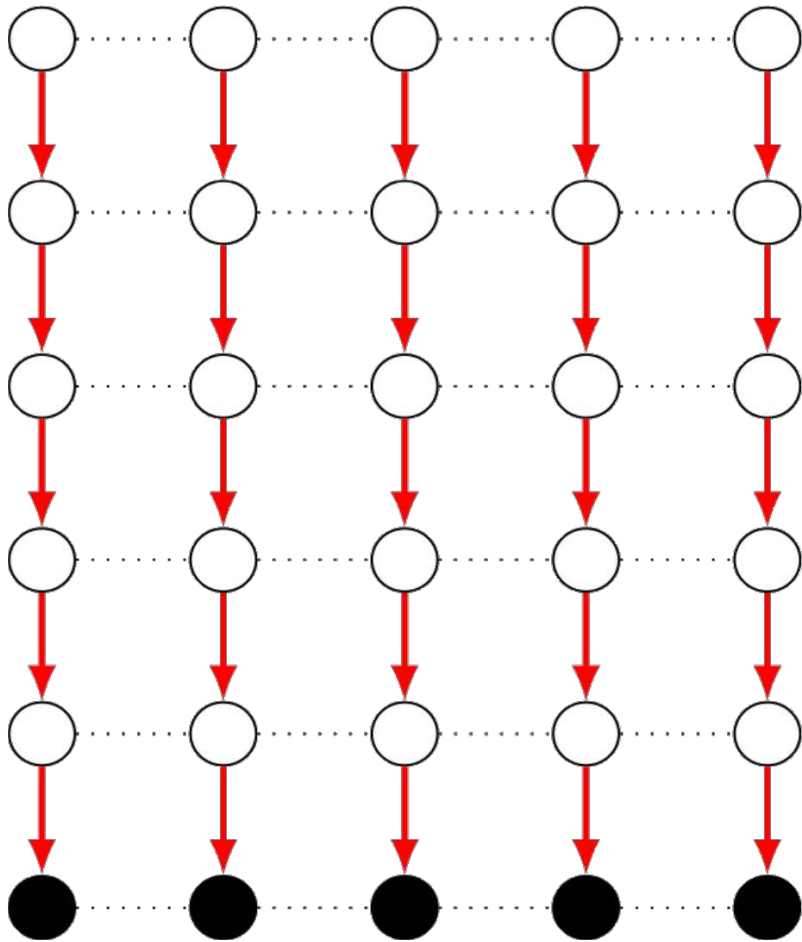
Our method — Translations identification — Sketch of the proof



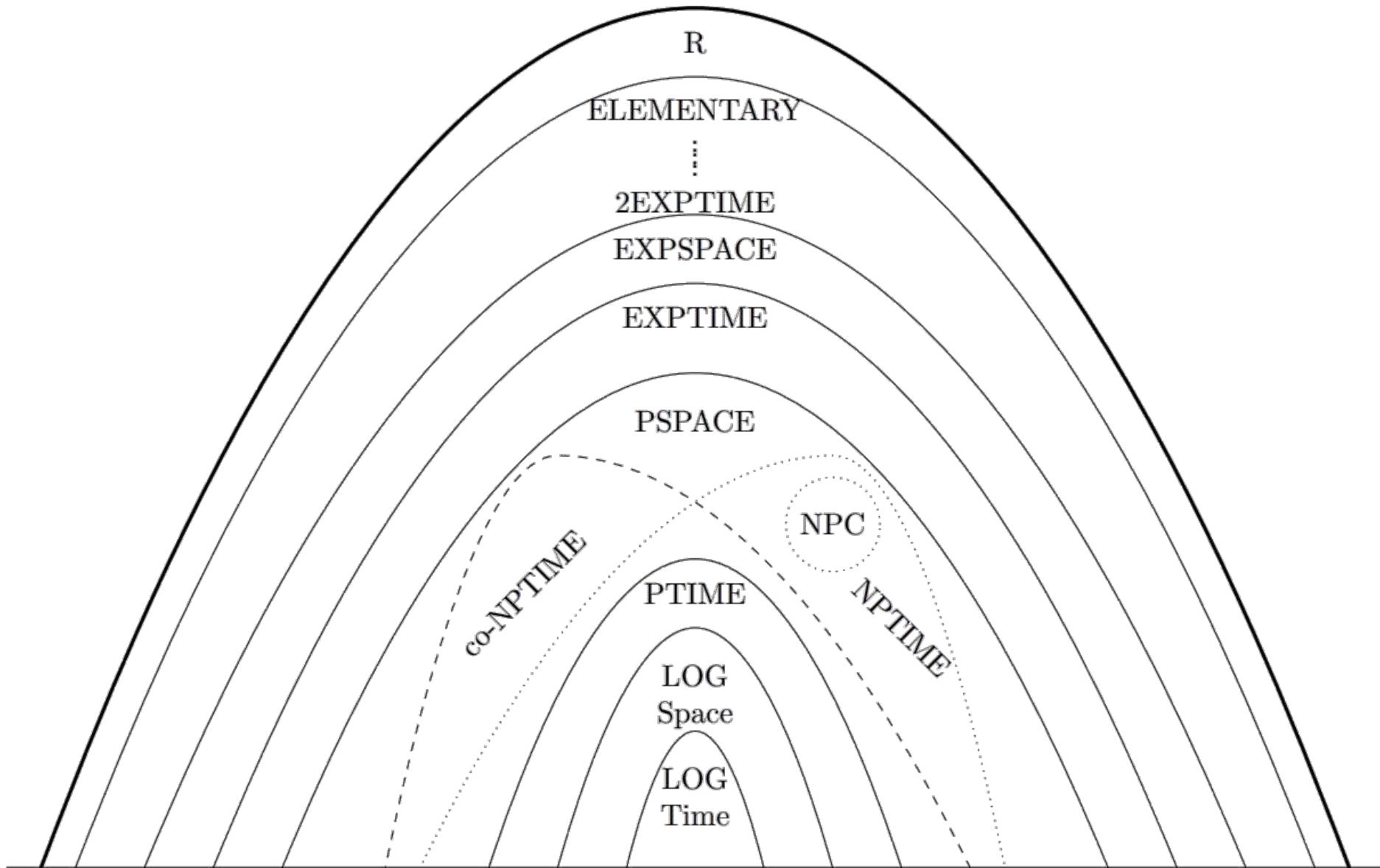
Our method — Translations identification — Sketch of the proof



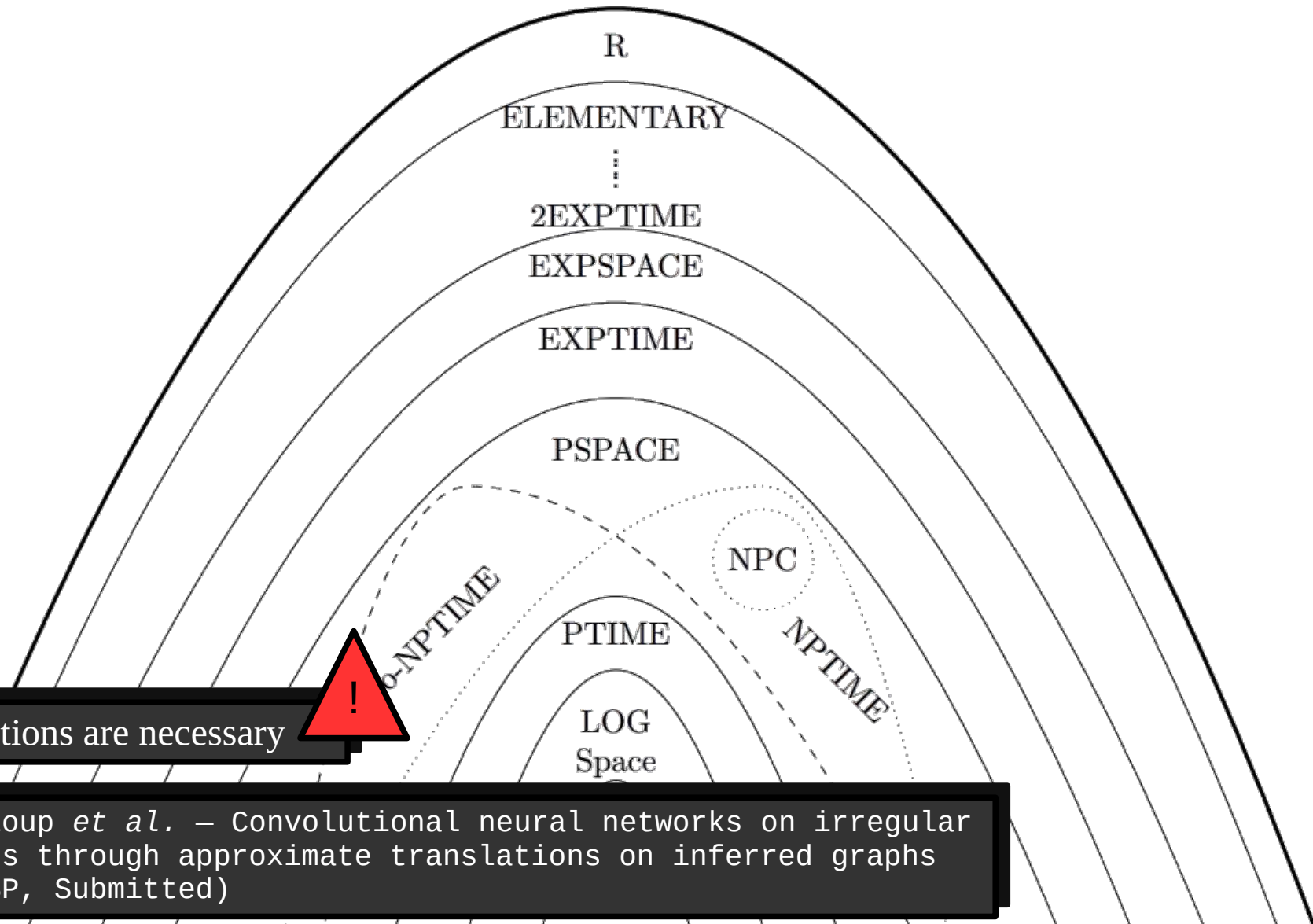
Our method — Translations identification — On the grid graph



Our method — Translations identification — A difficult problem



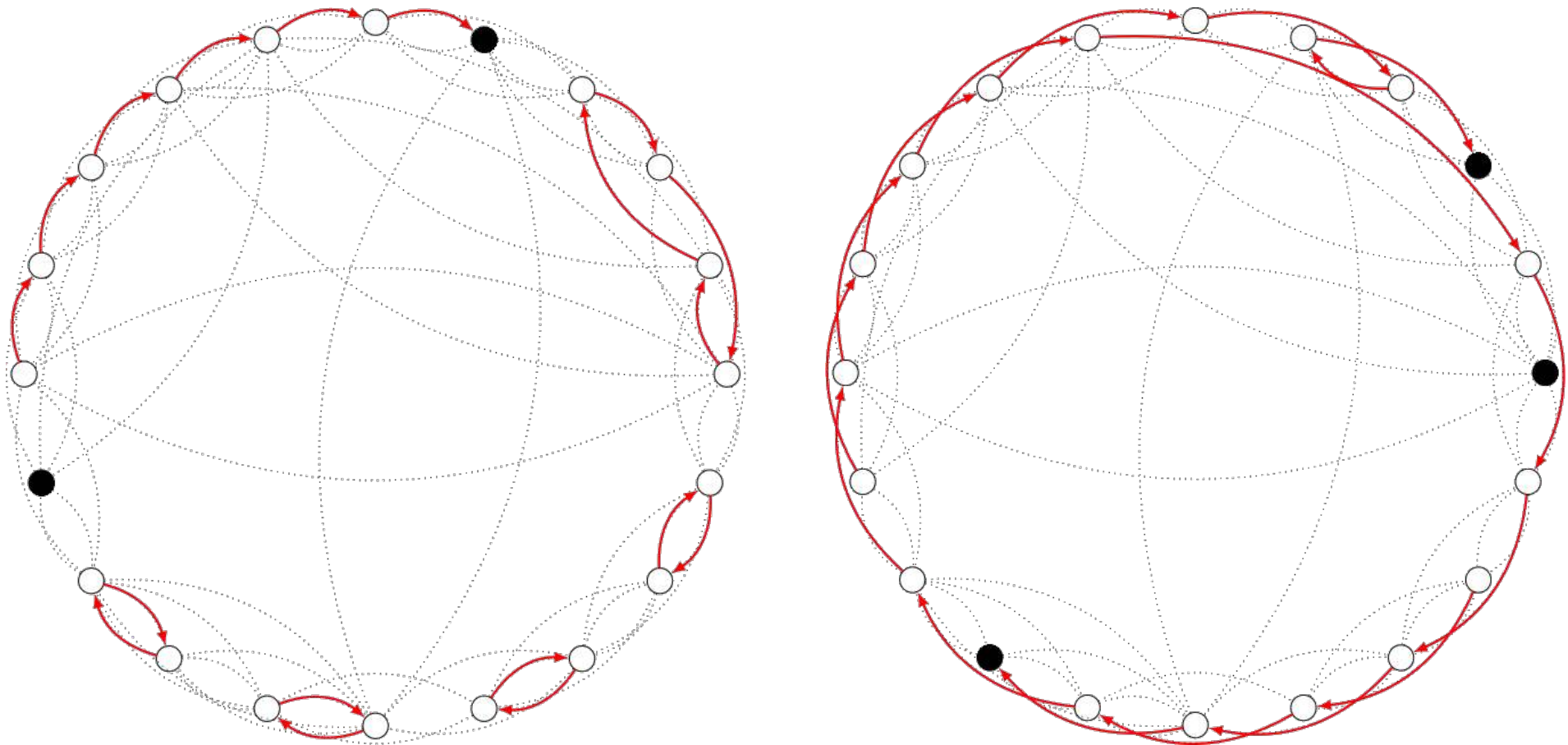
Our method — Translations identification — A difficult problem



Relaxations are necessary

Pasdeloup *et al.* — Convolutional neural networks on irregular domains through approximate translations on inferred graphs (ICASSP, Submitted)

Our method — Translations identification — Watts-Strogatz model



Outline

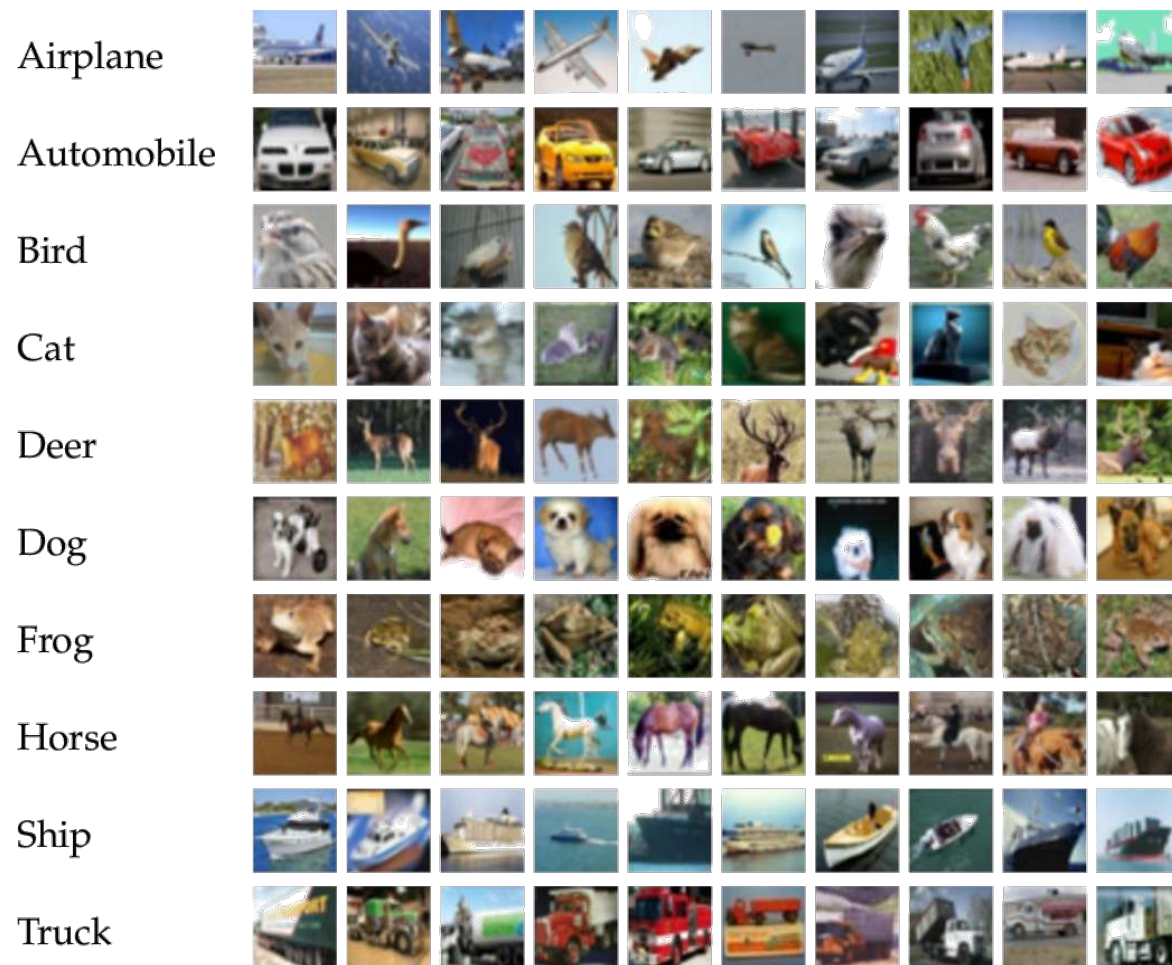
Existing methods & required notions

The method we propose

- Global idea of the method —
- Graph inference from data —
- Identification of translations —

Results

Results — The CIFAR-10 dataset



Dense
72.7 %

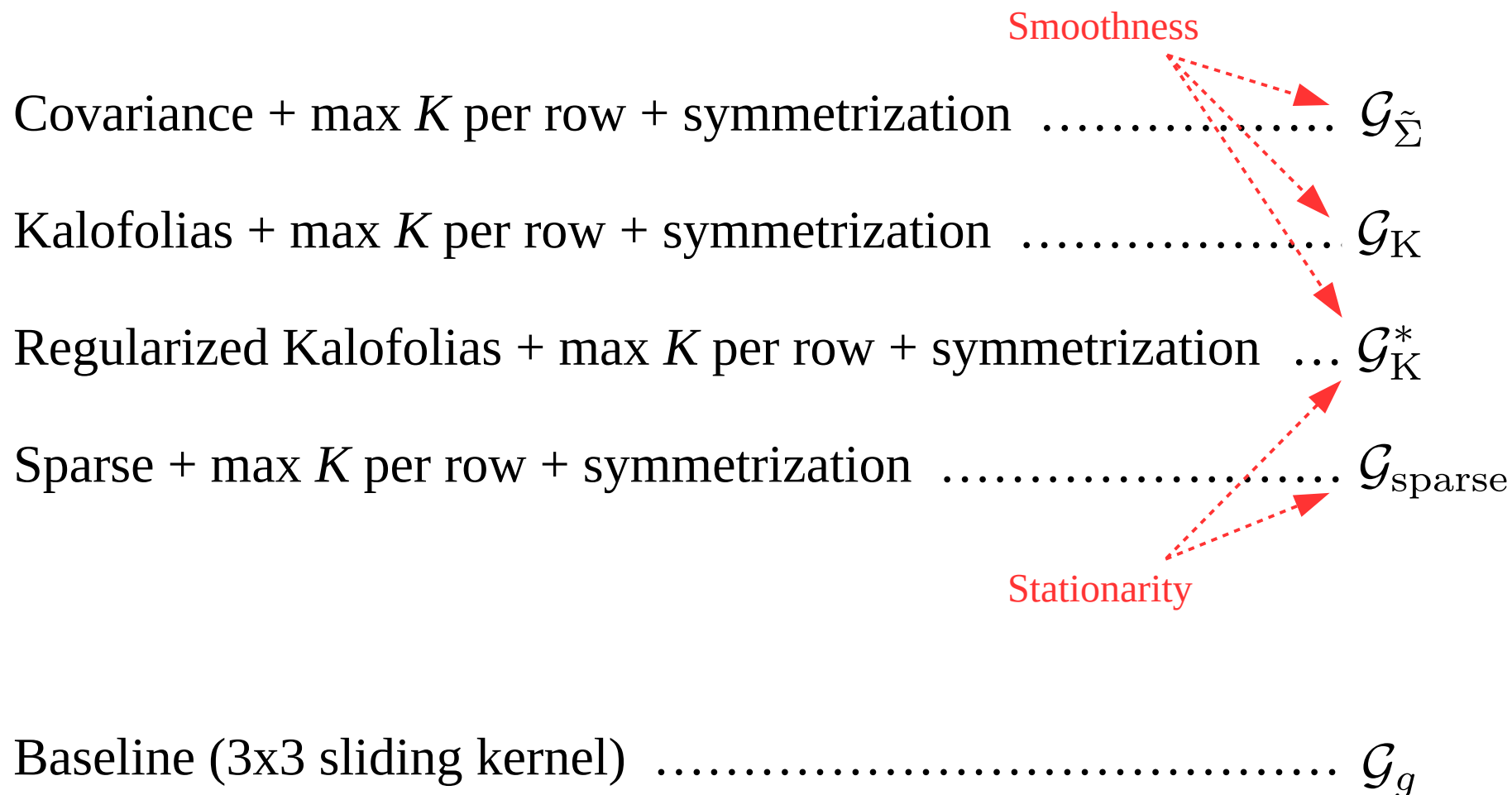
Convolutional
94.81 %

Data augmentation
97.28 %

Results — The graph inference methods used

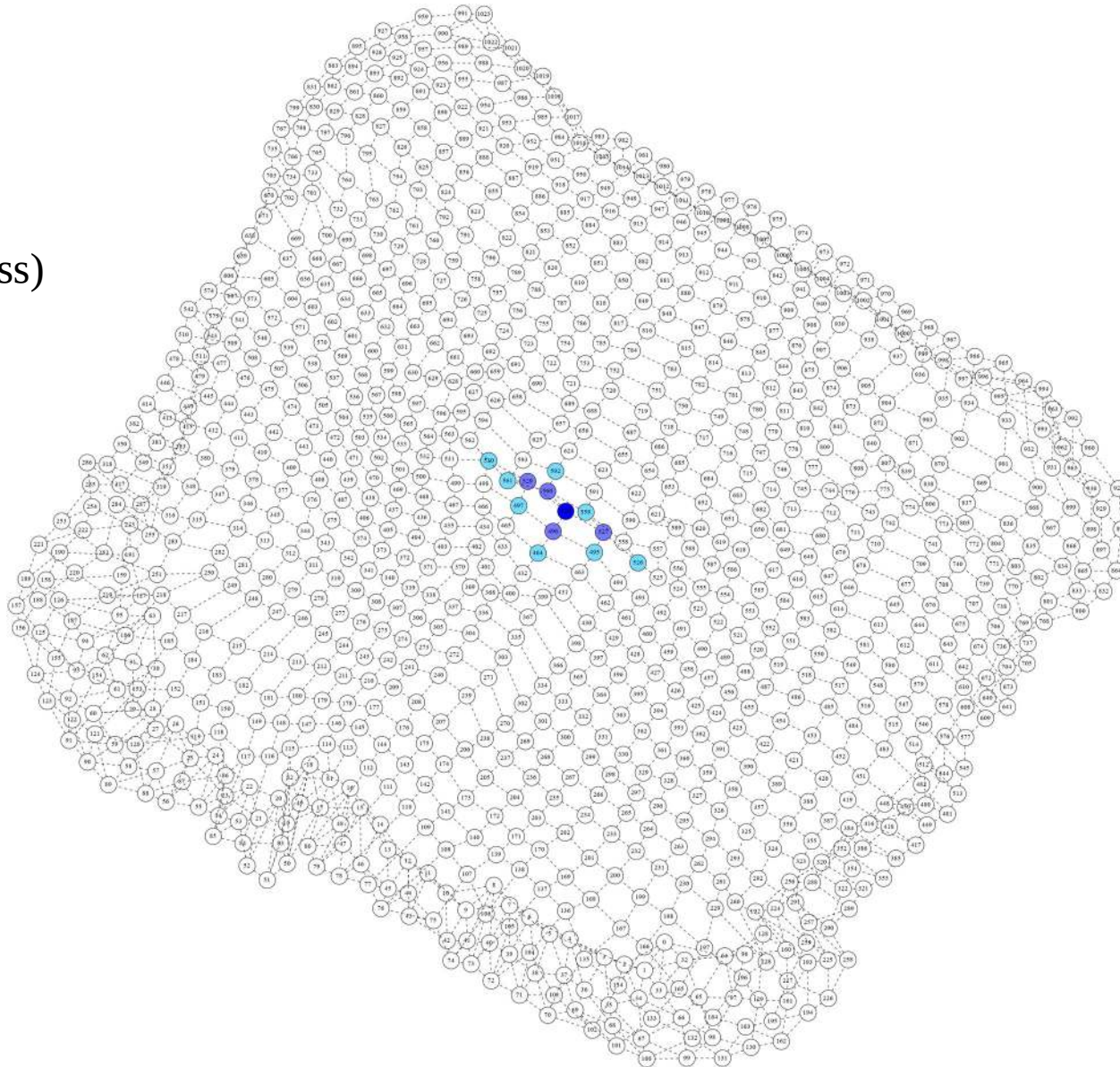
Covariance + max K per row + symmetrization	$\mathcal{G}_{\tilde{\Sigma}}$
Kalofolias + max K per row + symmetrization	$\mathcal{G}_K$
Regularized Kalofolias + max K per row + symmetrization ...	$\mathcal{G}_K^*$
Sparse + max K per row + symmetrization	$\mathcal{G}_{\text{sparse}}$
Baseline (3x3 sliding kernel)	$\mathcal{G}_g$

Results — The graph inference methods used



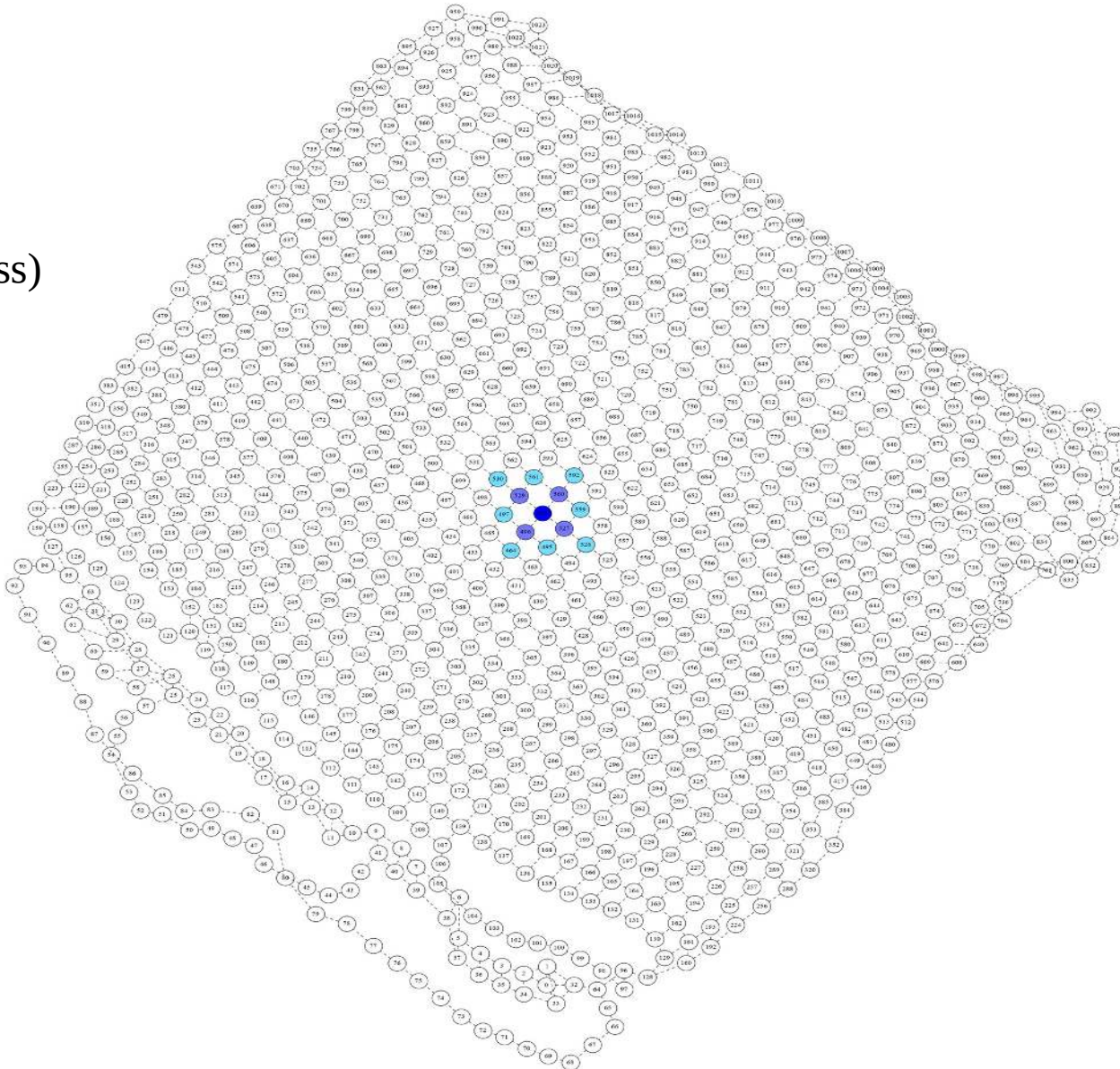
Results — Some of the inferred graphs

$\mathcal{G}_{\tilde{\Sigma}}$
(smoothness)



Results — Some of the inferred graphs

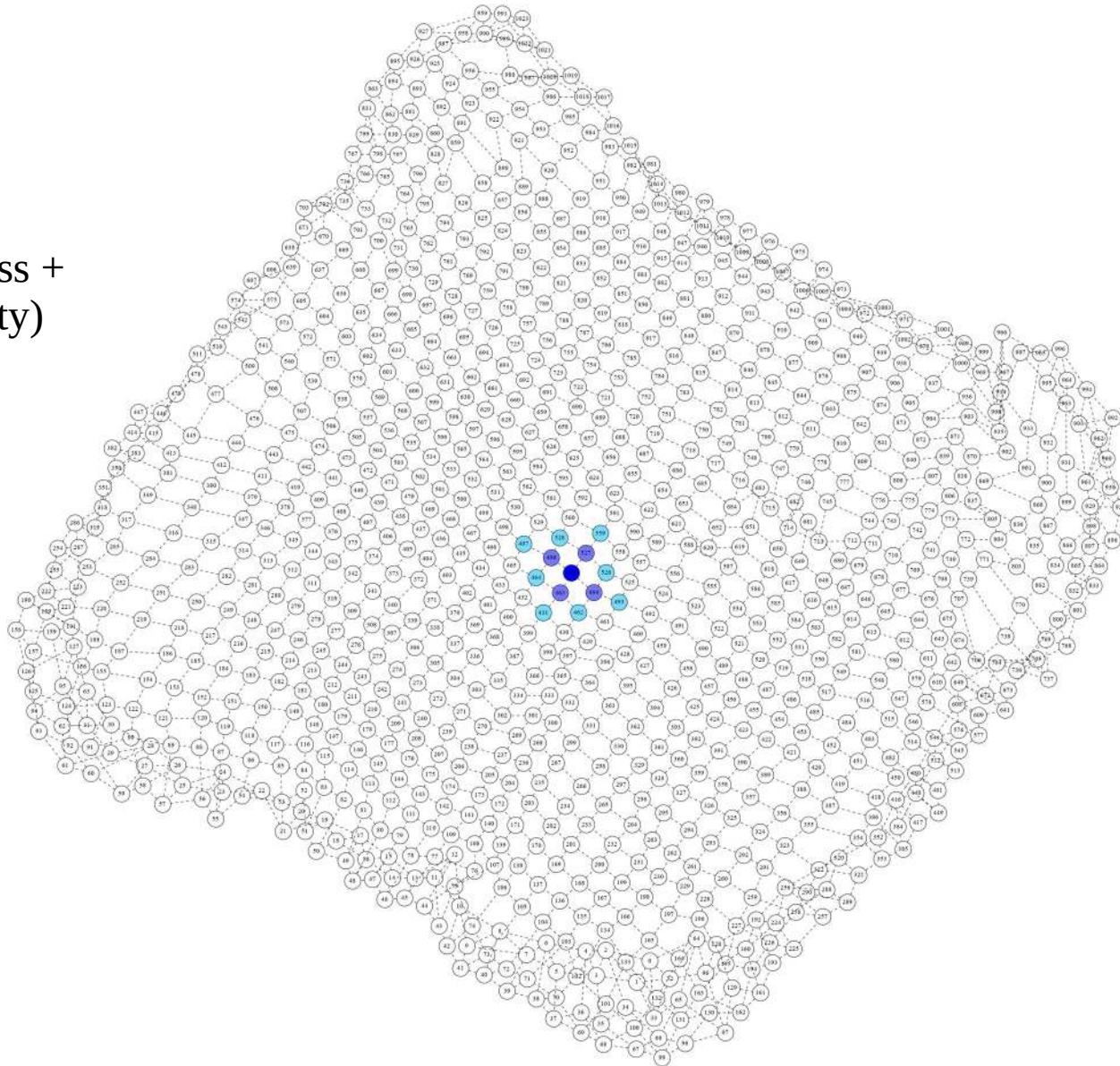
$\mathcal{G}_K$
(smoothness)



Results — Some of the inferred graphs

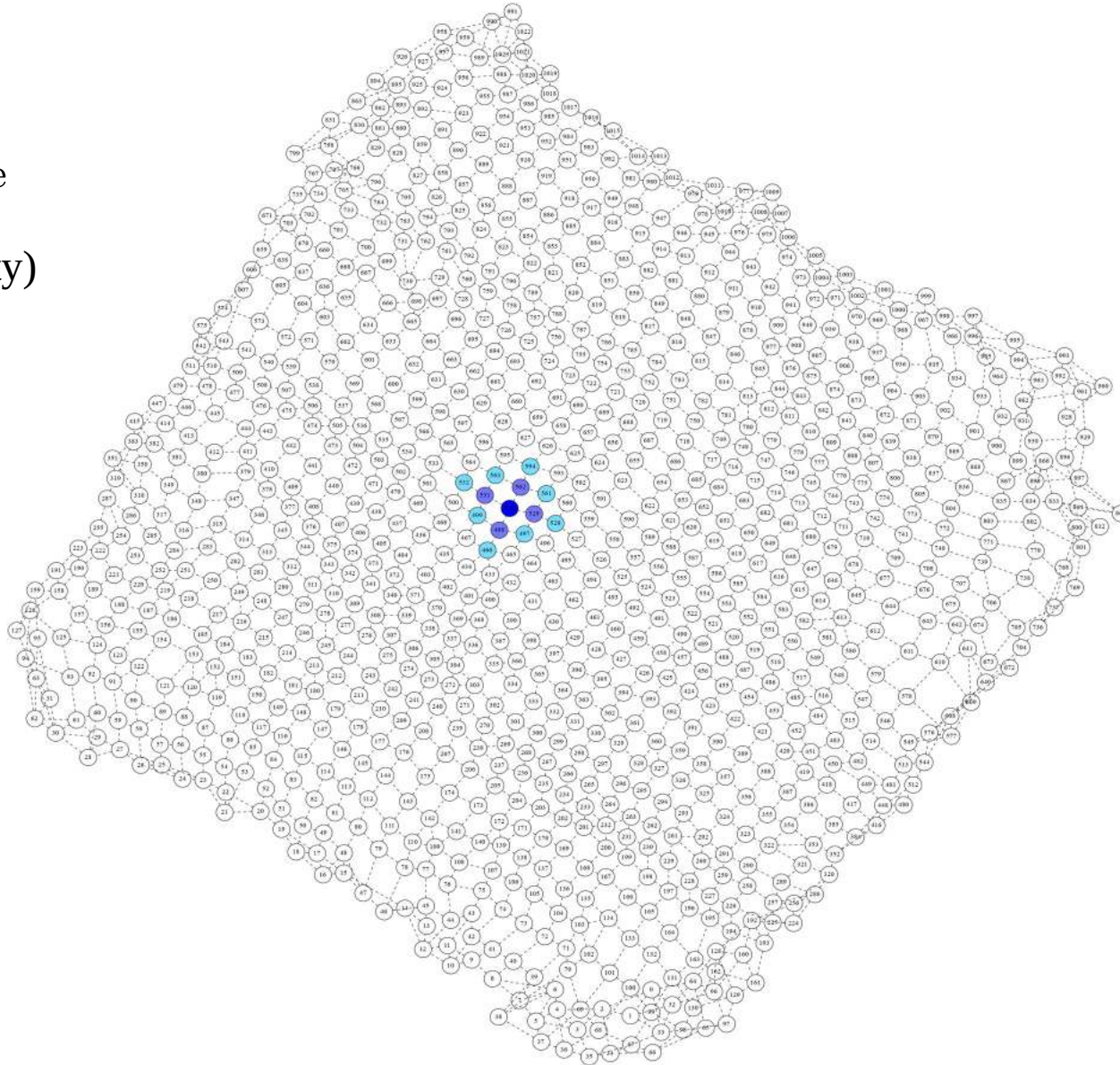
$\mathcal{G}_K^*$

(smoothness +
stationarity)



Results — Some of the inferred graphs

$\mathcal{G}_{\text{sparse}}$
(stationarity)



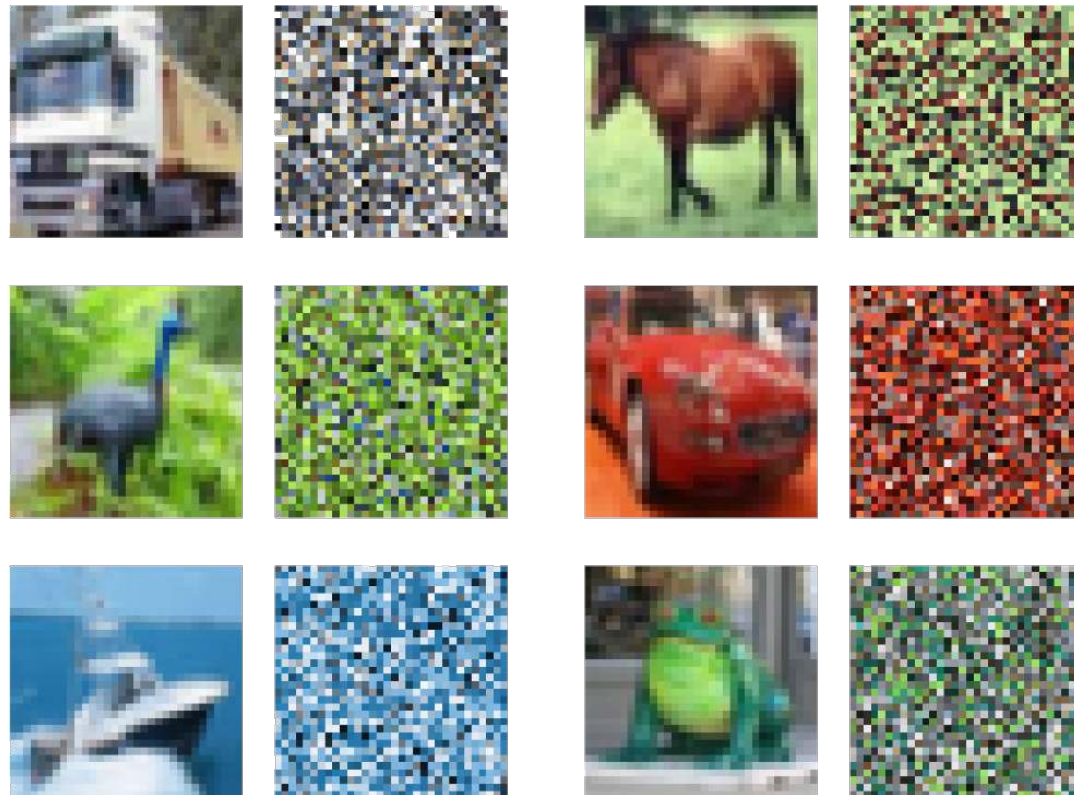
Results — Obtained results

$\mathcal{G}_g$	$\mathcal{G}_{\tilde{\Sigma}}$	$\mathcal{G}_K$	$\mathcal{G}_K^*$	$\mathcal{G}_{\text{sparse}}$
85.98%	85.03%	85.33%	85.81%	77.54%

Best without
using knowledge
72.7 %

Results — On the scrambled version of CIFAR-10

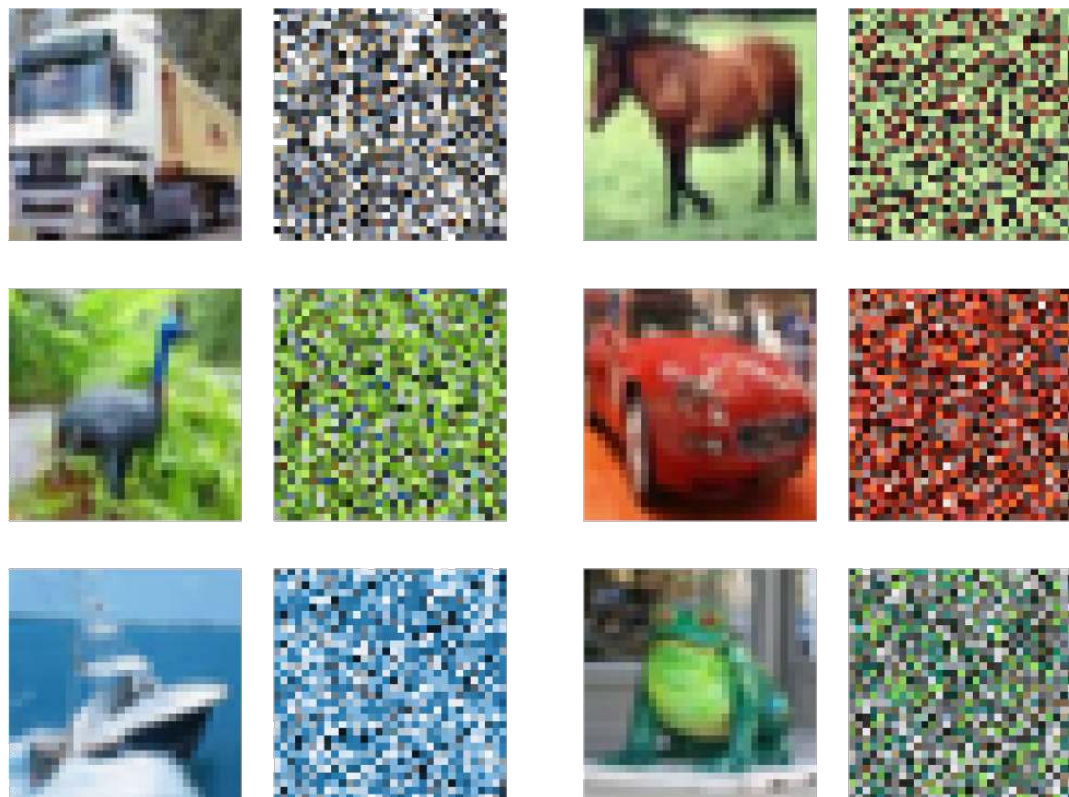
$\mathcal{G}_g$	$\mathcal{G}_{\tilde{\Sigma}}$	$\mathcal{G}_K$	$\mathcal{G}_K^*$	$\mathcal{G}_{\text{sparse}}$
85.98%	85.03%	85.33%	85.81%	77.54%



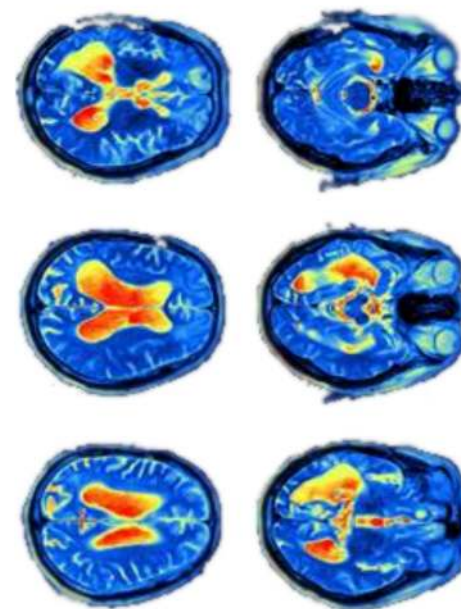
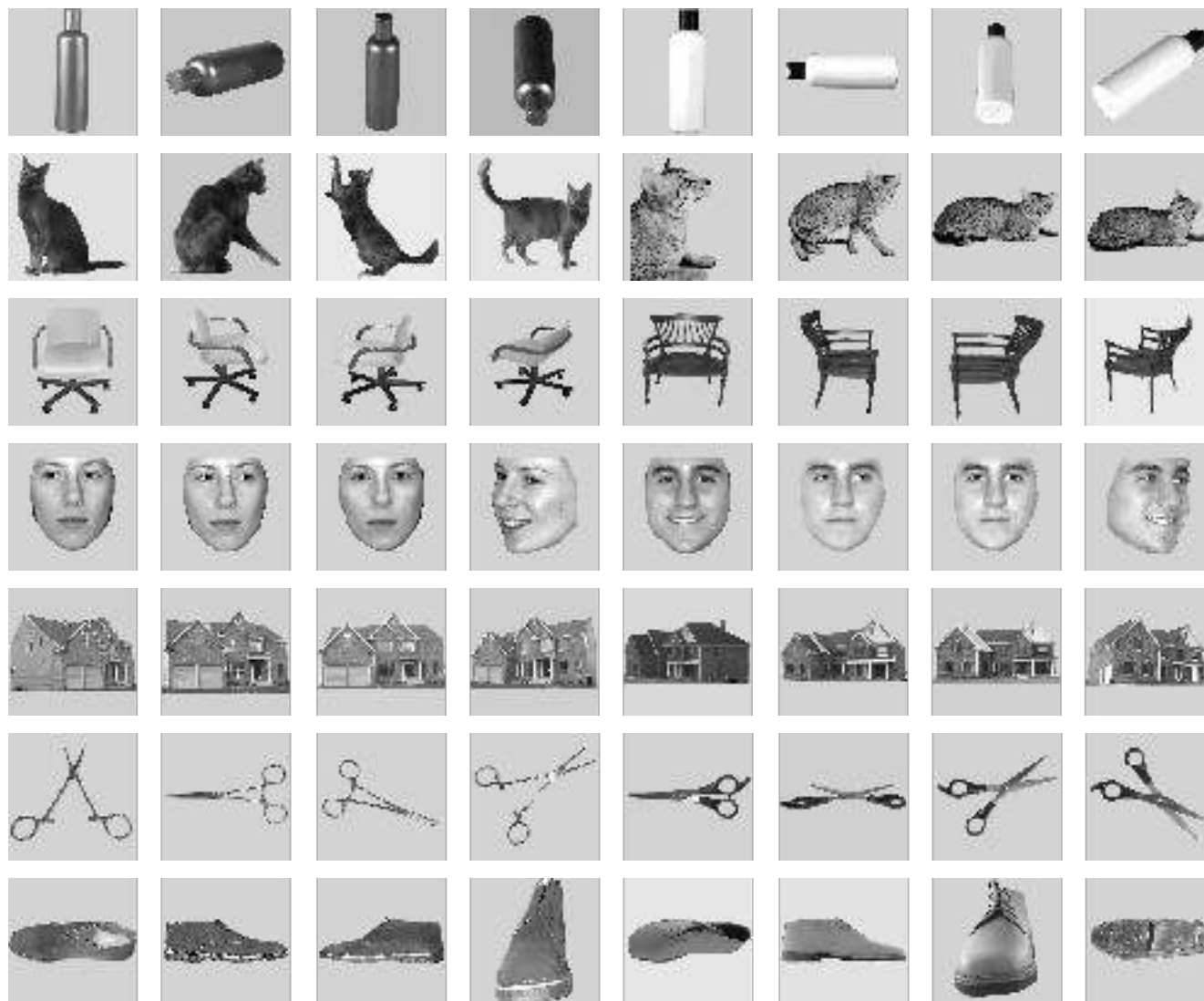
Results — On the scrambled version of CIFAR-10

$\mathcal{G}_g$	$\mathcal{G}_{\tilde{\Sigma}}$	$\mathcal{G}_K$	$\mathcal{G}_K^*$	$\mathcal{G}_{\text{sparse}}$
85.98%	85.03%	85.33%	85.81%	77.54%

56.74 %

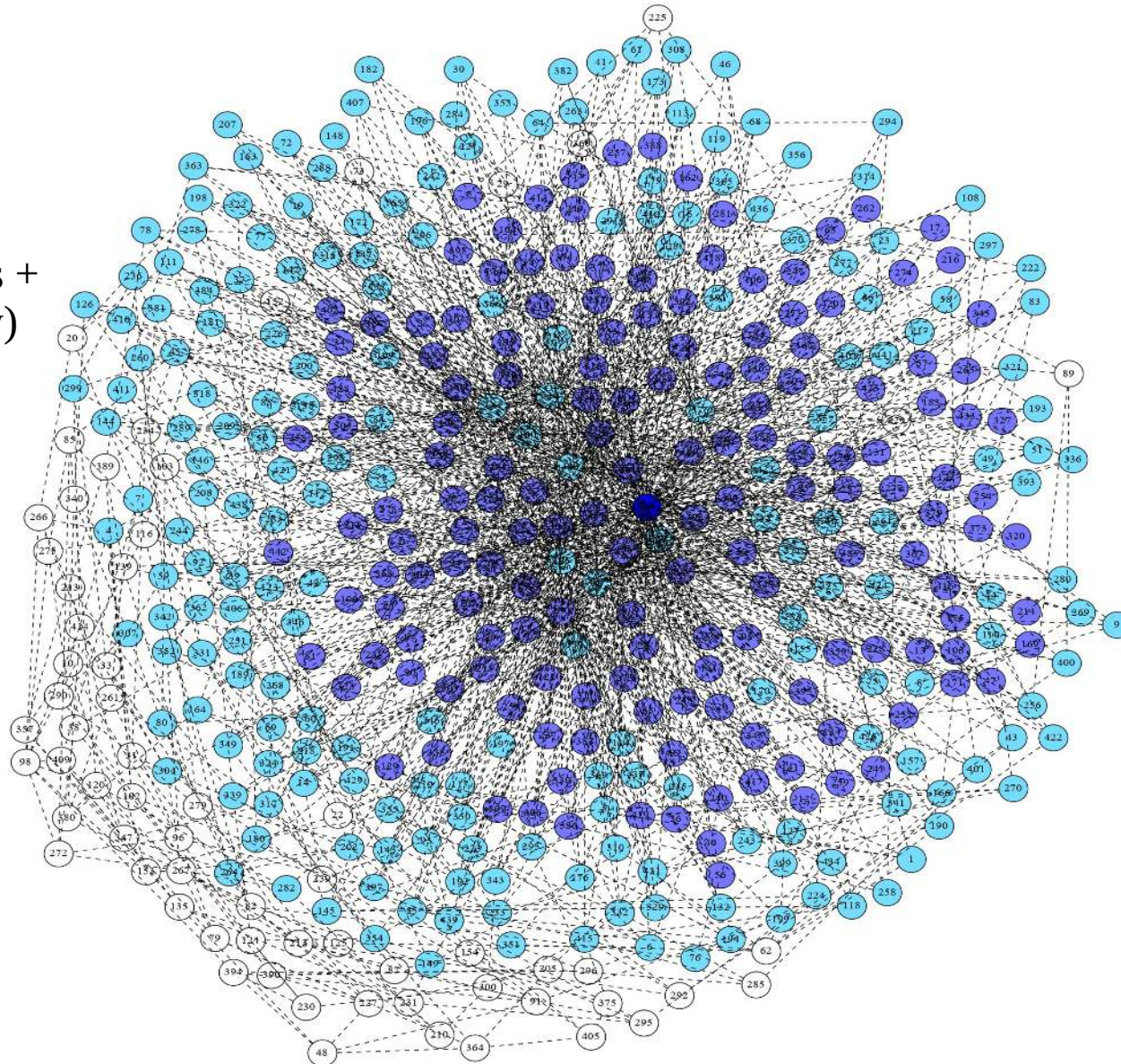


Results — The Haxby dataset



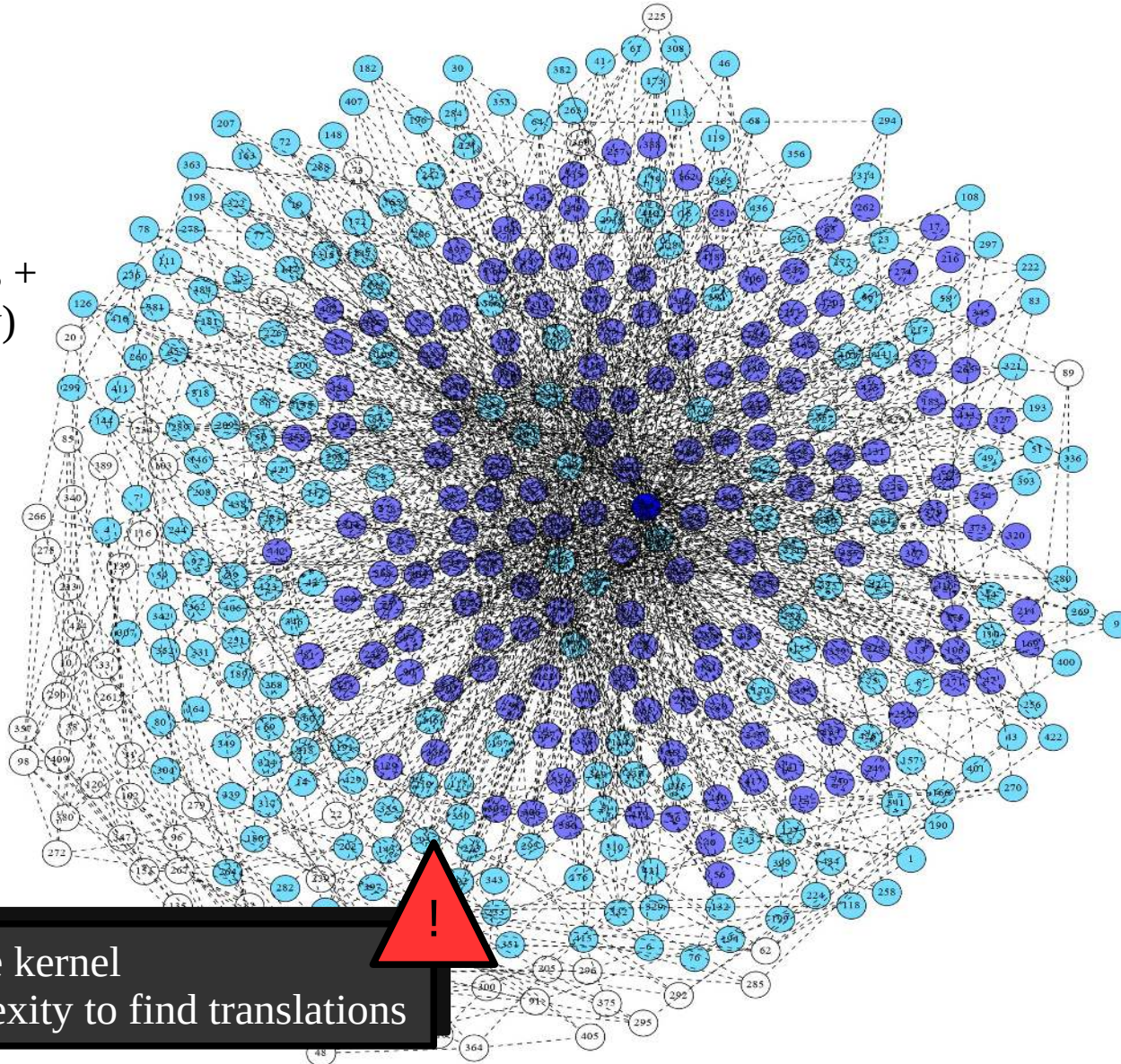
Results — The Haxby dataset

$\mathcal{G}_K^*$
(smoothness +
stationarity)



Results — The Haxby dataset

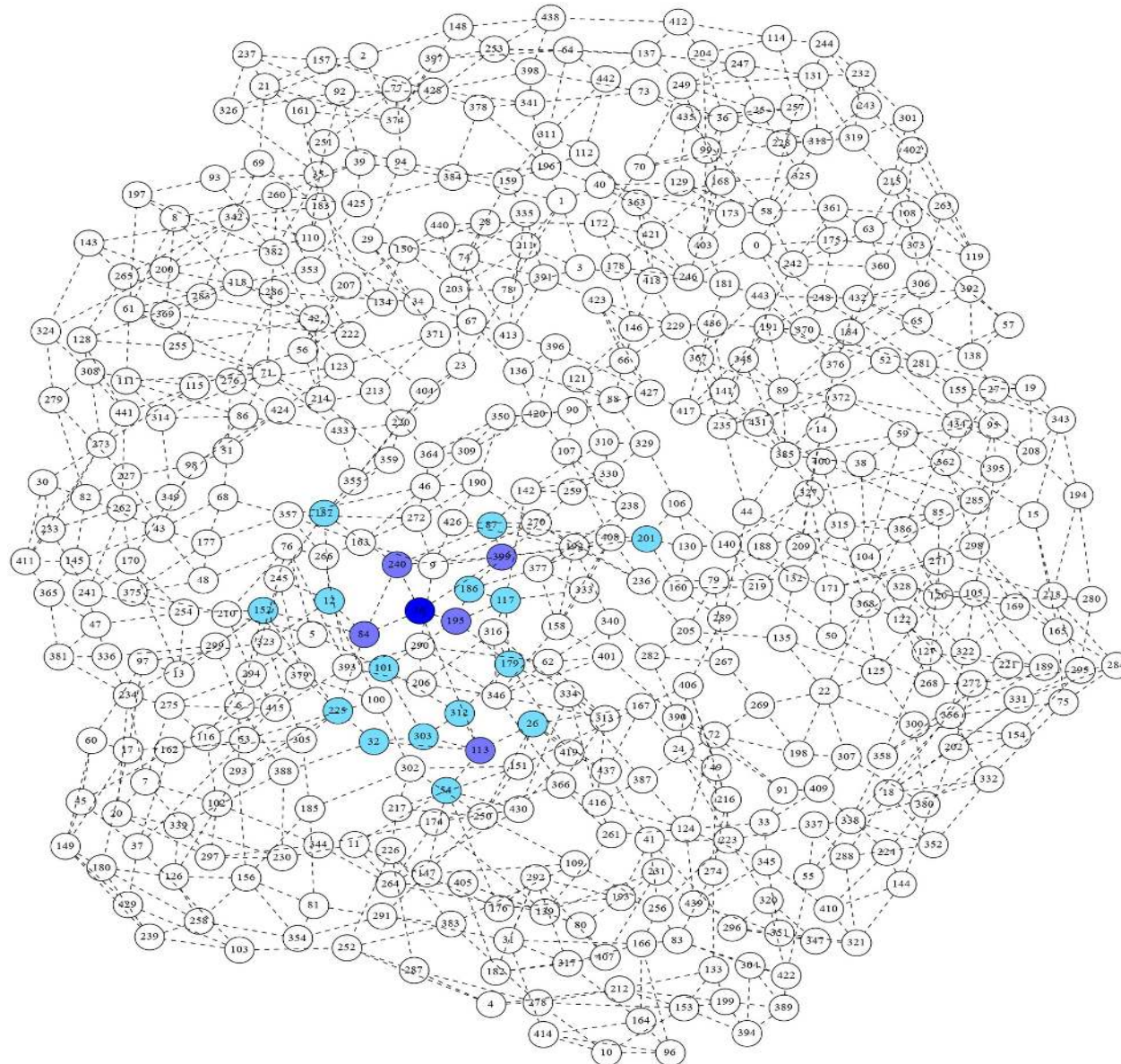
$\mathcal{G}_K^*$
(smoothness +
stationarity)



No locality of the kernel
Very high complexity to find translations

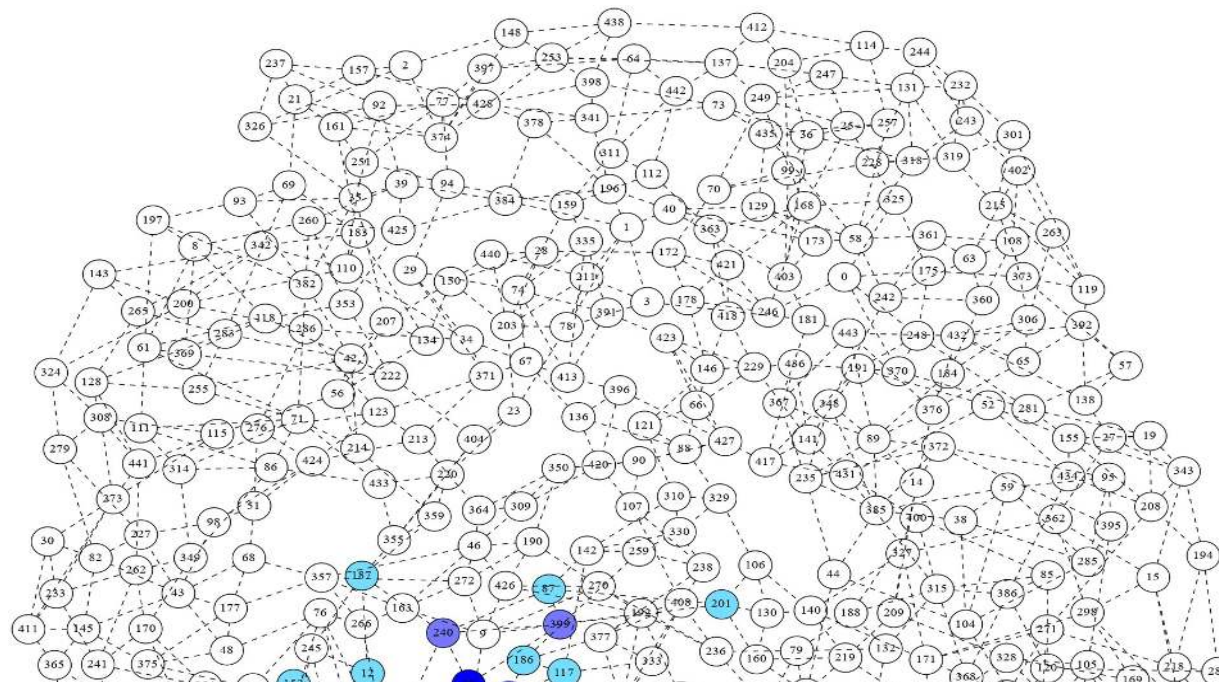
Results — The Haxby dataset

$\mathcal{G}_{\text{geo}}$
(geometric
coordinates)



Results — The Haxby dataset

$\mathcal{G}_{geo}$
(geometric
coordinates)



	$\mathcal{G}_{geo}$	MLP	SVM	LR ($C = 1, l_1$)	LR ($C = 50, l_1$)	(LR $C = 1, l_2$)	Ridge
Subject 1	56%	54%	50%	51%	47%	49%	35%
Subject 2	52%	44%	42%	48%	47%	47%	34%
Subject 3	47%	48%	35%	44%	38%	41%	28%
Subject 4	53%	42%	32%	34%	28%	34%	22%
Subject 5	42%	39%	31%	42%	34%	31%	23%
Subject 6	49%	49%	40%	42%	40%	38%	31%
Average	49.9%	45.9%	38.2%	43.6%	38.9%	39.7%	28.8%

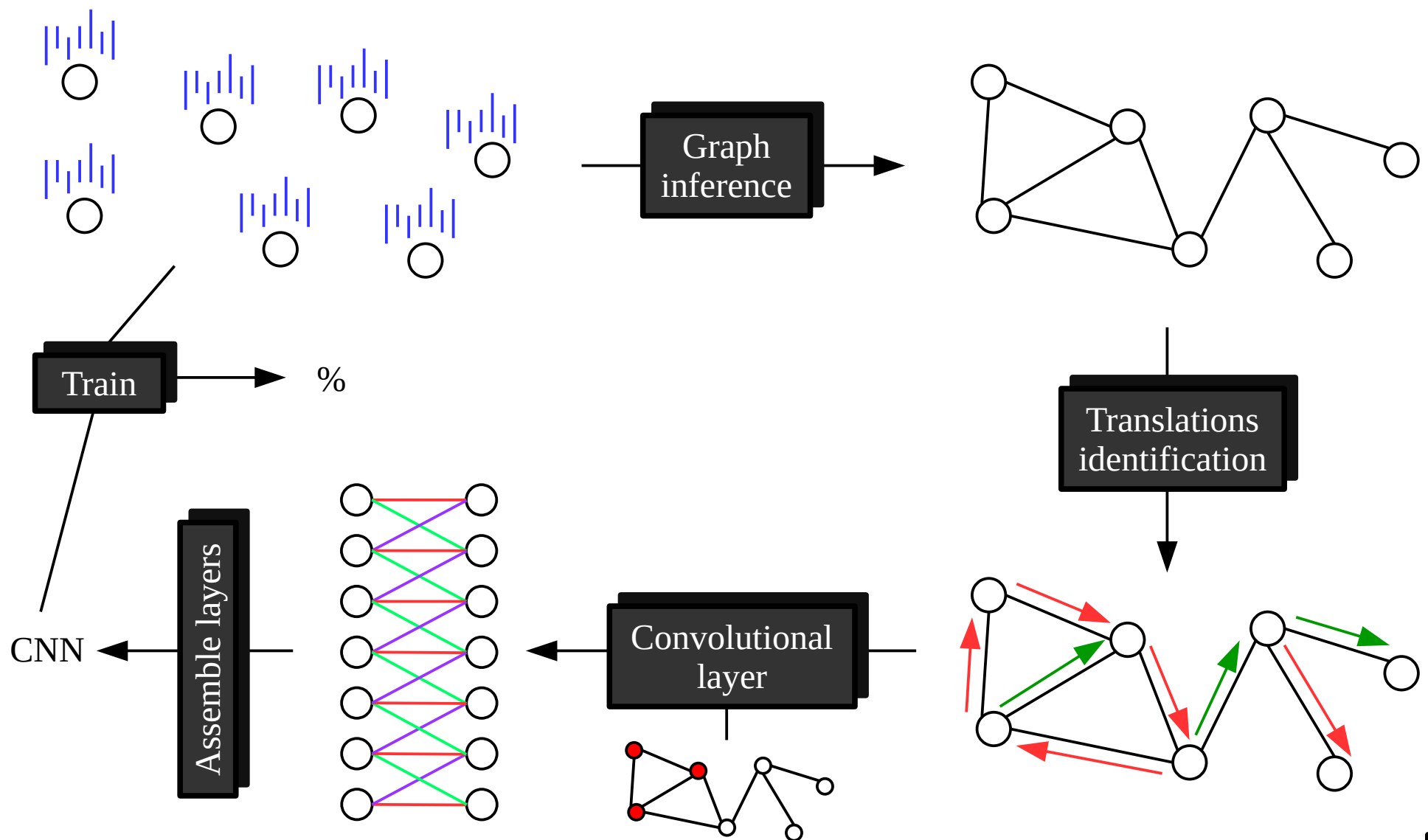


Outline

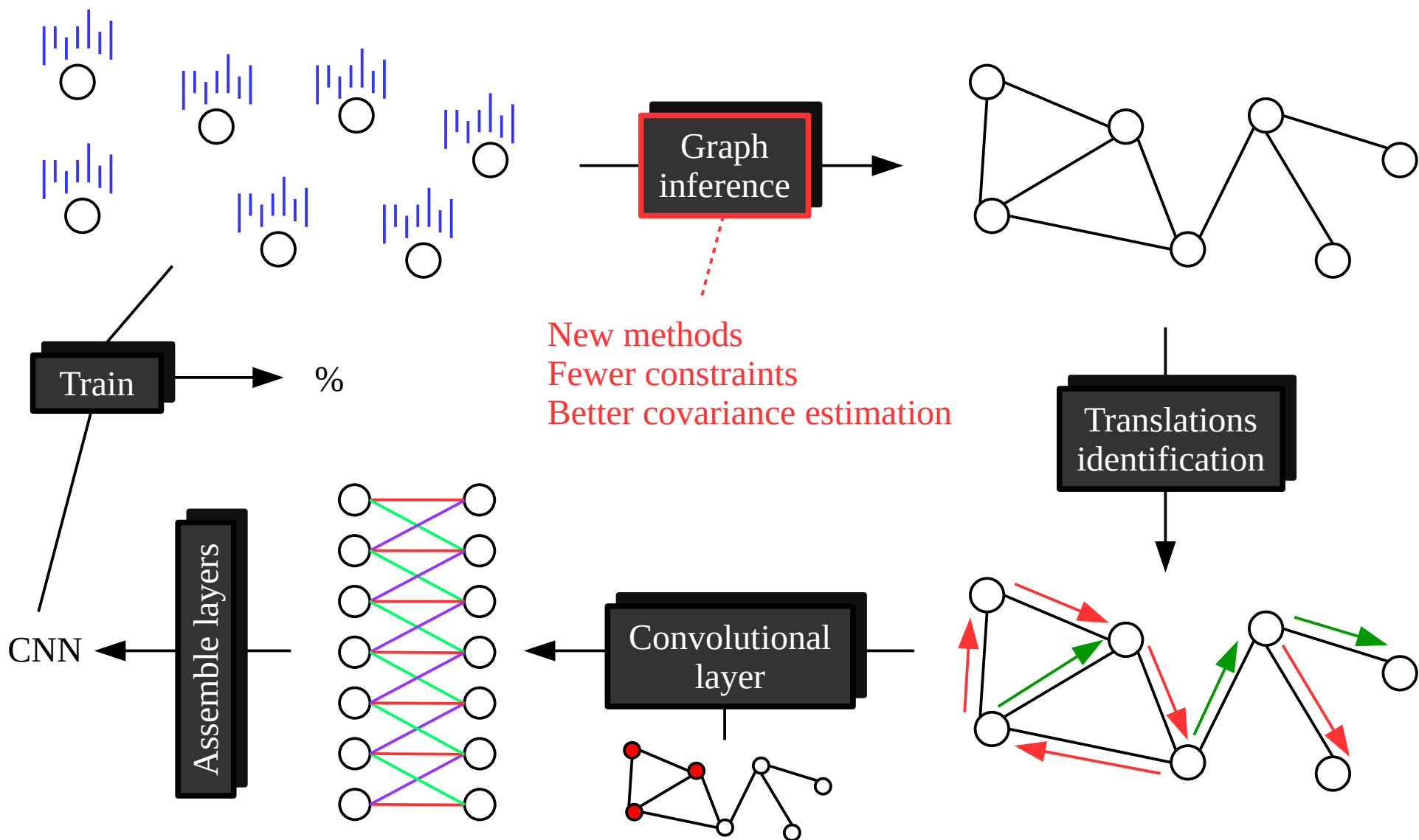
Conclusions & perspectives



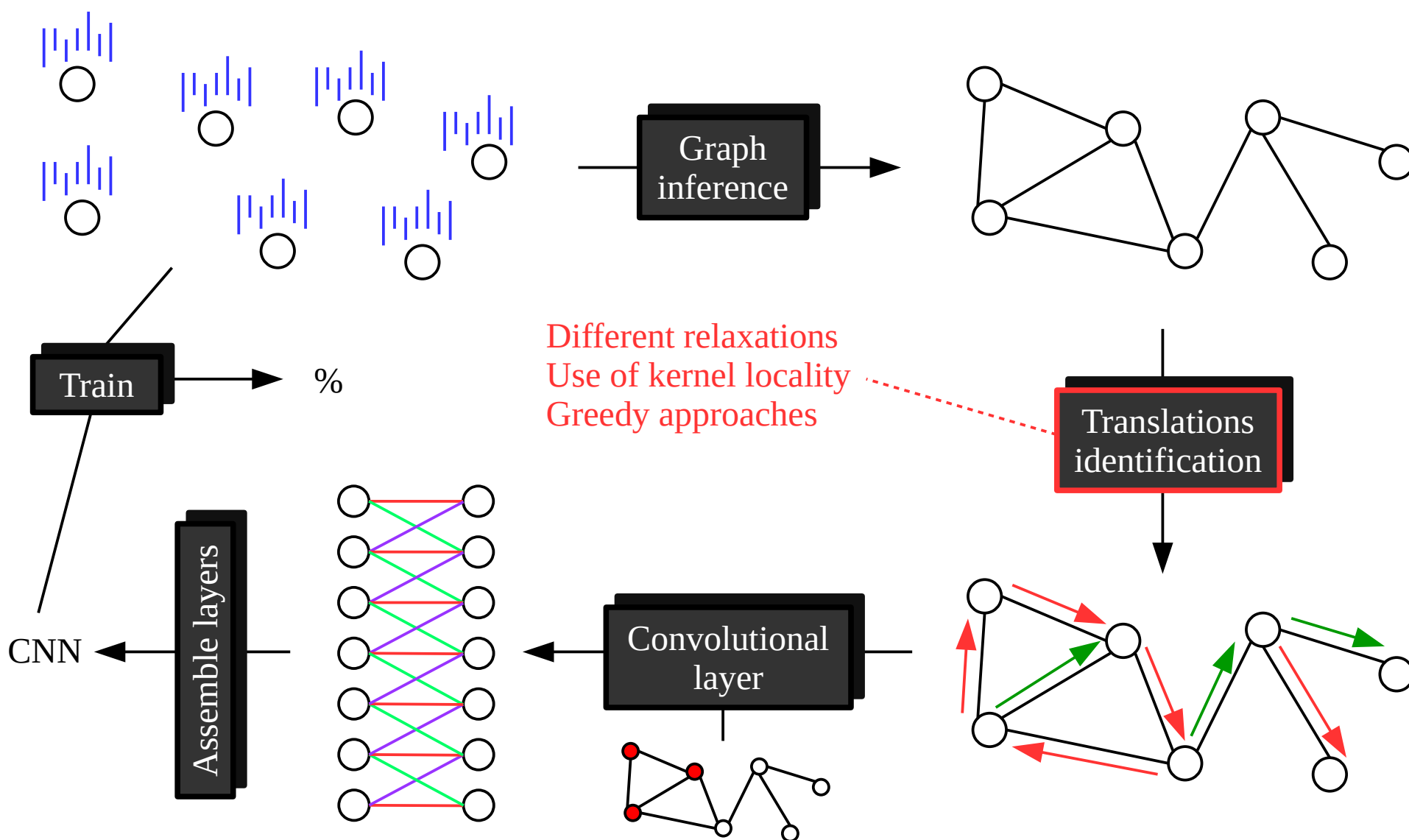
Conclusion & perspectives — Summary of the method



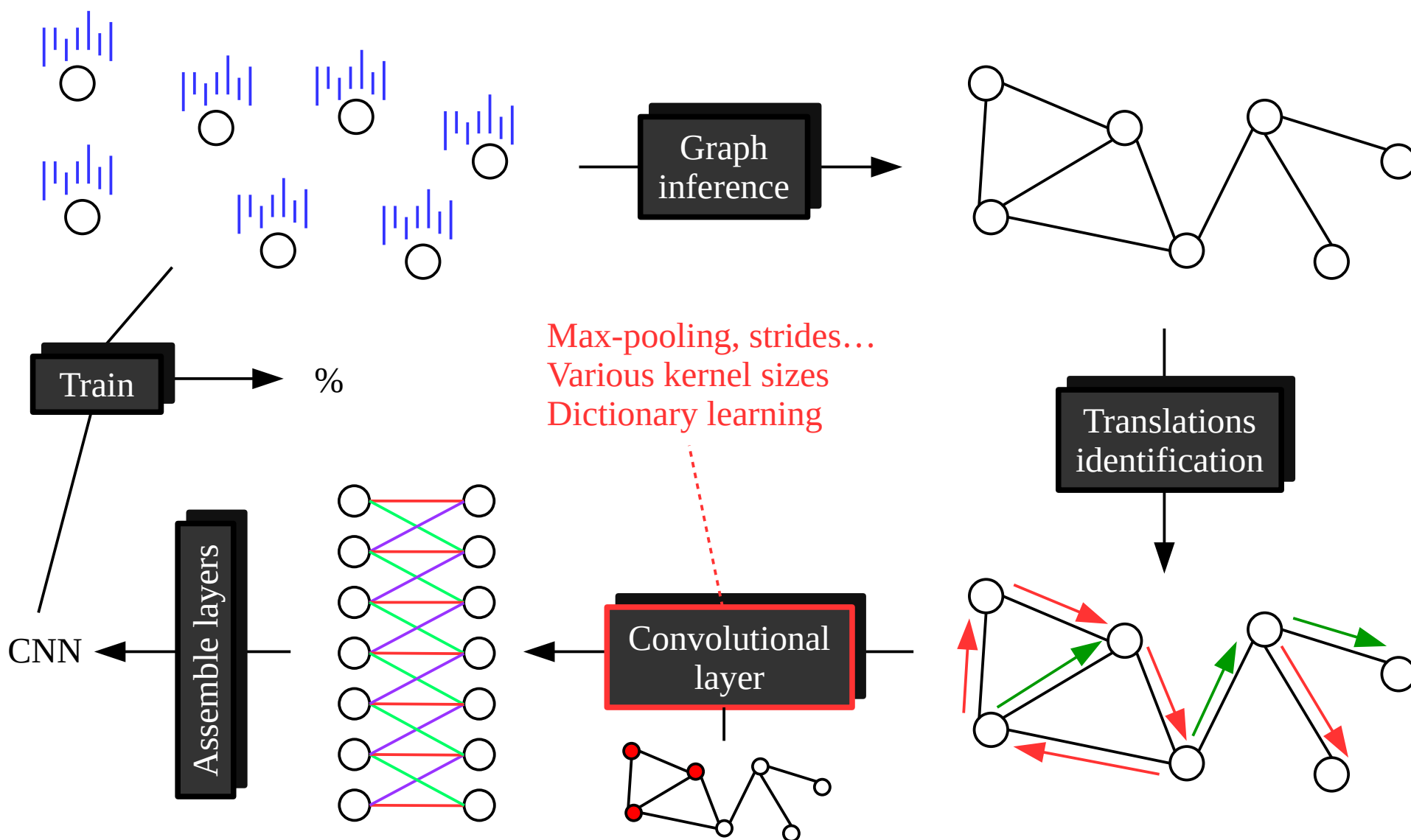
Conclusion & perspectives — Graph inference



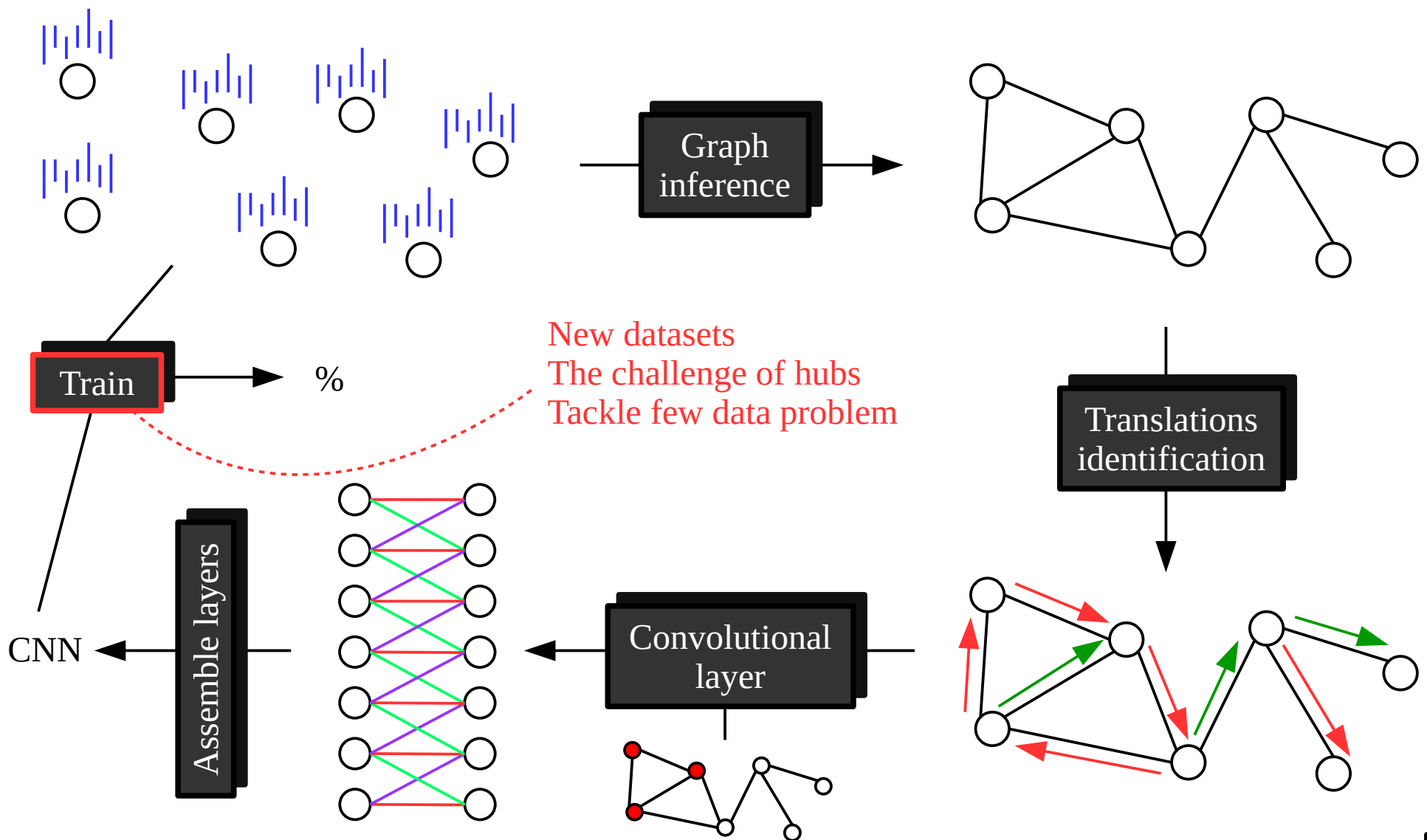
Conclusion & perspectives — Translations identification



Conclusion & perspectives — Convolutional layer



Conclusion & perspectives — CNN training



Conclusion & perspectives — Journals

Pasdeloup *et al.* – Translations on graphs with neighborhood preservation (IEEE Transactions on Signal Processing, Submitted)

Pasdeloup *et al.* – Characterization and inference of graph diffusion processes from observations of stationary signals (IEEE Transactions on Signal and Information Processing over Networks, 2017)

Conclusion & perspectives — Conferences

Pasdeloup *et al.* – Convolutional neural networks on irregular domains through approximate translations on inferred graphs (ICASSP, Submitted)

Ménoiret *et al.* – Evaluating graph signal processing for neuroimaging through classification and dimensionality reduction (GlobalSIP, 2017)

Grelier *et al.* – Neighborhood-preserving translations on graphs (GlobalSIP, 2016)

Pasdeloup *et al.* – Towards a characterization of the uncertainty curve for graphs (ICASSP, 2016)

Pasdeloup *et al.* – Graph reconstruction from the observation of diffused signals (Allerton Conference on Communication, Control and Computing, 2015)

Pasdeloup *et al.* – Toward an uncertainty principle for weighted graphs (EUSIPCO, 2016)

Conclusion & perspectives — Workshops

Grelier *et al.* — Translation sur graphe avec préservation du voisinage (Gretsi, 2017)

Pasdeloup *et al.* — Characterization and inference of graph diffusion processes from observations of stationary processes (Bellairs workshop on graph signal processing, 2017)

Pasdeloup *et al.* — Characterizing graphs from diffused signals (Graph signal processing workshop, 2016)

Pasdeloup *et al.* — Characterizing graphs from diffused signals (Workshop on data driven approach to networks and languages, 2016)

Pasdeloup *et al.* — Vers une caractérisation de la courbe d'incertitude pour des graphes portant des signaux (Gretsi, 2015)

Conclusion & perspectives — Education

Conferences

Pasdeloup *et al.* — PyRat — Un jeu sérieux pour l'enseignement de l'informatique (QPES, 2017)

Workshops

Pasdeloup *et al.* — PyRat — Un jeu pour l'apprentissage de l'informatique avec Python (Journées UTICE, 2016)

Pasdeloup *et al.* — PyRat — Un jeu pour l'apprentissage de l'informatique avec Python (PyCon, 2016)



Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

Bastien Padeloup

bastien.padeloup@imt-atlantique.fr

Ph.D. director: Dominique Pastor

Ph.D. advisor: Vincent Gripon

December 12, 2017



UNIVERSITE

BRETAGNE

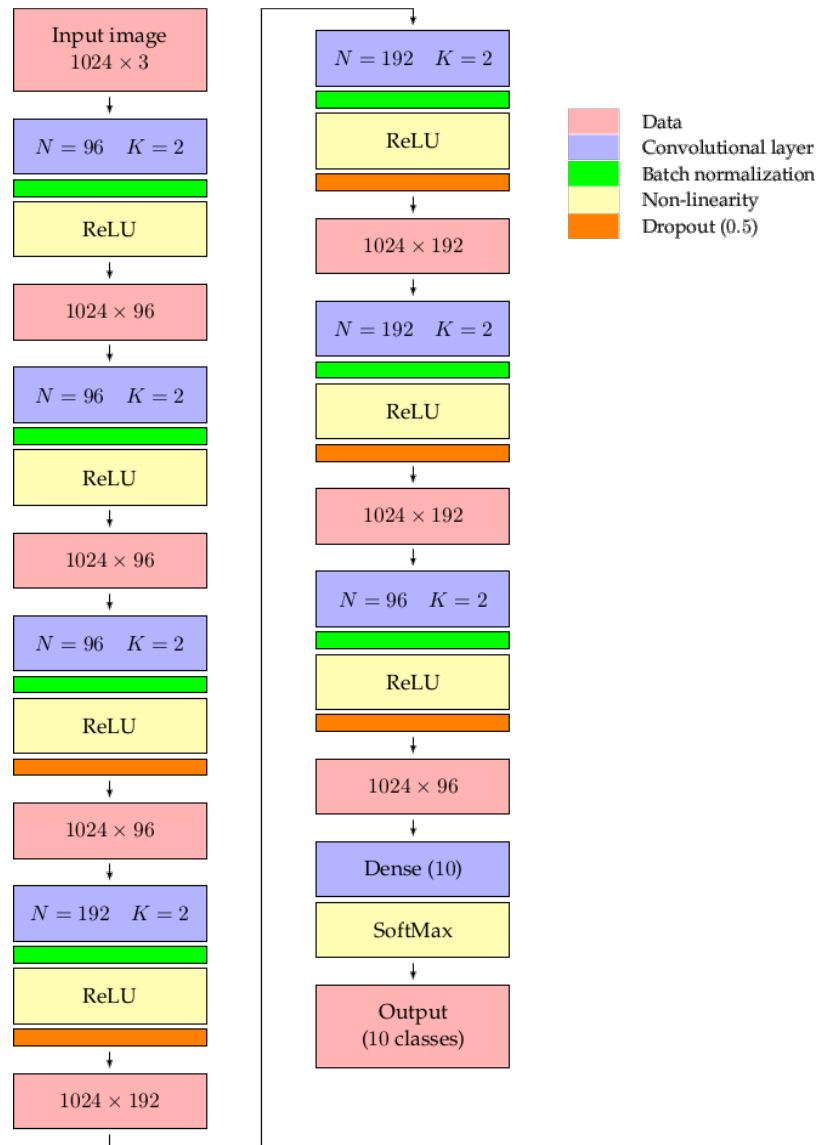
LOIRE / MATHSTIC

Outline

Supplementary slides

— CNN architecture used for CIFAR-10 —

Supplementary — CNN architecture — Details



7 convolutional layers

1 kernel size (2 hops)

Dropout (avoid overfitting)

~~Max-pooling, strides...~~

Outline

Supplementary slides

- Optimization problem to approximate translations —

Supplementary — Approximate translations — Optimization prob.

$$\mathbf{A}_\psi^* = \underset{\mathbf{A}_\psi \in \{0,1\}^{N \times N}}{\operatorname{argmin}} \|\mathbf{A}\mathbf{A}_\psi - \mathbf{A}_\psi\mathbf{A}\|_1$$
$$s.t. \quad \left\{ \begin{array}{l} \forall i, j \in \llbracket 1, N \rrbracket : \mathbf{A}_\psi[i, j] \leq \mathbf{A}[i, j] \\ \forall j \in \llbracket 1, N \rrbracket : \sum_{i=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \forall i \in \llbracket 1, N \rrbracket : \sum_{j=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \|\mathbf{A}_\psi\|_1 = C \end{array} \right.$$

Supplementary — Approximate translations — Optimization prob.

$$\mathbf{A}_\psi^* = \operatorname{argmin}_{\mathbf{A}_\psi \in \{0,1\}^{N \times N}} \|\mathbf{A}\mathbf{A}_\psi - \mathbf{A}_\psi\mathbf{A}\|_1$$

Minimize
SNP violations

$$s.t. \quad \left\{ \begin{array}{l} \forall i, j \in \llbracket 1, N \rrbracket : \mathbf{A}_\psi[i, j] \leq \mathbf{A}[i, j] \\ \forall j \in \llbracket 1, N \rrbracket : \sum_{i=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \forall i \in \llbracket 1, N \rrbracket : \sum_{j=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \|\mathbf{A}_\psi\|_1 = C \end{array} \right.$$

EC

Injective

Scale factor

Supplementary — Approximate translations — Another possibility

$$\mathbf{A}_{\psi}^* = \underset{\mathbf{A}_{\psi} \in \{0, 1\}^{N \times N}}{\operatorname{argmin}} \quad \|\mathbf{A}\mathbf{A}_{\psi} - \mathbf{A}_{\psi}\mathbf{A}\|_1 + \alpha \|\mathbf{A} - \mathbf{A}_{\psi}\|_1$$

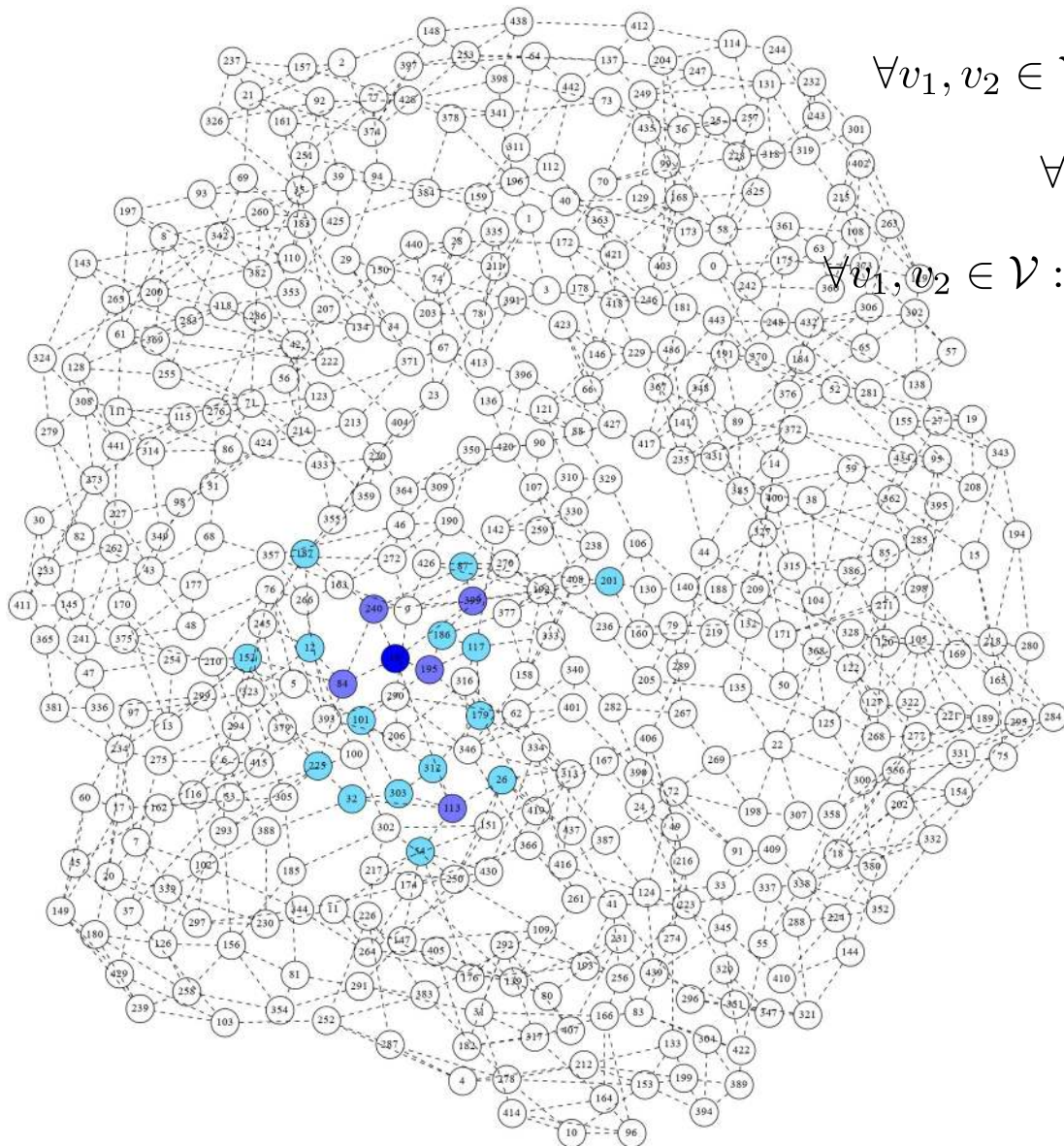
$$s.t. \quad \left\{ \begin{array}{l} \forall j \in \llbracket 1, N \rrbracket : \sum_{i=1}^N \mathbf{A}_{\psi}[i, j] \leq 1 \\ \forall i \in \llbracket 1, N \rrbracket : \sum_{j=1}^N \mathbf{A}_{\psi}[i, j] \leq 1 \\ \|\mathbf{A}_{\psi}\|_1 = C \end{array} \right. ,$$

Outline

Supplementary slides

— ICASSP 2018 submission —

Supplementary — ICASSP 2018 — Limits of our definitions

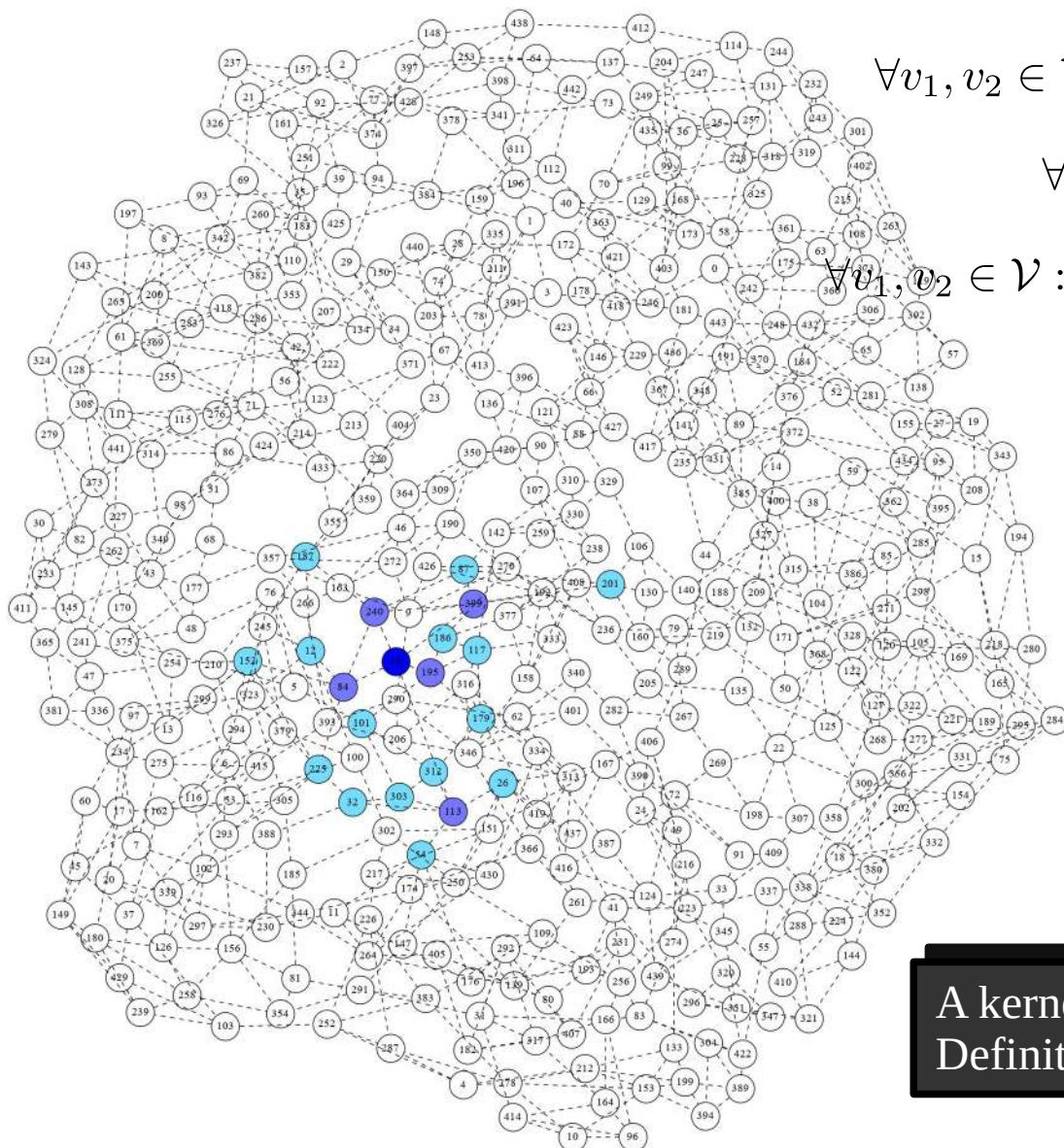


$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

$$\forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \\ \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp)$$

Supplementary — ICASSP 2018 — Limits of our definitions



$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

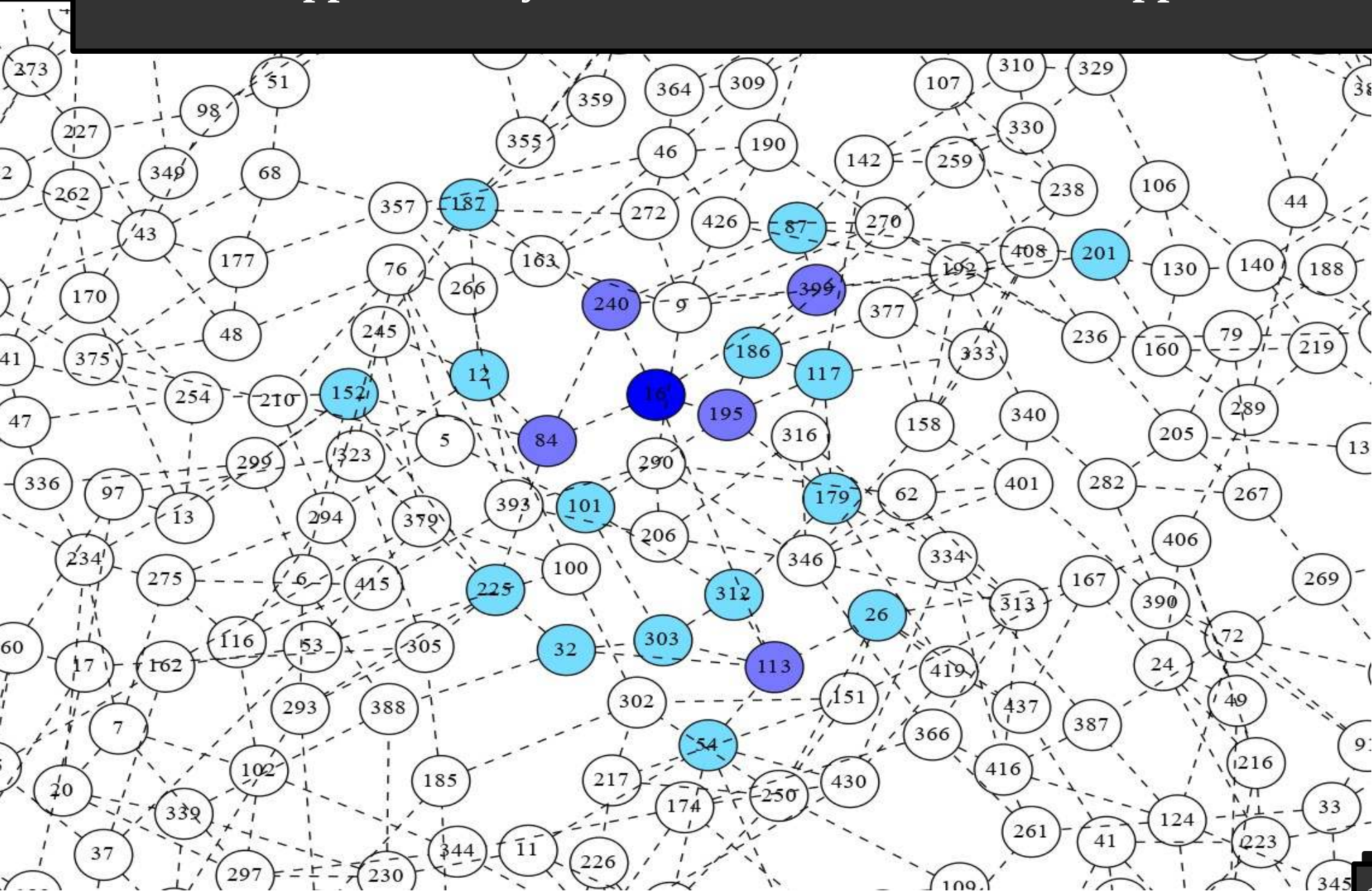
$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

$$\forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \\ \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp)$$

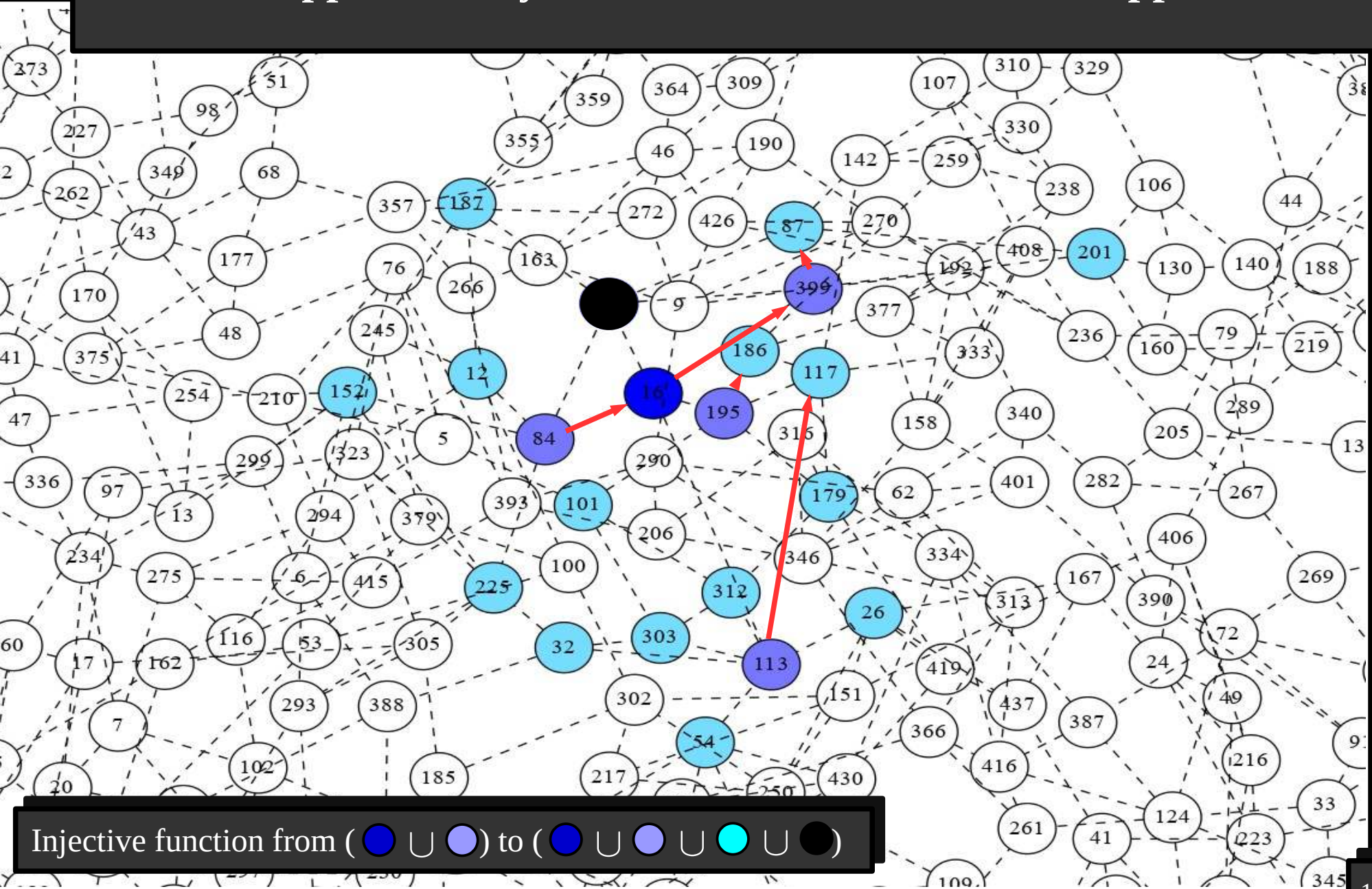
A kernel is a localized signal
Definition of translations is very strict



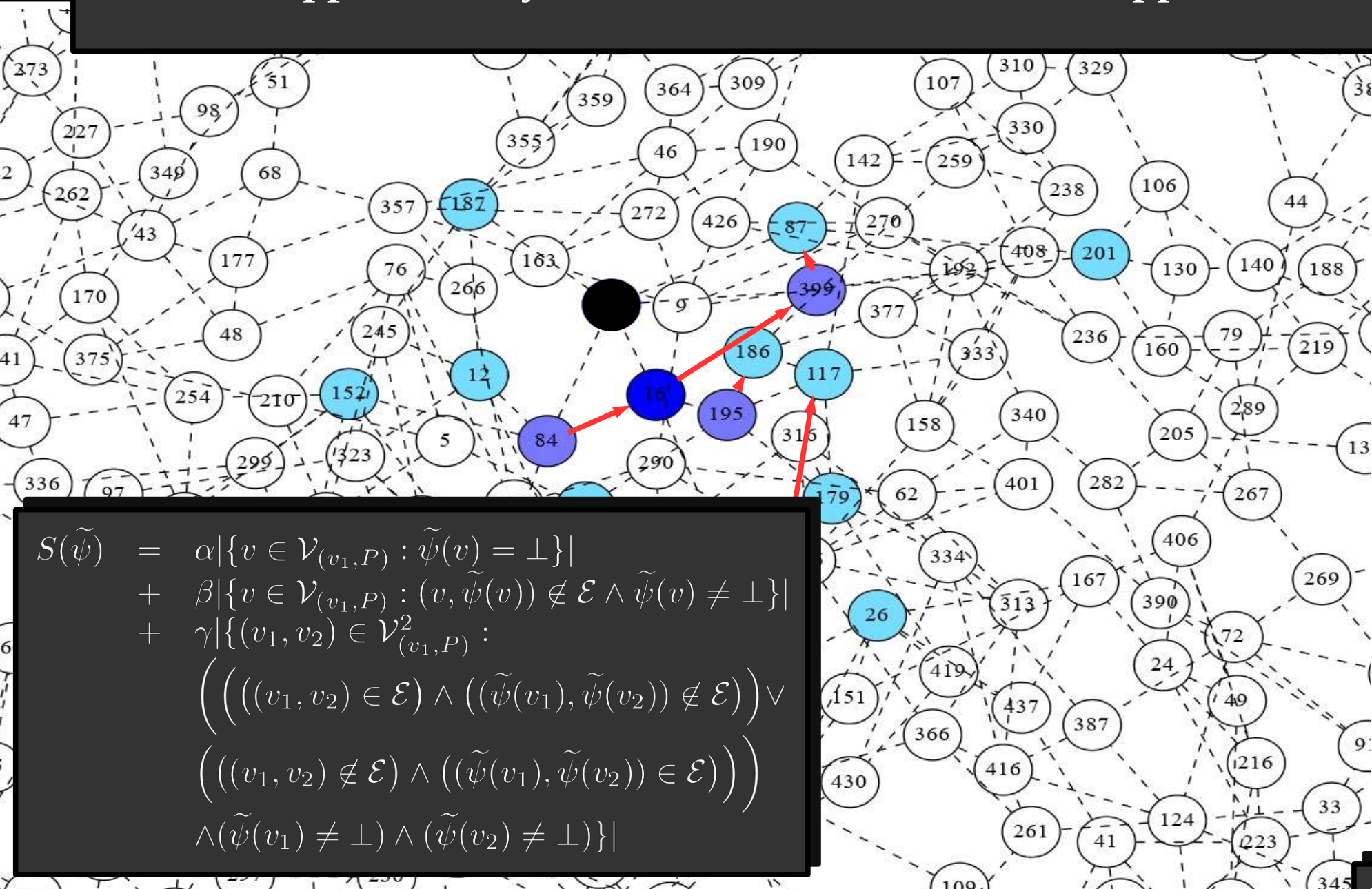
Supplementary — ICASSP 2018 — A local approach



Supplementary — ICASSP 2018 — A local approach

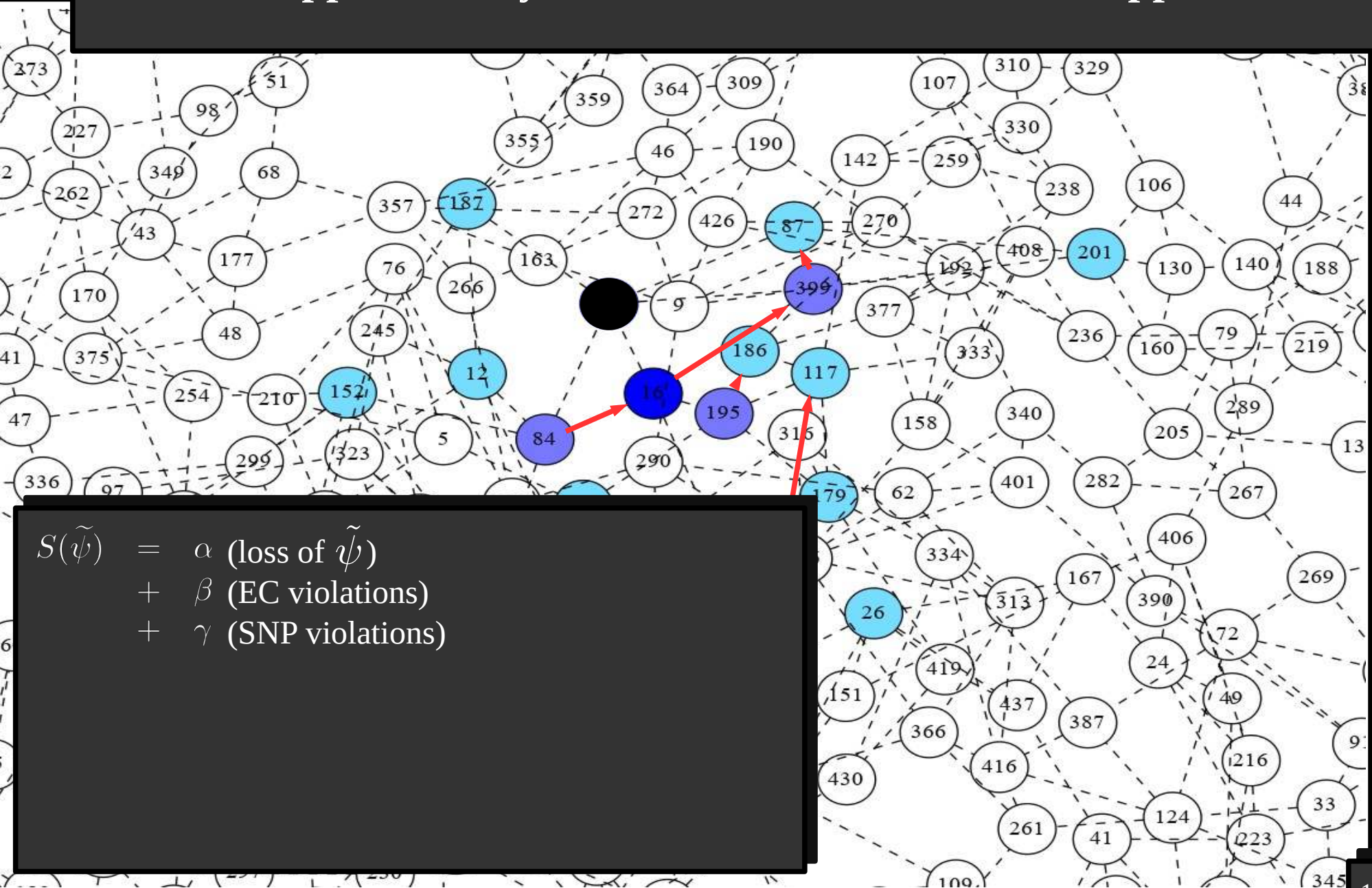


Supplementary — ICASSP 2018 — A local approach

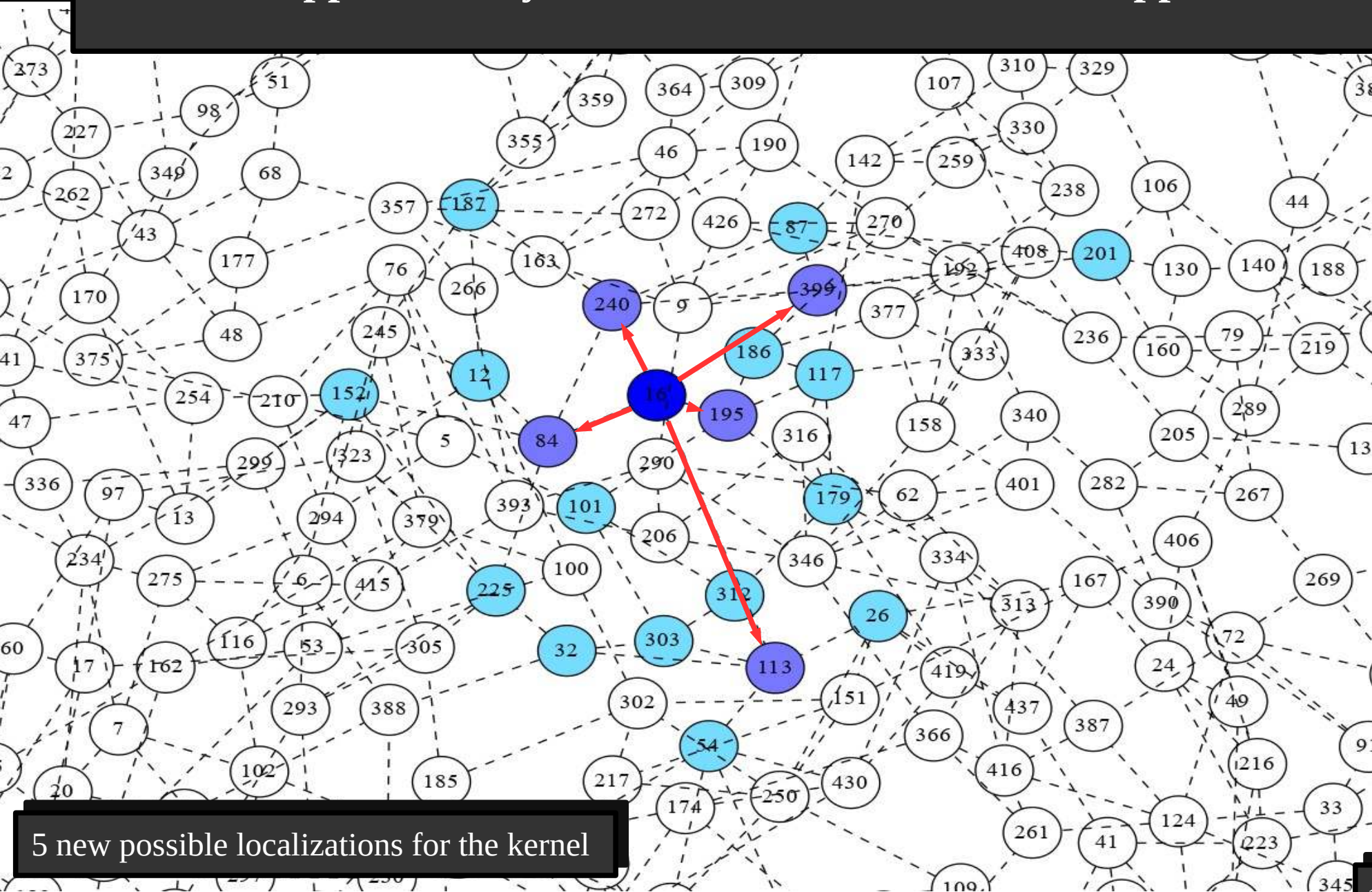


$$\begin{aligned}
 S(\tilde{\psi}) &= \alpha |\{v \in \mathcal{V}_{(v_1, P)} : \tilde{\psi}(v) = \perp\}| \\
 &+ \beta |\{v \in \mathcal{V}_{(v_1, P)} : (v, \tilde{\psi}(v)) \notin \mathcal{E} \wedge \tilde{\psi}(v) \neq \perp\}| \\
 &+ \gamma |\{(v_1, v_2) \in \mathcal{V}_{(v_1, P)}^2 : \\
 &\quad \left(\left(((v_1, v_2) \in \mathcal{E}) \wedge ((\tilde{\psi}(v_1), \tilde{\psi}(v_2)) \notin \mathcal{E}) \right) \vee \right. \\
 &\quad \left. \left(((v_1, v_2) \notin \mathcal{E}) \wedge ((\tilde{\psi}(v_1), \tilde{\psi}(v_2)) \in \mathcal{E}) \right) \right) \\
 &\quad \wedge (\tilde{\psi}(v_1) \neq \perp) \wedge (\tilde{\psi}(v_2) \neq \perp)\}|
 \end{aligned}$$

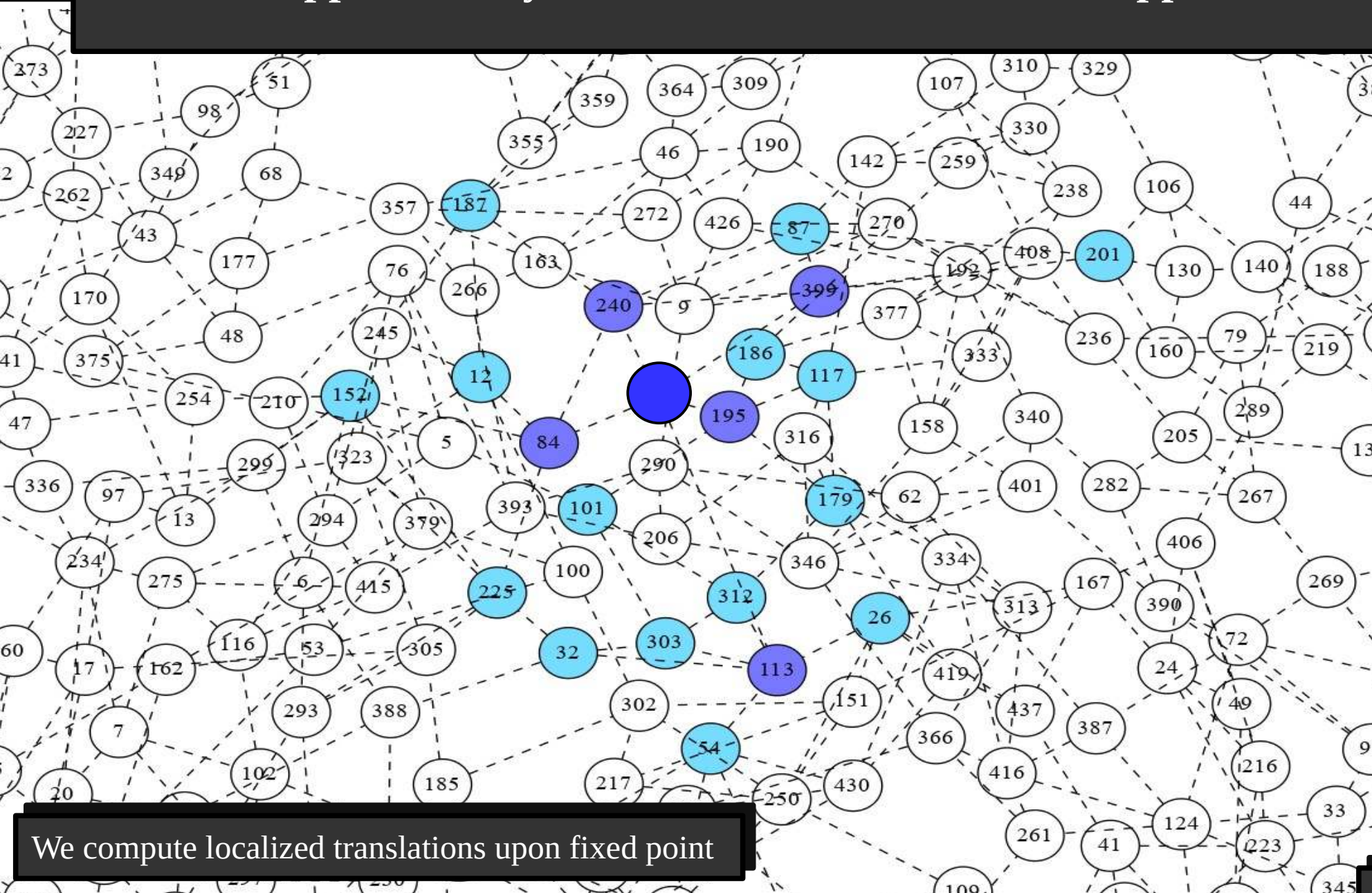
Supplementary — ICASSP 2018 — A local approach



Supplementary — ICASSP 2018 — A local approach

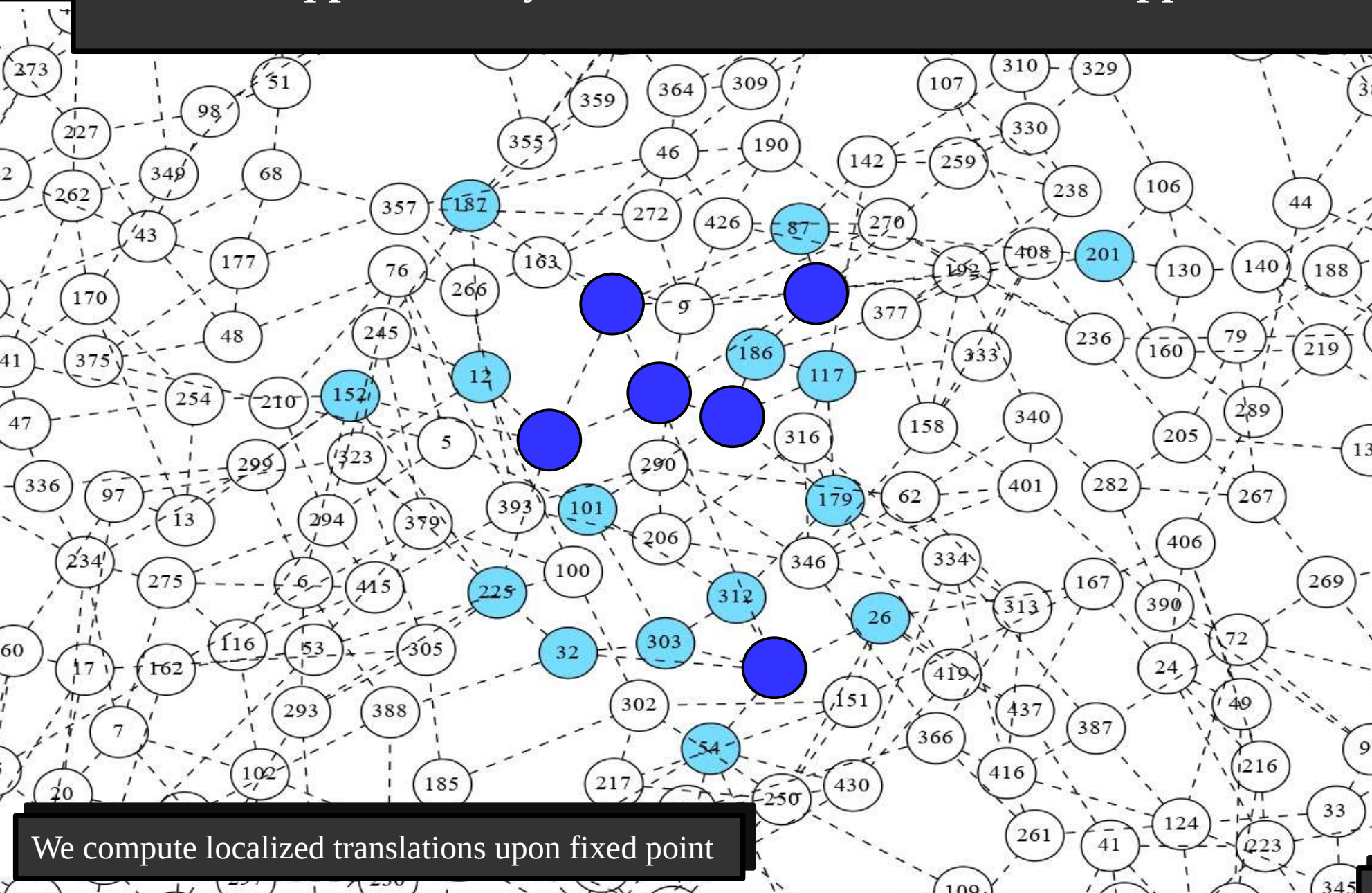


Supplementary — ICASSP 2018 — A local approach



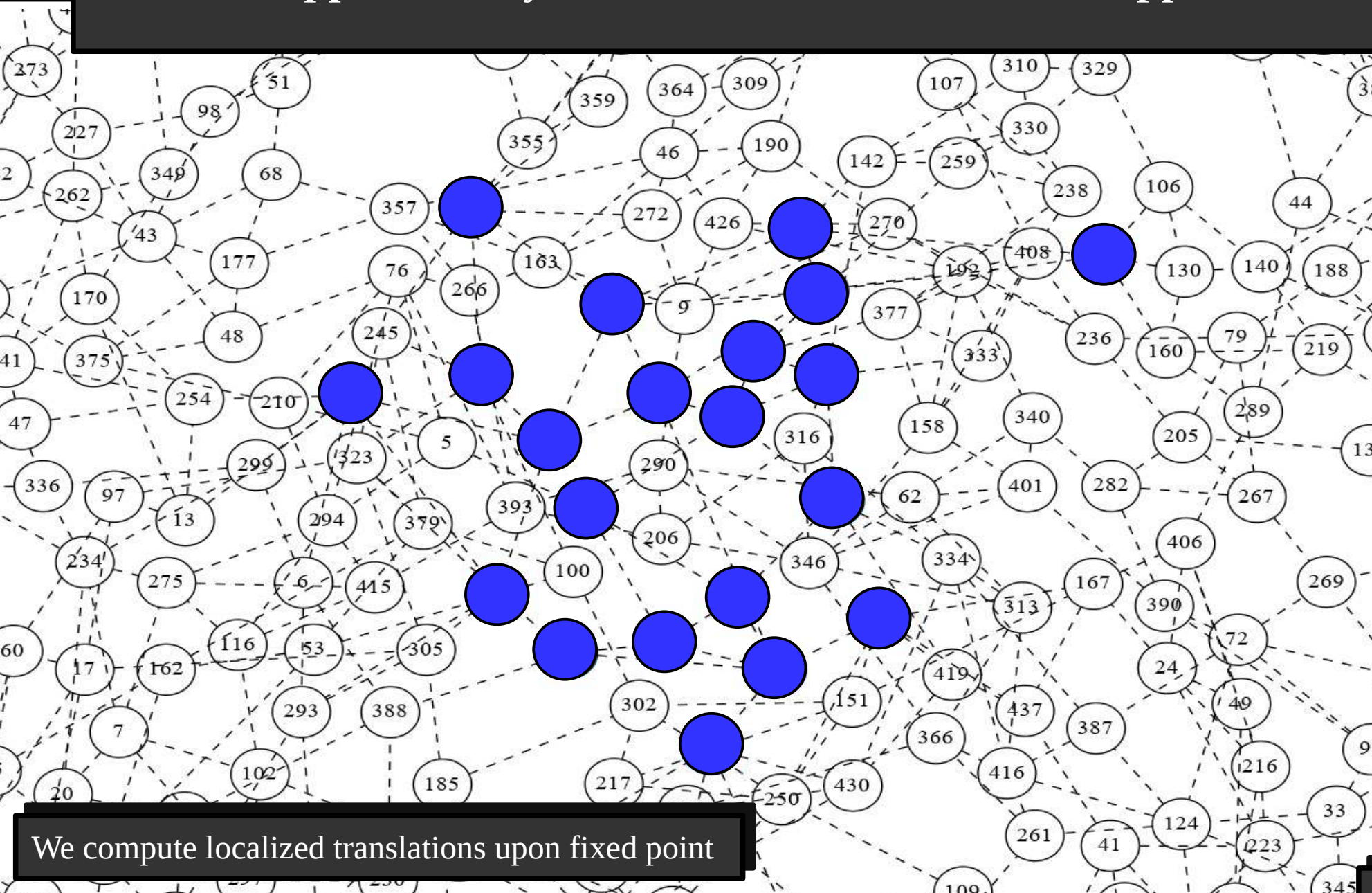
We compute localized translations upon fixed point

Supplementary — ICASSP 2018 — A local approach



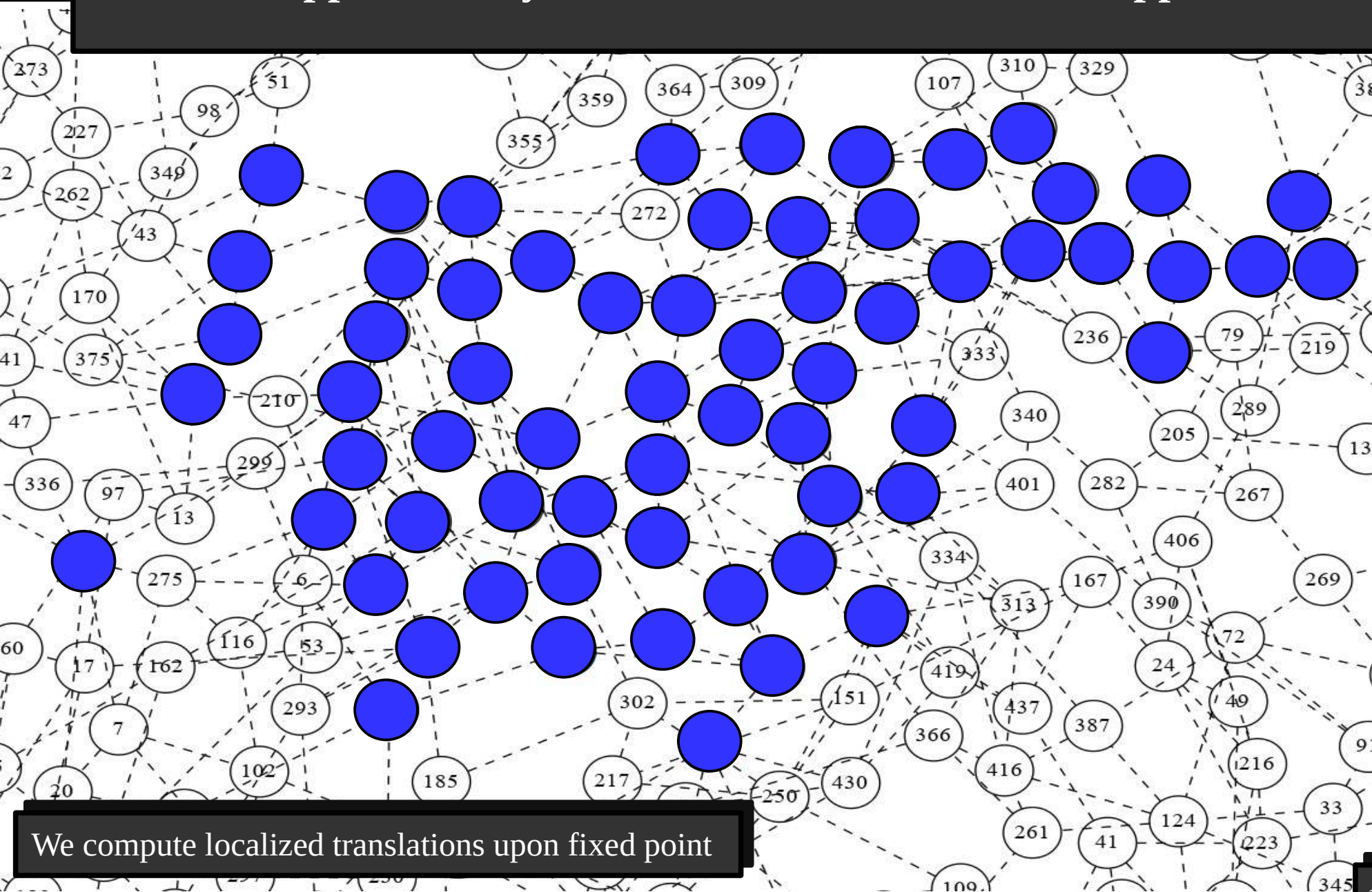
We compute localized translations upon fixed point

Supplementary — ICASSP 2018 — A local approach

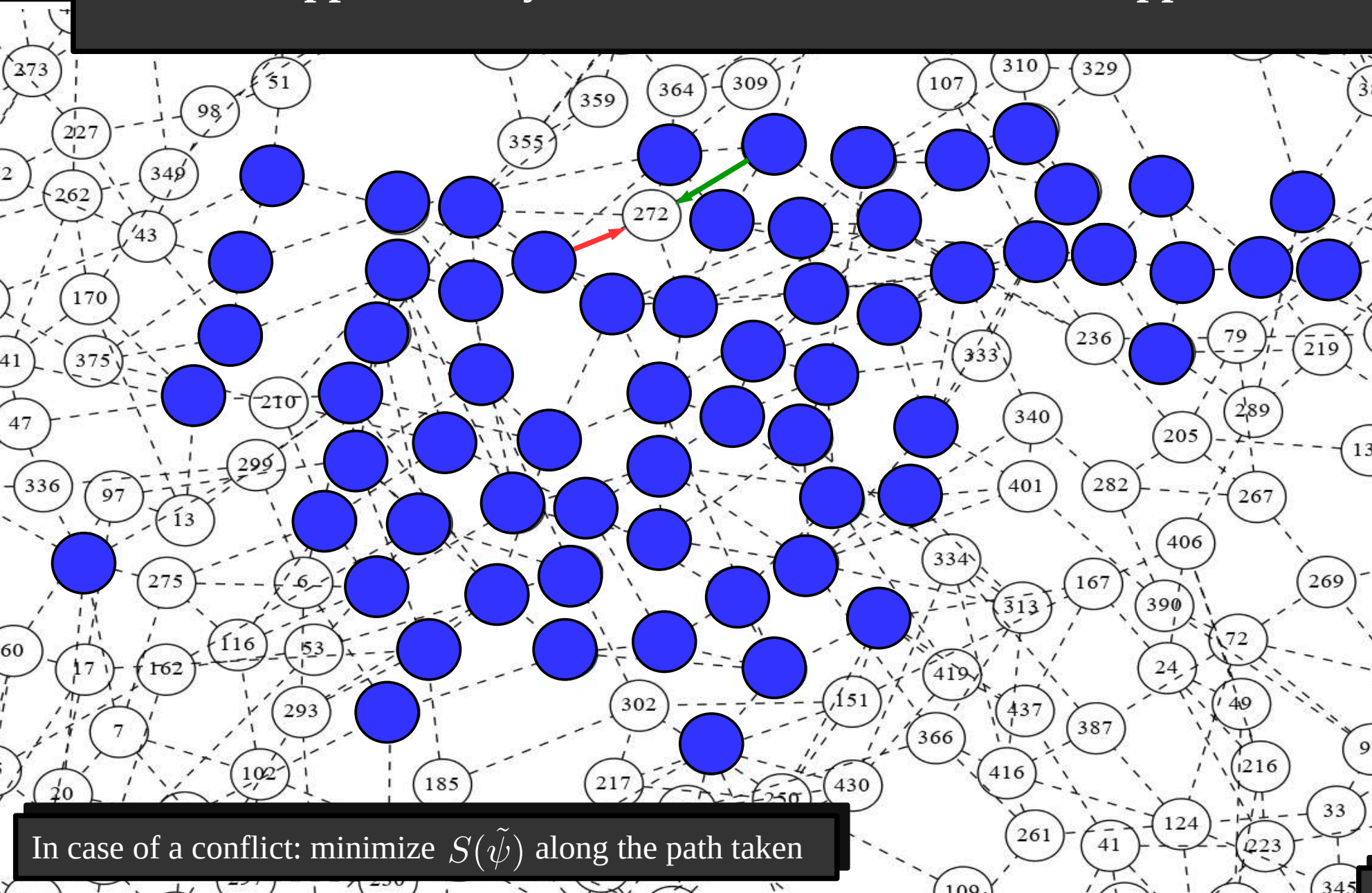


We compute localized translations upon fixed point

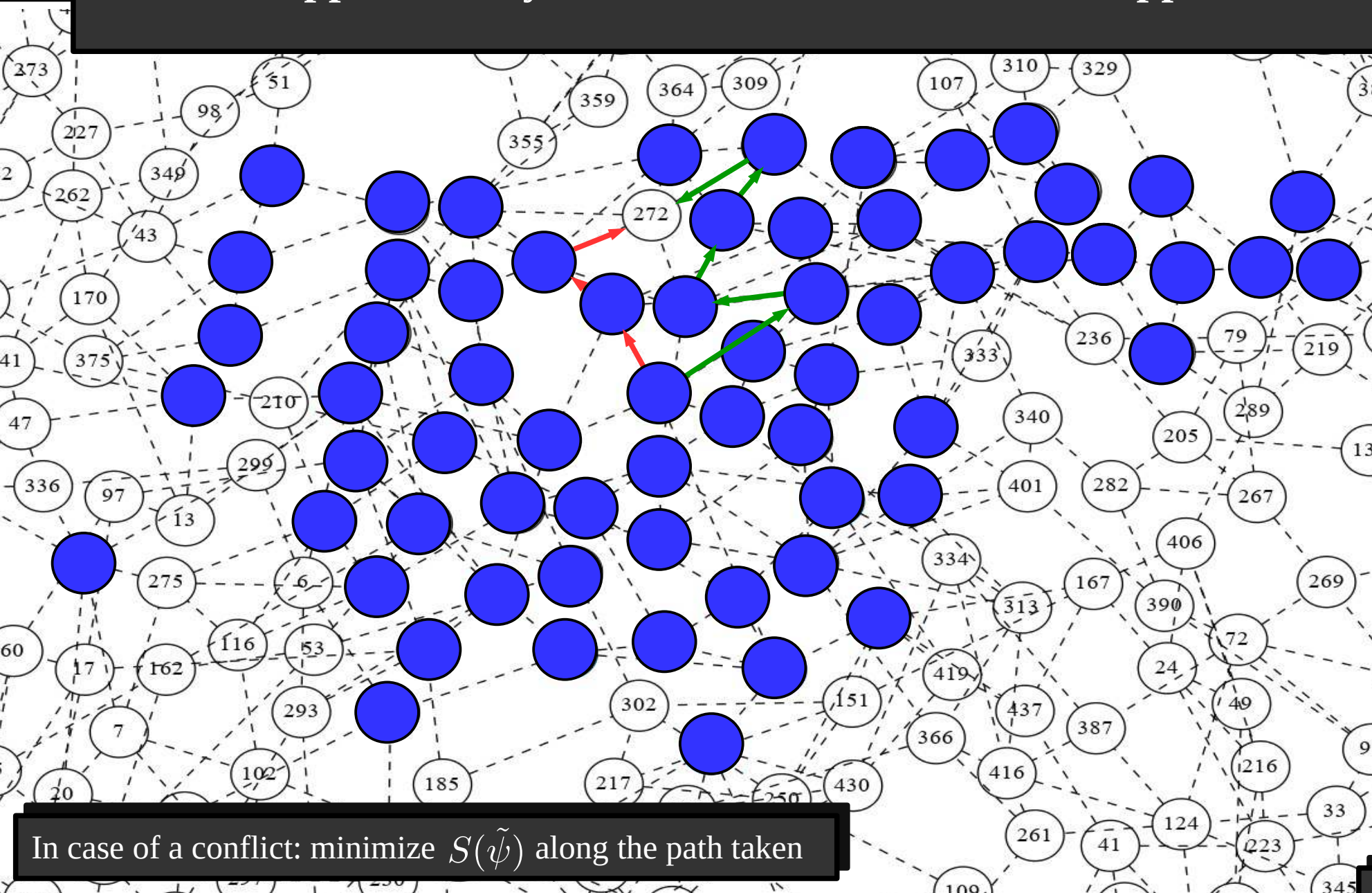
Supplementary — ICASSP 2018 — A local approach



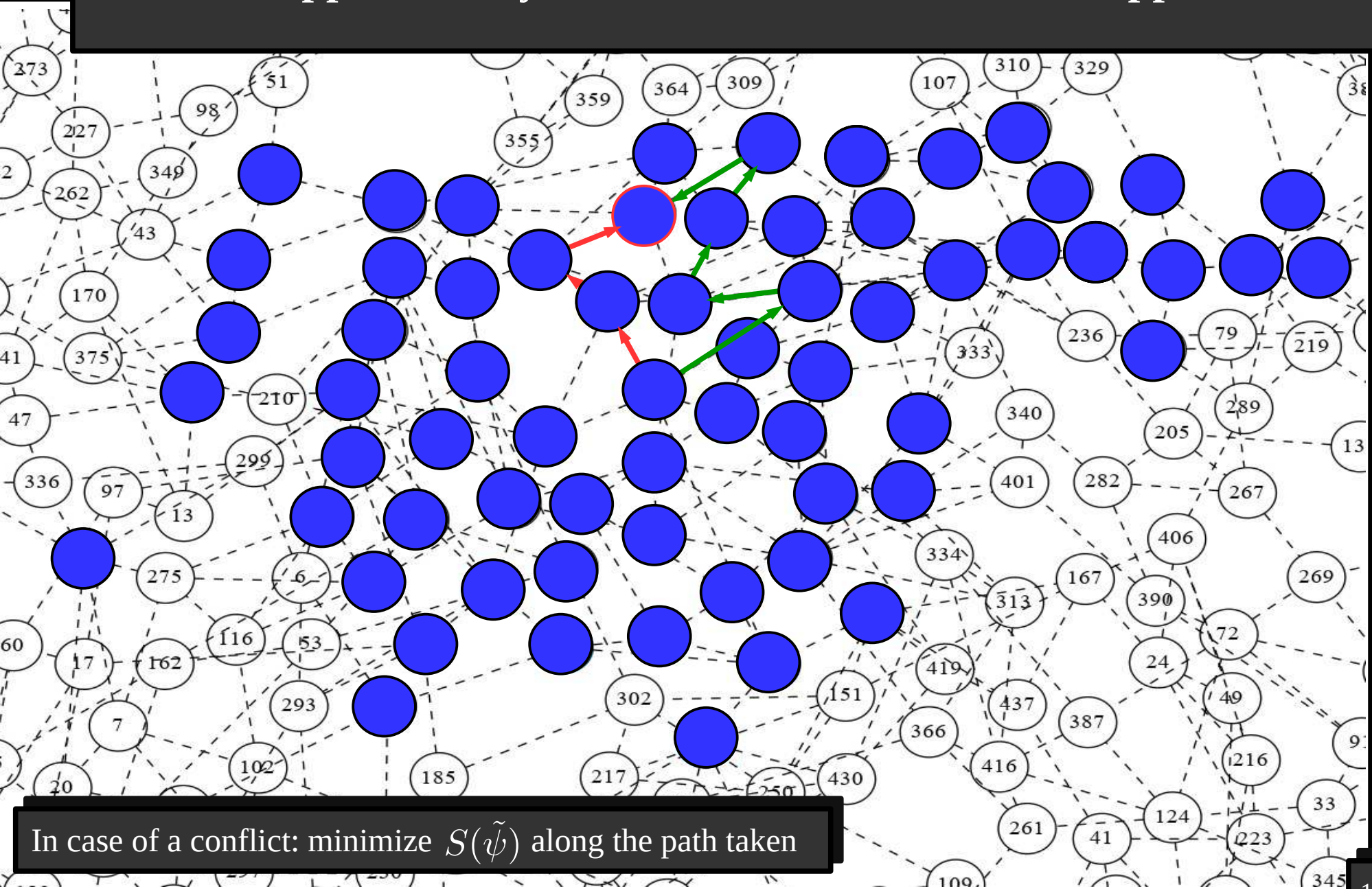
Supplementary — ICASSP 2018 — A local approach



Supplementary — ICASSP 2018 — A local approach



Supplementary — ICASSP 2018 — A local approach

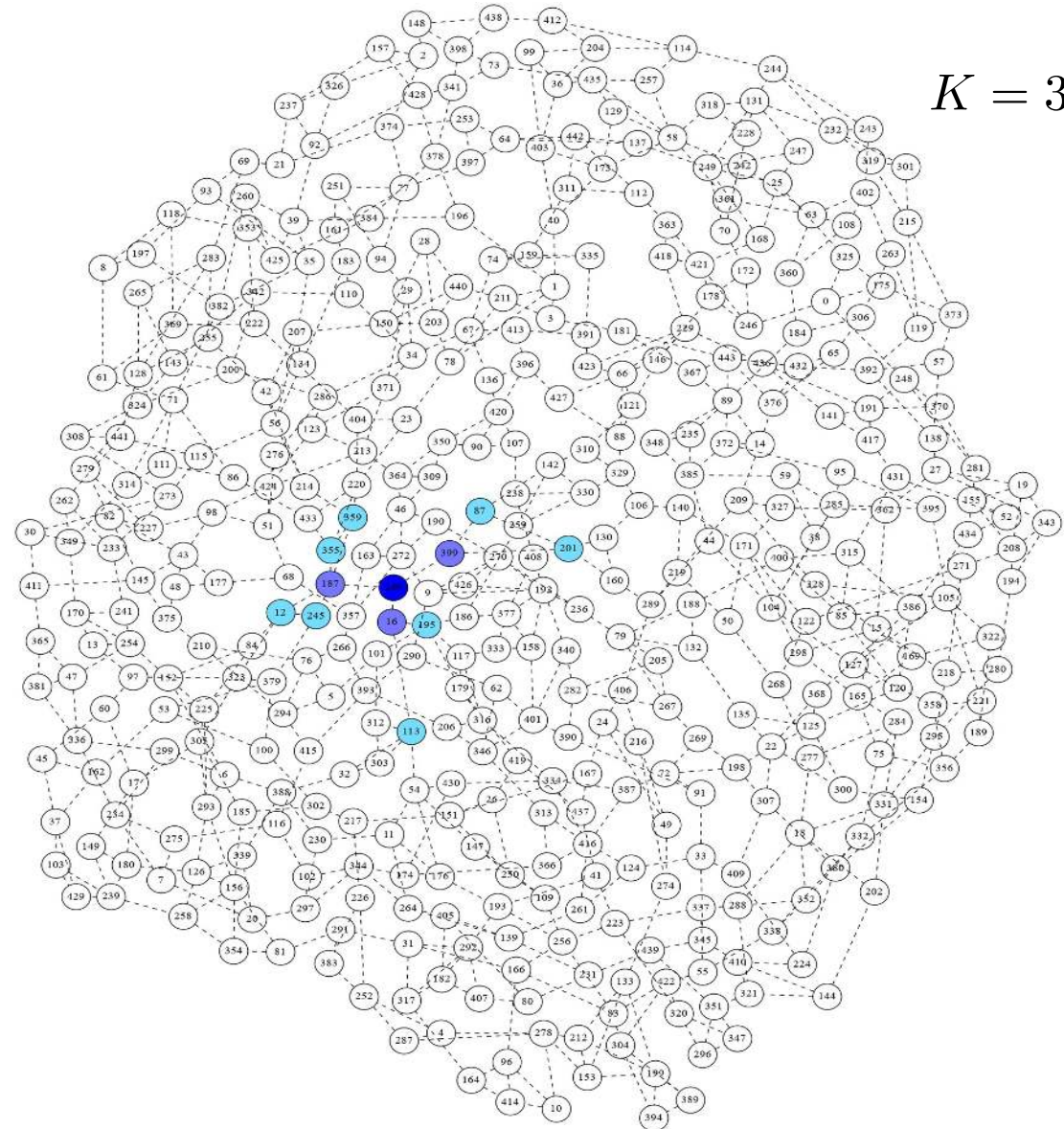
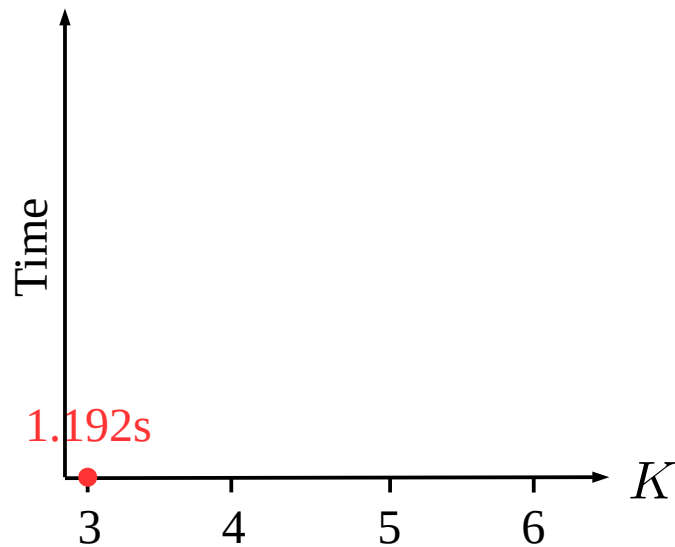


Outline

Supplementary slides

— Time measurements on Haxby —

Supplementary — Time measurements — Function of K

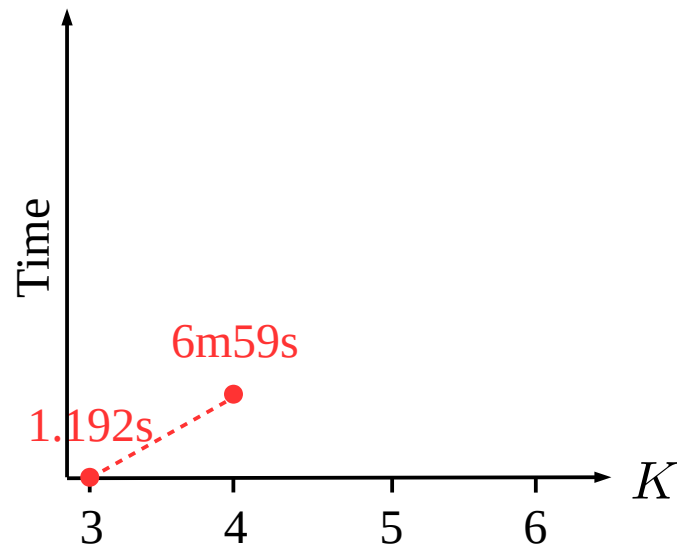


$\mathcal{G}_{\text{geo}}, \alpha = 1.0, \beta = 0.7, \gamma = 0.5$

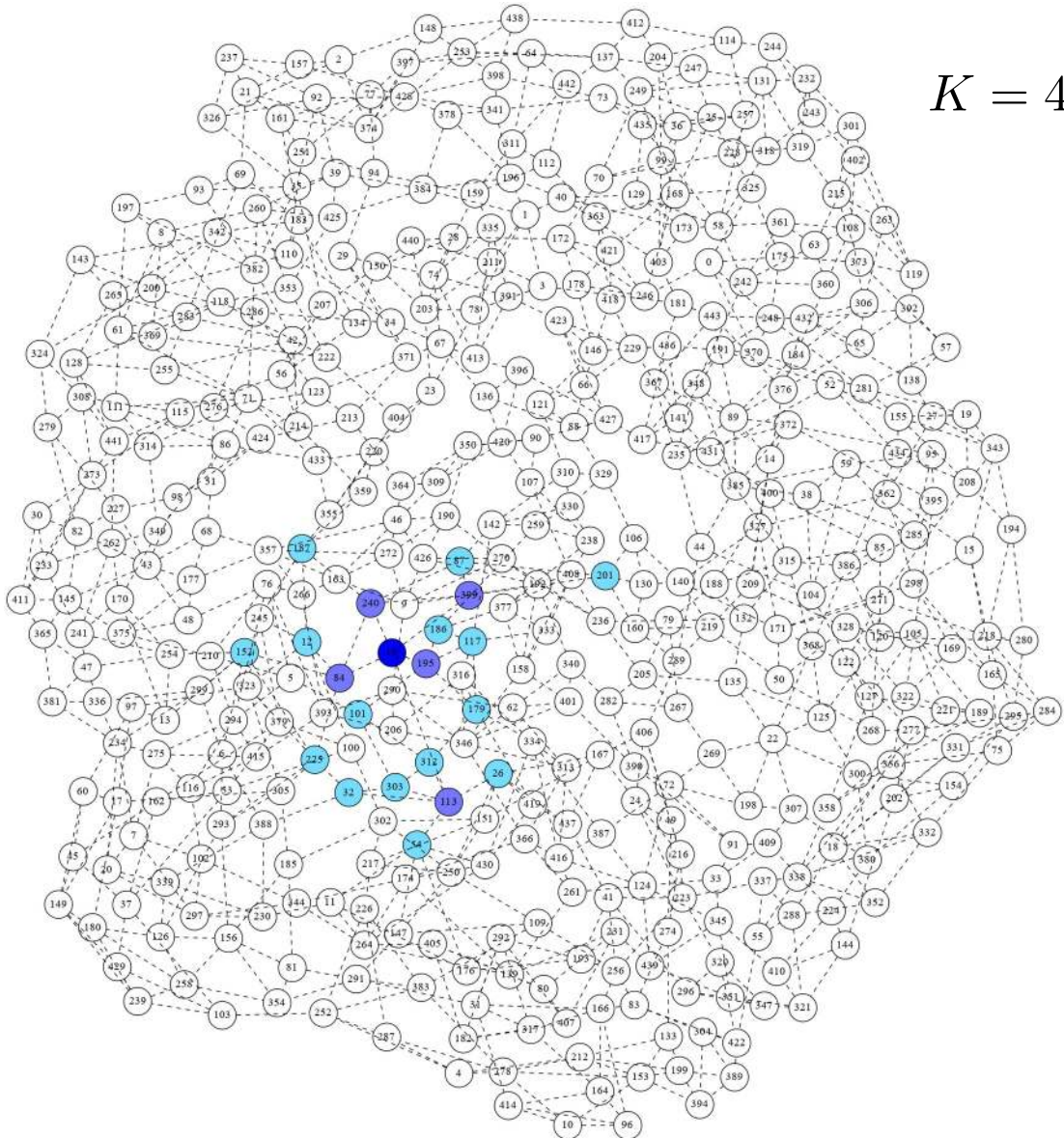
444 vertices

C++ implementation w/ backtracking

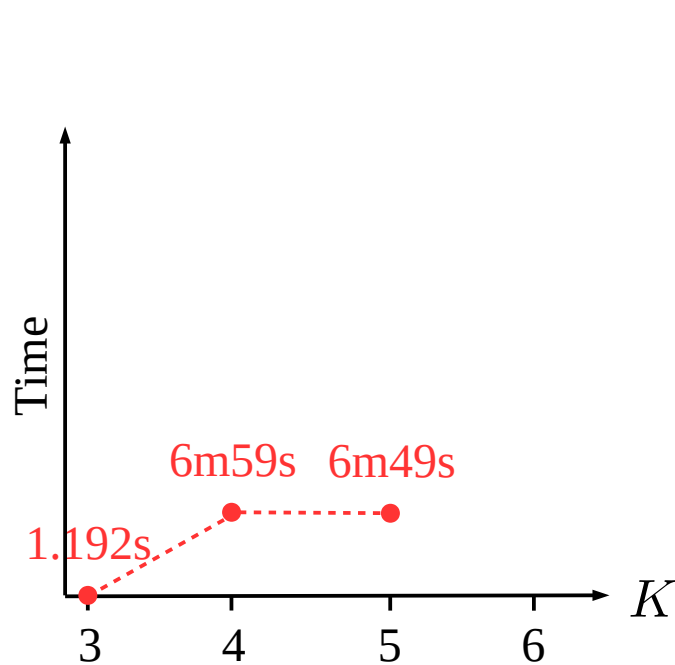
Supplementary — Time measurements — Function of K



$\mathcal{G}_{\text{geo}}, \alpha = 1.0, \beta = 0.7, \gamma = 0.5$
444 vertices
C++ implementation w/ backtracking



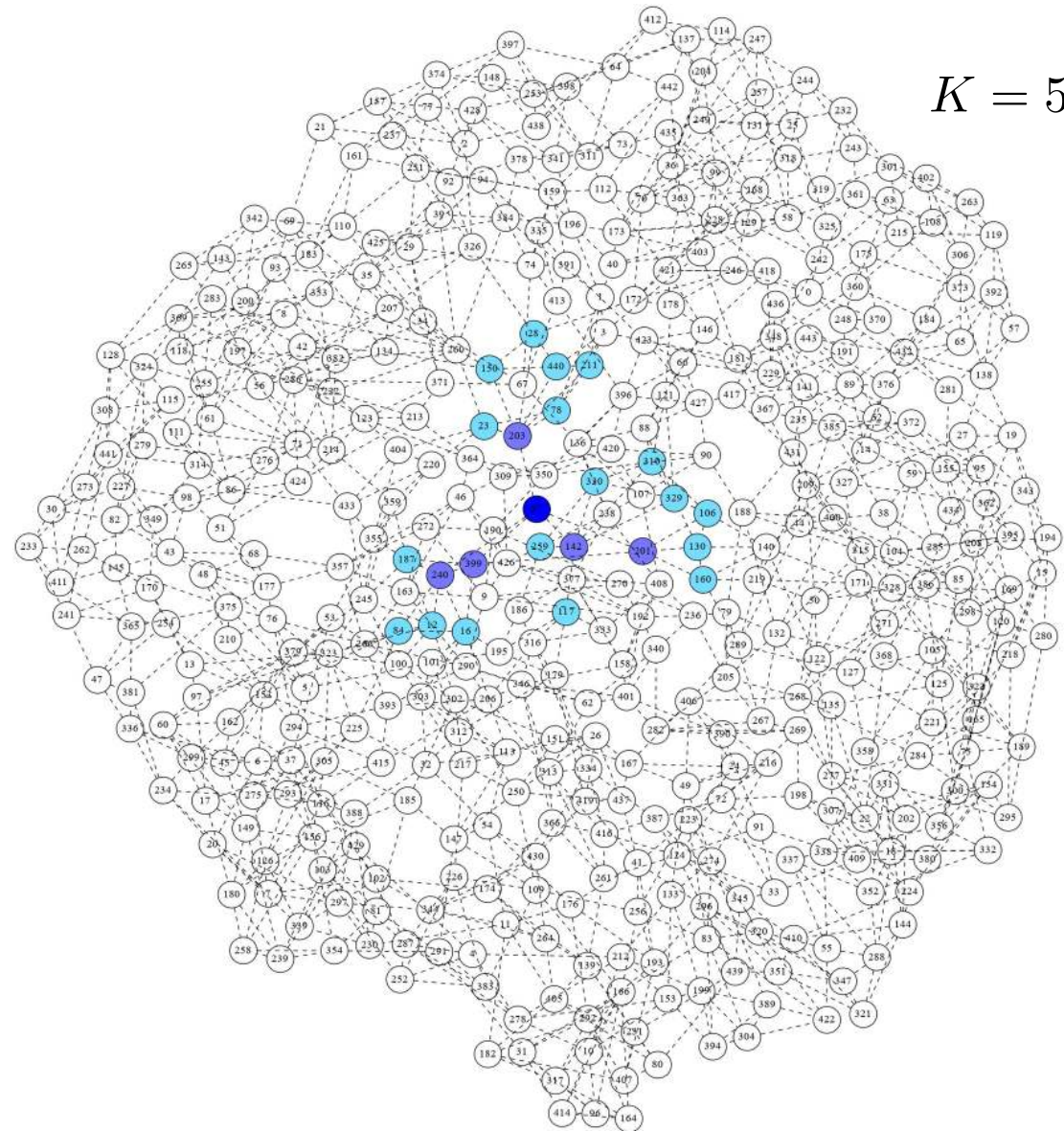
Supplementary — Time measurements — Function of K



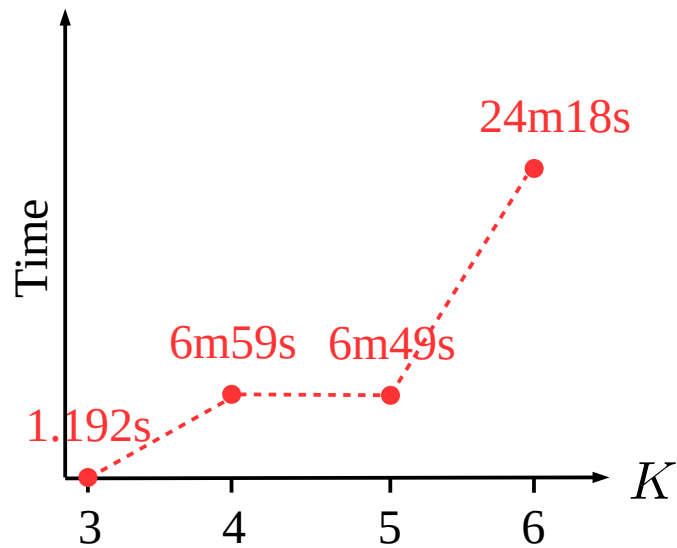
$\mathcal{G}_{\text{geo}}, \alpha = 1.0, \beta = 0.7, \gamma = 0.5$

444 vertices

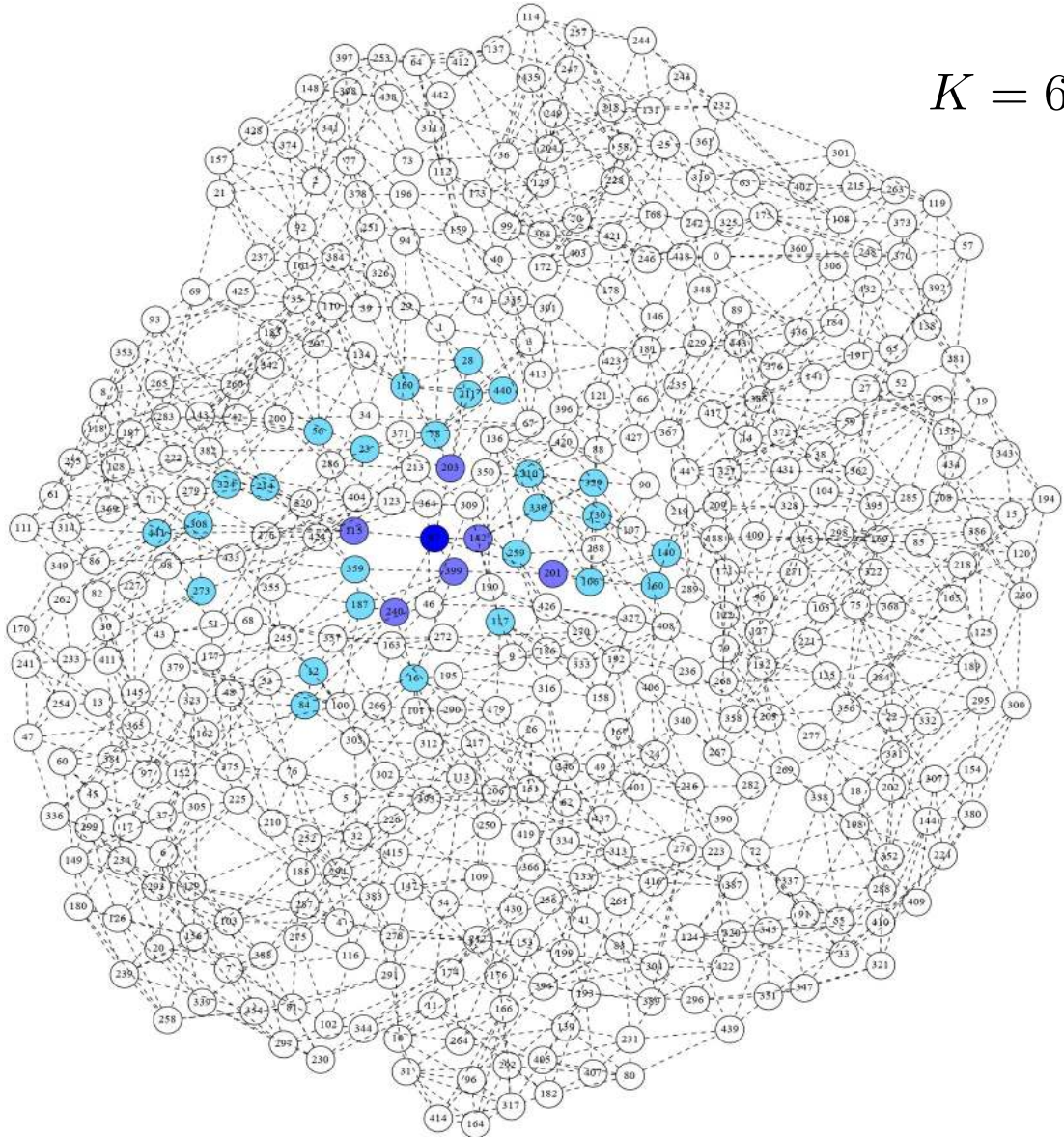
C++ implementation w/ backtracking



Supplementary — Time measurements — Function of K

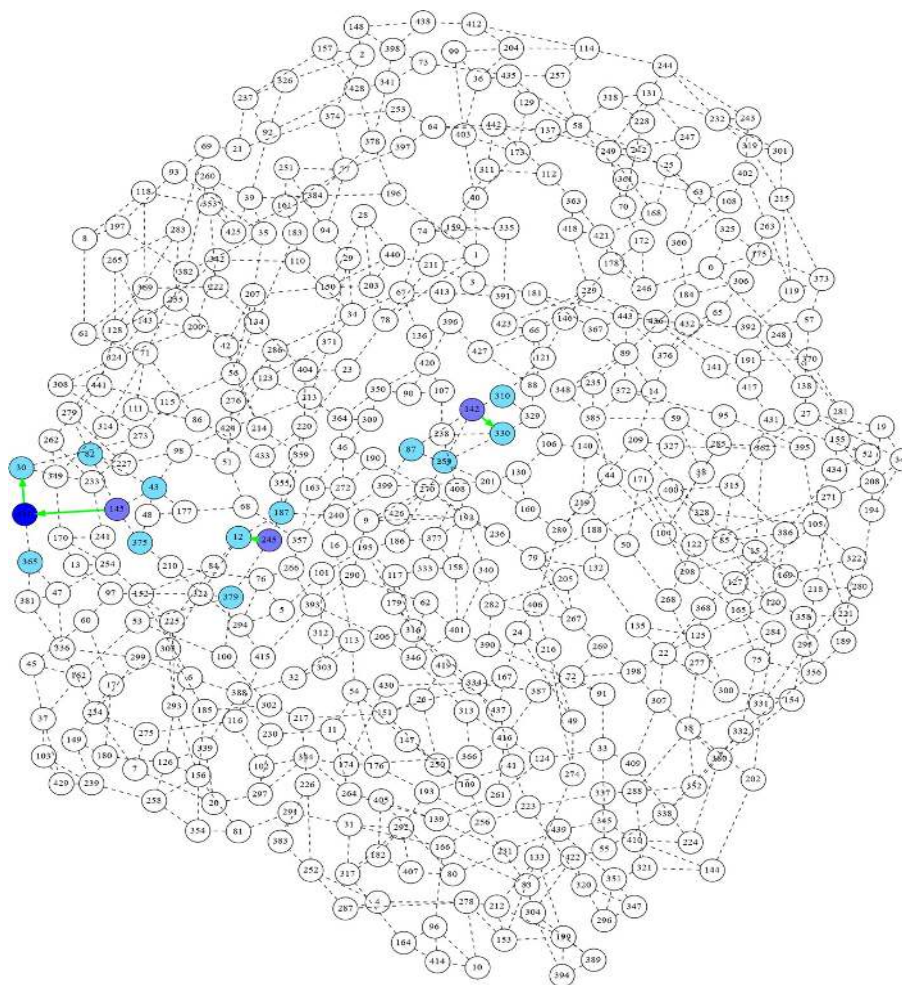
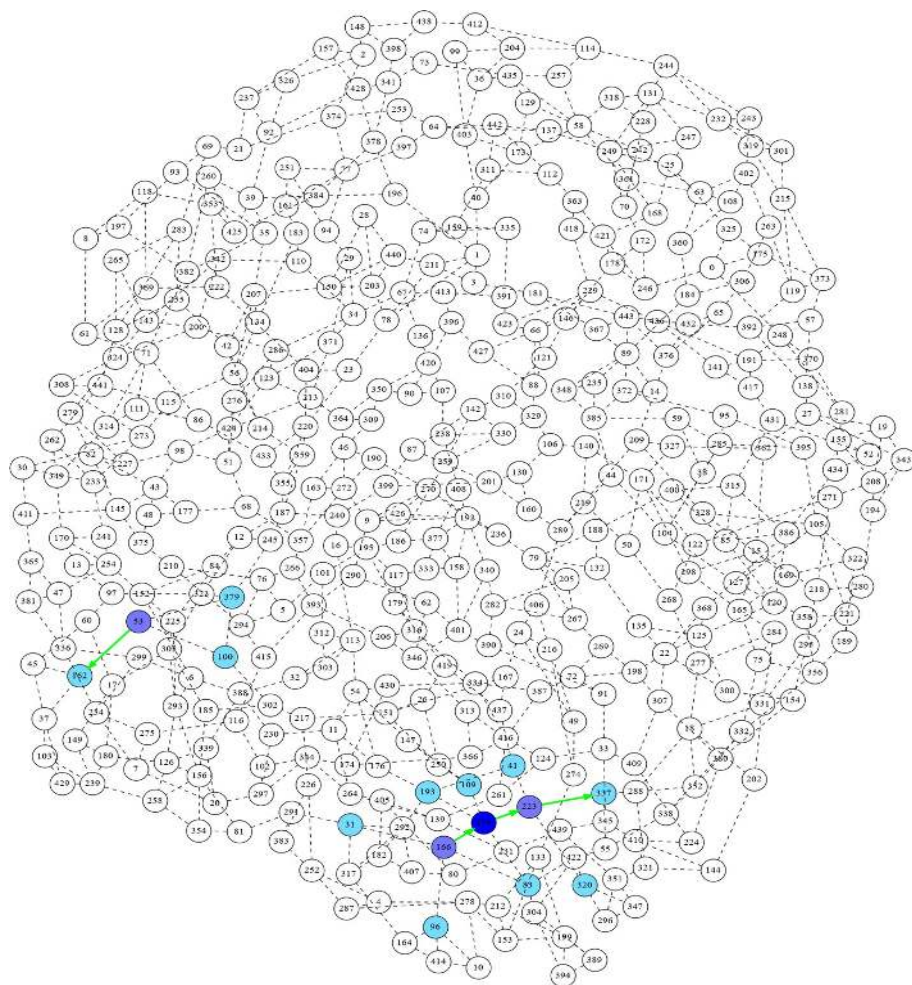


$\mathcal{G}_{\text{geo}}, \alpha = 1.0, \beta = 0.7, \gamma = 0.5$
444 vertices
C++ implementation w/ backtracking



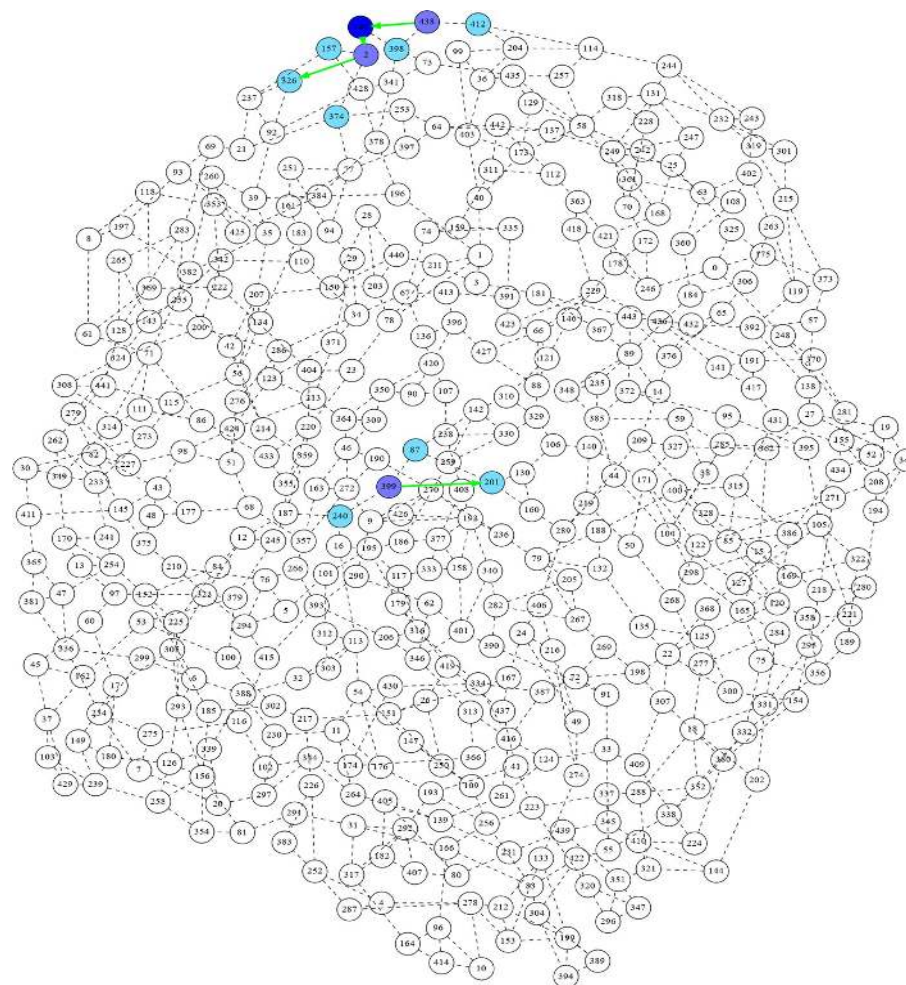
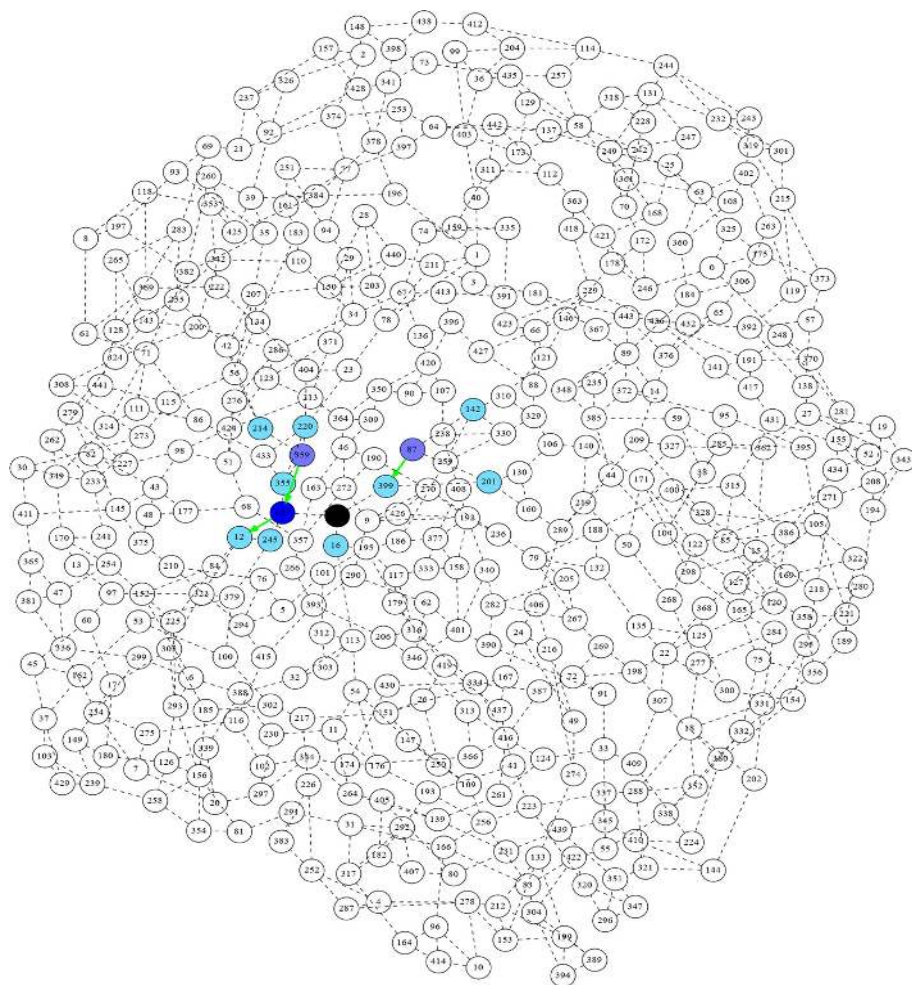
Supplementary — Time measurements — Impact of parameters

$$\mathcal{G}_{\text{geo}}, K = 3, \alpha = \infty, \beta = 0.7, \gamma = 0.5$$



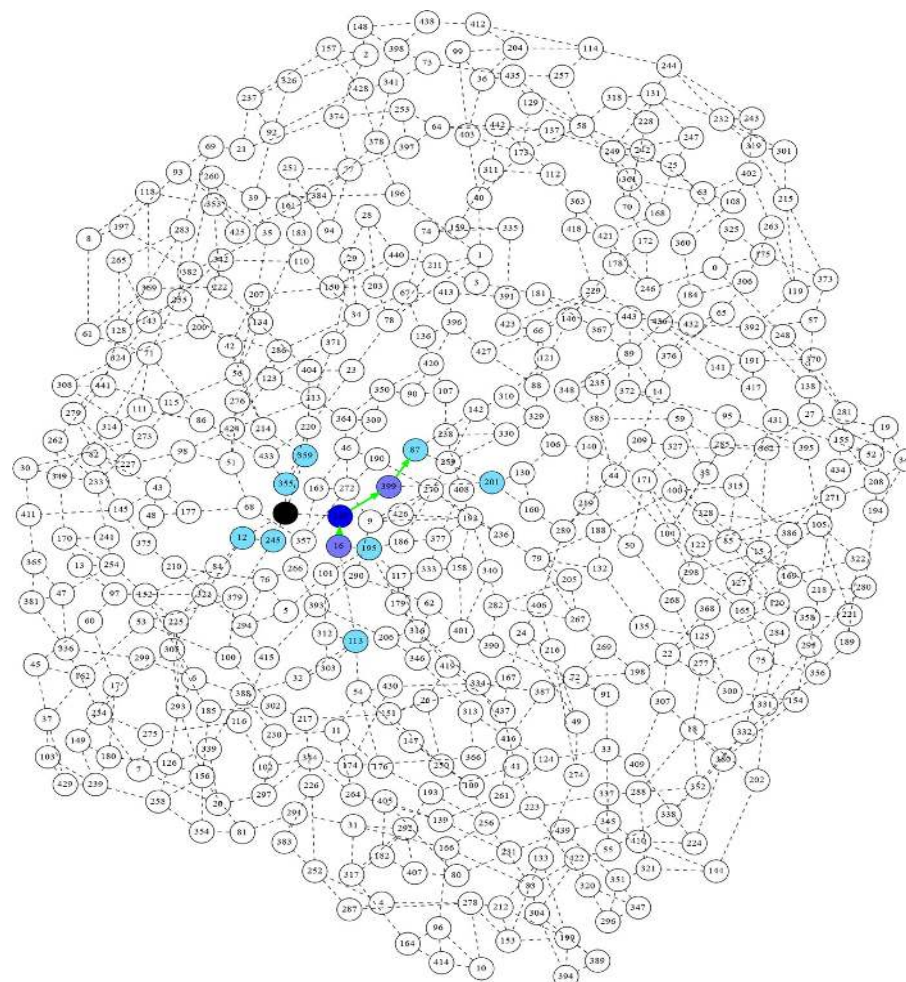
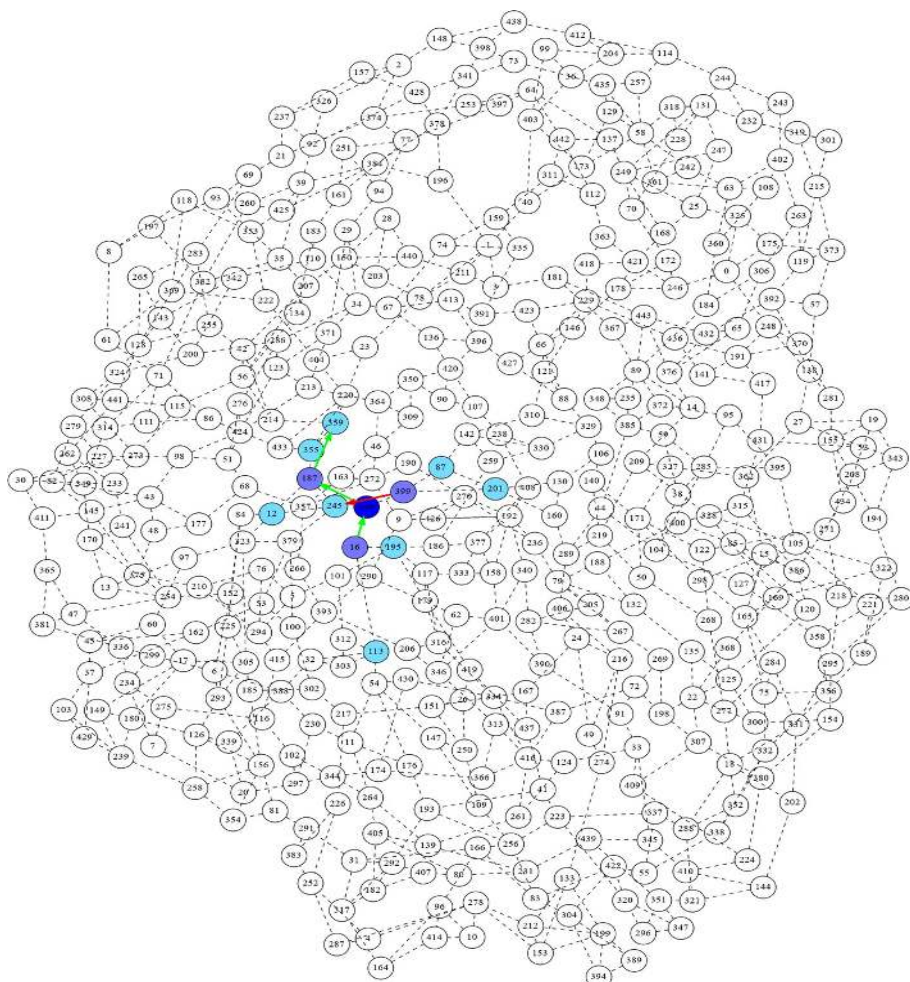
Supplementary — Time measurements — Impact of parameters

$$\mathcal{G}_{\text{geo}}, K = 3, \alpha = 1, 0, \beta = \infty, \gamma = 0.5$$



Supplementary — Time measurements — Impact of parameters

$$\mathcal{G}_{\text{geo}}, K = 3, \alpha = 1, 0, \beta = 0, 7, \gamma = \infty$$

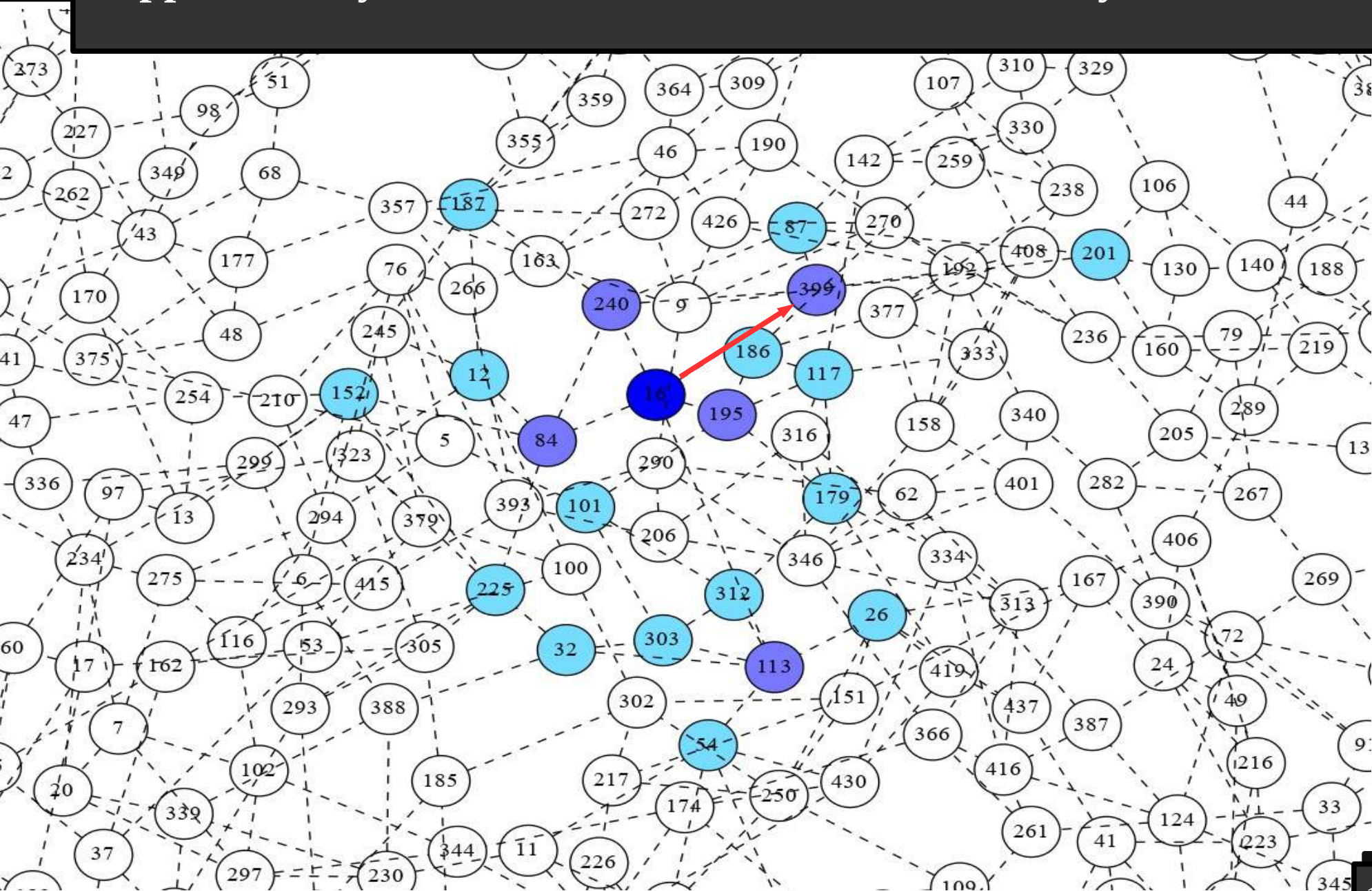


Outline

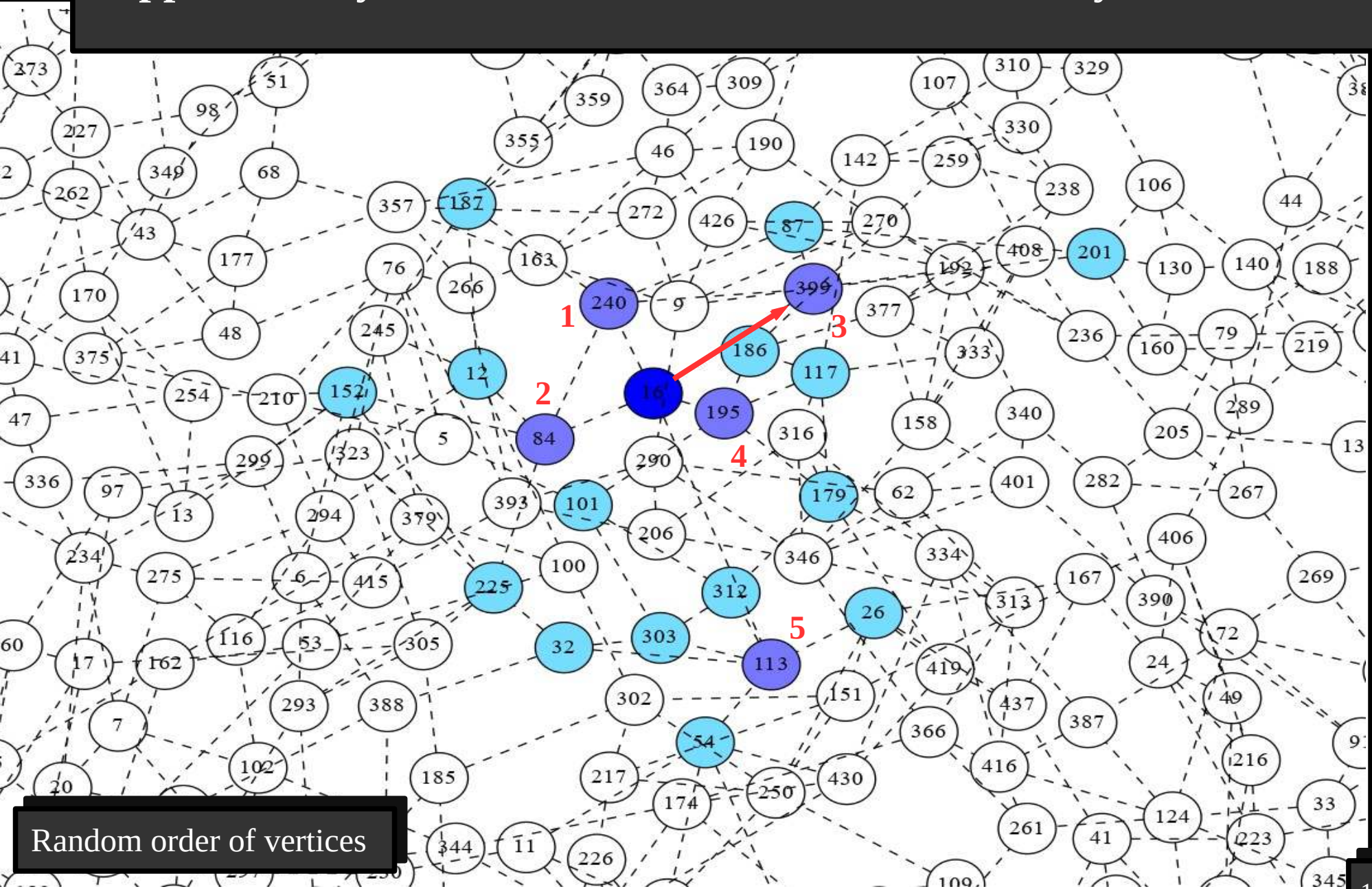
Supplementary slides

— Stochasticity to find translations —

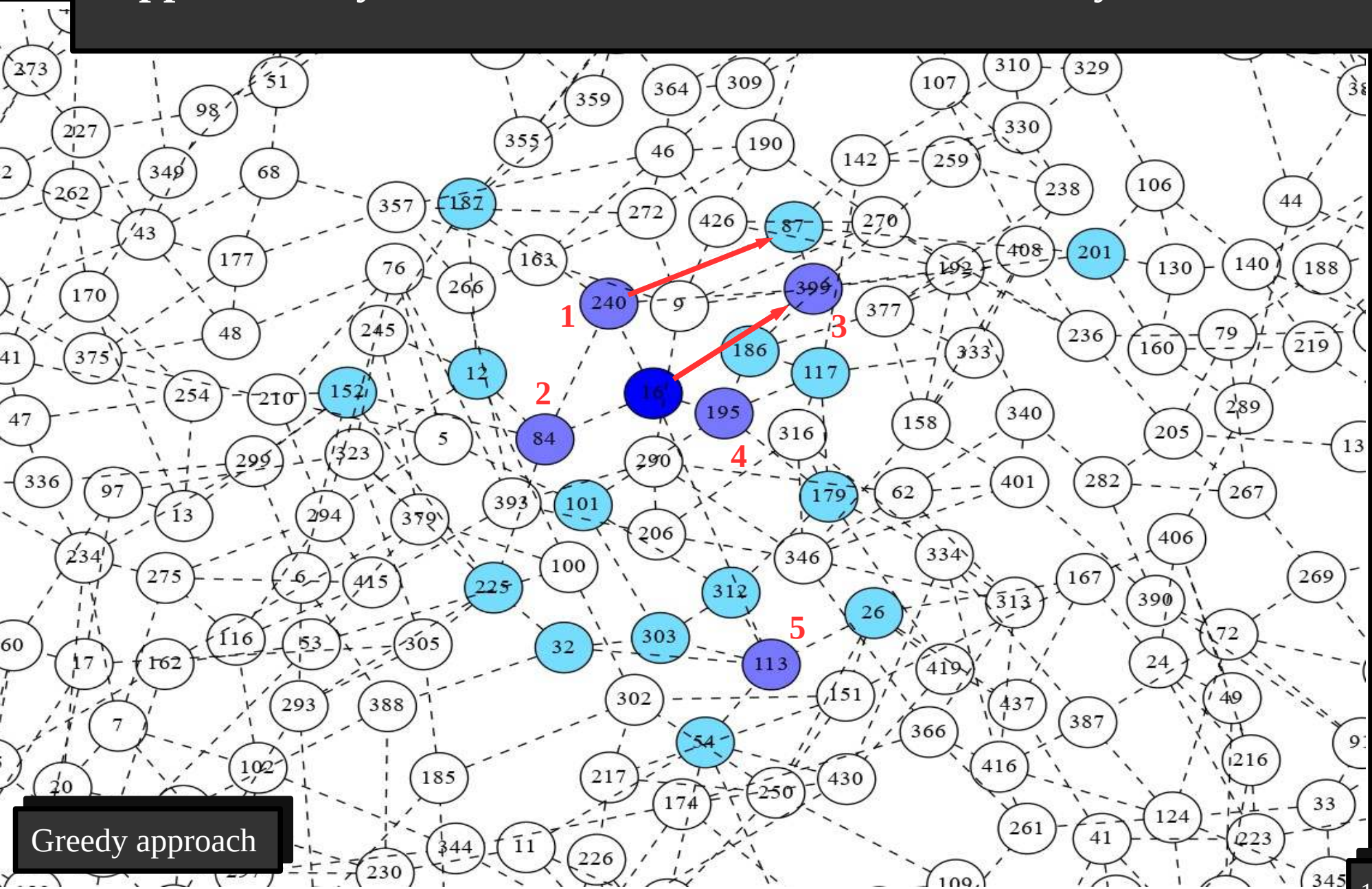
Supplementary — Stochastic translations — Greedy w/ random order



Supplementary — Stochastic translations — Greedy w/ random order

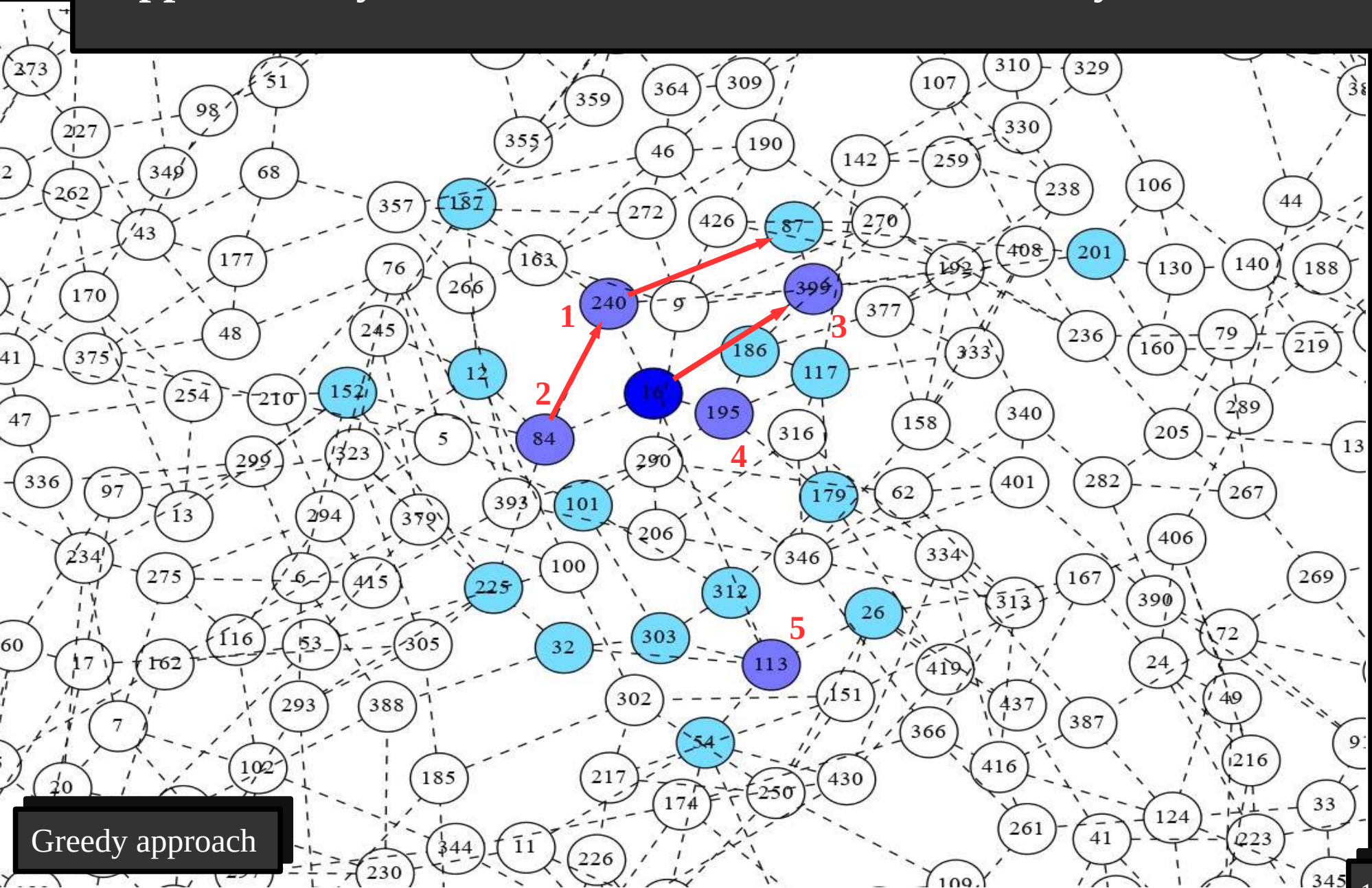


Supplementary — Stochastic translations — Greedy w/ random order

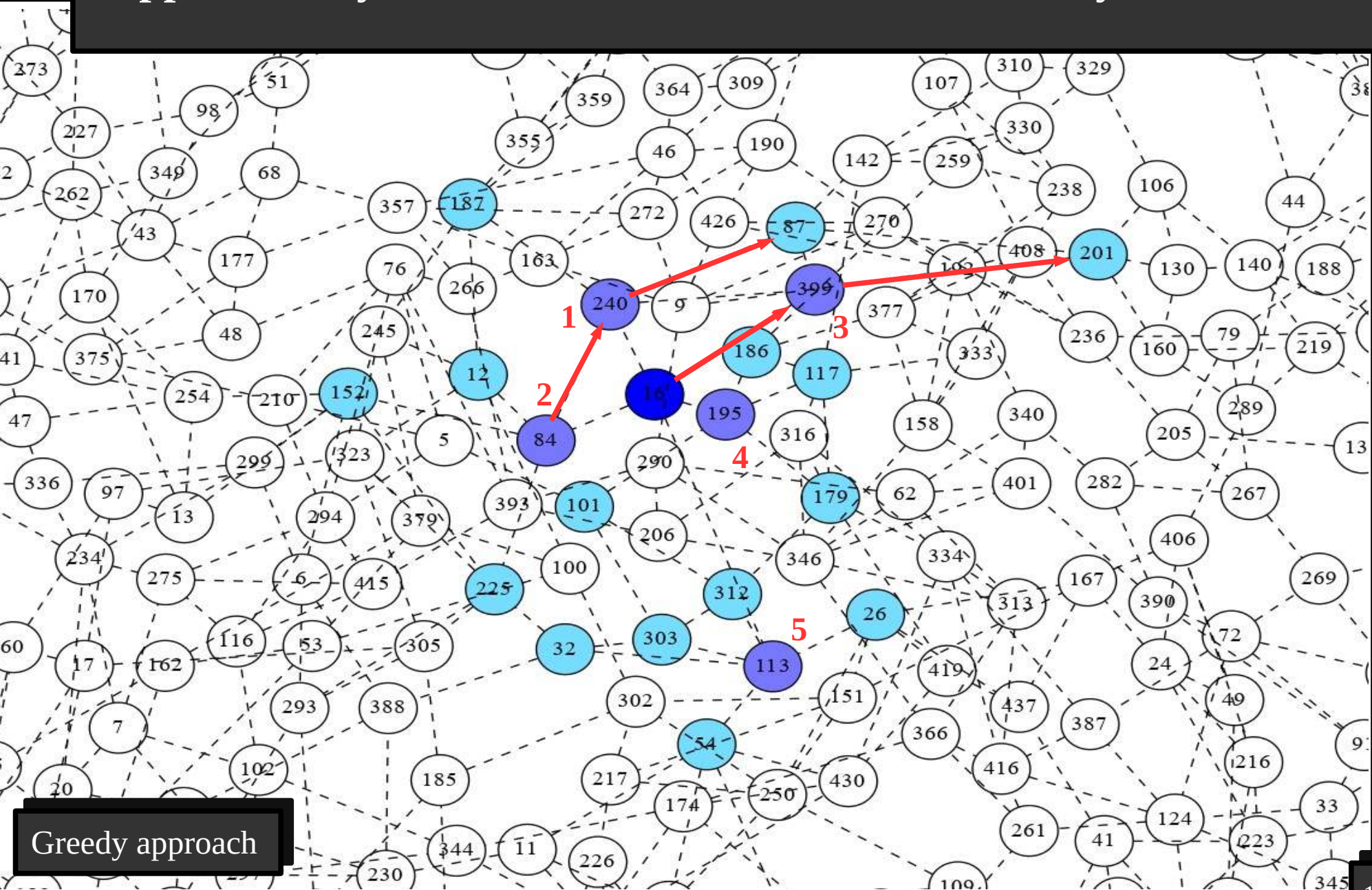


Greedy approach

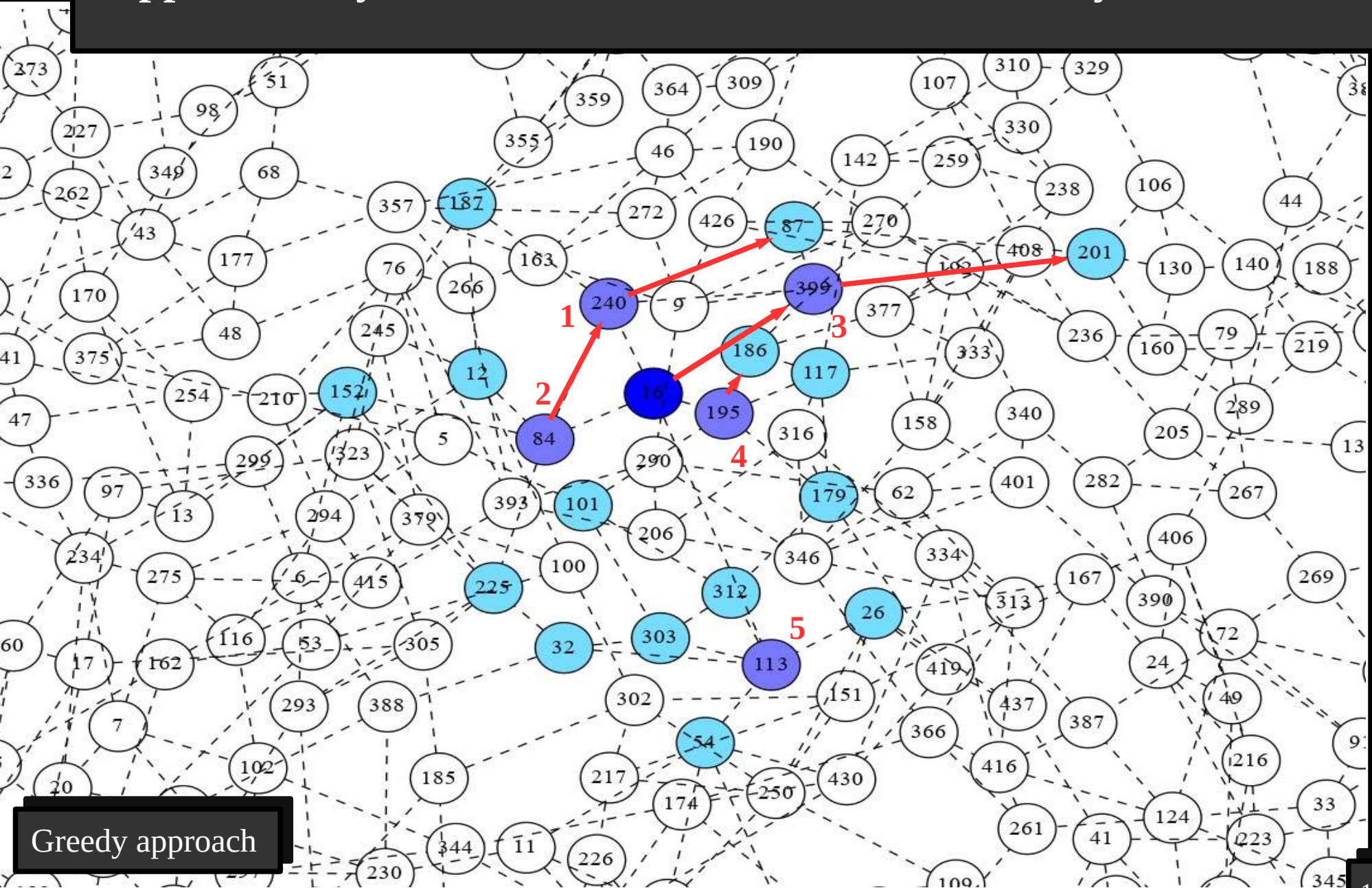
Supplementary — Stochastic translations — Greedy w/ random order



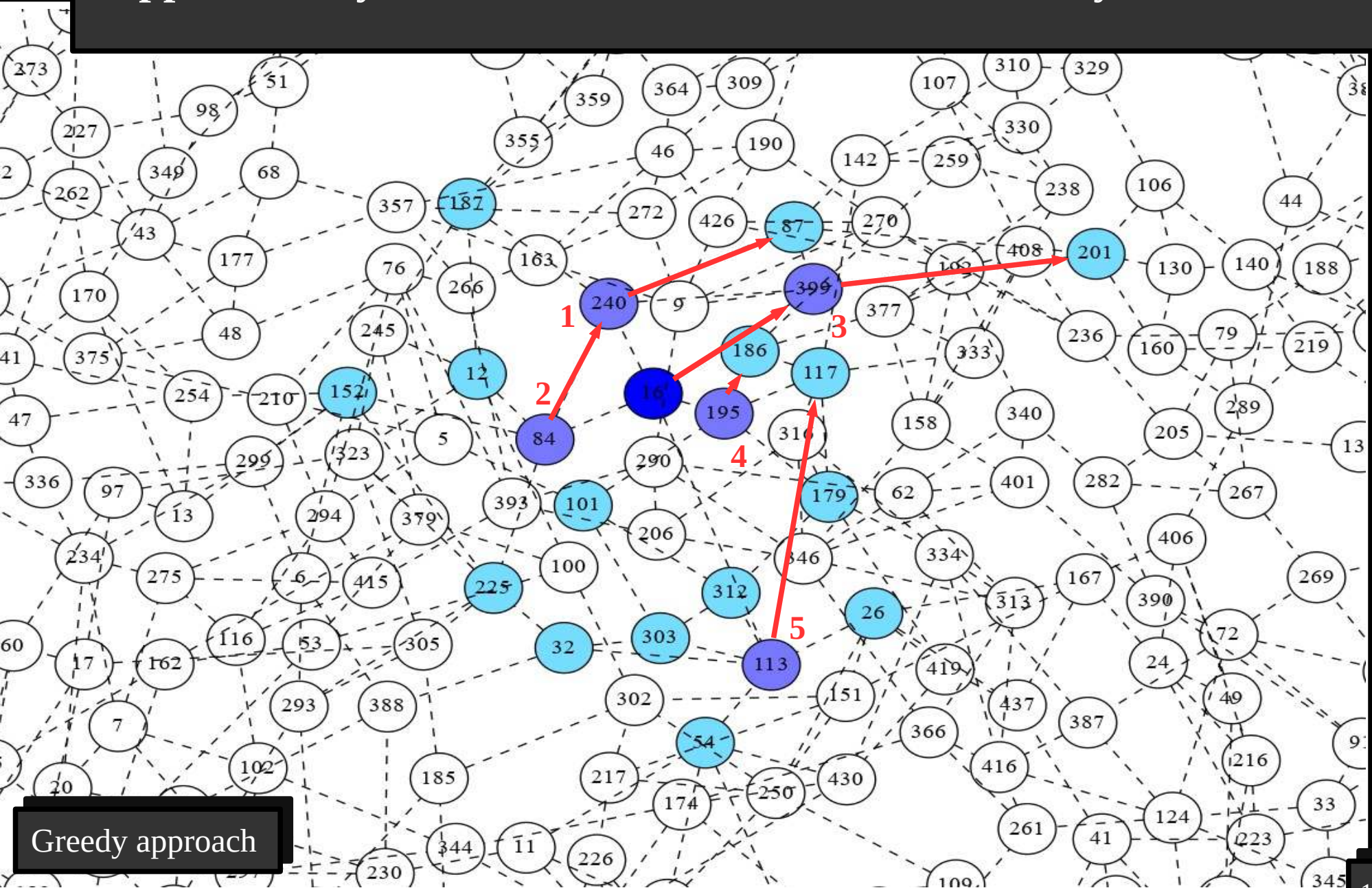
Supplementary — Stochastic translations — Greedy w/ random order



Supplementary — Stochastic translations — Greedy w/ random order

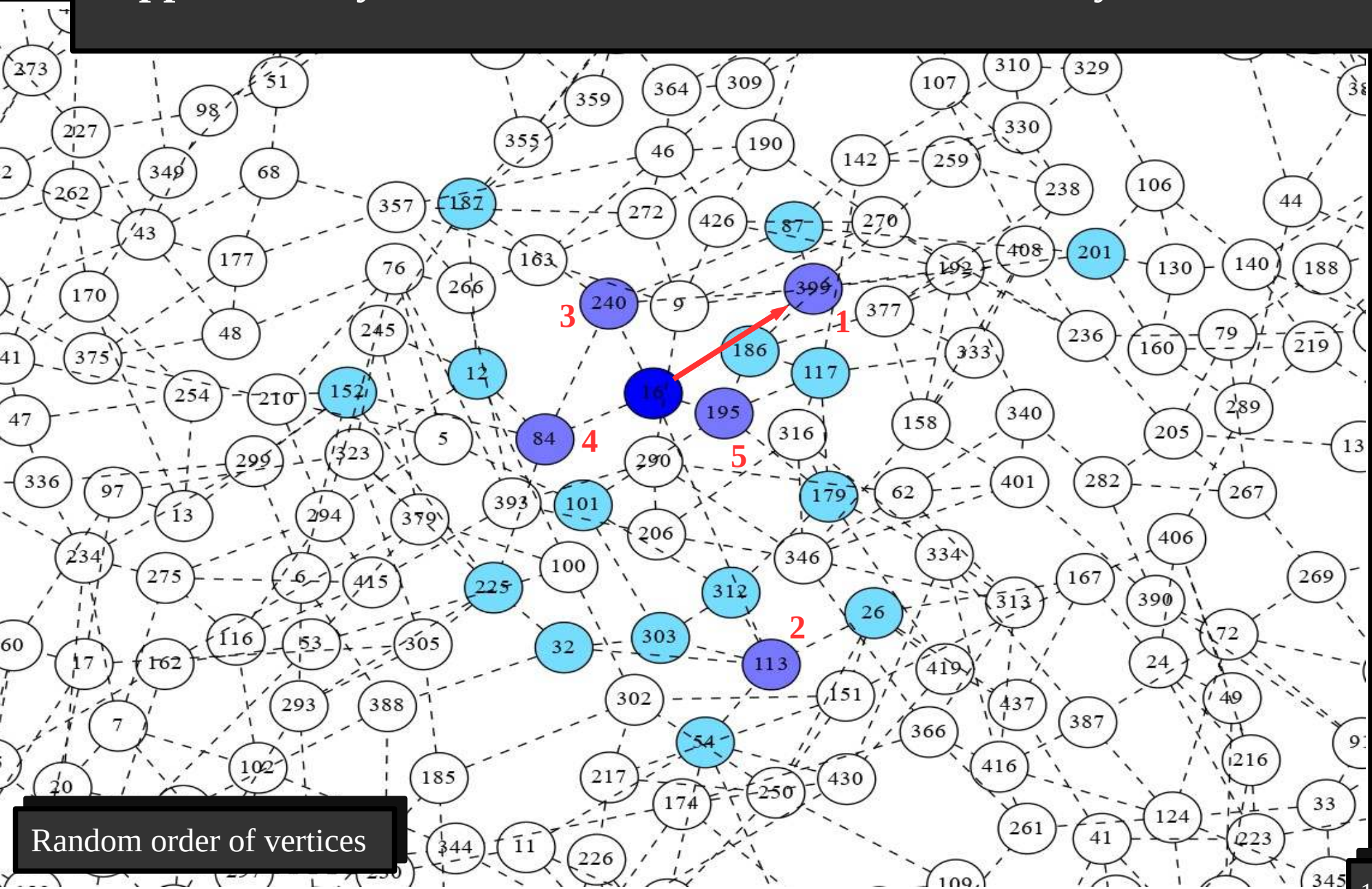


Supplementary — Stochastic translations — Greedy w/ random order



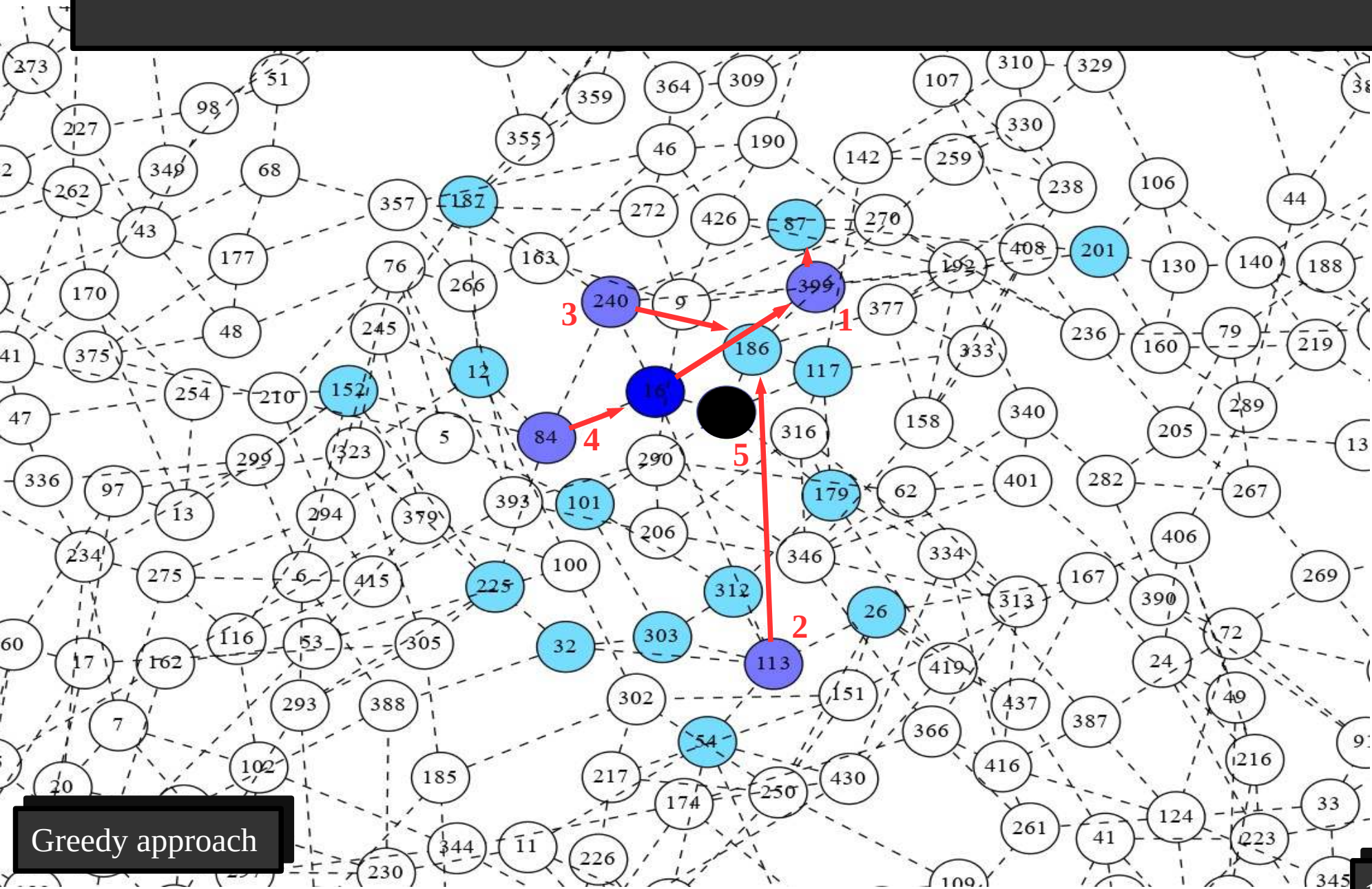
Greedy approach

Supplementary — Stochastic translations — Greedy w/ random order

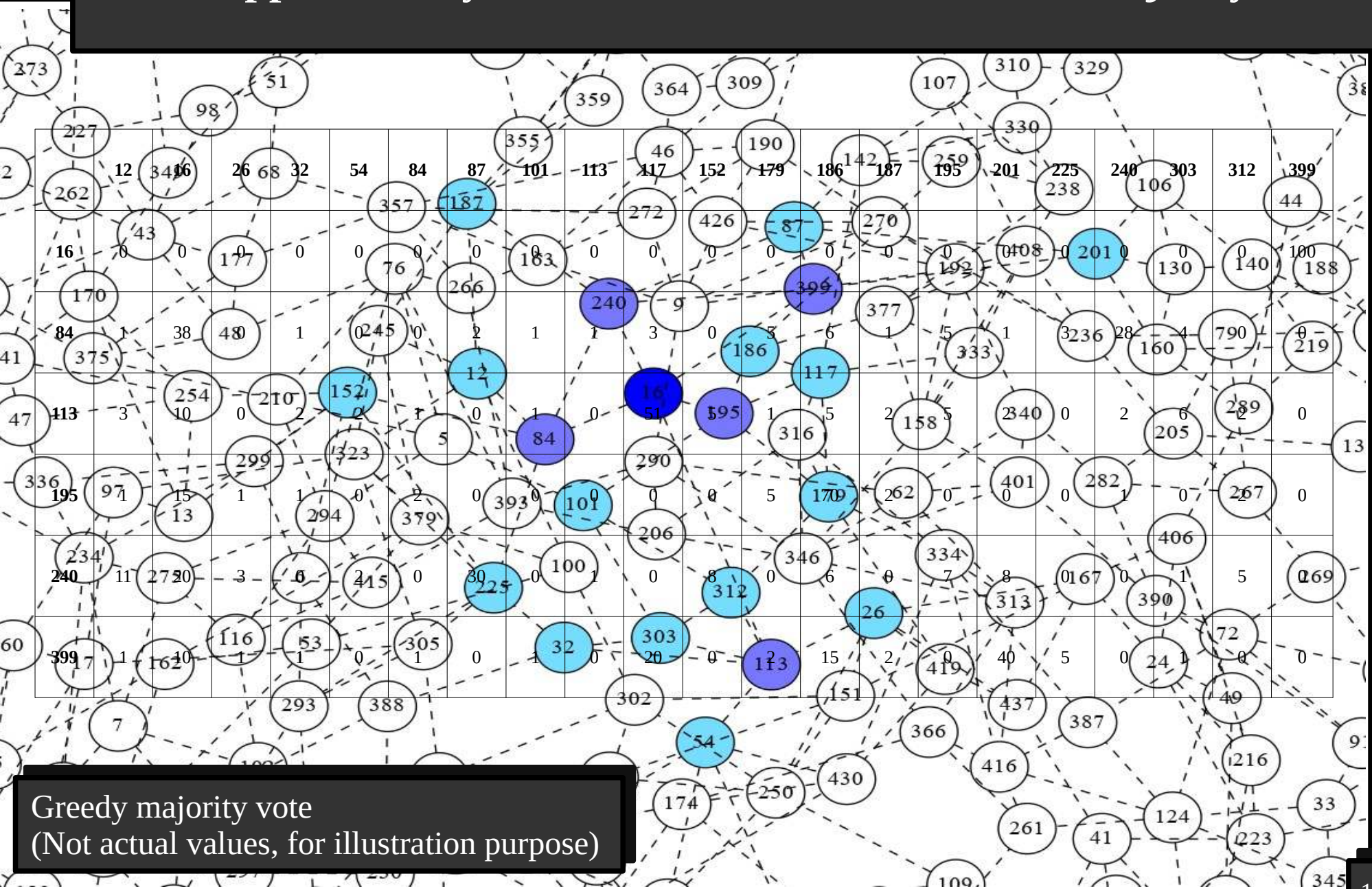


Random order of vertices

Supplementary — Stochastic translations — Greedy w/ random order

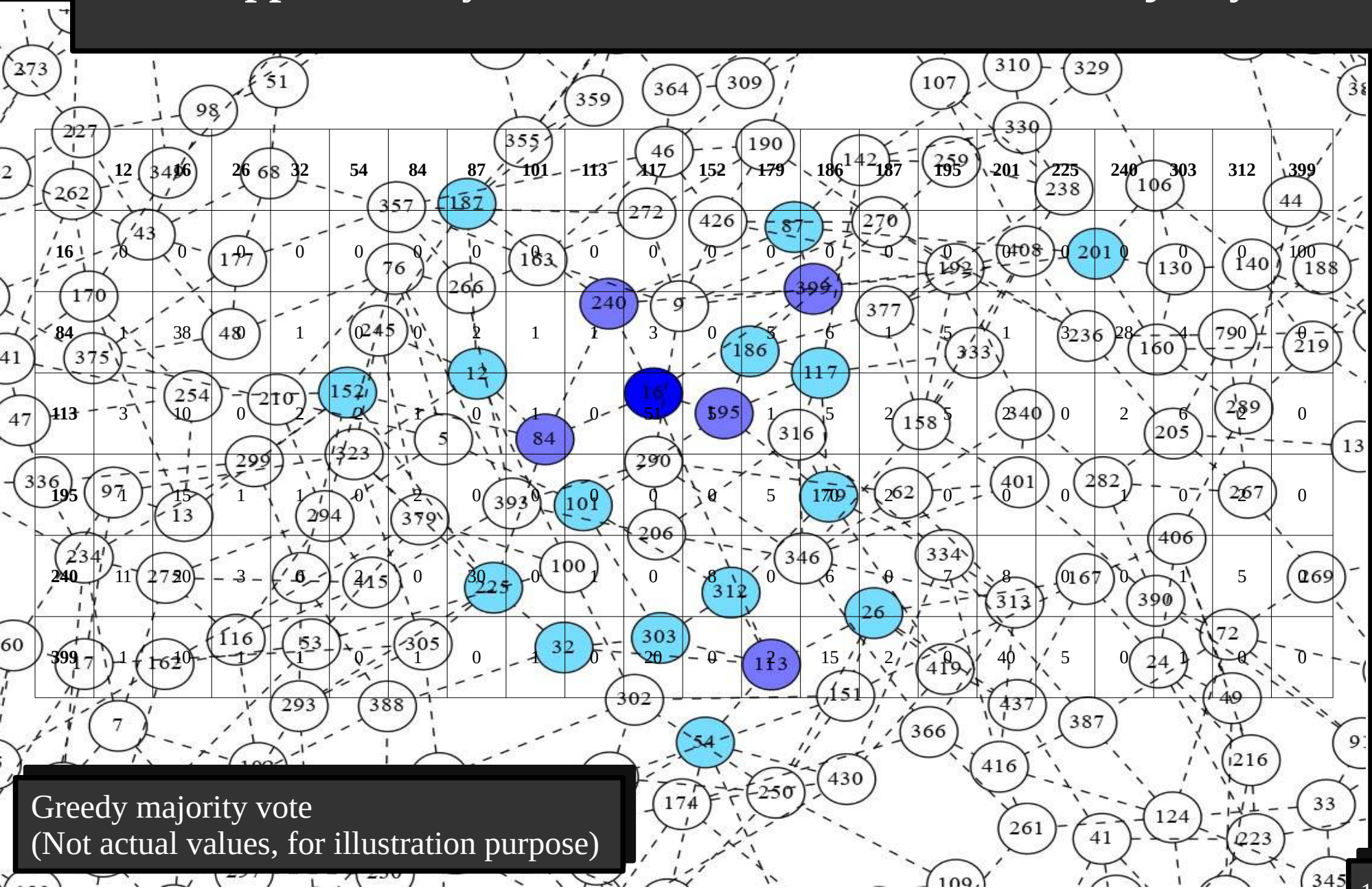


Supplementary — Stochastic translations — Majority vote

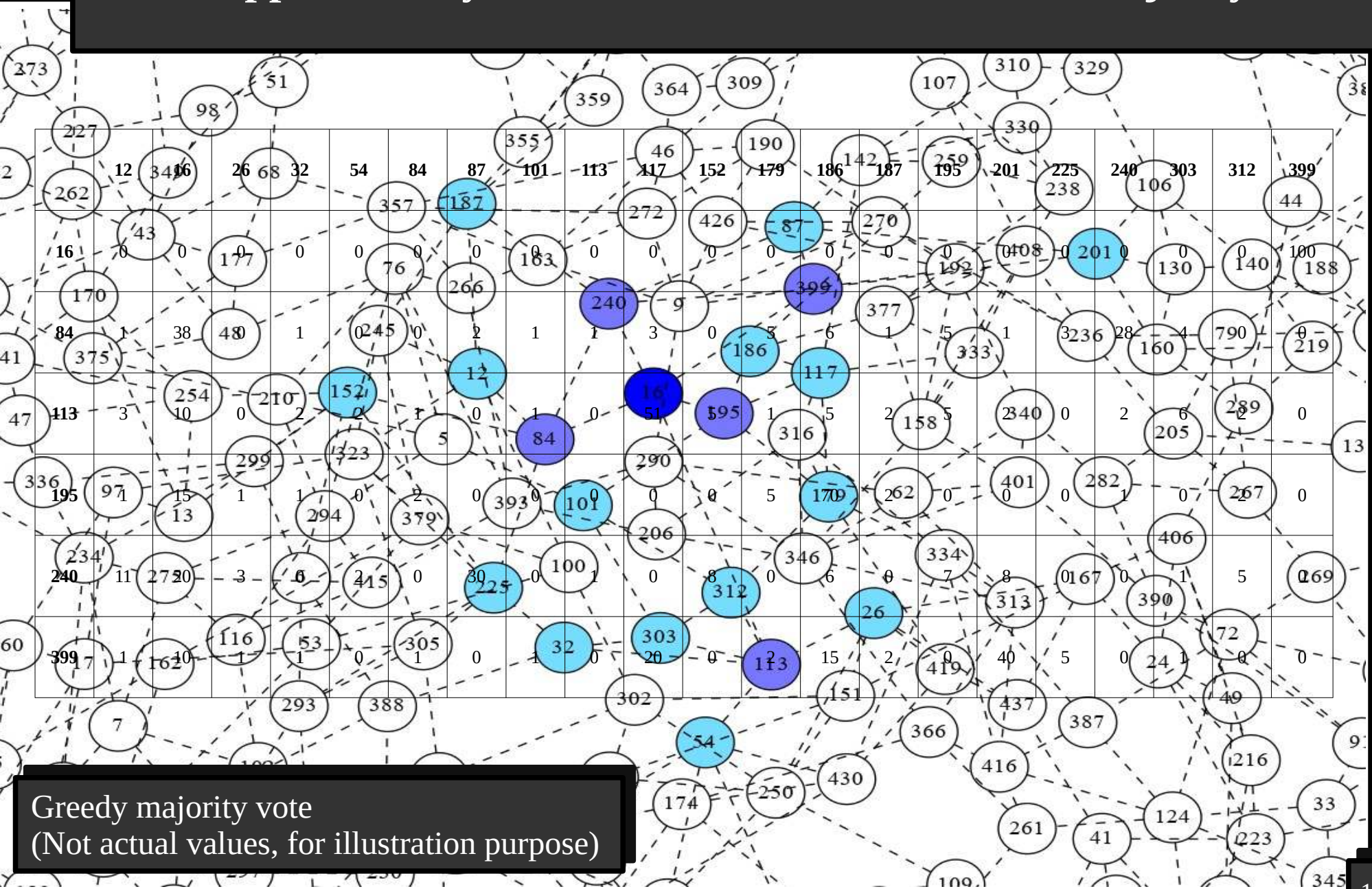


Greedy majority vote
(Not actual values, for illustration purpose)

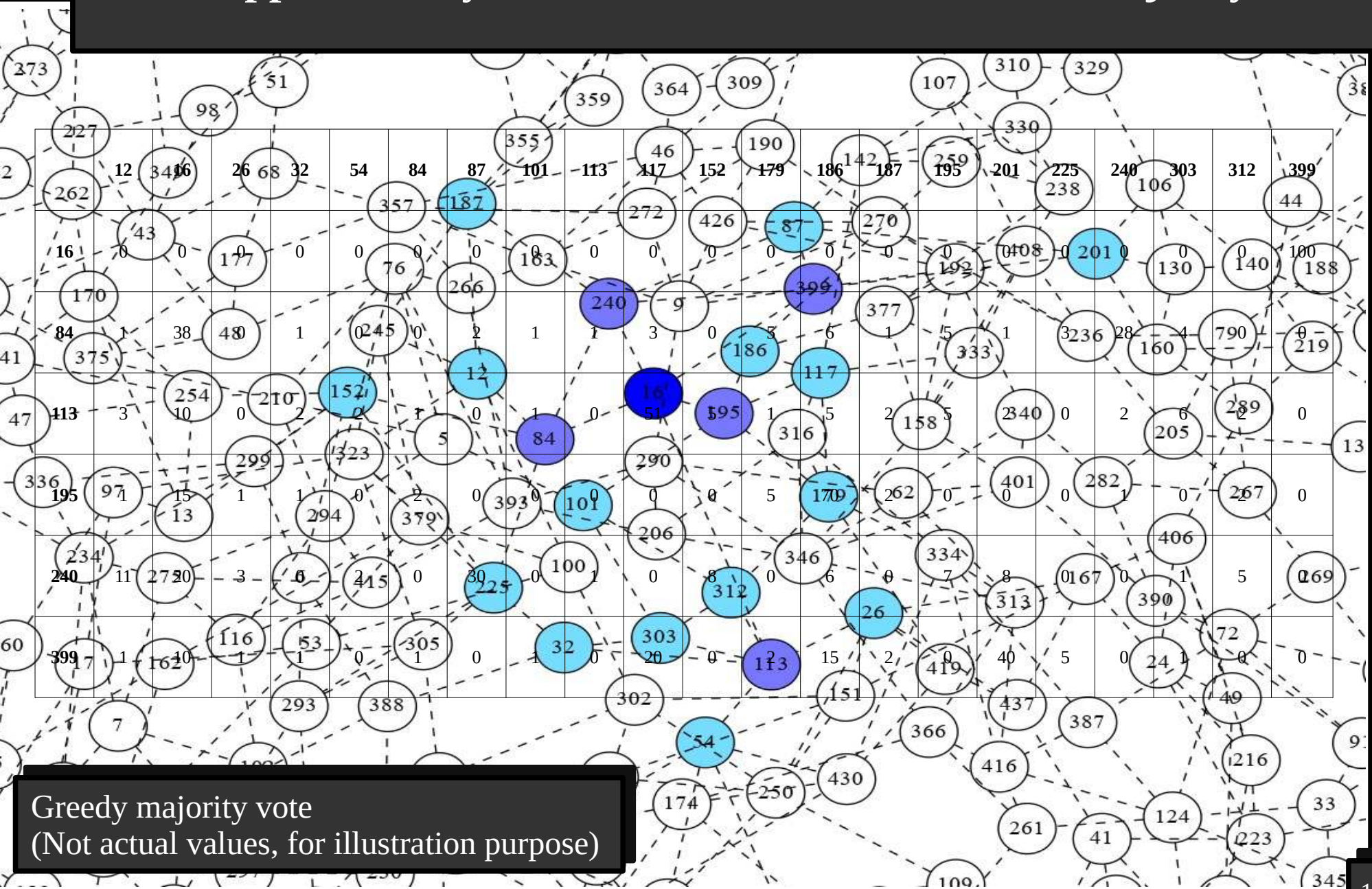
Supplementary — Stochastic translations — Majority vote



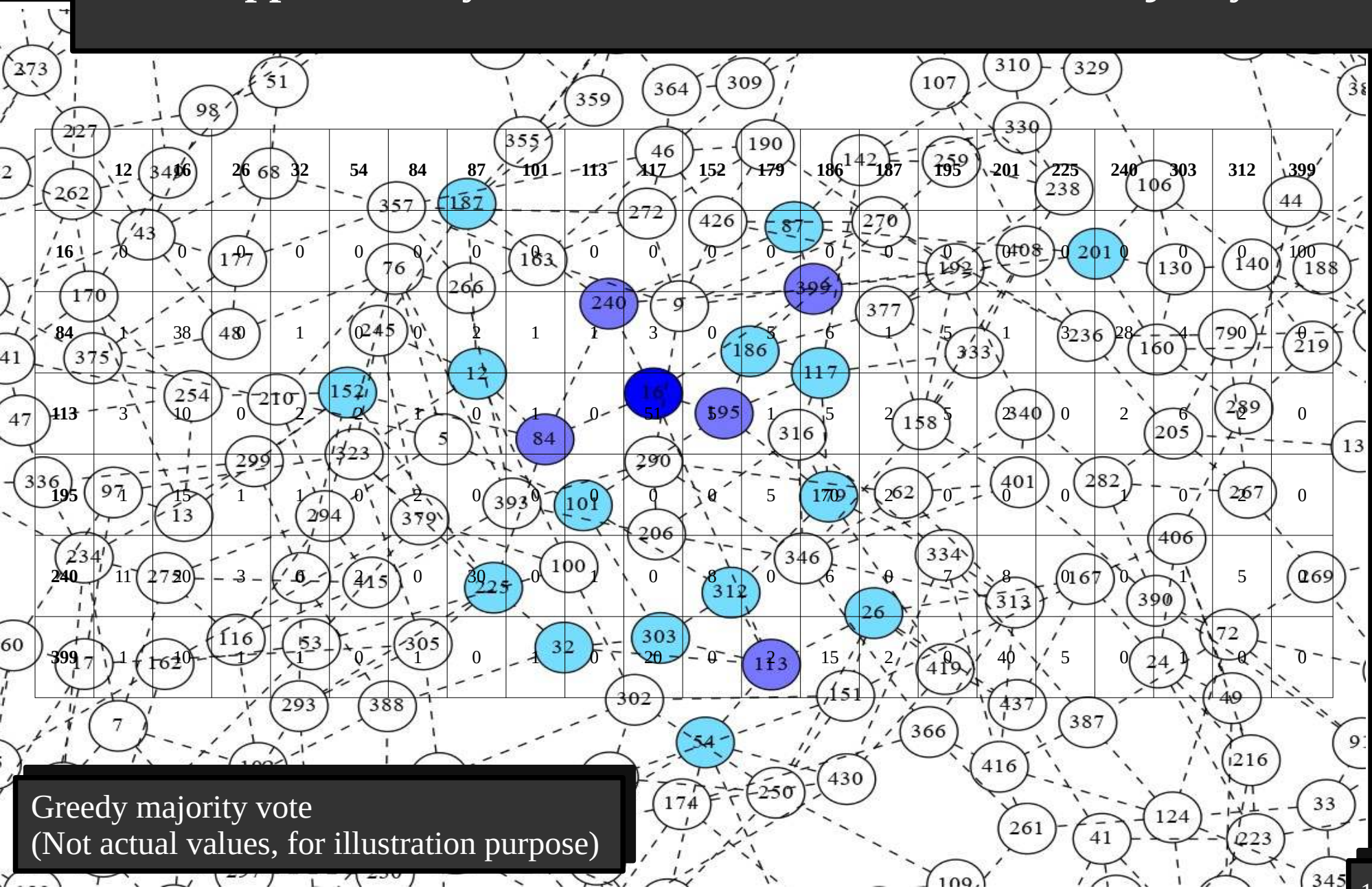
Supplementary — Stochastic translations — Majority vote



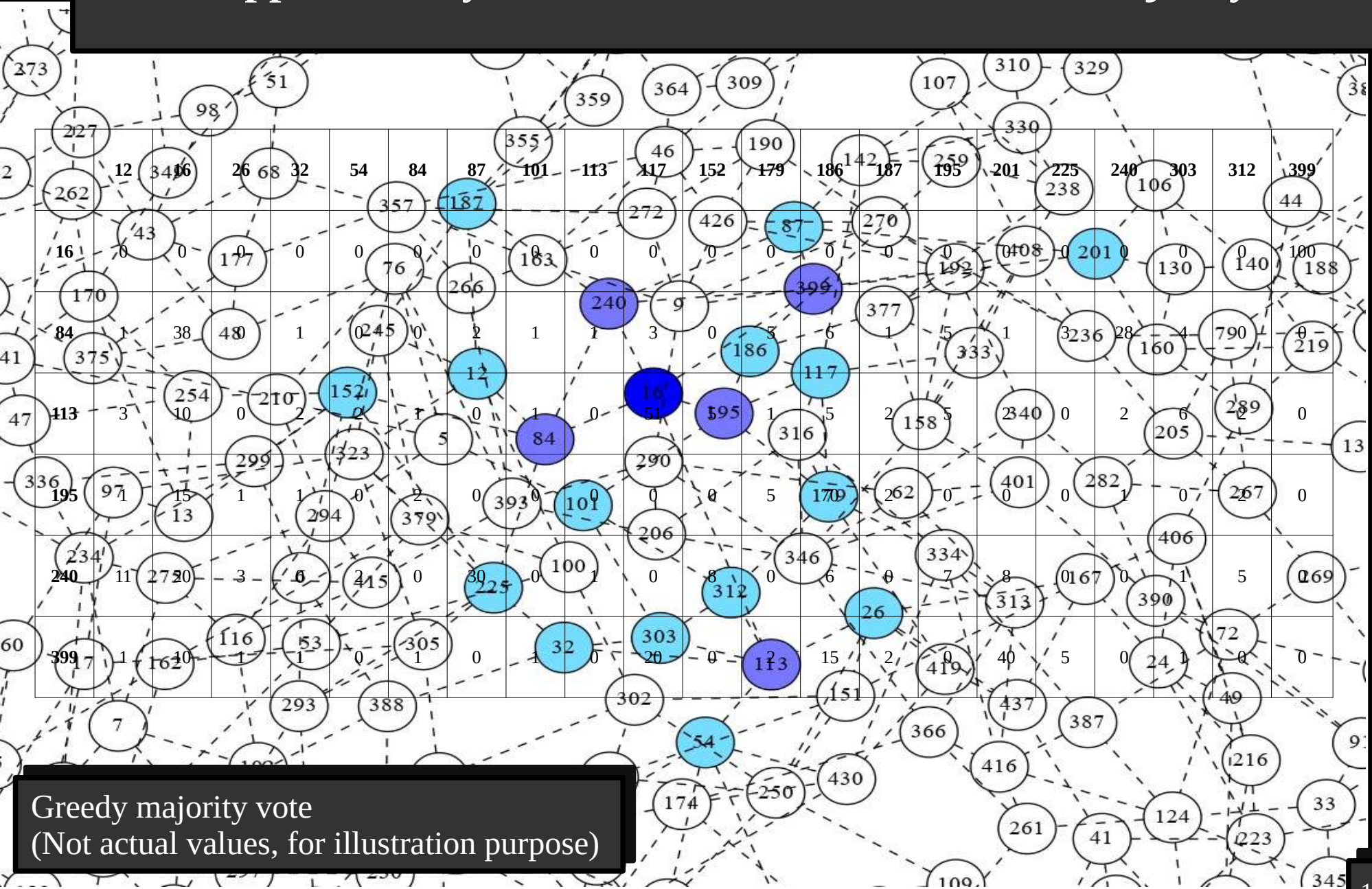
Supplementary — Stochastic translations — Majority vote



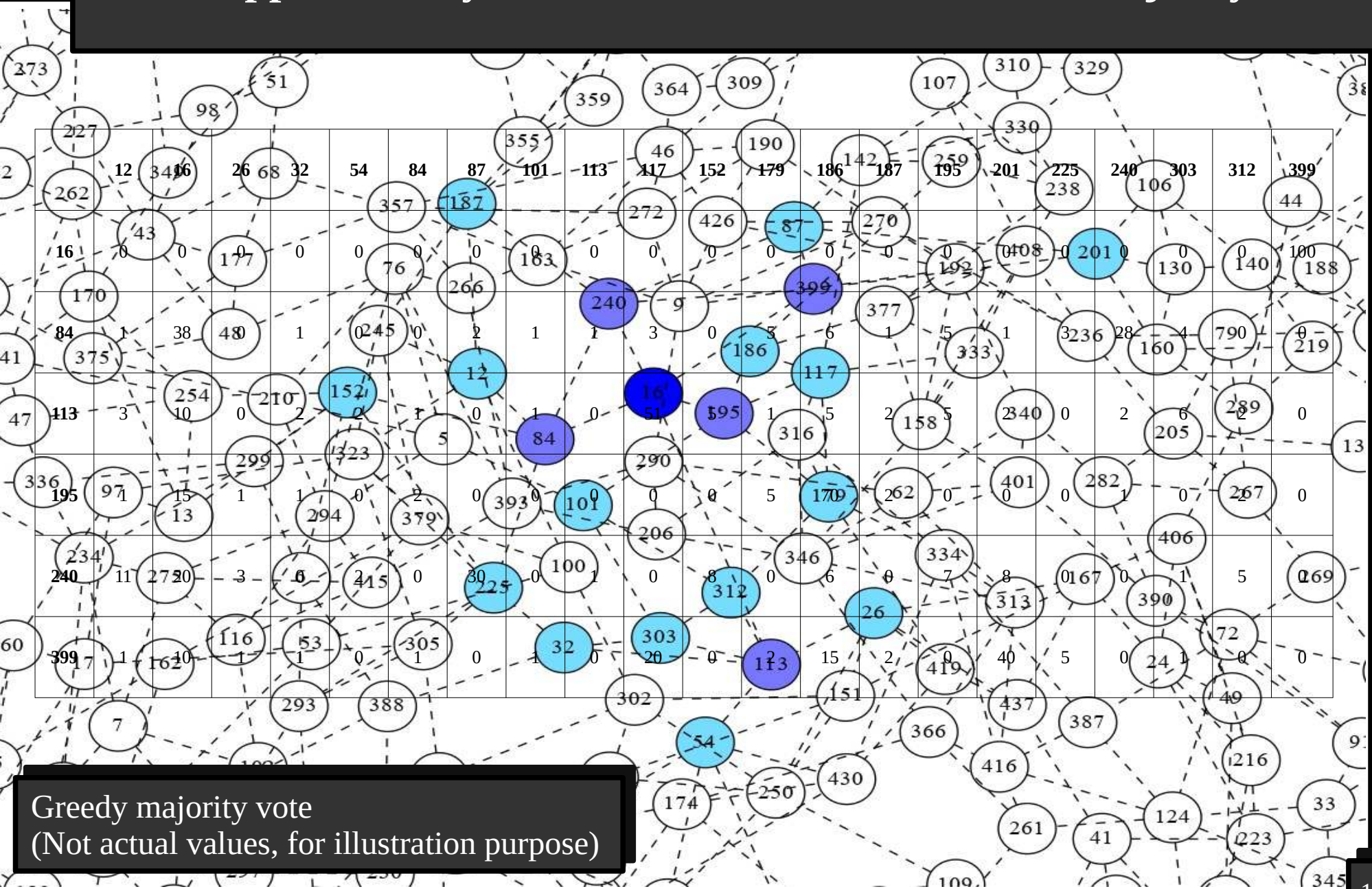
Supplementary — Stochastic translations — Majority vote



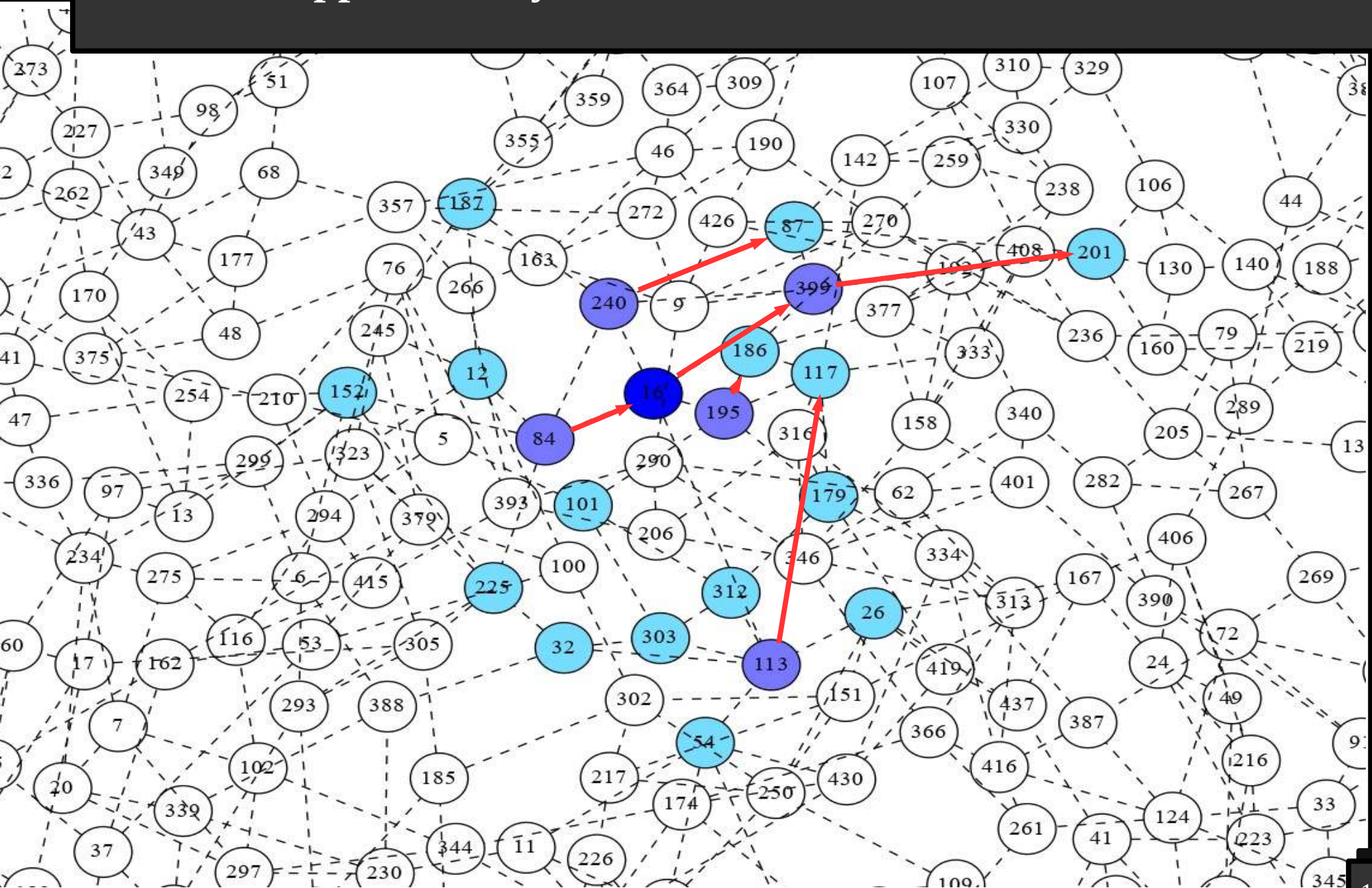
Supplementary — Stochastic translations — Majority vote



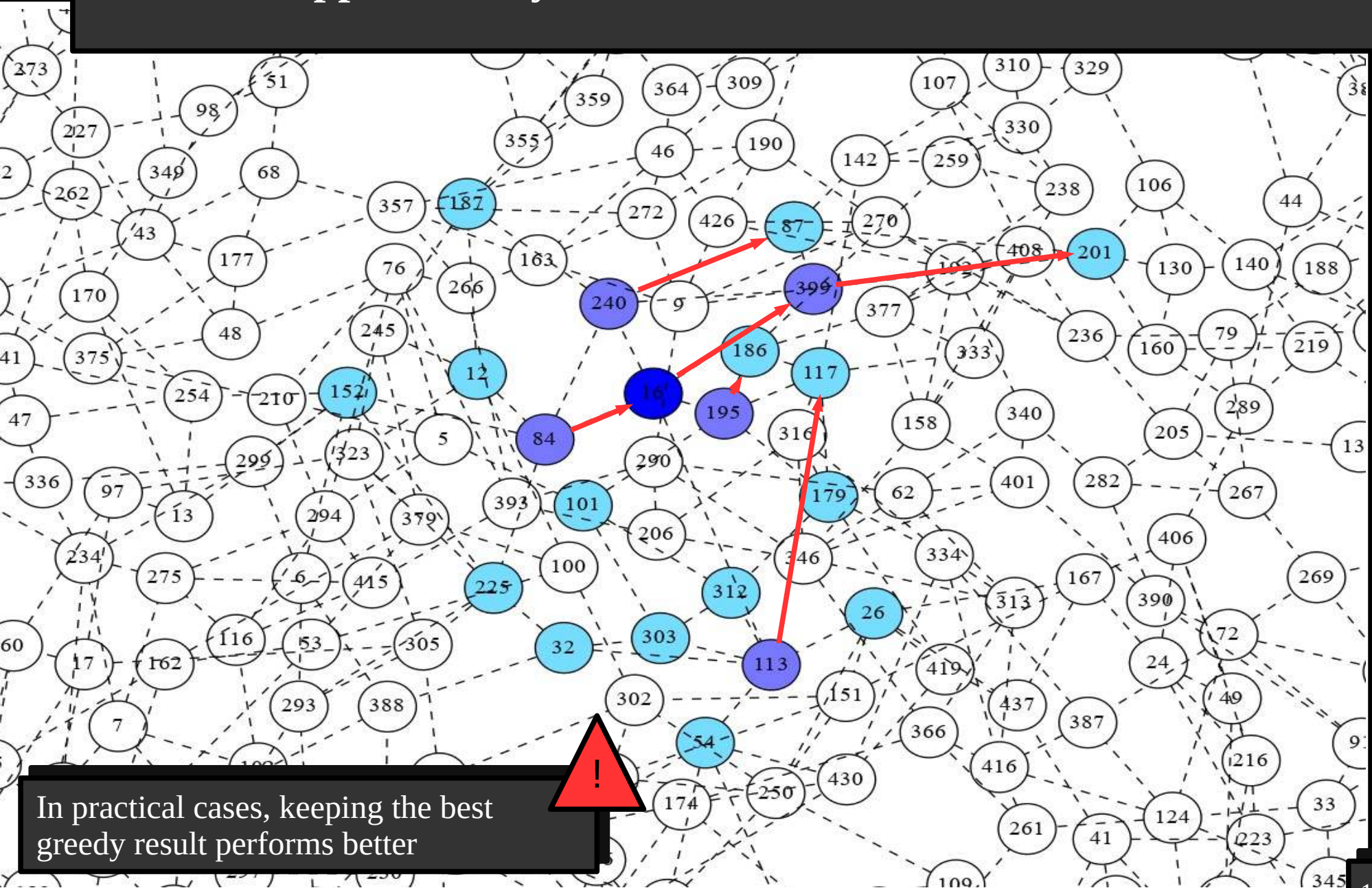
Supplementary — Stochastic translations — Majority vote



Supplementary — Stochastic translations — Result



Supplementary — Stochastic translations — Result

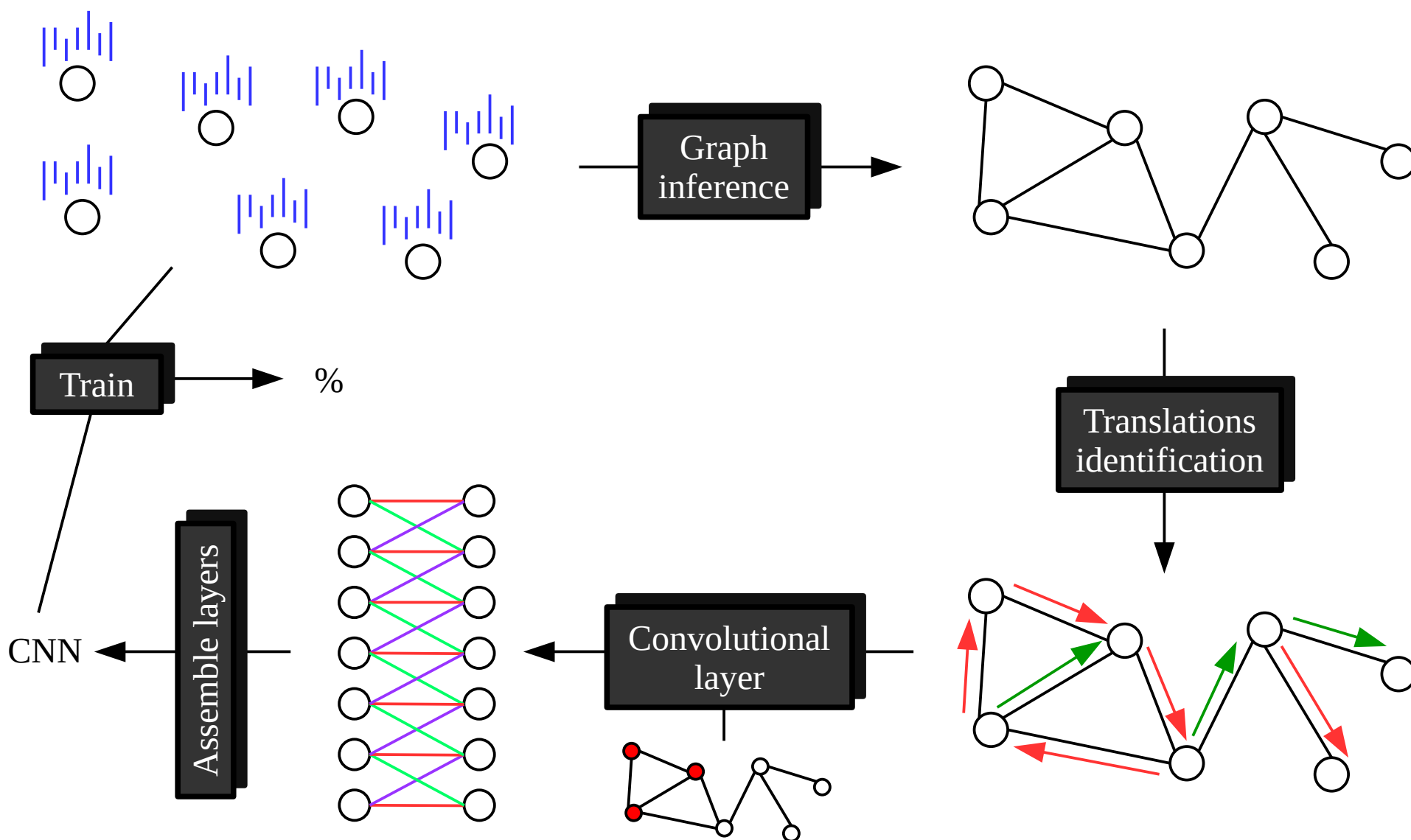


Outline

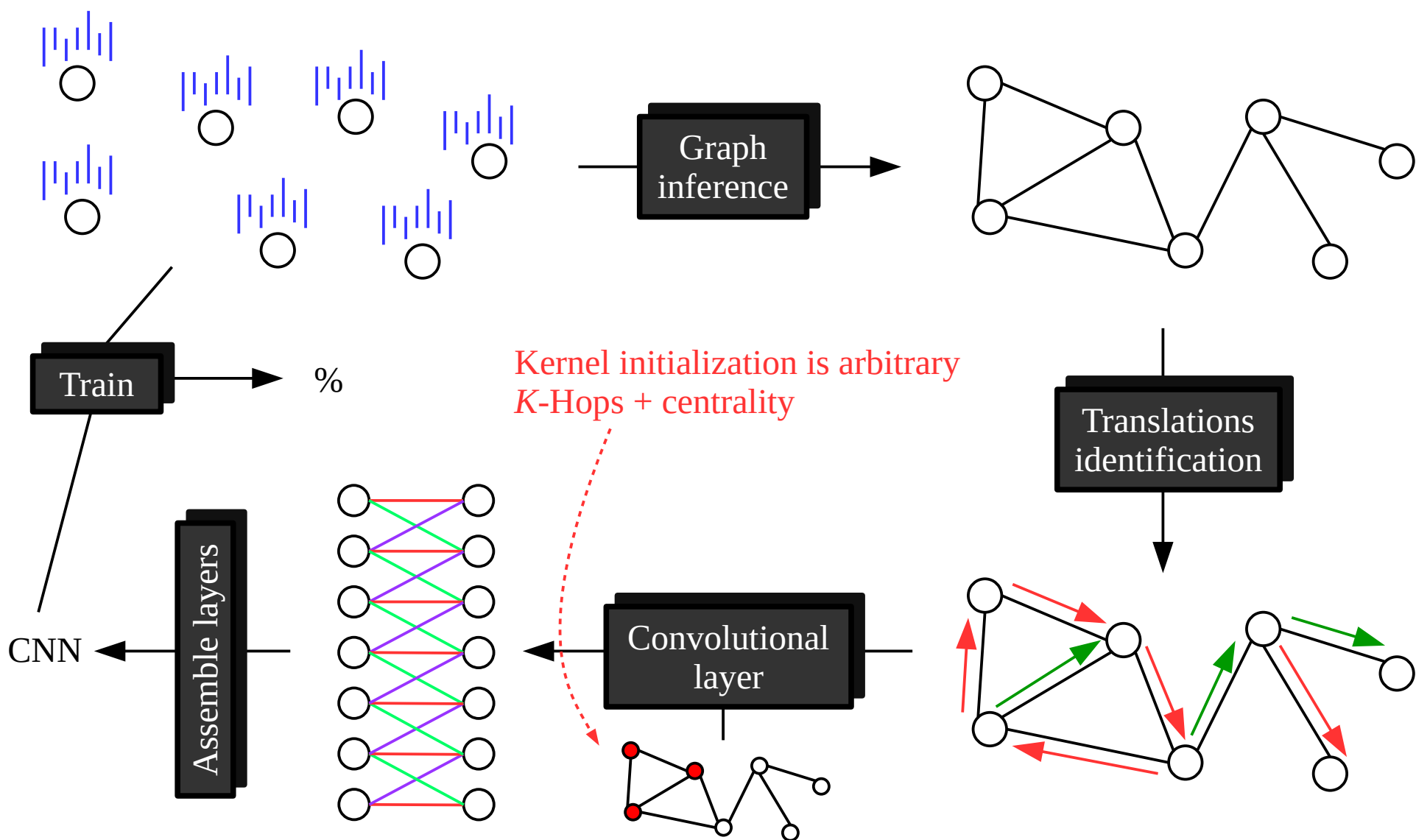
Supplementary slides

— Ideas on dictionary learning —

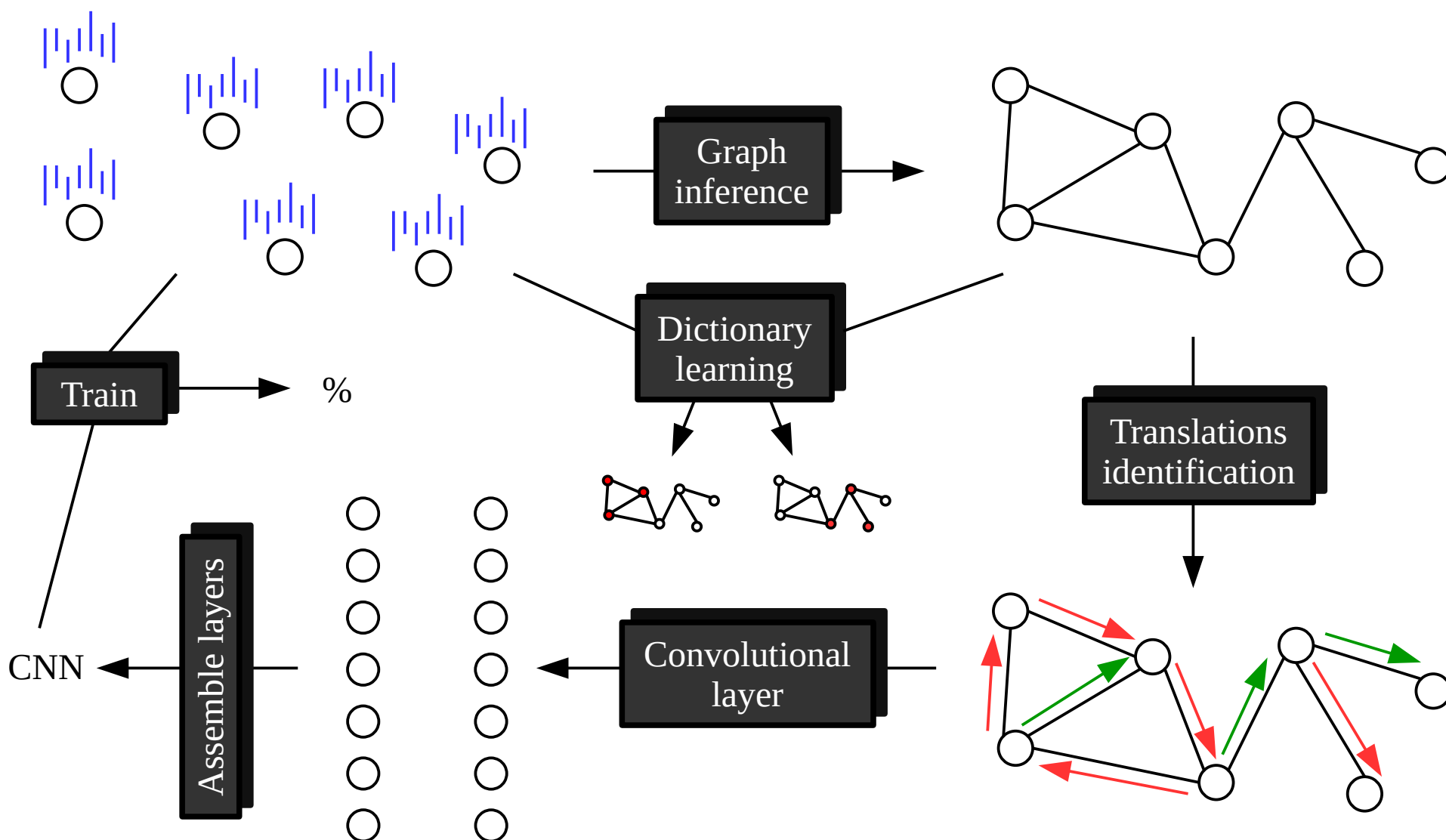
Supplementary — Dictionary learning — The idea



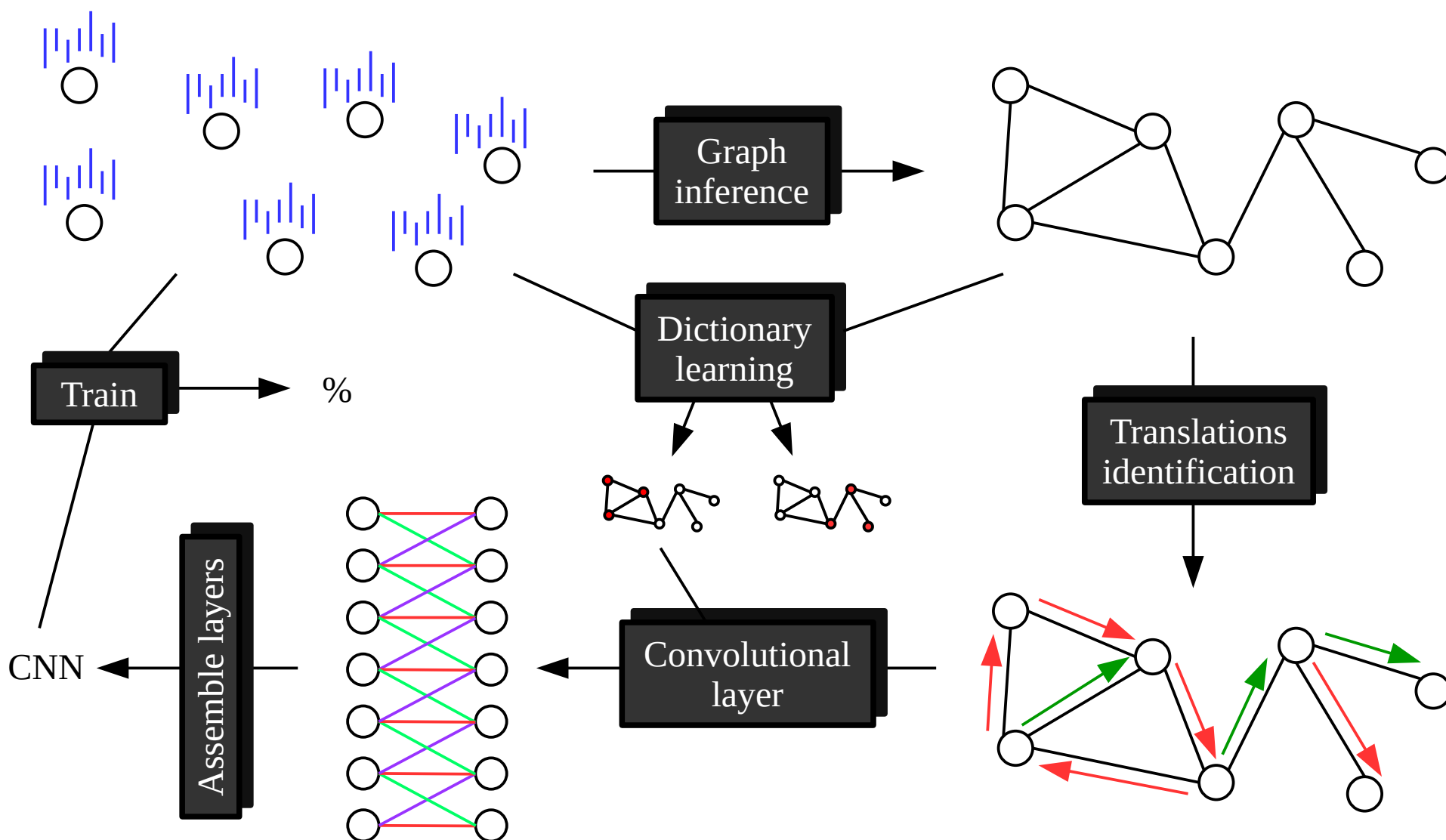
Supplementary — Dictionary learning — The idea



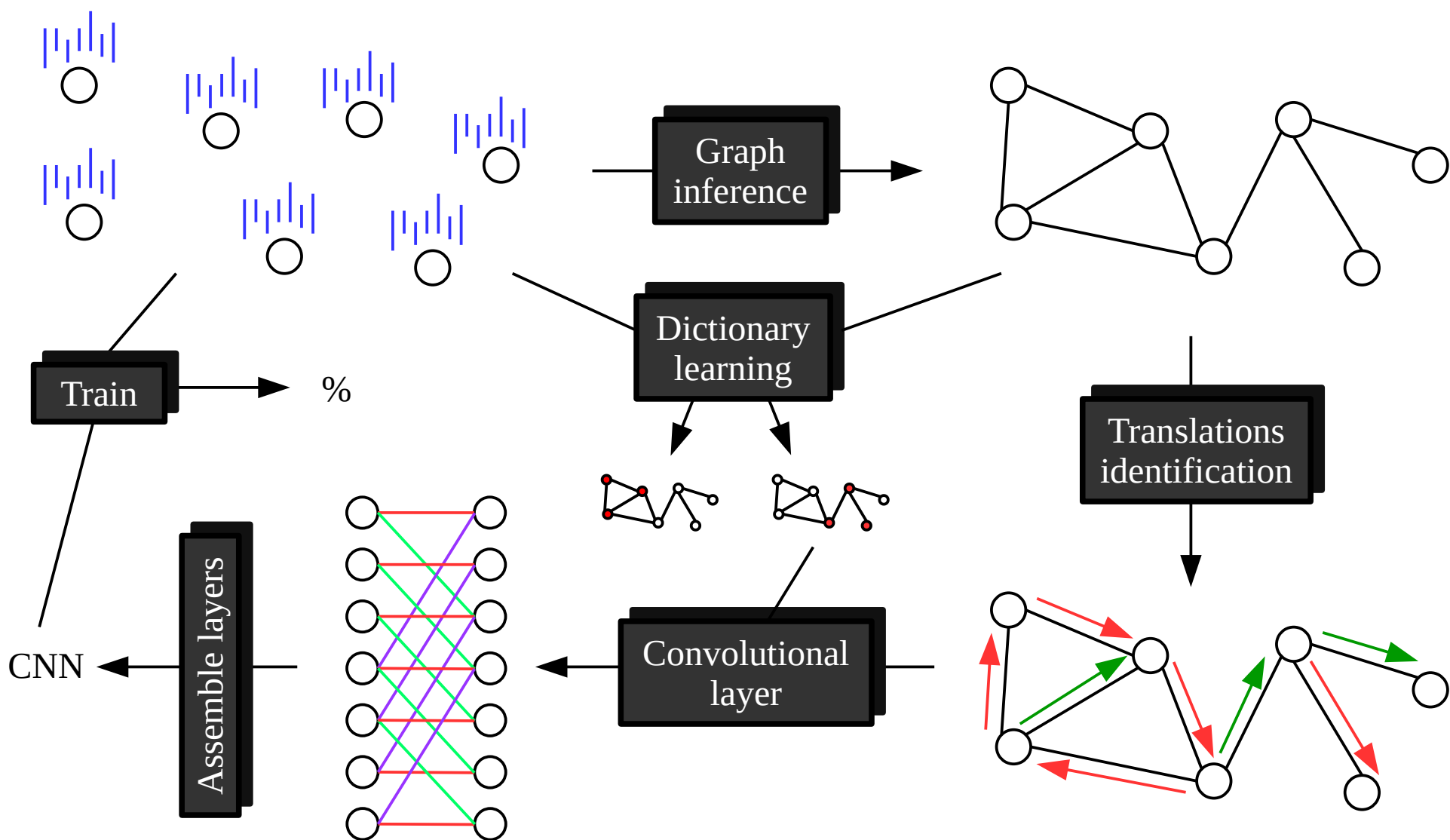
Supplementary — Dictionary learning — The idea



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Supplementary — Dictionary learning — The idea

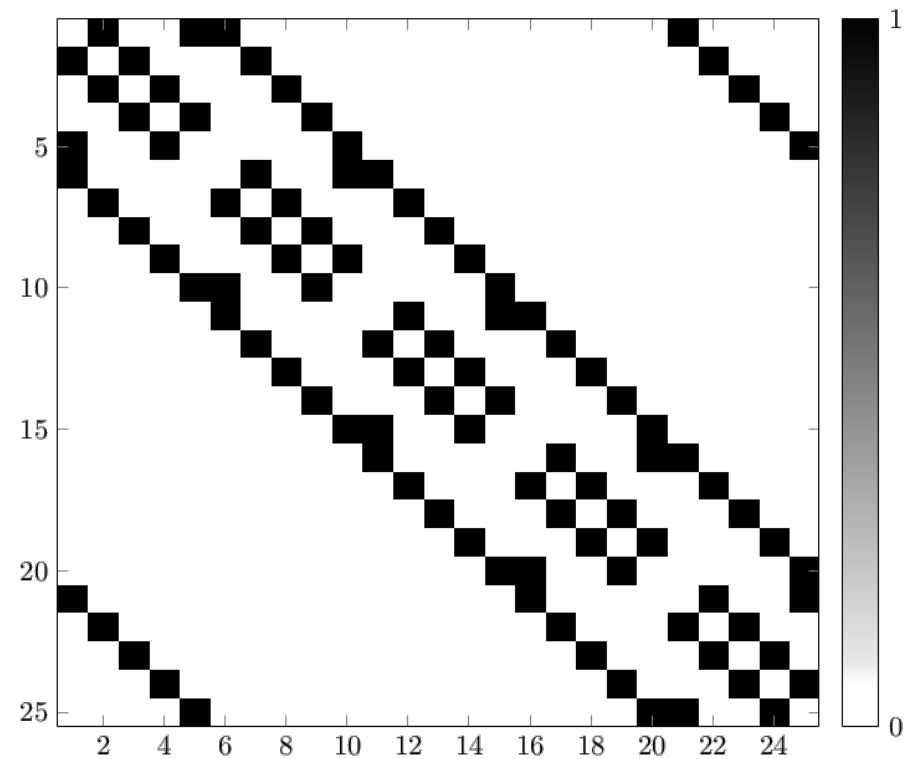
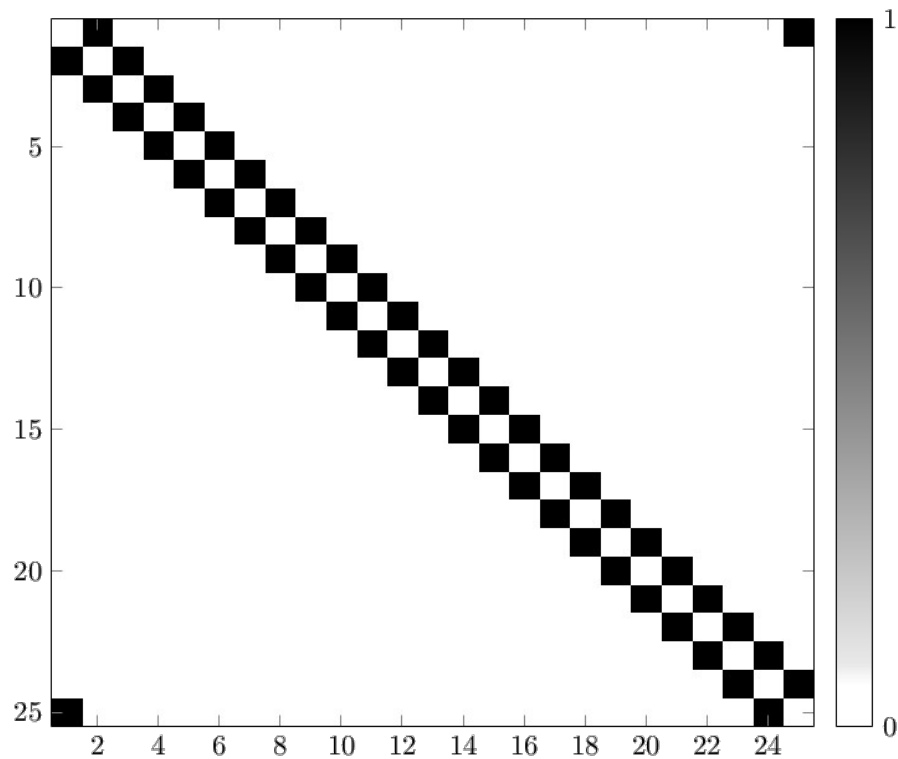


Outline

Supplementary slides

— Ideas on circulant matrices —

Supplementary — Circulant matrices — Particular graphs



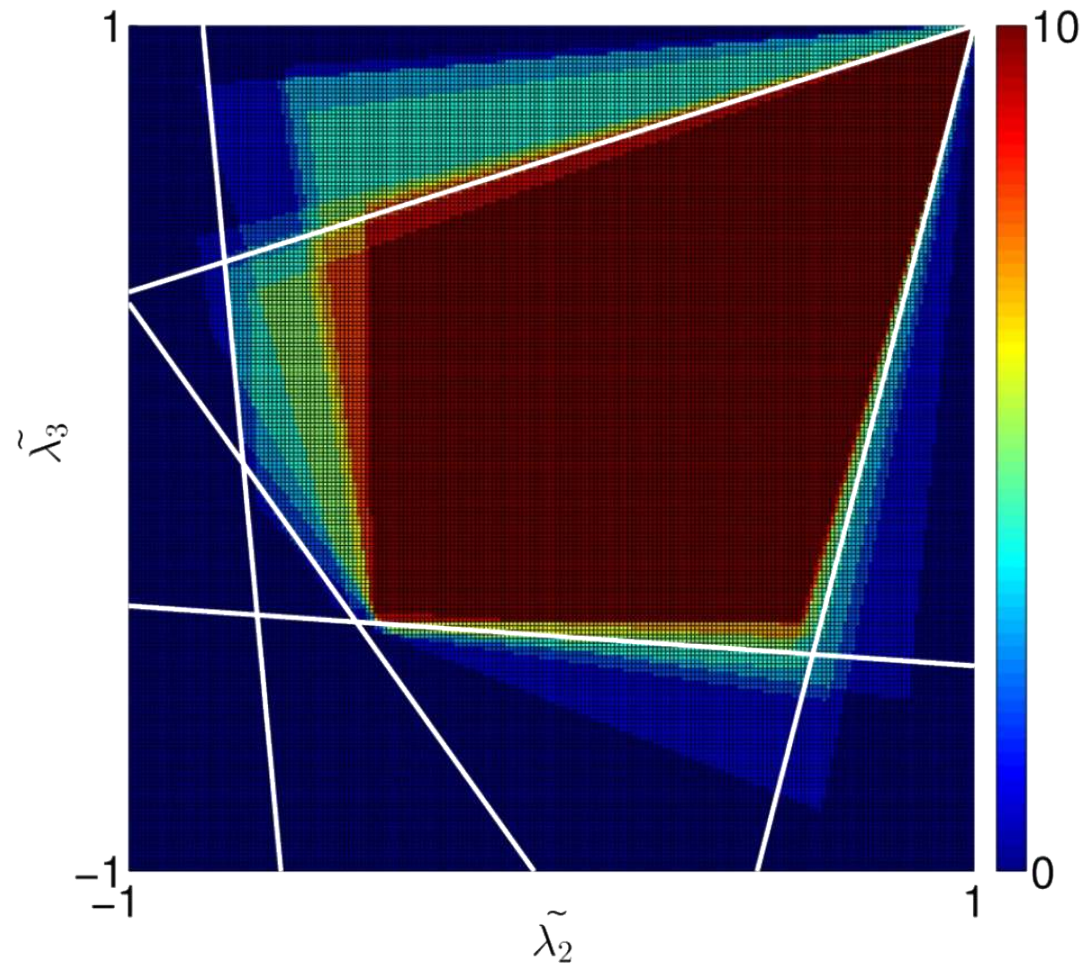
Outline

Supplementary slides

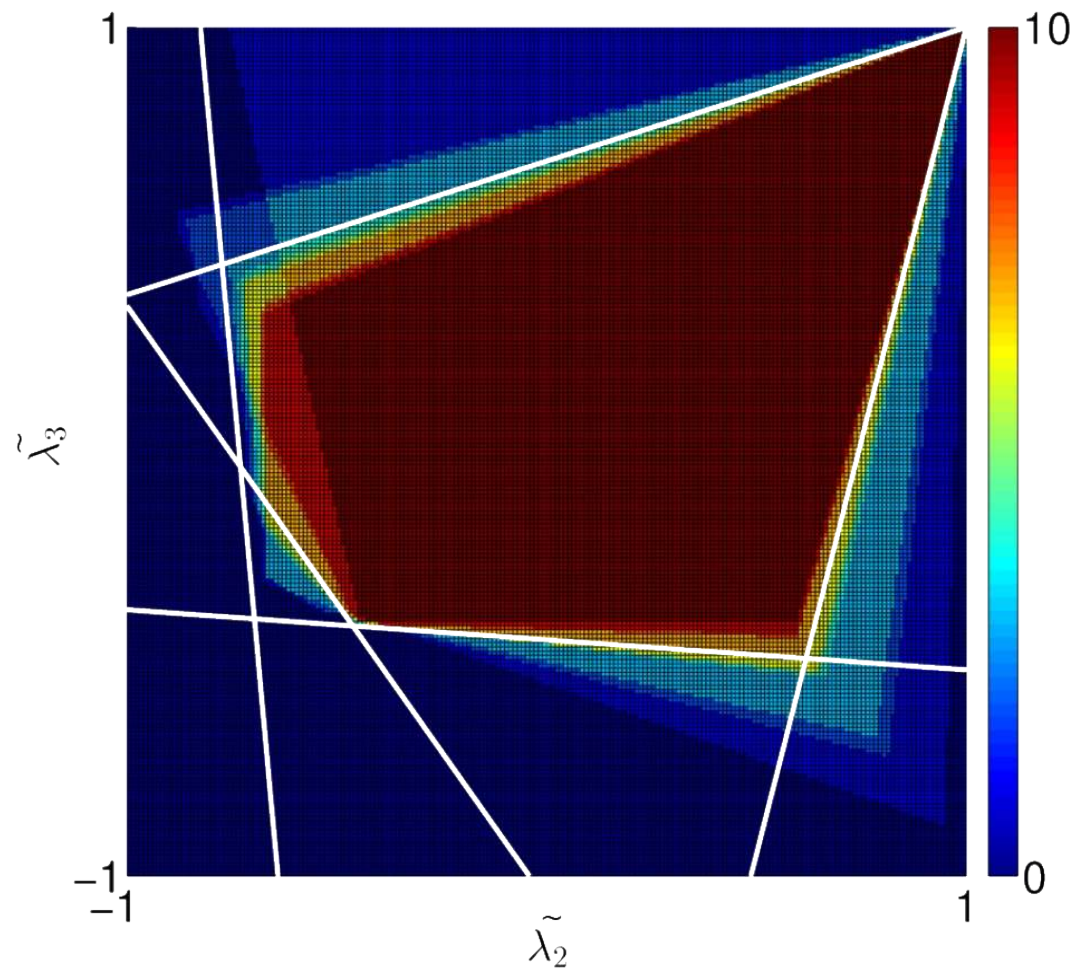
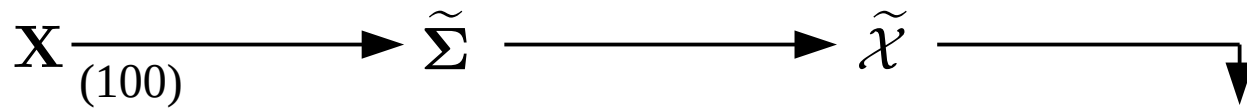
— Convergence of the eigenvectors of $\tilde{\Sigma}$ —

Supplementary — Eigenvectors of $\tilde{\Sigma}$ — Experimentally

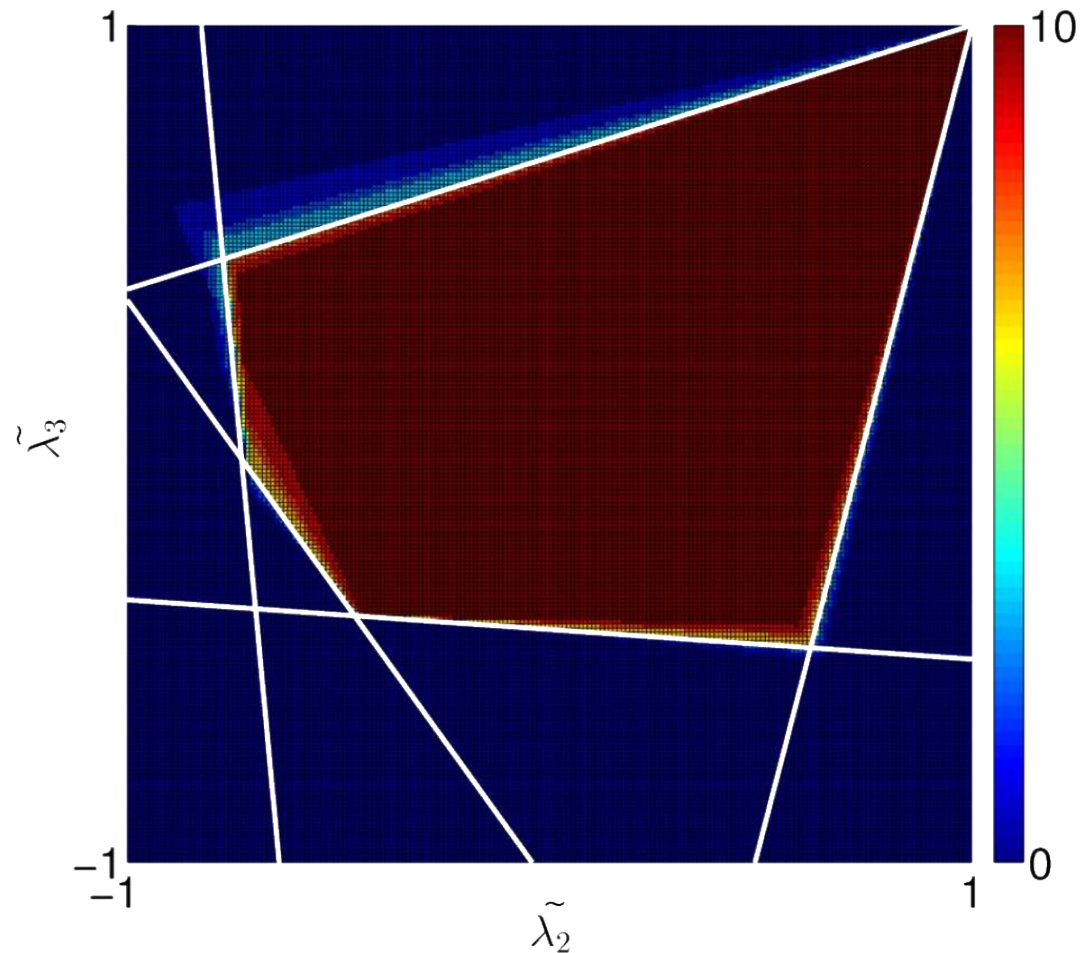
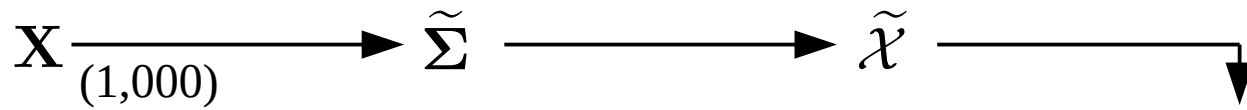
$$\mathbf{X}_{(10)} \longrightarrow \tilde{\Sigma} \longrightarrow \tilde{\chi} \longrightarrow$$



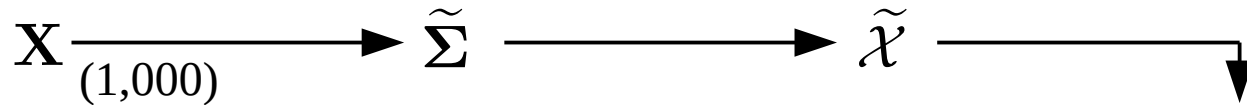
Supplementary — Eigenvectors of $\tilde{\Sigma}$ — Experimentally



Supplementary — Eigenvectors of $\tilde{\Sigma}$ — Experimentally



Supplementary — Eigenvectors of $\tilde{\Sigma}$ — Formally

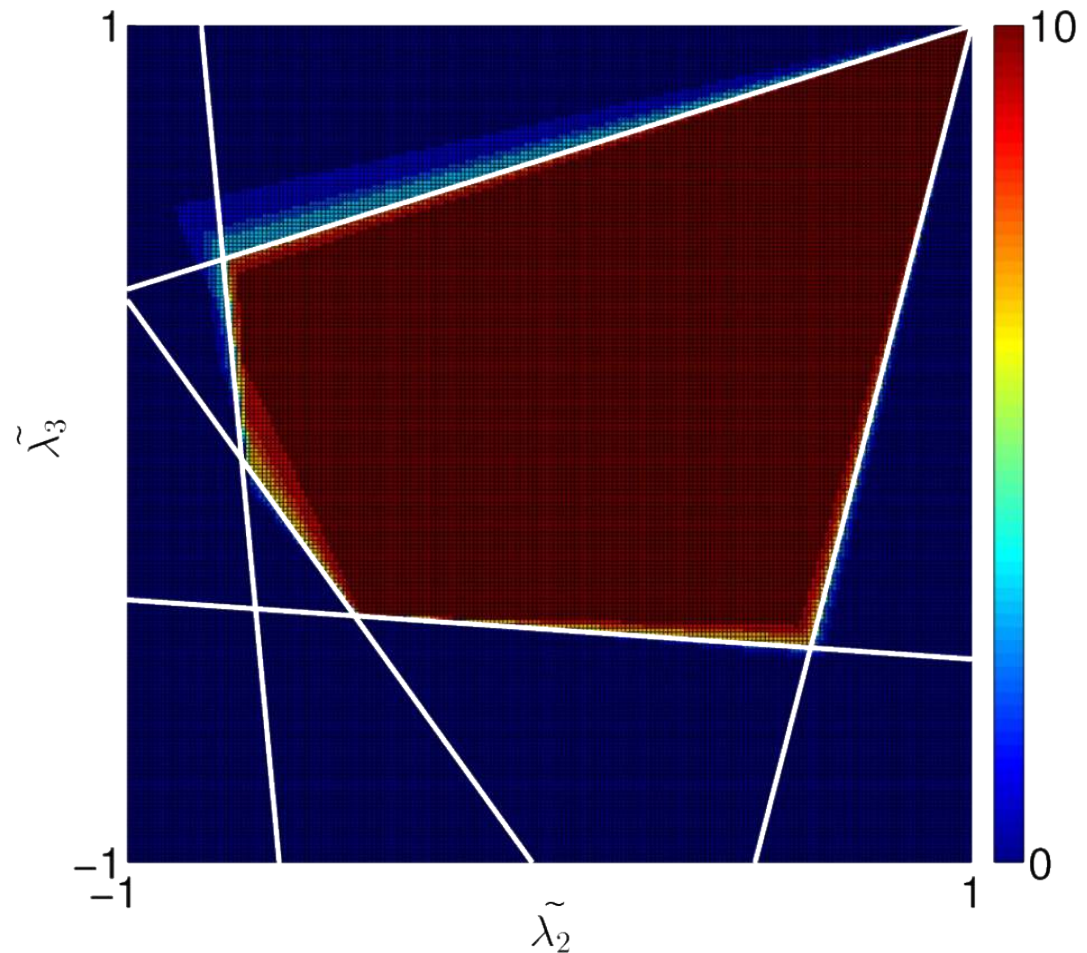


Variance of inner product
between $\mathcal{X}_i$ and $\mathcal{X}_j$:

$$\frac{\lambda_i \tilde{\lambda}_j}{(\lambda_i - \tilde{\lambda}_j)^2}, \lambda_i \neq \lambda_j$$

Best estimator (for distinct λ):

$$\tilde{\lambda}_i^* = \frac{M-1}{M} \tilde{\lambda}_i$$

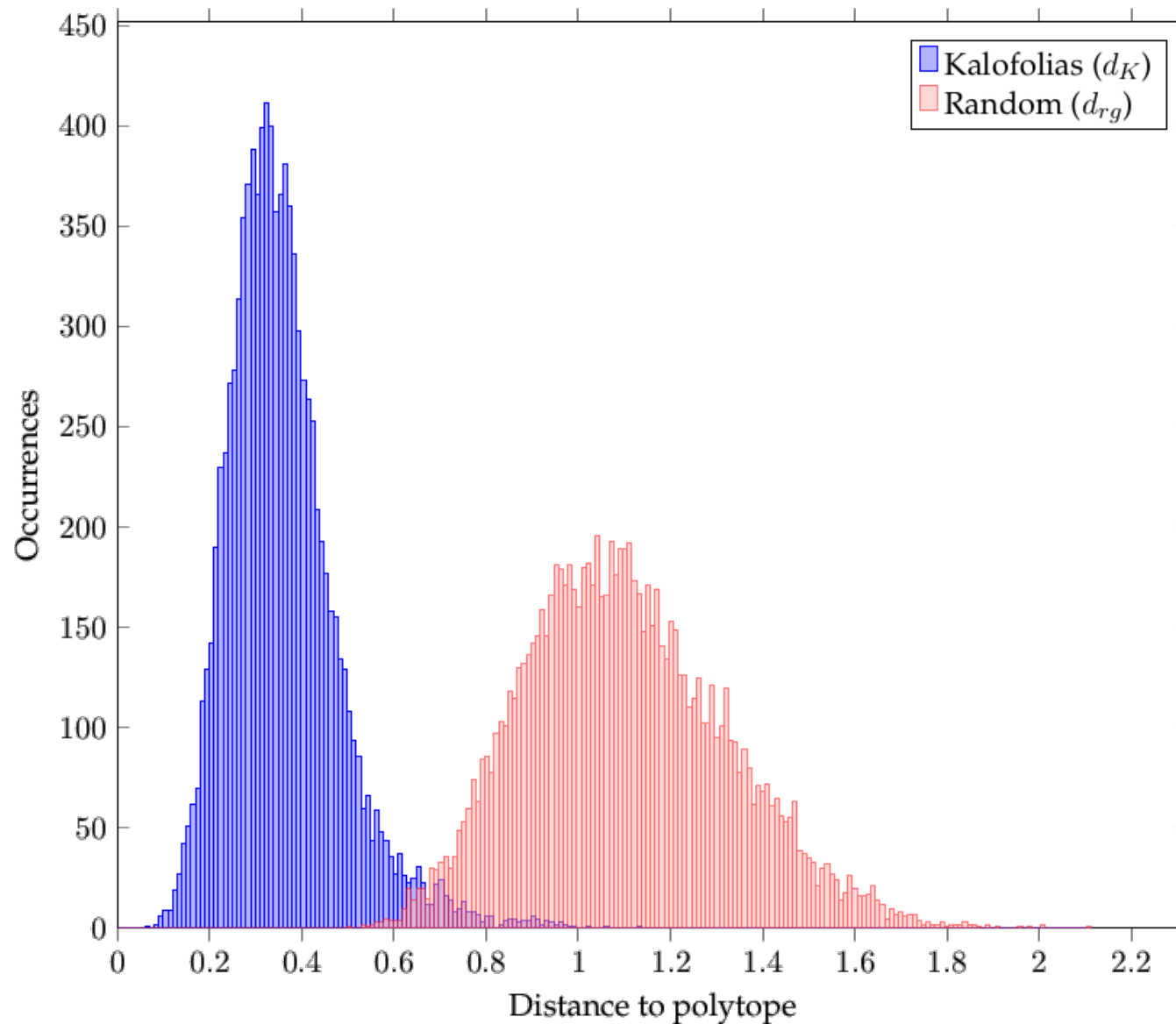


Outline

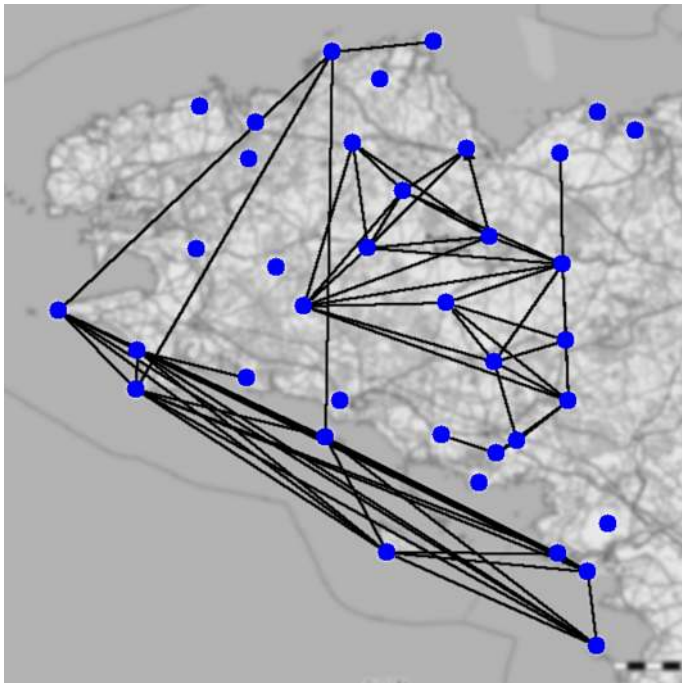
Supplementary slides

— Regularization of a graph inference method —

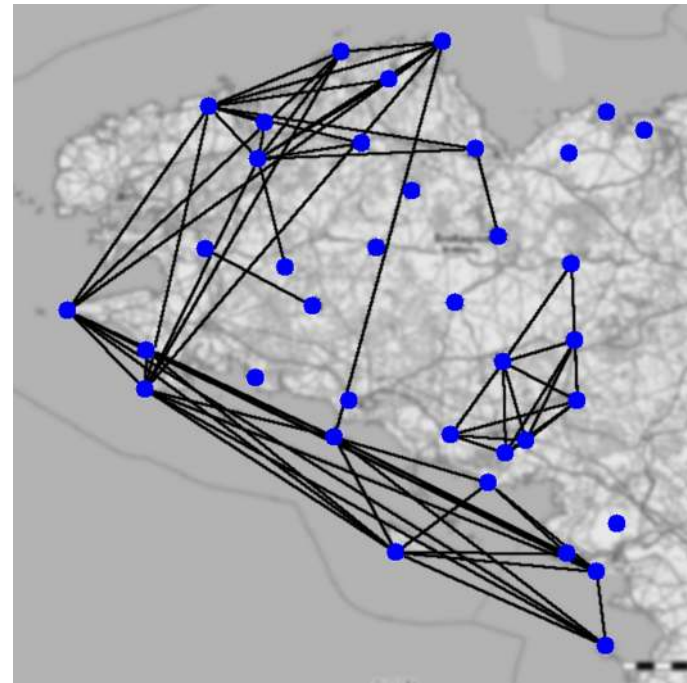
Supplementary — Regularization — Distance to the polytope



Supplementary — Regularization — Temperatures in Brittany



$\mathcal{G}_K$



$\mathcal{G}_K^*$